SECOND EDITION

COMPLEXITY THEORY AND THE SOCIAL SCIENCES

The State of the Art

David Byrne and
Gillian Callaghan

ROUTLEDGE

Complexity Theory and the Social Sciences

This expanded and updated edition of *Complexity Theory and the Social Sciences: The State of the Art* revisits the use of complexity theory across the social sciences and demonstrates how complexity informs approaches to various contemporary issues in the context of the COVID-19 pandemic, widening social inequality, and impending social and ecological catastrophe wrought by global warming.

The book reviews complexity theory in the practice of the social sciences and at their interface with ecological science. It outlines how social theory can be reconciled with complexity thinking and presents a review of the way research can be done using complexity theory. The book suggests how complexity theory can be used to understand and evaluate governance processes, particularly with regard to social inequality and the climate crisis. The impact of the COVID-19 pandemic is also examined through a complexity lens, reviewing how complexity thinking has been employed in relation to the pandemic and how implementing a complexity framework can transform health and social care. The book concludes with a call to action and the use of complexity theory to inform critical thinking in the education system.

This textbook will be immensely useful to students and researchers interested in social research methods, social theory, business and organization studies, health, education, urban studies, and development studies.

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To our respective children and grandchildren who will have to live through the coming crises.

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Abbreviations

ABM	Agent-based model
ANT	Actor Network Theory
BBSRC	Biotechnology and Biological Science Research Council
CA	Cellular automata
CAS	Complex adaptive system
CBSS	Case-based scenario simulation
CDP	Community development project
CECAN	Centre for Evaluating Complexity Across the Nexus
CEP	Cultural emergent powers
ESRC	Economic and Social Research Council
FS/QCA	Fuzzy set/Qualitative Comparative Analysis
GACS	Generalized auto-catalytic sets
GVA	Gross value added
HMRC	Her Majesty's Revenue and Customs
IAM	Integrated Assessment Model
IMF	International Monetary Fund
LA	Local authority
LEP	Local economic partnership
MQCA	Multi-value Qualitative Comparative Analysis
NE	North East (usually of England)
NERC	National Environmental Research Council
NGO	Non-governmental Organization
NPM	New public management
OECD	Organisation for Economic Co-operation and Development
OFSTED	Office for Standards in Education
PEP	People's emergent powers
PEST	Political, economic, social, technological, and environmental
PR	Proportional representation
QCA	Qualitative Comparative Analysis
RELU	Rural Economy and Land Use
RCT	Randomized controlled trial
SEP	Structural emergent powers
SES	Socio-ecological systems
SWOT	Strengths, weaknesses, opportunities, and threats
TSPN	Territory, scale, place, network
WHO	World Health Organization
WISERD	Welsh Institute for Social and Economic Research

Introduction

Since the publication of the first edition of this book, the importance of understanding not only the social world through a complexity framing but also the intersections of the social and the natural has become absolutely evident. This is not just a matter of how complexity thinking is being deployed in the social and ecological sciences within the academia in the production of knowledge. The growth in work framed through complexity within the social sciences has been massive. The complexity frame dominates thinking and research at the intersection of the social and the ecological. Complexity thinking has become prominent in all domains of governance, including that of health systems so crucial given the COVID-19 pandemic, and in confronting impending climate catastrophe in consequence of global warming. The reason is straightforward – it would be wrong to say simple! All aspects of governance, that useful term which covers all the ways in which intentional interventions in the social and ecological world are managed through means other than 'pure' market relations,[1] have started to deploy complexity thinking in the development of policies and their implementation. This is because the crucial issues at all levels faced by governance in the 21st century are wicked issues to use the terminology introduced by Rittel and Webber in the 1970s.

> The search for scientific bases for confronting problems of social policy is bound to fail because of the nature of these problems … Policy problems cannot be definitively described. Moreover, in a pluralistic society there is nothing like the indisputable public good; there is no objective definition of equity; policies that respond to social problems cannot be meaningfully correct or false; and it makes no sense to talk about 'optimal solutions' to these problems … Even worse, there are no solutions in the sense of definitive answers.
>
> (Rittel and Webber 1973)

The key issues facing governance systems globally are increasing internal social inequality, notably but not exclusively in high-income countries, and the consequences of global warming. We disagree with the assertion that: 'the search for a scientific basis for confronting problems of social policy is bound to fail', because it relied on a definition of science which does not stand if we deploy Morin's (2008) conception of general complexity as the basis for scientific practice and policy development. It is through thinking in complexity terms *and* through the recognition of the nature of socio-ecological structures and the potential for agency to transform them within the range of their possibility space that governance can confront wicked problems and overcome them.

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While writing this second edition, we have been living through the impact of the COVID-19 pandemic. COVID falls exactly into the category of Rumsfeld's 'known unknowns', something which will happen but we don't know when. Pandemic disease has been a feature of human society for 10,000 years since the agricultural revolution, especially in urban systems, now the dominant form of life globally. COVID-19 has been relatively mild in terms of its demographic impact. It has been no Black Death, which destroyed European Feudalism, or even Spanish flu in terms of death rates of productive adults. What it has done is first demonstrate the inadequacy of neoliberal governance of health systems, and second remind all of us of the real potential of nature for challenging our social order. Healthcare after the health transformation – one of the greatest social transformations of human history since the agricultural revolution from a pattern of mortality which peaked in infancy and childhood and persisted throughout adult life into one in which the great majority of people live into old age – has become one of the most important social *and* economic activities in our systems. Neo-liberals who recognize its potential for the generation of profit have pushed hard for lean systems with minimal redundancy which give access to tax-based health systems to for-profit providers. It is this absence of redundancy, of the fever hospitals and staff for them to separate the actually infectious COVID patients from the general hospital population, which has challenged neoliberal systems and led to so much excess non-COVID deaths during the COVID pandemic.

In writing this second edition, we took the decision to review the state of the art in complexity science in relation to the major issues facing the human race on this planet in the 21st century. Central to this was our reflection on the meaning of the word 'crisis' in the complexity frame of reference. Crisis is a word which is much older than complexity itself. It describes the condition of a system when it is in a state which cannot continue as is but must resolve through radical change, originally to refer to the system of a patient with an acute deadly infectious disease where the immune system (pre- and possibly post-antibiotics) would either overwhelm the infection and the patient would live or the infection would prevail and the patient would die. Our human social world is in crisis. Complexity thinking is what we need to guide us towards good outcomes in relation to that crisis. We focus on COVID and health systems, and on climate crisis and all governance system because one of us (Byrne 2021) has written a book specifically on the crisis of inequality from a complexity frame of reference. We make no apology for putting crises front and centre. Anything else would be pointless academic indulgence.

Byrne's (1996) 'Introduction' to complexity for the social sciences drew on the turn to complexity in the physical sciences and on the mathematical vocabulary of complex dynamics. That reflected the way in which it was the turn to complexity in the physical sciences which brought complexity back into the domain of science. There was nothing wrong with that, but now we can get beyond it. First, as noted in the first edition of this book, complexity does not belong to the physical sciences alone but rather is an ontological position about the nature of systems in general and has philosophical origins dating back to Lewes's (1875) reflection on Darwin in the mid-19th century. Second, we can see that the methods and ways of thinking of the social sciences, and especially those which emphasize the role of agency in the construction not only of the social world but of the intersection of that world with the natural world, are confidently deploying complexity thinking in a way which breaks out of the limits of restricted complexity and takes a more radical turn. In the first edition, we identified many forms of complexity modelling as still within the restricted set. The turn in modelling to co-production involving the tacit knowledge of key social actors, usually described in policy modelling as stakeholders,[2] brings agency not only into the formulation of models but into interpretation and necessarily into the actual actions which are guided by the models. When the agency of actors within the systems being investigated is present, things are different, and the passivity of conventional science, including a lot of social science, no longer applies. We are working with general complexity. This is how complexity can be made to work, and *must* be made to work, to deal with the crises we face as a species/civilization.

This second edition is organized much on the lines of the first edition but with significant changes and additions. First, there are three chapters which deal with the place of the complexity frame of reference in science in general and the social sciences in particular. We begin with a *manifesto* setting out the intellectual and institutional journey towards the complexity frame of reference in the social sciences. This draws on the Gulbenkian Report *Open the Social Sciences* (1996). It focuses on complexity as a foundation for interdisciplinary work in general. Next is a chapter dealing with 'The complexity frame of reference and action'. This establishes the ontological status of the complexity frame of reference and identifies complex realism as a programme within general complexity. The chapter reviews philosophical discussion of complexity and complex systems. It lays out the character of complex realism as a methodological programme in relation to the history of complexity-focused philosophical debate as this has developed in the late 20th and early 21st centuries. We emphasize complexity as a frame for praxis, i.e., for knowledge which must be deployed in action. This first part concludes with a chapter addressing 'Doing complexity science'. This focuses on the difference between a developed complex realist frame of reference and both restricted complexity understood in terms of micro-emergence and currents in meta-theoretical thinking which have some relationship with complex realism but are less than satisfactory both as social meta-theory and as a basis for praxis. We review micro-emergent complexity, the discussion of models as a basis for understanding, the character of Actor Network Theory (ANT), and what in summary are called 'the new materialisms'.

The next part examines the relationship between complexity as a frame of reference and the traditions of social theory. It reviews key areas of social theory to consider how social processes, historically described in rather different terms, can be shown to demonstrate complexity-consistent approaches. Its aim is to integrate existing theory into a complexity frame. The chapters will use the works of Bourdieu in particular as a way of building upon the coherence of body of social theory that has an explicit ontological/epi stemological/methodological approach that is consistent with complexity. It is a body of work that appreciates the world as both constructed and real. First is a chapter addressing 'Evolutionary theories'. This works through system theoretic explanations and so will look at Parsons, Luhmann, and Spencer. The idea of punctuated equilibria after Gould forms a link to the discussion of macro-system transformation in the last chapter in this part. Next is a chapter dealing with 'Structure and agency'. This examines social theories that focus on the mutual shaping of agents and structures within systems. The primary focus is on Bourdieu because he recognizes the immanence of each in the other in his articulation of habitus and field. There will also be some discussion of Archer on Morphogenesis as explicitly engaging with critical realist ontology. This is followed by a chapter dealing with *Time and Place*. This focuses on theoretical work rethinking the ontological status of each, away from absolutes in a Newtonian frame, to contingent and recursively constructed in social practice. Understanding the role of space and time is fundamental to understanding complex social systems, and this chapter will provide a basis for appreciating these dimensions of multiple causality and emergence. The last chapter in this part is new and addresses *macro-system transformation*. This develops a complexity-informed discussion of macro-social transformation. It reviews accounts of transformation from Weber, Polanyi, and Marx. The use of Weber's application of 'elective affinities' will elaborate the causal discussion that links to multiple realization. The chapter reviews the introduction of a world systems frame by Wallerstein and considers how the concept of Capitalocene, in its rejection of the nature/society dualism, develops engagement with complexity theory.

Next is a section dealing with the application of the complexity frame of reference to issues of methodology and methods in the social sciences and at the intersection of natural and social science. The emphasis is on how we can use a range of approaches, both quantitative and qualitative, to produce narratives of change – dynamism is central – and methods for exploring causality through case comparison but also by extending ideographic accounts of single cases at the macro-social level beyond description towards causal reasoning. The first chapter in this part deals with *complexity-informed qualitative work*. Normally, discussions of method in complexity start from quantitative approaches including modelling, simulation, and social network analysis. This ordering is deliberately inverted here to emphasize the centrality of the case at all social levels and intersection of the social and natural as the basis of social investigation. Narrative is crucial because we have to describe systems as they are but even more importantly attempt to understand how they came to be as they are (History) and what potential for transformation is embodied in their present condition. Much existing thinking underpinning qualitative methods of research is complexity consistent because qualitative research, by its nature, is non-reductionist. Rather than generalization, a focus on mechanisms of operation and interaction in the construction of settings provides the basis for generating accounts. The chapter examines how this kind of approach changes our general conception of the role of qualitative work in science.

Next is a chapter reviewing *complexity-informed quantitative work – modelling, simulation, social network analyses*. All of these approaches derive from the opportunities offered by computer manipulation of data (the argument will be dismissive of modelling approaches which are wholly abstract without any empirical foundation) and are widely employed in complexity-informed work. The empirical illustrations come from their deployment in relation to the COVID 19 pandemic and at the interface between the natural and the social in relation to ecology and human health. There is also a discussion of how time-ordered data series can be deployed in generating descriptions of change at every level. Classification is an important part of this repertoire together with systematic comparison as exemplified by Haynes's development of 'Dynamic Pattern Synthesis'. The chapter shows how the quantitative programme as a whole, including straightforward data description and exploration, can work within the complexity frame of reference.

The final chapter in this part deals with *mixed and Multi-Method Work* as an innovatory programme. It examines the shift in not only social science but also at the interface between the social and natural sciences towards multi-method-based approaches to exploration as a basis for policy development and implementation. The chapter explores the background to this shift and notes that it is connected both to the emphasis on applied research for developing policy and practice and the increasing involvement of the human subjects/agents in the research field through participation/co-production – processes which demonstrate a radical shift in the relationship between science as investigation and the subjects of that investigation. It draws on examples from studies of land use in the UK, work in relation to the establishment of what works in addressing environmental crime, and the historical experience of area-based community development. A key aspect of much mixed method work is that it is action research – research conducted from within the systems being examined *and* with an explicit objective of changing the character of those systems. In an era of both inequality and climate crisis, this is the mode of social research which is necessary and appropriate. Here we outline a complexity take on the construction of scenarios to inform action directed towards achieving desired futures.

The next part deals with *governance and evaluation* – understanding settings and in order to effect change. The first chapter examines 'Modes of governance'. It reviews the literature on thinking about governance, as the management of complex systems draws on empirical examples across policy domains. It demonstrates how policy and practice (implementation) in these domains can be integrated to form a coherent whole. There is an extended discussion of levels of governance and the role of participation, subsidiarity, and partnership with economic and civil society actors. The implications of differential power are central to this chapter. Again, the issues of inequality and impending climate crisis and the modes of both governance structure and governance process necessary to deal with these form a backdrop to discussion. The next two chapters deal with evaluation as a mode of informing governance practice. The first 'Evaluation in a complexity frame' reviews methodological and method discussions of evaluation in relation to the complexity frame of reference. It draws on the extensive discussion of complexity in evaluation. The primary issues are presented in terms of black box problems, framing the setting for research into an intervention and defining system boundaries as well as the nature and application of findings in complexity terms. It reviews work dealing with participation and agency and uses of complexity's weak and strong ontologies in current evaluation research. The next examines *complexity-informed evaluation in practice*. This describes how complexity is actually being used in evaluative work drawing on examples of practice from a range of domains across the whole spectrum of social intervention. In the 21st century, more social science, and most interesting and innovative social science, is being done in application, much of it outside the academia by social science trained and aware social researchers working in and around governance. Here we outline that area of work and review its approaches and significance.

The next part examines the deployment of the complexity frame of reference across the disciplines and fields of the social sciences with an emphasis on what is now called translational work – that is the deployment of knowledge in use. It begins by reframing issues and problems, outlining how complexity is driving a transdisciplinary agenda generally applied social science understood as empirical investigation in the terms presented by the Gulbenkian report. The first chapter in this section deals with 'Complexity theory – shaping research and policy'. The corresponding chapter in the first edition conducted a review of how complexity was deployed across a range of disciplines and fields. In the revised edition, we are proposing to do this in an integrated fashion. This allows us to frame presentation with complexity theory being the core and treatments and approaches in both the academy and practice being described in thematic rather than bounded terms. Next is a chapter entitled 'Beyond disciplines and fields – through and after COVID'. Here, we want to develop the account of globalized capitalism in the era of the Capitalocene through applying a complexity frame lens to the COVID pandemic in relation both to what happened during it and to complexity-framed accounts of that experience. COVID should have been a game changer for how not just social but all science informs governance. We consider to what extent this happened.

We *conclude* with a reflection on the implications for the social sciences *and* their application of what we have said in the book as a whole. Here, we include a section on the pedagogy of complexity because how we teach people is central to developing their capacity to understand and act in relation to increasing destabilizing inequality within nation states and the implications of potential climate catastrophe. This allows us to integrate the discussion of social theory in part one of the book with the reviews of both the ways in which we understand the social world in disciplines and fields and how that understanding can inform social action to address fundamental social problems which actually challenge the global social order itself.

Notes

1 Much of governance in this neoliberal era has been devoted to asserting that unfettered competitive markets are the ideal mode for all resource allocation and service delivery. In reality, we have massive imperfect competition with oligopoly in most areas of privatized or emergent (IT-based) public utilities, including social care. This is particularly important in the urban context in relation to the urban infrastructure including transport.

2 Just who should have a stake is a matter of power relations. When participation is more than a vacuous rhetoric, the range is very wide indeed.

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1 A manifesto for interdisciplinarity founded on the complexity frame of reference

Context is fundamental to any account based in a complexity frame of reference. We begin here by situating the arrival of complexity theory into the crowded theoretical world of the social sciences. This underpins our decision, in this second edition, to extend the scope of our review of the state of the art into a range of explicitly practice-related areas. Complexity implies new ways of framing problems, deploying methods and interpreting data. It necessarily means revisiting some old debates to bring to the surface assumptions underpinning current approaches to gathering and assessing evidence and to developing theory-based accounts. The rejection of divisions that led to different trajectories for natural and social sciences across the last two centuries is a greater challenge for those in the natural sciences than in the social sciences where we have faced the problem of understanding a world for the most part not amenable to the development of laws and prediction. Nonetheless, there is a considerable body of work in the social sciences that has sought to understand the world within a positivist frame, through simplification and reduction. In the first edition of this book, we commented on the opportunity complexity theory offers to overcome barriers to the pursuit of social science knowledge through intellectual factions and in academic silos. These silos are not merely matters of theory but of the social construction of the social sciences themselves. The pursuit of knowledge is not wholly a matter of advancing the subject. It is also importantly a place of career development and social closure established, in the current political and economic climate, in a competitive context. Our interest here lies in the different theoretical framing that complexity offers. We will use that to acknowledge all science as both constructed and real so that any attempt to maintain a separation between knower and known fails.

The context for the emergence of complexity theory can be established by considering the future role for and structure of the social sciences through an exploration of their past. Our approach is through a diagnostic examination of the source of current troubles and hence situating complexity theory as having the potential to inform a radically different and more relevant approach to social science.

Historically, the foundational separation was that made between humanity and nature and the development of human exceptionalism through religious and humanist approaches in the 18th century. This gained ground in the 19th century and became a notable feature of the discussion of evolution. In anthropology and sociology, Boas and Durkheim explicitly denied any shared ground between social and natural worlds and hence between natural and social facts. This was the ground for cultural determinism, behaviourism, and debates over the roles of heredity and environment in the early 20th century. At the same time, in the adoption of evolutionary theory, biologists were working with an interactionist paradigm more amenable to a complexity understanding. The

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shift from typological to population thinking in evolutionary thought was significant here. The social/biological split has been long debated. Washburn (1968 20) pointed to inseparability in terms of the feedback relation that existed between the two. More recently, reuniting has developed in the conceptualizations of Anthropocene, Capitalocene, and Chthulucene. Not only have the paradigms developed over two centuries broken down in significant ways, but the problems faced in a world of climate crisis and pandemic have shown that no single discipline is equal to the task of understanding. To address these issues, we need to be able to comprehend the interpenetration of social, individual, and ecological forces within systems. It is here that both the need and the foundation for inter- or postdisciplinarity became clear.

The social sciences emerged and were institutionalized through debate about foundational knowledge, theory and methods, their role in the academy, and their potential utility for informing future action. Reviewing how we got here enables us to recognize the constructed nature of disciplines. From that standpoint, we can reassess the current position and prospects. This was the logic underlying the approach of the Gulbenkian Commission report (1996) into the future of the social sciences. Wallerstein et al.'s historical account tells of the shifts and divorces, initially between philosophy and science and subsequently between natural and social science that demonstrates the constructed nature of divisions. This happened when the academy made a commitment to a single ontological frame as a basis for authoritative scientific knowledge. The rise of Newtonian science, accorded an absolute level of authority, resulted in both a separation in areas of study and a hierarchy of ways of knowing in which the place of the social sciences was diminished. The majority of intellectual work and academic research in the social sciences was treated as methodologically deficient. The Gulbenkian report argued that these social constructions have become obstacles to serious intellectual work. In the context of disciplinary institutionalization and enculturation, its dominant actors and organizations have been blind to, or rather sought to ignore, the active and situated nature of the knowledge produced. The political and institutional context of academic study prompted the social sciences to reassert their place through a search for 'social facts', description and analysis through quantification, although often with minimal attention to the constructed nature of statistics. Issues of objectivity and bias, the emphasis on universal truth, and generalization led to ahistorical accounts.

The social sciences also came under structural attack from a liberal hegemony which separated fields internally according to a division of the public sphere into state, market, and civil society (Wallerstein et al. 1996). In post-war social science, the movement in this direction was powerful, and although not uncontested, it came to dominate. Variants of positivist thinking formed the basis of philosophy of science while the value accorded to quantitative evidence reinforced modernist rationalist basis of policy.[1] The establishment of a hierarchy of evidence such as that defined by the Cochrane Collaboration in the field of health, in which RCTs are accorded 'gold' standard and qualitative approaches and case studies regarded as inferior, continues to exercise huge influence. This is what we need to challenge to be able to undertake the reorientation required to understand a complex world.

Wallerstein et al summed it up:

> Most of the nomothetic social sciences stressed first what differentiated them from the historical discipline: an interest in arriving at general laws that were presumed to govern human behavior, a readiness to perceive the phenomena to be studied as

> cases(not individualities), the need to segmentalize human reality in order to analyze it, the possibility and desirability of strict scientific methods (such as theory-related formulation of hypotheses to be tested against evidence via strict, and if possible, quantitative, procedures), a preference for systematically produced evidence (e.g., survey data) and controlled observations over received texts and other residuals.
>
> (1996 31)

The significance of this was huge because in social science it was not enough to observe and describe the world in various ways. If everything was knowable through reduction to its component parts, rational policy planning and action could be possible when the right data was available. Newtonian explanations provided the basis for a modernist understanding and led to a view of social science as in its infancy, requiring more robust methods to become truly useful. The implication of this is again made clear in Wallerstein (1999)

> The link between liberal ideology and the social science enterprise has been essential and not merely existentialist. I am not saying simply that most social science were adherents of liberal reformism. This is true, but minor. What I am saying is that liberalism and social science were based on the same premise – the certainty of human perfectibility based on the ability to manipulate social relations, provided that this be done scientifically (that is, rationally). It is not merely that they shared this premise but that neither could have existed without it, that both build it into their institutional structures.
>
> (Wallerstein 1999 147–148)

This set of modernist beliefs provided a powerful underpinning to the growing interest in the potential of social science for much of the 20th century. Wallerstein argues that it was the inevitable disappointment of these expectations that created the later disillusionment and marginalization of social scientific knowledge.

This is a broad brush account of the history of social scientific inquiry. While the positivist and the quantitative prevailed, it encountered strong and sustained opposition. History was not accorded scientific status being placed on the ideographic side, but the divide occurred within sociology, anthropology, and other social sciences. The assertion that these were 'soft science' failed because social scientists were dealing with a world in which 'putting nature on the rack' palpably changed 'nature' itself. Throughout the 19th and 20th centuries, many social scientists asserted that much of the subject of study could not be appropriately framed through quantitative accounts which, in their inevitable simplification, did violence to the richness of the social world. Examining and exploring real social processes and events led to a recognition that we could only know many things about the world in ways not amenable to Newtonian mechanical approaches. Qualitative researchers argued that their methods were the only ones that allowed genuine explanations of cause to be rendered, albeit in very different terms. Rather than searching for causal variables, rich contextual accounts allowed the 'why' and the 'how' to emerge. However, issues of the relationship of observer to observed, generalization, and of power were necessarily problematized in research in a way that posed fewer problems for those working on the quantitative side of the divide.

The emphasis on evidence yielded through experimental and quantitative methods was reinforced through institutional and cultural reproduction. Some social sciences have continued to cling to that programme, notably those economists reliant on rational action theory and quantification with experimental methods favoured in psychology, while other disciplines were internally riven, but reached some accommodation in the practice of social research through mixed method approaches. The paradigm wars were fought on multiple fields with internal debates about what could be known and externally in application as to how such claims could be relied upon.

The story that specifically grounds the development of social science provides an incomplete basis for understanding the emergence of complexity. Here we need to look to the paradigm shift in evolutionary biology in the 19th century, initially in a close conversation with social science. Evolutionary biology was inevitably a theory of interaction and that laid the ground for an interest in systems as set out by Von Bertalanffy (1968). The objectivity of scientific knowledge was fundamentally undermined by cybernetics which sees knowledge as 'merely an imperfect tool used by intelligent agents to help it to achieve personal goals' (Heylinghen et al. 2006 123). Systems theorists debated the role of objectivity/subjectivity, system autonomy, and social constructions of reality alongside the problem of hierarchical and heterarchical relationships operating in systems. These debates have been part of the movement towards a conception of complex open systems which have structure and are relatively permanent. We now see systems as open and interacting with their environments. In this the very definition of system boundaries is problematized and becomes a matter for the researcher to frame within the setting researched. The possibility of other boundary framings is always open.

The ontological discussion reflected in this history has engaged with complexity ideas for some considerable time. In a form which is readily recognizable in terms of the contemporary discussion, we can find the idea developed in the late 19th century by G.H. Lewes and others in response, in large part, to the emergence of evolutionary theory based on the work of Darwin and Russell. Warren Weaver's statement of the 1940s (1948) deploys the word complexity itself and is often regarded as the foundation of a broad scientific (in the widest sense of that word) interest in complex systems in general. Attention to complexity precedes the development of the mathematical formalisms of catastrophe theory and chaos theory although the publication of popular science books stimulated by those developments played an important role in the resurgence of interest in the field. It certainly drew the attention of one of us to it and resulted in the publication of *Complexity Theory and the Social Sciences – An Introduction* (Byrne 1998).

The Gulbenkian report makes explicit reference to non-linear science and complexity in describing the world social science must fit itself for and argues that the shift in the natural sciences allows for a reassessment of non-positivist forms of knowing.

> new developments in the natural sciences emphasized nonlinearity over linearity, complexity over simplification, the impossibility of removing the measurer from the measurement, and even, for some mathematicians, the superiority of qualitative interpretive scope over a quantitative precision that is more limited in accuracy. Most important of all, these scientists emphasized the arrow of time.
>
> (1996, 61)

We have made extensive reference to the Gulbenkian report because its timing was far from accidental, while its framing and the account it develops give weight to the promise of complexity theory as a way of reframing ways of knowing. We have some sympathy with Wearne's identification of the lack of contextualization (1998) in its description of current social science. While Wallerstein et al tracked earlier developments in concert with the role of church and state, they appear to ignore the role of business and the market as controllers of contemporary knowledge and the overarching frame of its current funding. Social science has been characterized as both a creature and a critic of liberal capitalism, and this tension explains much of the current debate about its nature and purpose.

Even at the time of the Gulbenkian Report's publication, there were domains of social science practice other than the geographically defined area studies which had moved in an interdisciplinary direction. These were located in the 'fields' not only of business and organizations, but also in health, education, and urban studies. All of these have the full academic apparatus of university departments, under- and postgraduate degrees, and specialist journals. Fields of this kind have proliferated even more in the 21st century. A constant theme in this book will be that not only is much of the best complexity-informed work conducted in these fields as part of the academia itself, but also it is at the interface between academic work from these fields and public policy formation and governance practice that the complexity frame of reference has the most relevance for the contemporary crises confronting human civilization on this planet.

The historical account has brought us to an understanding of how the social sciences became constituted and institutionalized over 200 years. We now turn to the core terms that this book is concerned to explore, those of complexity and theory and their implications for doing social science.

Defining complexity and complexity 'theory'

Complexity is a property of systems. The Oxford English Dictionary Online gives us rather a lot of definitions of that word: system. The following is a selection.

1. An organized or connected group of objects.

   A set or assemblage of things connected, associated, or interdependent, so as to form a complex unity; a whole composed of parts in orderly arrangement according to some scheme or plan; rarely applied to a simple or small assemblage of things (nearly = 'group' or 'set').
2. *Physics*. A group of bodies moving about one another in space under some particular dynamical law, as the law of gravitation; *spec.* in *Astron.*, a group of heavenly bodies connected by their mutual attractive forces and moving in orbits about a centre or central body, as the *solar system* n. at solar adj. and n.[1] Compounds 3 (the sun with its attendant planets, etc.), the *system* of a planet (the planet with its attendant satellites).
3. *Biology*. A set of organs or parts in an animal body of the same or similar structure, or subserving the same function, as the nervous, muscular, osseous, etc. systems, the digestive, respiratory, reproductive, etc. systems; also, each of the primary groups of tissues in the higher plants.

   With *the* or possessive: The animal body as an organized whole; the organism in relation to its vital processes or functions.
4. In various scientific and technical uses: A group, set, or aggregate of things, natural or artificial, forming a connected or complex whole.

   of natural objects or phenomena, as geological formations, mountains, rivers, winds, forces, etc.; also of lines, points, etc. in geometry.

   of artificial objects or appliances arranged or organized for some special purpose, as pulleys or other pieces of mechanism, columns or other details of architecture, canals, railway lines, telegraphs, etc. With reference to business and social organizations and the operations or interactions they involved.

Some of these definitions include the word 'complex', but some instead describe systems which are merely complicated. Organisms and most of the social world are complex, but a mechanism or communications system (as machinery) is merely complicated. A complicated thing can be taken apart into its bits and reassembled from those bits. Another way of saying that is to say it can be analysed and integrated which means that it can be described by a mathematical system founded on linearity. A complex one can't be analysed and integrated either in reality or in mathematical representation. The physics definition above appears to describe a complicated system which can be understood in terms of Newton's Laws which depend exactly on the mathematical process of analysis (differential calculus) which he devised for that purpose. However, once we have three bodies in play, then Newton's methods, as Poincaré demonstrated in the early 20th century, do not work. An interesting word in the *OED* definitions is assemblage. We will be coming back to that. For the moment let us agree that a system is a set of inter-related elements and that a complex system is one in which, in plain English, the whole is greater than the sum of its parts.

Mathematical description is important but not primary. There have been a number of books written by mathematicians and physicists, for example, Strogatz's *Sync* (2003) and Johnson's *Simply Complexity* (2007b), which start from the mathematical formalisms describing the emergence of *Order out of Chaos*, the title of Prigogine and Stenger's seminal book (1984). First, this kind of approach[2] tends to assert, in the fashion which has bedevilled Western thought since Plato formulated the myth of the cave, that mathematics give us a way of describing the ideal transcendental world which exists beyond the imperfect reality we know through experience. Second, it leads to what Morin (2008) called restricted complexity, complexity understood only as the emergent product of interaction among simple agents – as micro-emergence. This is more than a matter of conceptual argument because one of the most important methods of constructing (a word we use very deliberately) descriptions of complex systems, simulation in the form of agent-based modelling, is predicated on the notion that the complex emerges from rule-based interactions among simple elements. We deny neither the reality of restricted complexity nor the value of agent-based modelling. Rather we assert with Morin that there is another form of complexity, both in the nature of reality and in the way we can approach reality, in other words both ontologically and epistemologically, which he calls general complexity. The world of general complexity is composed of complex systems which are not just the product of simple interactions but have properties which are not to be understood in those terms and have to be addressed as real in and of themselves. So here while noticing and using mathematical formalisms we yield no precedent to them.

The idea of complexity precedes the development of mathematical formalisms describing catastrophe and chaos, and complex systems can be described other than through such formalisms. While measurement and quantitative description are absolutely valid tools in the development of a social science of the complex, the development of systems of equations, however sophisticated, may well have not that much to tell us about the social world. We do not dismiss them out of hand because work has been done using them which is useful and informative. However, they are at best part of the toolkit, not an ideal standard to which social scientists must aspire. Here we want to stamp firmly on one criticism which is often advanced in relation to discussions of complexity in the social sciences. That is that the use of these ideas is 'merely metaphorical'. Of course, they are metaphors, but to paraphrase the title of Lakoff and Johnson's book (2003), metaphors are what we live by. Let us quote from two mathematicians (and an experimental biologist in the form of Cohen) who understand this very well:

> Mathematical descriptions of nature are not fundamental truths about the world, but models. There are good models and bad models and indifferent models, and what model you use depends on the purposes for which you use it and the range of phenomena which you want to understand … reductionist rhetoric … claims a degree of correspondence between deep underlying rules and reality that is never justified by any actual calculation or experiment.
>
> (Cohen and Stewart 1995 410)

> Mathematics is also seen by many as an analogy. But it is implicitly assumed to be the analogy which never breaks down.
>
> (Barrow 1992 21)

A model is a metaphor. Any description of reality is metaphorical. There are two, and it is very important to stress this, *different* ways in which such descriptions are of value to us. One, which Crutchfield (1992) calls the engineering mode, is whether or not such a description works in prediction, an essentially pragmatic position. The other which he calls the mode of science is whether or not it is an accurate description of reality, which in more formal terms we can say is isomorphic with reality. Both approaches seek to establish models which can be used for prediction, but while the scientist requires fundamental underlying laws, the engineer wants the bridge to stay up and bear loads. The issues underlying these different purposes will be revisited throughout this text.

Our position dismisses as irrelevant the kind of criticisms advanced by Sokal and Bricamont (1998). Certainly, the term non-linearity became current in consequence of the forms of description of physical and chemical systems developed by applied mathematicians attempting to deal with phenomena which could not be dealt through conventional linear modelling. However, these approaches can be considered to be developments of a Newtonian mind set, although the inability of that approach to provide more than local accounts (see Nicolis 1995) is a fundamental challenge to that argument. Regression models which have so dominated quantitative social sciences of a non-experimental form, which means the overwhelming majority of quantitative social science other than in psychology, are predicated on straightforward linear modelling. Non-linear modelling based on non-linear differential equations is a different matter, and simulations based on this are enormously important in climate science, although largely so far at the level of atmospheric physics and chemistry. There is important work being done here in ecology and in social ecology which we will discuss in relation to mixed methods. The blunt point is that non-linearity is a product of emergence. We need to start from emergence and develop a science that fits that crucial aspect of complex reality.

We have stated our position, at least in a preliminary fashion, in relation to what we mean by complexity and complex systems. Let us turn our attention to 'theory'. Again, it is useful to begin with definitions from the *OED*. The following are relevant for our purposes:

3. A conception or mental scheme of something to be done, or of the method of doing it; a systematic statement of rules or principles to be followed.

4. a. A scheme or system of ideas or statements held as an explanation or account of a group of facts or phenomena; a hypothesis that has been confirmed or established by observation or experiment, and is propounded or accepted as accounting for the known facts; a statement of what are held to be the general laws, principles, or causes of something known or observed.
   b. That department of an art or technical subject which consists in the knowledge or statement of the facts on which it depends, or of its principles or methods, as distinguished from the *practice* of it.
5. In the abstract (without article): Systematic conception or statement of the principles of something; abstract knowledge, or the formulation of it: often used as implying more or less unsupported hypothesis (cf. 6): distinguished from or opposed to *practice* (cf. 4b). of in theory (formerly in the theory): according to theory, theoretically (opp. to *in practice* or *in fact*).

Here none of these quite fits what we will mean by theory when we talk about complexity 'theory'. We agree that theory will cover 'the formulation of abstract knowledge', but when we talk about complexity theory, we mean something other than the formulation of hypotheses and want to challenge the distinction between theory and practice. Of course, these various meanings of 'theory' are wholly valid, and we will pay particular attention to what Merton (1968) called 'theories of the middle range', although not always in precisely his terms. However, what we mean by 'theory' when we refer to 'complexity theory' is really 'ontology', which the *OED* defines thus:

a. *Philos*. The science or study of being; that branch of metaphysics concerned with the nature or essence of being or existence.

Castellani and Hafferty put it like this and we agree with them:

> Social complexity theory is a more a conceptual framework than a traditional theory. Traditional theories, particularly scientific ones, try to explain things. They provide concepts and causal connections (particularly when mathematicised) that offer insight into some social phenomenon … Scientific frameworks, in contrast, are less interested in explanation. They provide researchers effective ways to organize the world; logical structures to arrange their topics of study; scaffolds to assemble the models they construct. When using a scientific framework 'theoretical explanation' is something the researcher creates, not the other way around.[3]
>
> (2009 34)

When we say complexity 'theory', we mean a framework for understanding which asserts the ontological position that much of the world and most of the social world consists of complex systems, and if we want to understand it, we have to understand it in those terms. We will work with an ontology defined by Reed and Harvey (1992) as complex realism based on a synthesis of critical realism after Bhaskar (1979) as a philosophical ontology, and complexity as a scientific ontology. For us complexity theory is an ontologically founded framework of understanding and not a theory of causation, although it can generate theories of causation. The foregoing discussion represents a preliminary shot at saying what complexity theory is. We have a good deal more to say on this topic including a development of the argument that there is not one 'complexity theory' but rather two sets described in Morin's terms as restricted and general. We also have to say what complexity theory is doing and how it is doing it.

The question posed by Bhaskar was what must the world be like for science to be possible? We can begin to make explicit some foundational ideas in answering this question that will be developed later in the book. We have shown that the 20th century has seen the failure of a straightforwardly progressive model of knowledge development aligned with modernist views of science. From confidence in reductionism and determination, we have moved to recognition of open systems and contingency. This has allowed the original divorce between natural and social to be reviewed, bringing awareness of the continuity between natural and social systems to the fore. The divorce between philosophy and science created problems that could not be addressed within a Newtonian frame. Historically, the division proceeded along the lines of a 'claim that we could only know what was true in science and that philosophy would deal with what was good. … Science insisted it had the monopoly on the search for the true' (Wallerstein et al. 1996 187).

It was claimed that while 'hard' sciences could operate free of a concern with the 'good', it was more difficult for social science to maintain this stance. The resolution of this division however comes from the direction of the 'natural' sciences recognizing an inherent value base rather than from social sciences shedding theirs. Indeed, it presents fewer problems for the social sciences where the inextricable relation between social science and culture has long been acknowledged: 'Social science is not a scholarly arcanum, but an organised part of culture which exists to help man in continually understanding and rebuilding his culture' (Lynd 1939 ix).

It is worth taking a little time to think about the role of values because the notion of objectivity has been used to delineate a line between hard and soft science and has become an implicit, if problematic, standard. Longino (1990) helps us to distinguish between the role of values in science as constitutive and contextual, where constitutive values of objectivity and rationality are those that underpin 'good practice' in science while contextual values are often implicit but have shaping power. Pointing to the range of assumptions of female inferiority underpinning much research, she establishes the socially constructed nature of contextual values. In terms of objectivity she asks, 'how can the contextualist analysis of evidence, with its consequent denial of any logically guaranteed independence form contextual values accommodated within a perspective that demands or presupposes the objectivity of scientific inquiry?' (1990 65–66). She resolves the debate between objectivity and social construction in terms of science as practice, and practice undertaken in the main by social groups rather than individuals. This is particularly true of science as currently produced and evaluated through processes of peer review, through which funding decisions are made. Longino argues that far from being inherent in scientific method it is the social nature of scientific knowledge that allows it to be objective. 'A method of inquiry is objective to the degree that it permits *transformative criticism*' (1990 76). This social construction creates both the challenge to and the possibility for objectivity.

As an acknowledged creature of liberal capitalism, the social sciences have accommodated this apparent contradiction. Both Comte and Durkheim worked within a positivist frame, prizing objectivity in overcoming ideological and other sources of bias. Nonetheless, they saw the accrual of knowledge as for the purpose of social improvement, through observing reality and establishing social laws. While classical science was founded in an attempt to achieve its values of objectivity and rationality, the ability of these to comprehend the whole universe of knowing has begun to be questioned from both sides. We have identified discontent with ideas of objectivity and the separation of knower from known as an abiding awareness that has found expression within the social sciences. Morin, whose whole academic endeavour resisted categorization in nomothetic or idiographic terms, puts it succinctly, 'there is magic as well as rationality in our society and even within our rationality' (2008 91).

Discontent with the use of traditional 'value-free' methods has similarly been an abiding concern for some, aware of the disconnection that those methods force on complex social phenomena.

> The more exact an experiment is – that is, the more elementary and isolated the phenomenon, and the more constant the conditions – the greater is its artificiality, and the greater its distance from the study of the individual.
>
> (Lynd 1939 110)

This all implies a significant rethinking of the nature of social science in terms of the intricate and mutual implication between society and its subject itself. It leads us on to a further reflection on the nature of knowledge about ourselves because it relates to not only the content of knowledge but our frames of knowing, not as through the modernist post-enlightenment shift to rationality but in a more mutable and complex interaction with our representations of ourselves. We have seen that over 200 years of development in science within a positivist frame has not enabled the social sciences to comprehend a complex world. Despite significant (and continuing) pressure to work in this vein, social scientists have continued to document the inconsistencies and even violence that enforcing it does to the social subject at every level. If a new ontology of complexity is to be adopted, a new approach must be taken to how we frame our questions, make observations, record, and draw implications. Exploring how this is already happening in social science practice is a major objective of this book.

> So the question becomes how is the nature of social science in complexity different, remembering first of all that complexity is not … a rejection of science as a mode of knowing. It is a rejection of science based on a nature that is passive in which all truth is already inscribed in the structures of the universe.
>
> (Wallerstein 1999 189)

What is the promise of the new science? Prigogine sets out a discussion of dialectic syntheses of categories hitherto regarded as mutually exclusive: order/disorder, determinism/probability, being/becoming, necessity/freedom, determination/unpredictability, and constraint/possibility, by changing the epistemic status of disorder, unpredictability, and uncertainty into inherent structural elements (Condorelli, 2016 427). This indicates an epistemological shift, particularly away from previous confidence in prediction. Uncertainty is no longer a failure but an acknowledged feature of human and natural history. Rather than imposing frames that torture it, uncertainty, contingency, and emergence allow the world to be described as it is.

What also enters this new approach is the role of agency and ethics. As Heylinghen et al tell us,

> Complexity theory (and postmodernism), of course, cannot devise a better ethical system, or at least not a system that will solve the problem. What it can do however, is to show that when we deal with complexity – and in the social and human domain we always do – we cannot escape the moment of choice, and that we are *never* free

of normative considerations. Whatever we do has ethical implications, yet we cannot call on external principles to resolve our dilemmas in a final way.

(2006, 130)

The implications of this refiguring become more evident when we consider practical social science. In the 'old science', we would expect to be able to characterize our study in terms of the set of assumptions of rational action, well-defined problems giving rise to equilibrium that can be described as socially optimal. Yet, as Arthur (1999) points out, players in the economy don't know how other agents will react – the world is fundamentally one of uncertainty, problems are ill defined, and making sense is muddling along. There is no equilibrium, and we are reacting to a situation we don't understand. Systems that are non-linear are necessarily sensitive so that what might bring about minor change in a Newtonian model can bring changes in magnitude, quality, and type in a complex system. Moreover, open systems, whether natural or social, are inherently evolutionary systems. We will argue that these understandings frame an approach to knowing that, while it rejects prediction, can nevertheless produce valuable knowledge about the effects of interventions in their interaction within the system. To do so, we need to consider how handling these new ways of looking shapes our approach to the new science? How do we conceptualize and talk about systems differently?

One of the foremost methods for conceptualizing and seeking to understand social systems has been through models. Human and Cilliers point to early (failed) attempts to deal with complex problems through a process of prioritizing selected elements regarded as 'natural and sufficient to explain the phenomenon under consideration' (28 2013). This approach fitted with the 'atomistic and rationalistic tendency of western metaphysics' (2008 28). It allowed those matters excluded to be regarded as 'noise'. They distinguish between difference and heterogeneity, arguing that while classical science can deal with difference, its models cannot cope with heterogeneity. Noting the inadequacy of rule-based understandings for complex systems, they focus on processes as relational rather than mechanistic, in which specifying context is required while, unlike the postmodernists, not rejecting reality. Referring to the work of Derrida and Bataille as well as Morin, they propose the notion of an economy to counter the tendency to view systems as having an inside and an outside. Clearly, in complex systems exclusion is impossible because the excluded may have real impact. They use the restricted/general complexity distinction defined by Morin (2008) as a basis for defining a new science approach. The new science must necessarily use models that are simplifications but it must do so in the awareness of imposing 'disjunctions, separations'. This must be done in the awareness of a 'general economy (of analysis which) aims to keep heterogeneity in mind while granting us the use value of a restricted economy' (2008 32).

This requires, avoiding the reification of systems and, further, it means changing the very language we use from that of natural science.

The established language of science serves complexity badly as it brings with it deeply embedded sets of epistemic commitments. Bastardas Boadas (2016) has suggested we need to develop a language of 'complexics' more appropriate to a 'meta-transdisciplinary field concerned with giving us suitable cognitive tools to understand the world's complexity' (2016 2). This includes radical shifts, using verbs rather than nouns to emphasize observation of relations and processes in interaction rather than static, disconnected elements. The figurational approach of Elias provides a worked ground for this change, and we can see it in the work of other sociologists, including that of Pierre Bourdieu. In developing social science on the basis of complexity theory, we are required to be conscious of these issues in the application of methods and in the prescription of solutions in the highly applied fields such as evaluation and governance. The ability to acknowledge 'noise', the awareness of how change occurs in non-linear ways and of the frequent unintended consequences that arise, become part of the investigation rather than regrettable interference with our neat models.

Conclusion

'We are not at the end of science. We are at the end of linear science's potential, but we are at the beginning of a new science' (Prigogine, quoted in Condorelli 2016 423).

In this second edition, we have sought, in an entirely complexity-consistent fashion, to introduce complexity theory into the context of contemporary social science as a whole. Our interest in the Gulbenkian report lies in its concern to identify the intellectual and the institutional context for the development of social science to date. In doing this, it also identified the ontological space emerging for the social sciences to engage with complexity theory. This resonates with important discontents in the natural sciences with mechanistic and reductionist understanding and with a growing interest in systems understood as open and interacting with their environment. In this way, the investigation moves from substances to include the relations between them. What is clear from that account is that rather than adapting existing structures of knowing, we need to reframe through a shared ontology based on an ongoing dialogue with issues of standpoint and value. The need for inter-/postdisciplinarity to answer questions, formerly jealously guarded as the turf of natural or social science, has forced its way onto the agenda. The answer, a new science and a new ontology that allows coexistence rather than opposition.

In the first edition of this book, we discussed the significance of participation in complex systems, not only in terms of understanding the subjects of enquiry but more fundamentally in terms of the production of knowledge. Most importantly, it requires no simple separation of observer from observed but rather the recognition of social science as practice. We can ask, how will complexity theory change the power relations across the social sciences? The answer has to be that in normal times to expect such radical transformation would be unrealistic. Institutional processes persist, and careers are forged through processes of in-group/out-group exclusions. *But these are not normal times*. Arguments will rage over the character of complexity in a way that is already happening and perhaps is necessary to the dynamic process of academic enquiry. Complexity theory can, however, take us beyond paradigmatic difference in a way that allows previously warring practitioners of quantitative and qualitative methods to find reconciliation, assuming their place as contributing to understanding a world that cannot be reduced to its constituent elements. It is a world in which we need to be able to recognize emergence as a fundamental feature of social life and so central to policymaking and implementation. It raises wider issues of pedagogy, not only in opening the social science but a much more wholehearted broadening of education. Setting out this context has raised some of the issues in gathering evidence that complexity theory practitioners must overcome. We need to not only understand the world differently but to *practice* our understanding.

In a context of a social world in crisis, in large part because of the impact of the processes of the Capitalocene on the natural world, the times have to be a changing if science as a whole is to have any constructive role in addressing crises of inequality and impending climate catastrophe. Radical transformation must happen – that has to be the praxis for the future.

Notes

1 In philosophy itself as an academic discipline positivism was challenged, even its most sophisticated forms. The analytical tradition, much influenced in the UK by Wittgenstein, was strong, but what social scientists got was largely crude positivism taught in a methods context.
2 Although this accusation cannot be levelled at Prigogine and Stengers given the considerably greater degree of philosophical sophistication which informs their text. It is worth noting that those schooled in the Prigogine tradition, and particularly Peter Allen (1997), are notably more cautious and open minded in relation to the implications of social complexity than is the case with those coming from a purely Anglo-Saxon scientific background.
3 The same point was made by Von Bertalanffy in the introduction to the British edition of *General Systems Theory* when he asserted the necessity for a '*systems philosophy*, i.e. the reorientation of thought and world view ensuing from the introduction of "system" as a new scientific paradigm (in contrast to the analytic, mechanistic, oneway-causal paradigm of classical science' (1968 xix) and went on to develop this by identifying both a specific systems-oriented ontology and epistemology.

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2 The complexity frame of reference and action

This is the inscription on Karl Marx's tomb in Highgate Cemetery in London:

> 'Philosophers have hitherto only *interpreted* the world in various ways; the point is to *change* it'. – Thesis XI on Feuerbach (written in 1845)

The way in which complexity can inform and indeed must inform change is a central theme in this book. It is conventional to talk about *Complexity Theory*, and we have used that term as the title of this book. However, complexity is not a theory in the sense of a grand social theory as developed by Marx, Durkheim, Weber, Elias, or Bourdieu. Nor is it a descriptive theory of the middle range after Merton and as deployed by Pawson and Tilley (1997) in evaluative mode, although it can inform investigation at that level. Rather as Castellani and Hafferty (2009) put it, it is an ontological frame of reference. They were writing primarily with reference to the social world, but now we have to extend it to all of the interrelationships of the social and the natural. To say it is a frame of reference is to say that it is a general way of thinking. To say that it is ontological is to say that it is a way of thinking about what the world is made from, both in terms of the entities which constitute the world *and* the relationships among those entities. It is a realist ontology because it asserts that it is possible to know the world, not simply and directly as with crude positivism, but to know it nonetheless. It is a materialist ontology because it rejects all idealist framings which assert that the world itself is separate from our knowledge of it and which inevitably reduce ontology – what the world is – to epistemology, how we articulate our accounts of it. Epistemology is necessary, but it has been intimately linked with ontology in a combination best described as meta-theory – a specification of how we understand what we are doing when we develop scientific accounts – scientific in the sense defined by the Gulbenkian Commission – of reality in whatever aspect. For us that creation of account must inform purposeful action to confront the crises facing humanity in the 21st century.

Understanding of the world which can inform what we might do within it towards an unequivocally clear set of objectives – teleology as we will see is back up and running – should not be derived from philosophical principles developed in the abstract. Philosophy can inform and help, but that is all. The primary engagement with philosophical accounts here will be with 'continental philosophy' and much less with the analytical Anglo-American tradition. Whitehead was very much a Brit, but he stands outside the analytical tradition. British Marxists like Thompson and Raymond Williams are very different both from it and from variants of what Thompson called 'the French flu'.[1] Latour is an ANTer (although he playfully would deny that label) who has some useful things to say. One of

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them is in his outstanding review of Stengers's (2005) exposition of Whitehead's whole pattern of thought where he says that if we are looking for a great thinker of the 20th century whose name begins with W, we should look to Whitehead rather than Wittgenstein. To say that is to light an explosion under the analytical project. Language matters, but we will get more from socio-linguistics, and in particular the synthesis of linguistics, evolutionary psychology, and Merleau-Ponty's conception of the role of perception proposed by Lakoff and Johnson (1999), than from the dominant trends in the analytical tradition.

Complex realism elaborated

This book draws on the innovative synthesis of Bhaskarian realism and complexity proposed first by Reed and Harvey (1992). What complexity implies is that the world as a whole including the social world and its intersection with the natural world is best understood as composed of complex, open, far from equilibric systems, which cannot be understood as the merely complicated but can be comprehended by reduction to components and explanation in terms of the properties of those components. This is not a holistic principle. It does not mean that everything derives from the properties of the system as a whole. Rather, explanation has to be based on an understanding of the relations among the parts with each other with the system as a whole with causal powers running in all directions. *And* relations with other systems which compose the environment of any complex system also have causal powers. Morin puts this well:

> Two capital consequences flow from the idea of an open system: the first is that the laws of organization of the living [and the social Byrne and Callaghan] are not laws of equilibrium, but rather of disequilibrium, recovered or compensated, stabilized dynamics. … The second consequence, perhaps more important still, is that the intelligibility of the system has to be found, not only in the system itself, but also in its relationship with the environment, and that this relationship is not a simple dependence: it is constitutive of the system. … *Reality is therefore as much in the connection (relationships) as in the disconnection between the open system and its environment*. This connection is absolutely crucial, epistemologically, methodologically, theoretically, and empirically. Logically, the system cannot be understood except by including the environment. The environment is at the same time intimate and foreign: it is part of the system while remaining exterior to it.
>
> (2006 11; original emphasis)

This resonates absolutely with the emphasis on the importance of relations found in Whitehead and Stengers's exposition of this thought (2011), and in Emirbayer's (1997) manifesto for a relational sociology. However, it goes beyond them in recognizing that while relations constitute systems, including of course the internal relations of system components with each other and with the system as a whole, relations are not everything. They are crucial, but systems themselves matter because the entities of interest to us are complex systems in their own right.

This conception of systems differs in important respects from that of Maturana and Varela (1980) and through their influence of Luhmann (1995). Autopoiesis, a concept developed in consideration of the character of a biological cell, emphasizes self-organization within boundaries and draws a distinction between assemblages, a word to which we will return, which rely on components external to the self-organizing system, and the autopoietic system as such, although of course as Maturana and Varela always realized cells require energy and materials from their own environments to survive. Cilliers in an important article discusses 'Boundaries, Hierarchies and Networks in Complex Systems' (2001). He reviews the idea of autopoiesis drawing on Zeleny (1996). Zeleny defined an autopoietic system thus:

> A system that is generated through a closed organization of production processes such that the same organization of processes is regenerated through the interaction of its own products (components) and a boundary emerges as a result of the constitutive processes.
>
> (1996 123)

The crucial word here is 'boundary' because if we follow Maturana and Luhmann, then the boundary sets the complex system apart from the rest of the world. However, Cilliers picks on the way in which Zeleny developed the notion of boundary:

> All social systems, and thus all living systems, create, maintain, and degrade their own boundaries. These boundaries do not separate but intimately connect the system with its environment. They do not have to be just physical or topological, but are primarily functional, behavioral, and communicational. They are not 'perimeters' but functional constitutive components of a given system.
>
> (1996 133)

This is an ontological statement which defines boundary in a way that allows systems precisely to be open and asserts that the boundaries are actually the elements in a system which enable and maintain that very openness.

Cilliers presents his own take on the nature of boundaries:

> I have a certain unease with the strong influence of ideas based on the concept of autopoiesis. A strong emphasis on the 'operational closure' of complex systems is at odds with the central insight that complex systems are open systems which constantly interact with their environment in rich ways.
>
> (2008 31)

> Boundaries are simultaneously a function of the activity of the system itself, and a product of the strategy of description involved. In other words we frame the system by describing it in a certain way (for a certain reason) but we are constrained where the frame can be drawn. The boundary of the system is therefore neither purely a function of our description, not is it purely a natural thing.
>
> (2001 141)

The second statement is both ontological and epistemological. It is ontological because it asserts that systems are both real in that they exist and simultaneously (particularly social systems) have a reality which is constituted by our actions in defining them. It is epistemological because it asserts that we know systems by defining them in terms of boundaries, but that reality has a voice in setting those boundaries and constrains our definition. Reality has a voice. And these very descriptions then have a constitutive role in social reality going forward.

This is Cilliers writing in a way which corresponds to the essence of Bhaskar's (1979) ontological and epistemological position. Bhaskar conceptualized three levels of reality. These are the Real: the deep level of generative processes. The generative processes of the real produce the Actual – what is there in experience. Our social systems as we live in and through them are products of the core social relations of the generative social system. The Empirical stands at the interface of ontology and epistemology. It is the level of our descriptions of the Actual in relation to the generative powers of the Real. Note that its components too have causal powers. As the Thomases (1928) asserted, things are real if they are real in their consequences. Scientific descriptions have very real consequence through their recursive impact on social and now as we must understand natural reality at the level of the Actual.[2]

The word *cause* has entered with force into our exposition. We are going to brutally and summarily bypass for the moment the enormous philosophical debate on what causes are, if they are indeed anything (although we are hard adherents both of their reality and of our ability to know them), by dealing with a sort of synonym of cause: *determine*. Raymond Williams (1980) conducted a careful analysis of the Marxist diktat: Base determines superstructure. He paid particular attention to that word 'determine' and pointed out that while it can mean: if A then exactly B; it can also mean to set the limits for – to construct a boundary of possibility. This corresponds with complexity theory where there are multiple possible states for the future of a complex system, but these are limited in the possibility (phase) space. What will happen depends on a precise configuration of complex causes derived both from the internal characteristics of the system and from its whole environment, including systems with which it is interpenetrating and with which it shares subsystems.

Byrne (2011) noted that while a great deal of attention had been paid to the nature of cause, much less had been paid to the nature of the other word intimately associated with it: *effect*. We can consider the nature of a complex system as passing through a set of states. First, it comes into being – often by a radical qualitative change of a pre-existing complex system as with the transition from feudalism to capitalism. Then it stays much the same. Complex systems are relatively stable for long periods of time. Then it undergoes other qualitative changes to become something different – undergoes a phase shift. It may even cease to exist in any recognizable form – it may terminate. The key point is that the effect is the general state of the system, not just at any single time point but through a duration of relative stability, of stable trajectory. Another way of saying this is that an effect is the position of the system within an attractor in the state/possibility space for that system. Stability is position within something like a Torus attractor – moving around but not changing essential nature.

> The terms trajectory and history are synonyms when we are considering complex systems. The implication of this proposal about the nature of effects is that we explore causality by working backwards from specific different effects. … In other words we should engage with processes of retroduction to explain what has happened and in terms of applied social science develop a retrodictive approach to guiding actions towards the achievement of desired outcomes. This is very much in accord with the general critical realist programme of explanation. We are dealing with effects understood as system states and understand these system states to be the product of complex *and* multiple generative mechanisms.
>
> (Byrne 2011, 89)

Byrne (2019) notes that individual human beings are themselves complex systems with the nature of their trajectory through their life course, being determined (in Raymond Williams' sense) by both their own characteristics which are themselves continually modified by interaction with other surrounding systems in their environment and the social world at all levels which surrounds them. Now we must also include the natural systems of disease, pollution, and potential climate catastrophe. The implication of this way of thinking is that when we look to the future, we are not looking at what will happen *but what will be made to happen*. Human agency has effective causal powers, not only in relation to the social but in the era of the Anthropocene/Capitalocene for the state of the natural world in interaction with the possibility space of that very social.

There is a resonance between Cilliers' account of systems and their relationships and the notion of assemblage. We might say all complex systems have assemblage characteristics although there is more to them. Not all assemblages, particularly as the term is deployed in cultural theory, are complex systems. With explicit reference made to social systems, and in particular to the way in which social systems can be constituted across space and do not necessarily have a common social location, Cilliers remarks:

> non contingent subsystems could be part of different systems simultaneously. This would mean that different systems interpenetrate each other, that they share internal organs. How does one talk then of the boundary of the system under these conditions?
>
> (2001 142)

If we think of systems as somewhat like assemblages, the possibility of interpenetration seems obvious. Elements of different systems can be in multiple systems and can join or leave those systems. We can see this at the micro-level in relation to the shifting group affiliations and actions of individual human beings. On another scale, this perspective can inform the macro-historical interests of international relations.

Let us turn to the idea of structure. Westergaard in a thoughtful reflection on the historical trajectory of intellectual positions in sociology in the UK remarked that '"structure" is only a metaphor, but useful to denote persistence and causal force' (2003 2). We agree completely about persistence and causal force. Structure is of course a metaphor, as are all our descriptions of reality of whatever form, but that there is something real which we are describing. Cilliers identifies structure as an emergent but allows it no prior reality:

> Structure is the result of action in the system, not something that has to exist in an a priori fashion.
>
> (2001 14; original emphases)

We agree that structure results from action but think that here Cilliers is missing two important points. The first is that systems have a history. Structure in any social system is the product of past actions and has causal potential to the scope for future actions. The important notion of path dependence expresses this idea in a straightforward fashion. The second relates to the way in which any system, interpenetrating with other systems, has to take account of the nature of those other systems and of its own relationship with them. In the social world, one way of describing the overarching environment within which all systems operate and within which they are contained is social structure. Of course, such structures are malleable and subject to transformation, not least to transformations driven by human agency which may be individual – think charisma – or more generally have a collective character – think revolution. So we want to allow both a historicity to structure within systems and that the structure of containing systems, generally expressed in discussions of complexity as nested systems, has an influence on the trajectory of systems contained within it. Here we are dealing with determination exactly in the sense proposed by Raymond Williams (1980).

When we talk about systems being nested, it might imply a hierarchy of systems. We do not agree. Here we need to think about both subsystems within any system, constituted by both reality and ourselves, and about the way systems relate each other. With regard to the issue of internal hierarchy within any given system, we find Cilliers's position, articulated in an argument with the classic conception of hierarchy within systems expressed by Simon (1962), persuasive. Cilliers opts for a middle way between Simon's argument that hierarchical systems contain substantial redundancy and can therefore be modelled in ways which are simpler than a complete reconstruction of the complexity of the system and the view, often associated with autopoiesis, that complex systems have no central control systems. The essence of his position is that systems must have structure, however temporary and plastic that structure is:

> The classical understanding of hierarchies tends to view them as being nested. In reality however, hierarchies are not that well structured. They interpenetrate each other, i.e. there are relationships which cut across different hierarchies. These interpenetrations may be fairly limited or so extensive that it becomes difficult to typify the hierarchy accurately in terms of prime or subordinate parts.
>
> (2001 143)

Systems can be nested, so long as we allow for interpenetration in Cilliers's terms, and do not specify a unidirectional downward pattern of causation. So when Reed and Harvey say:

> Such ontological layerings and their dialectical interactions produce a nested reality that is composed of layered entities. Moreover, these reciprocal determinations give the hierarchies an overall irreducible complexity. The world as a whole and its hierarchical structure are both open and greater than the sum of their parts. Were it otherwise, science as we know it would be impossible.
>
> (1992 358)

with specific reference to Bhaskar's critical realist ontology, they open up the idea that systems can be layered both internally and with reference to other systems but without imposing any unidirectional determination or control. Hierarchy matters in complex systems, but interpenetration, layering, and multi-directional causality mean that we cannot describe causal processes in terms of any one direction of cause.

Malcolm Williams' *Realism and Complexity in Social Science* (2021) contains much with which we agree, given that he too explicitly endorses complex realism as a frame of reference.[3] He is a supporter of Popper's views on causality. For Popper, frequency is a property of the case and outcomes are specific to the properties of the case. Hence, he and Williams (M) argue that we can deal in single-case probabilities. The frequency understanding of probability ignores this and smooths out 'the complexity of the generating conditions of case characteristics that make up the aggregate' (2021 40). We agree.

Williams develops his treatment of this by considering the meaning of contingency which he uses:

> to indicate a particular form of dependency, in so far as event B (in time) is dependent upon event A, which has a degree of ontological probability. We should not think of this as a linear dependency, whereby A's probability is itself determined by a singular prior event, but rather as a nesting of prior probabilities.
>
> (2021 41)

For us this is compatible with Gould's (1990) conception of contingency in a historical sense. We have some points of argument, not so much in principle as in relation to language. So Williams M. (2021 56) describes determination as synonymous with exact specification in contrast with our use after the other Williams (R) of setting limits, as constituting a possibility space. This is fundamental to what we mean by cause. Put exactly, we always mean determine in the sense of setting the limits of a possibility space at whatever level and the nature of possible attractors within it. For us this gets us past much of the anguished debate about causation. So when Malcolm Williams says, 'the probabilistic character of the world is real and is not an artefact of lack of knowledge of it' (2021 66), we do not agree. Probability applies in relation to the trajectory of cases at whatever level but not in the sense of the future position of the case – the effect/outcome – being inherently probabilistic. If we think about the individual case, using what we make of Popper's term propensity, then if we knew enough then the outcome would be exact. Knowing enough is not simply a matter of knowing all possible causal 'variables' and all possible interactions at multiple levels among them which would be the basis of a complex and even non-linear statistical model (although such models have their uses). That kind of model would be a way, albeit a clumsy and difficult way, of addressing the causal powers of all internal components of the case as a complex system in its own right *and* all causal powers of interwoven other complex systems which have an impact on that case as a system.

At the micro-level, using the kind of survey, administrative, or even 'Big' data about cases, we can deal with those kinds of models, but we deal with them in relation to not individual cases but ensembles of cases. So our probabilistic explanations relate to producing accounts for enough of the individual cases to be meaningful – a model that fits the data. *But* the model never fits all the data.[4] One of the advantages of Qualitative Comparative Analysis (QCA) is that it allows for multiple configurations in the form of sets of case attributes and hence for multiple specifications of cause which are also inherently complex.

However, we are not always dealing with lots of cases and especially not dealing with lots of cases from a sample. Often, we have all relevant cases, particularly when we have meso-level data for institutions or spatial entities such as city regions. Sometimes when we deal with the world system, we have only one case – this planet. The essence of determine understood as setting limits is that it defines the range of possible future outcomes – system states – and our pursuit of causality is about understanding what it is which will determine which of those future states has come or will come into being. With multiple cases, whether from a sample or a complete population, individual cases may and very probably will end up in different locations in that possibility space although the determinant power of casual factors may mean variation even then is minimal. Provincial English city regions under the policy regimes and capital logics of neo-liberalism and austerity are very similar in terms of contemporary character and the consequences of rentier developer capital specification of new urban forms. For us, cause is always about determination understood as bounding future possibility space and defining potential attractors within it.

Castellani (private communication 2012) has suggested that we can regard social systems in general as negotiated orderings at different scales which have an assemblage character in that additions to and/or deletions from the assemblage 'rework' the negotiated order. We find this argument entirely persuasive, and it helps us in grasping the character of social systems and in framing our understanding of boundaries in systems.

Restricted versus general complexity

Morin distinguishes between two forms of complexity: restricted and general

> Restricted complexity made possible important advances in formalization, in the possibilities of modelling, which themselves favour inter-disciplinarity. But one still remains within the epistemology of classical science. When one searches for 'laws of complexity', one still attaches complexity as a kind of wagon behind the truth locomotive, that which produces laws. A hybrid was formed between the principles of traditional science and the advances towards is hereafter. Actually, one avoids the fundamental problem of complexity which is epistemological, cognitive, paradigmatic. To some extent, one recognizes complexity, but by decomplexifying it. … In opposition to reduction, [Generalized] complexity requires that one tries to comprehend the relations between the whole and the parts. The knowledge of the parts is not enough, the knowledge of the whole as a whole is not enough, if one ignores its parts; one is thus brought to make a come and go in loop to gather the knowledge of the whole and its parts. Thus, the principle of reduction is substituted by a principle that conceives the relation of the whole-part mutual implication.
>
> (Morin 2006 6)

There is a form of complexity work in relation to social systems which employs modes of explanation with which contemporary scientistic science is comfortable and attempts to develop accounts of emergent social reality which can be expressed in terms either of mathematical formalisms as these develop in an equation-driven simulation or sets of rules governing the behaviour of agents in agent-based simulation. Morin sees this, we might say, as not straying beyond a comfort zone. However, in his specification of 'restricted complexity', he misses out one key word. That word is ontology. A key distinction between restricted and general *social* complexity is ontological.[5] We must examine how conventional 'restricted' complexity really does restrict our potential for understanding *and acting in* the social world. There are simplistic versions of restricted complexity, generally expressed by 'hard' scientists who assert the validity of their approaches for understanding the social world,[6] particularly when they do so in abstraction without any empirical referent.

So, for example, we find Nicolis[7] saying:

> Nonlinear Science('s) … aim is to provide the concepts and the techniques necessary for a unified description of the particular, yet quite large, class of phenomena whereby simple deterministic systems give rise to complex behaviours with the appearance of unexpected spatial structures or evolutionary events.
>
> (Nicolis 1995 xiii)

Note the phrase 'simple deterministic systems'. Likewise, Holland in a book with the title *Emergence* asserts that 'It is the thesis of this book that the study of emergence is closely tied to [the] ability to specify a large complicated domain via a small set of "laws"' (1998 123). Here he does not mean laws in the sense of invariant nomothetic scientific laws in the classical sense but rather rules for the behaviour of entities. It is from the interaction of these entities with each other in terms of these rules that we see emergence. Johnson's *Simply Complexity* defines complexity as 'the study of the phenomena which emerge from a collection of interacting objects' (2010 3–4). Yes, that is simple complexity (Byrne 2005) which is the same thing as what Morin (2006, 2008) calls restricted complexity, but not much of the social world can be dealt with in that frame of reference. There are sophisticated versions of restricted complexity which do have a social sensibility but remain committed to an ontological position which denies an independent reality to the social itself. This is not enough.

One key issue is the degree to which complexity should proceed by the development of mathematical formalism. Cilliers deals with this well:

> In an editor's note to a short review article by Corning (1998), the following statement is made: 'Until the "complexity science" researchers can develop a formal notation in symbols and syntax, while at the same time respecting its subjective nature [sic], it will not really be a science' (197). If this strict, formal and quantificatory attitude remains the way in which science is defined, then there will be no 'science' of complexity. However, our knowledge of complex systems is, to my mind at least, undermining such a strict understanding of science. It forces us to consider strategies from both the human and the natural sciences, to incorporate both narratives and mathematics – not in order to see which one is best, but in order to help us explore the advantages and limitations of them.
>
> (2001, 136–137)

We agree absolutely and will argue for a synthesis of measurement and qualitative work, noting on the way that the 'hard' sciences often remain trapped in a primitive and inadequate understanding of what they are actually measuring when they venture into the realm of the complex.[8] That said we think that ideas from applied mathematics are of value to us. We agree with Cilliers (1998) that the vocabulary of chaos has had too much influence on our approach to complex systems, not least because most complex systems are actually remarkable robust. However, the description of the complex world contained in the idea of phase or possibility space and of attractor states within that space is very useful. It is a metaphor, both in any mathematized formalism of it and as we deploy it. We make no apologies for the use of this sort of language, quite the contrary.

The Gulbenkian Commission defined science as 'systematic secular knowledge about reality that is somehow validated empirically' (1996 2) and that is what we mean when we talk about science in this book. One of the great promises of complexity as a frame of reference is that it offers the possibility of transcending the sterile arguments between quantity and quality, between the 'hard' and the 'human' sciences and opens up the possibility of a unified approach to understanding and action.

Complex systems and modelling

In his 2010 essay 'Difference, identity and complexity', Cilliers begins by reference to Saussure's theory of language and Freud's early neurological model of the brain, showing how both depend absolutely on an understanding of difference as the source of the substantive output of each system – meaning for language and memory for the brain.

> Such a system of differences can also be used to describe how a complex system works … Such systems consist of a number of components which interact non-linearly. The complexity of the system does not reside in the components but is a result of these interactions. If these interactions were ordered, homogenous and symmetrical, no interesting behaviour would arise. There has to be asymmetry. This is another way of stating that the relationships between the components are relationships of difference.
> (2010 7)

Descriptions of complex systems are models of the systems. Relationships are crucial to the form of systems. Understanding them as relationships of difference is a powerful aid to thinking and modelling. However, to say this, as Cilliers points out, is not to say enough. It would allow for an analytical programme based on an understanding of such systems as merely complicated rather than complex. Cilliers considers that this corresponds to Morin's understanding of restricted complexity. However, we do not agree since we think restricted complexity allows for emergence and hence for true complexity, whereas a structuralist account of complicated systems does allow for analytic reduction as a mode for establishing full understanding of a system. Certainly, Cilliers is right when he says that difference is not enough and his turn to Derrida's treatment of trace and difference takes us forward. Cilliers understands trace as standing for the individual differences among a system's components whereas difference can stand for the actual dynamics of the system (2010 7).

How Cilliers develops this argument is crucial for understanding how complexity informed accounts can make any claims towards being representations of reality. Cilliers does not forbid us to make representations. Rather he tells us to be careful, to be modest:

> The point to be emphasized is that an abundance of difference is not a convenience, it is a necessity. Complex systems cannot be what they are without it, and we cannot understand them without making profound distinctions. Since the interactions in such systems are non-linear, their complexity cannot be reduced. The removal of relationships, i.e. the reduction of difference in the system, will distort our understanding of such systems. A failure to acknowledge this leads to error, an error which is not only technical, but also ethical. When we pretend that we can understand or model a complex system in its full complexity, such pretence is not only hubristic, it is also a violation of that which is being modelled, especially when we are dealing with human or social systems. Trying to understand complex systems involves a certain modesty. … A complex system is constituted through the relationships of differences. These relationships are non-linear. If the complexity is reduced, i.e. some of the difference is removed, it distorts our understanding of the system. Nevertheless, we have to reduce the complexity in order to be able to say something about the system at all. Because of the non-linearity, the magnitude of the resulting distortion cannot be predicted. Since we know this beforehand, we have to accept responsibility for these distortions.
> (2010 8)

This is *not* a renunciation of modelling. Indeed in his 1998 book Cilliers says: 'I suggest that complex systems can be modelled' (1998 ix). Rather it is a recognition that exact representation cannot be achieved by anything which is less complex than the system itself *and* with the added implication that the representation would have to possess all the dynamic potentials of the original system. Like Poincaré, Cilliers was an engineer and he came to complexity in part through his work on neural networks in an engineering context. Crutchfield tells us something important about the engineering mentality:

> the epistemological problem of nonlinear modeling can be crudely summarized as the dichotomy between engineering and science. As long as a representation is effective for a task, an engineer does not care what it implies about underlying mechanisms; to the scientist though the implication makes all the difference in the world. The engineer is certainly concerned with minimizing implementation cost … but the scientist presumes, at least, to be focused on what the model means vis-a-vis natural laws. The engineering view of science is that it is mere data compression; scientists seem to be motivated by more than this.
>
> (Crutchfield 1992 8)

This is a pragmatist position, but it is more than that. Engineers still cannot model turbulence, the expression in their world of chaos. They work their way around it through models which they scale up to reality, having tested them for ship hulls in wave tanks and for airframes and bridges in wind tunnels. The turn to the material is a recognition of the limitation of mathematical description. However, engineering is more than just a way of pragmatically describing. Engineers model in order to make. The end result, say a ship or a bridge, is something made by human agency to do something and to do something which has an effect in the world. Human agency becomes material.

Models matter for understanding the world so we can act within it. Models are not ontologically innocent although many who deploy them do not address even implicitly the implications of that fact – not a word we use often, and when we use it we mean it. So we need to look carefully at what is being said both about what the world is like and how to understand it when models are made. We agree with Hedstrom's (2005 3) dismissal of the 'fictionalist temptation' of abstract theory-based models with no empirical connection with social reality. However, like Hedstrom, we can see potential in models which deploy real data – he would say quantitative but we would say of whatever form – which describes the social world as the foundation of investigation. To deploy real data constructed from the world is to allow reality a voice. The usual vocabulary for discussing emergence deals in the relationships between the micro and the macro, with the former considered as the constitutive base of the latter *even if* the latter, the macro, is allowed to have causal powers in relation to the former, the micro. In the social world, this argument is expressed in terms of the relationship between the micro in the form of individuals within society, and the social considered as the emergent form. Our first reservation in relation to this formulation is that it tends to ignore the fact that individuals are themselves complex systems, and certainly more complex in every way than the agents in agent-based simulations. In particular, they possess the power of discursive agency – agency which is not merely imposed rule bounded – both individually and collectively. To say that collectivities possess agency is to say that collectivities have a reality beyond the individuals who constitute them. And so we say, not only of formal institutions at any scale but also of social collectivities based on interests or identities or both. A corporation is real and has agentic power and so does a 'class for itself' or a social movement. The issue of the ontological reality of nested and interpenetrating complex social systems beyond individuals is fundamental to our argument. The demonstration of this central point is a crucial function of this chapter and of this book as a whole. This is not just a matter of the relationship between the social micro and the social macro, but also and even more crucially of the relationship between agency and social structure.

Sawyer (2005) identifies three waves of social systems theory: Parsonian systems theory, the development of a social version of general systems theory, and a third wave which:

> grew out of developments in computer technology. … In the 1990s … computer power advanced to the point where societies could be simulated using a distinct computational agent for every individual in a society through a computational technique known as *multi-agent systems*.[9]
>
> (2005 2)

The technique of multi-agent simulation modelling is for Sawyer the foundation of a complexity programme in the social sciences since 'it offers theoretical concepts and methodological tools that have the potential to speak to unresolved core sociological issues' (2005 10). Here it is worth noting what Hayles has to say on the subject of simulation, indeed on the nature of scientistic description of reality in general when she identifies the dominant contemporary modes of scientistic abstraction as the old game of the Platonic backhand and the new one of the Platonic forehand:

> The Platonic backhand works by inferring from the world's noisy multiplicity, a simplified abstraction. So far so good: this is what theorizing should do. The problem comes when the move circles around to constitute the abstraction as the originary form from which the world's multiplicity derives. Then complexity appears as a 'fuzzing up' of essential reality rather than as a manifestation of the world's holistic nature. Whereas the Platonic backhand has a history dating back to the Greeks, the Platonic forehand is more recent. To reach fully developed form, it required the assistance of powerful computers. This move starts from simplified abstractions and, using simulation techniques such as genetic algorithms, *evolves* a multiplicity sufficiently complex that it can be seen as a world of its own. The two moves make their play in opposite directions. The backhand goes from noisy multiplicity to reductive simplicity, whereas the forehand swings from simplicity to multiplicity. They share a common ideology – privileging the abstract as the Real and downplaying the importance of material instantiation. When they work together, they lay the groundwork for a new variation on an ancient game in which disembodied information becomes the ultimate Platonic form.
>
> (1999 12–13; original emphasis)

Scientism always seeks to render its abstractions, its metaphors as real. To say this is not all do dismiss metaphor as a means of describing the world. As China Mievelle (2011) says in *Embassytown*, a metaphor is a lie which tells the truth. All descriptions and models of reality *of whatever form* are inherently metaphorical.

Notes

1 Originally Althusser's structuralism which he destroyed in his magisterial *The Poverty of Theory* (1978) but also of course of post-structuralism and 'post-modernisms' in general. He explicitly exempted Bourdieu.

2 Desrosières's (1998) discussion of the generation of all forms of social (including economic) statistics recognizes the reality and causal powers of the data, but as an ANT he is reluctant to assign reality itself any role in the original constitution of the data.
3 Williams's (2021) discussions of Cause and Mechanism are particularly valuable.
4 Always look for the R-squared or pseudo R-squared which gives a measure of how much of the variation in the actual data set is 'explained' by the model. It is often not very much.
5 In our view, this distinction also applies in relation to ecological complexity.
6 A notable exception is the work of Peter Allen, a physicist trained in the Prigogine tradition, whose work is acutely sensitive to the possibility and potential of human agency. It is worth noting here that those who actually do empirical work, for example, Allen and Batty (1997, 2007), are much more sensitive to agency and its implications than those who merely construct abstract models.
7 Nicolis's (1995) introductory guide to complex systems is excellent and a really good way into the mathematical approaches.
8 Things are improving, and physicists, for example, Boulton et al. (2015), are dealing with social complexity in terms which make sense.
9 We agree that the development of computing systems' ability to handle enormous amounts of data was crucial. Byrne (2002) called this the emergence of the macroscope likening this to the enhancement of visual range provided by the microscope. In a way, our brains and the computers are at the very least assemblages and possibly complex systems in and of themselves which give us access to a range of reality which we could not access merely by our normal senses. This of course includes the statistical techniques which already existed and which we handle now with computing power rather than log tables and slide rules. It also includes the core of so-called Artificial Intelligence – learning algorithms working through networked emulations of the mammalian brain.

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3 Doing complexity science

Price stated:

> The complexity view is, in its most general articulation, is that modern sociology (and all of science) is in need of modification. By correcting inadequacies in our scientific paradigm, we may appropriately and fruitfully continue to do 'science'.
>
> (1997 4)

We agree. Doing science involves doing research. Science has to have empirical foundations. A central argument in this book is that science, in the general sense of systematic empirical knowledge about the world, is possible, and that social science has to be founded on an ontological understanding of the social as composed of intersecting complex systems. What we want to do in this chapter is first, challenge the view that simulation is *the* necessary basis of a complexity-framed social science, and second, use the complexity frame to challenge important 'moves' at the interface of social science and philosophy.

Confronting the limitations of simulation

Macy and Willer (2002) described a move in quantitative sociology.

'From Factors [underlying causal variates generated by multi-variate analyses] to Actors [simulated agents].' DeLanda takes this into the realm of complexity:

> Simulations are partly responsible for the restoration of the legitimacy of the concept of emergence because they can state interactions between virtual entities from which properties, tendencies and capacities actually emerge. Since this emergence is reproducible in many computers it can be probed and studied by different scientists as if it were a laboratory phenomenon. In other words, simulation can play the role of laboratory experiment in the study of emergence, complementing the role of mathematics in deciphering the structure of possibility spaces.
>
> (2011 6)

Let us see what the issue is through deconstructing a crucial statement of DeLanda:

> while an ontology based on the relations between general types and particular instances is hierarchical, each level representing a different ontological category, an approach in terms of interacting parts and emergent wholes leads to a flat ontology, one made exclusively of unique, singular individuals, differing in socio-temporal scale

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> but not in ontological status. … It is unclear to what extent Deleuze subscribes to this idea of a flat ontology. Some parts of his theory seem to demand such an ontology … Yet, elsewhere, he does seem to talk of totalities. Thus, while I view the realm of the social as a flat ontology (made up of individual decision-makers, individual institutional organizations, individual cities, individual nation states) and thus would never speak of 'society as a whole' or 'culture as a whole', Deleuze does talk of 'society as whole' and specifically of a 'virtual multiplicity of society'.
>
> (2006 58 and 89)

We agree that all the entities which matter in the social world have a reality and that it is inappropriate to apply a *solely downward deterministic* hierarchical arrangement as a description of them. DeLanda seems to allow for this when he talks of real entities operating at different socio-temporal scales:

> In a flat ontology of individuals, like the one I have tried to develop here, there is no room for reified totalities. In particular there is no room for entities like 'society' or 'culture' in general. Institutional organizations, urban centres or nation states are, in this ontology, not abstract totalities but concrete social individuals, with the same ontological status as individual human beings but operating on larger spatio-temporal scales. Like organisms or species these larger social individuals are products of concrete historical processes having a date of birth and, at least potentially, a date of death or extinction. And like organisms and species, the relations between individuals at each spatio-temporal scale is one of parts to whole, with each individual emerging from causal interactions among the members of populations of smaller scale individuals.
>
> (2006 153)

We agree with the treatment of social entities as real 'social individuals' but disagree with the assertion found in many discussions of simulation (see, for example, Sawyer 2005) that such higher-order entities emerge *only* from interactions among lower-level entities.[1] Such interactions are constitutive of higher-order entities, *but* higher-order entities can be the emergent product of interactions at their own levels and/or across levels. Forms of governance are a typical example. Individual humans as decision-makers play a part in, let us say, a re-organization of a nation-state's structures of local governance, but existing patterns – path dependency, legal requirements of transnational political entities, the pressures, and interventions of political parties, all have a role. We want to write structure into the process, and that is a crucial issue for restricted complexity based on simulations. Human actions carry the actions of institutions, but those humans act within the terms of the institution, and the consequent action is best understood as derived from the institutional level. For us DeLanda's social entities are *too* discrete at any level. This is related to Deleuze's and hence DeLanda's conception of boundary establishment through territorialization. Our notion, after Reed and Harvey (1992) and Cilliers (2001), of nested but interpenetrating systems with causal powers running in all directions seems more appropriate. We also do consider that there is a 'social' at whatever level from the smallest collective assemblage of human beings to the level of the world system as a whole. We have to recognize in a Durkheimian mode that we exist within a social world and that all the entities which are social exist also within that world.[2]

Sawyer (2005) has an interesting take on how to get beyond the kind of atomistic individualism as the source of emergence, which underpins, for example, not only the equilibric theories of neoclassical economics but also the potentially more open systems potential of, say, rational action theory. Despite this, he is much more restrictive than DeLanda in assigning reality only to the level of individuals. His focus is on the micro-macro link with the micro defined explicitly as a level constituted of individual human actors. He frames the argument by a contrast between 'sociological realists' and methodological collectivists:

> Many accounts of the micro-macro link have explicitly used emergence to argue that collective phenomena are collaboratively created by individuals, yet are not reducible to individual action. … Most of these accounts argue that although only individuals exist, collectives possess emergent properties that are irreducibly complex and thus cannot be reduced to individual properties. Some of these accounts reject sociological realism and instead take the weaker position of methodological collectivism. … Others defend the stronger claim that emergence can be used to ground sociological realism. However, emergence has also been invoked by methodological individualists in sociology and economics. Methodological individualists accept the existence of emergent social properties yet claim such properties can be reduced to explanations in terms of individuals and their relationships.
>
> (2005 63–64)

Here we want to return to the discussion of assemblages to argue again against the version of flat ontology proposed as the basis for treating all entities as only assemblages. It is not that we disagree with that idea of assemblage in and of itself and in particular with the version of it developed in commentary by Harman (2010). Certainly, DeLanda is deeply and explicitly realist, in a way which acknowledges the influence of Bhaskar, and there is little to quarrel with in Harman's argument to the effect that the apparent differences between them on 'flatness' are illusory (2010 180) since they agree on the reality of 'countless layers of larger and smaller structures [with] equal ontological dignity' and reject the compression 'of all of reality into the single plane of empirical givenness to human perception'. This is not our issue. Rather it is DeLanda's denial of the reality of the social as a nested complex system containing other nested systems within it. We are happy with what Harman identifies as 'a vision of the world as chain of ascending and descending compounds, each of them autonomous from the pieces that create them and equally independent of the wider contexts in which they are enmeshed' (2010 181). Or rather we are happy with the idea of autonomy if not of independence. However, the notion of ascending compounds does not seem to us to square with DeLanda's explicit denial of a social real. That said, Harman's commentary on DeLanda's theory of emergence seems to suggest that this denial is actually deeply contradictory to his general position:

> an assemblage tends to have retroactive effects on its parts. He [DeLanda] credits Bhaskar with this point: In DeLanda's own words: '… although a whole emerges from the interactions among its parts, once it comes into existence it can affect those parts. … In other words … we need to elucidate … the macro-micro mechanisms through which a whole provides its component parts with <u>constraints and resources</u> placing limitations on what they can do while enabling novel performances.'
>
> (Harman 2010 185; original emphasis)

To develop the significance of this in a way which points forward to both our discussion of social theory and 'the methodology of complexity', it is useful to reflect on Harman's explicit contrast drawn between the positions of Latour[3] and DeLanda:

> If Latour is ultimate philosopher of relations, for whom a thing has no 'accidental' manifestations but is thoroughly bonded to all its specific features here and now, DeLanda is the upside-down Latour: multiplicities are <u>non-relational</u>, and robustly remain whatever they are, no matter what their relations might be.
>
> (Harman 2010 176; original emphasis)

The idea that the social is constituted in large part relationally is both important in social theory and a basis for one way in which we can seek to describe emergent social complexity. Note here Emirbayer's (1997) argument for a 'relational sociology' and the value of social network analysis as a mode of social inquiry. We can reject the shallow ontology of ANT while retaining an interest in networks. For us, Reed and Harvey's (1992) conception of nested systems is the way to resolve what Harman sees as contradictory positions.

Teleology is back

The *Oxford English Dictionary* gives three meanings of this important word *teleology*, and two are relevant here:

1. The branch of knowledge or study dealing with ends or final causes; the study of phenomena which may be explained in terms of intention, design, or purposiveness rather than by prior causes.
2. (The presence of) purposiveness, design, or final causality in nature; the fact of being directed towards a goal.

Juarro asserts the relevance of the relevance 'teleology knocking on the door' in the history of science very neatly when she observes:

> in Darwin's notion of natural selection … context was given a causal role for the first time in centuries. And to make matters even more complicated, the strict determinism of classical mechanics appeared to break down as a result of the self-organizing consequences of a spiraling reflexive causality.
>
> (2011 149)

There is a lot in this point. First 'context' – that is to say, causality can only be understood both in terms of descriptions which are located in particular spatial and temporal contexts *and* context itself has causal powers, an idea which resonates very strongly with the notion of complex systems as permeable assemblages. Second, we have 'a spiraling reflexive causality'. Reductionism which denies not only complexity but reflexivity cannot cope with that. As Juarro says, we had to go back to go forward:

> Complex systems became more tractable with the reintroduction of the Aristotelian concepts of formal and final cause by making it easier to account for the contextual embededness and goal-directedness of organisms and organizations.
>
> (2011 161)

The key phrase here is goal-directedness. We must add social to that set of bio-chemical-physical, and to add social in all its forms is to add the potential for intentional action towards ends. That is where governance comes into play.

So what about other ways of thinking?

So far we have been laying out our own position, derived from Reed and Harvey and coloured by Cilliers's approaches. Now we turn to reviewing other positions, most of which might be described as informed by 'the new materialism'. First, let us use Smith and Jencks and Cilliers to dispense with the relevance of much of what goes under the general heading of 'post-modernism'. Cilliers explicitly adopts the post-modernist label to describe his approach, but he does so in a very particular way. He constructs much of his argument around a discussion of the nature of language which for him is a complex system in and of itself. In a sense, he follows the linguistic turn which has been so important in 20th-century philosophy and has profoundly influenced both the humanities, especially literary theory, and 'the human sciences' approach to the social world. Let us be clear that 'the human sciences' modality is not defined by the linguistic turn. Rather its central proposition, which we can trace back to Vico in the 17th century, see in the humanistic Hegelian young Marx, find expressed particularly by Dilthey and those influenced by him, notably Weber, and identify explicitly in social ontology of MacIver (1942), is that human beings have agentic powers and that this agency has profound implications for our understanding of the nature of social reality. However, the linguistic turn is important, but there are two versions – the weak turn and the strong turn. The weak turn is the position which informs classical hermeneutic textual interpretation and underpins the deployment of that method in the social sciences. Pawson and Tilley call this 'Hermeneutics One' in contrast with the 'Hermeneutics Two', which informs the strong programme. Both programmes involve interpretation, but as Pawson and Tilley put it, the weak programme works

> by being witness to the day-to-day reasoning of their research subjects, by engaging in their life world, by participating in their decision making, the researcher would be that much closer to reality. The hermeneutic approach, it was assumed, almost literally placed one in touch with the truth.
>
> (1997 21)

This is the fundamental meaning of Weber's term *verstehen*, and it explicitly informs both ethnographic and historical sociological method.

Hermeneutic method is always a mode of interpretation, but to understand the contemporary mode of employment of 'Hermeneutics Two', we have to set it in relation to structuralism and the rejection of that approach in the form of post-structuralism. Structuralism was in some ways a positivist programme in that it argued for a privileged understanding of aspects of reality for those who could identify the structural relations underlying any aspect of it. Where it differed from, say, logical positivism is that it argued for an incommensurability among different structures, most notably – if in the most contradictory of fashions – between the structure of language and the structure of reality. While this poses few difficulties for the literary interpretation of texts, it seems deeply contradictory for structuralist description of any other aspect of reality, for example, the structuralist anthropology of, say, Leach (1961), who does seek to deploy the method to understand real human societies and how they change.[4] Be that as it may, we might see the post-structuralist turn which informs the strong programme of Hermeneutics Two as involving a renunciation of any claim to privileged understanding.[5] Pawson and Tilley put it like this remarking that Hermeneutics Two

> starts from the point of view that all beliefs are 'constructions' <u>but</u> adds the twist that we cannot, therefore, go beyond constructions. It insists, in other words, that there are no neutral/factual accounts to be made of the social world.
>
> (1997 21; original emphasis)

Ignorant pomo has argued that the ideas lying behind chaos theory, that is, the impossibility of prediction due to the extreme sensitivity of the future trajectories of systems to initial conditions, validate its argument that no knowledge system can describe reality. This is not the position of those who have deployed post-modernist ideas in a coherent way in relation to complexity theory. Smith and Jenks explicitly denounce such simplistic post-modernism:

> Post-modernism cannot be a full-blown doctrine which says 'meta-narratives have lost their credibility'. It would be less contradictory and certainly more accurate to say that post-modernism holds sway in the social spaces where meta-narratives are disbelieved. These spaces may be large or constitute the majority outlook, for example, in the study of culture by European scholars. Or it may be little more than peripheral variation. This is usually overlooked by academics and is one reason why the scant resemblance of post-modernism to complexity theory is entirely illusory. The possibility of a transformation from post-modernism to complexity theory is open, but so long as post-modernism is academically promoted as a general theory and not just as a reaction to Modernism, it remains just another over-simplified meta-narrative which lacks credibility. … The illusion is aural; it depends entirely on this gross simplification: society human being-in-the-world, culture is primarily constituted in language. This another version of human as content-free, the paradigm in which nothing is known and everything literally is learnable.
>
> (2006 135–136)

Cilliers is equally clear in his renunciation of the passivity of the post-modern as pomo:

> An argument will be presented against the view that a postmodern approach implies that 'anything goes'. Instead the suggestion will be that the approach is inherently sensitive to complexity, that is acknowledges the importance of self-organization whilst denying a conventional theory of representation.
>
> (1998 112–113)

What Cilliers wants is an understanding of knowledge as local, contextual, specific in time and space. His discussion of the way he uses (no other word will do) Lyotard makes this very clear:

> Lyotard rejects an interpretation of science as representing the totality of all true knowledge. He argues for a narrative understanding of knowledges, portraying it as a plurality of smaller stories that function well within the particular contexts where

> they apply. … Local narratives only make sense in terms of their contrasts and differences to surrounding narratives. What we have is a self-organizing process in which meaning is generated through a dynamic process, and not through the passive reflection of an autonomous agent that can make 'anything go'.
>
> (1998 114 and 116)

Keen as Cilliers is on Derrida, and the way Derrida employs the idea of difference is central to much of his argument and general understanding, he nonetheless is resolutely realist when he dismisses the argument that a crisis of knowledge is the result of 'the disruptive activity of "subversive" theoreticians like Neitzche, Heidegger and Derrida' (1998 121). No, it is the complexity of reality and in particular the complexity of contemporary post-modern society which generates the crisis in knowledge. Reality has a voice.

In clearing the ground here, let us conclude with an expressed agreement with Smith and Jenks when they say:

> complexity theory can be contrasted with humanism, however much their language and analytic ambition overlap. Modernist humanism, especially its political forms, is confounded by the complexity of outcome. Post-modernism equates this complexity with the open horizon of interpretation and so diminishes material cause to the point of insignificance. Complexity theory resolves this non-linear interactive organization in the form of auto-eco-organization. This is not an answer but rather a general condition for framing inquiry.
>
> (2006 151)

The new materialisms

Coole and Frost (2010) identify the new materialism as a response to and rejection of the dominance of radically constructivist discourses, which have predominated in the post-structuralist period of cultural studies.[6] They identify (2010 6–7) three themes in this turn:

1. First there is an ontological turn which resonates with current developments in the natural sciences in relation to chaos theory and post-classical physics. In their discussion complexity gets a look in but they reproduce the common confusion of post-structuralists of chaos with complexity.[7] The key element in this ontology is that 'it conceives of matter itself as lively or as exhibiting agency' (2010 7).
2. Second, there is an engagement with a set of biopolitical and bioethical issues which we might consider as about assigning a different status to all non-human nature.
3. There is a turn back to an engagement with the realities of political economy which bluntly was almost entirely lacking in radical constructivism.

It is the first turn which matters most for us. We will come back to engagement with the realities of political economy, noting firmly that in many domains of social science and general scientific practice, it never went away, not least in relation to inequality on all scales. We might say, and do say, so what else is new? Historical materialism had an interesting inter-relationship with the phenomenological materialism of Sartre, De Beauvoir, and we might argue Merleau-Ponty and even Camus.

One of the key exponents of the new materialism, in the particular form of an object-oriented ontology (OOO), is Harman (2018) who modestly asserts that it provides us with a new theory of everything.

> since reality is always radically different from our formulation of it, and never something we encounter directly in the flesh, we must approach it *indirectly* (original emphasis). This *withdrawal* or *withholding* (original emphases) of things from direct access is the central principle of OOO.
>
> (2017; original emphasis)

> OOO means 'object' in an unusually wide sense: an object is something which cannot be entirely reduced to the components of which it is made or to the effects it has on other things.
>
> (2010)

The latter statement is made in relation to a discussion of emergence, and with the anti-reductionist element of it we have no quarrel. Neither do we disagree with the notion that entities in the world are not directly accessible, but that does not mean we cannot know them and that we cannot act on them. Although Harman is in many respects a trenchant critic of Actor Network Theory, the most clearly developed 'new materialist' framing in the social sciences, to a considerable degree what he is doing, is extending the agentic potential of the instrumental non-human actants in the construction of science as knowledge sets to all matter.

These are the ontological propositions of ANT's main exponents:

> No one would deny that the world is complex, that it escapes simplicities. But what is complexity and how might it be attended to? How might complexities be handled in knowledge practices, nonreductively, but without at the same time generating ever more complexities until we submerge in chaos? And then again, is the contrast between simplicity and complexity itself too simple a dichotomy? These are the questions explored in this book.
>
> (Law and Mol 2002 1)

A good set of questions, but we agree entirely with Smith and Jenks (2006) that the 'semiotics of materiality', the underlying foundation of the Actor Network Theory (ANT) which informs the work of Law and Moll and the contributors to their edited collection entitled *Complexities*, does not provide a coherent set of answers to them. The introduction of 'the material' into our arguments about construction of knowledge is a move in the right direction, *but* ANT's failure to allow for, or actually it would be better to say ANT's rejection of, a relationship between the real and our knowledge of it renders the foundation of their accounts of the complex world fundamentally flawed. Let us allow the ANTs to define their approach in their own words:

> actor-network theory may be understood as a semiotics of materiality. It takes the semiotic insight, that of the relationship of entities, the notion that they are produced in relations, and applies this ruthlessly to all materials and not simply those that are linguistic. This suggests: first, that it shares something important with Michel

> Foucault's[8] work; second that it may be usefully distinguished from those versions of post-structuralism that attend to language and to language alone.
>
> (Law 1999 4)

For complexity theory relations are real, although our constructions of them in terms of scientific description are certainly made rather than found. What status does ANT give the voice of the real? For Law at least not much:

> The world as a 'generative flux' that *produces* (original emphasis) realities? What does this mean? … in this way of thinking the world is not a structure, something that we can map with our social science charts. We might think of it instead, as a maelstrom or a tide rip. Imagine that it is filled with currents, eddies, flows, vortices, unpredictable changes, storms, and with moments of lull and calm. Sometimes and in some locations we can indeed make a chart of what is happening round about us. Sometimes our charting helps to produce momentary stability. Certainly there are moments when a chart is useful, when it works, when it helps us to make something worthwhile: statistics on health inequalities. But a great deal of the time this is close to impossible, at least if we stick to the conventions of social science mapping.
>
> (Law 2004 6–7).

Compare and contrast this with Cilliers:

> Boundaries [of complex systems] are simultaneously a function of the activity of the system itself, and a product of the strategy of description involved. In other words, we frame the system by describing it in a certain way (for a certain purpose) but we are constrained in where the frame can be drawn. The boundary of the system is therefore neither a function of our description, nor is it a purely natural thing.
>
> (2001 141)

Law allows for a momentary fixing of position. Cilliers, and this is very important, allows us to frame 'for a purpose', that is to say he here allows in the potential of action, *and* he asserts that in our constructions (because the setting of boundaries on a system is a crucial part of any scientific construction), reality has a say.

There are positive aspects to ANT. The negative aspect is the ontology. As with DeLanda, we are faced with a shallow ontology, indeed in the case of ANT with an ontology reduced really to an epistemology, a theory of knowledge construction, not a theory of the real. Note very well that to say this is not to deny the constructed basis of scientific knowledge. However, and somewhat bizarrely, there seems to be little recognition of the fact that what is constructed becomes real in itself and then has agentic implications. One particular use of ANT, Desrosière's (1998) excellent account of the history and nature of social statistics, actually does assert the reality of the statistics as generated without admitting the role of the socially real in relation to the construction of these reflexively agentic measurements. In the end, ANT is just another, albeit a sophisticated and to some degree interesting, version of conventionalism. That will not do for us.

Let us turn to what is often called 'the new materialism'. We find much to agree with in Coole and Frost's discussion of 'Practicing Critical Materialism' (2010 24 and subsequently) although it presents the substantive framings of Western Marxism in historical materialist form without a careful historical exegesis and there really is nothing new about those framings as a basis for social action. They have an undue attachment to Althusser who tried very hard to write human agency out of history (see what Thompson 1978 made of that).[9] However, they get some things very badly wrong. So (2010) 29) they say that for new materialists:

> the capitalist system is not understood in any narrowly economistic way [fine Byrne and Callaghan] but rather is treated as a detotalized totality that includes a multitude of interconnected phenomena and processes that sustain its unpredictable proliferation and unexpected crises, as well as its productivity and reproduction.
>
> (2010 29)

'Detotalized totality' is the Sartre of *Being and Nothingness*, and frankly we are puzzled as to what this means in relation to capitalism as the real generative system for our era. However, it is the assertion as to 'unpredictable proliferation and unexpected crises' with which we take severe issue. There has been nothing unpredictable or unexpected about the financial crisis of 2008 or the crisis in the relationship between capitalism at this stage of development and COVID-19 as an expression of the natural world. Historically, materialist Marxism could not predict when the first would happen and had paid too little (but not no) attention to the latter but that both were immanent in capitalism was plain and it was a matter of not if but when.

For us the way out of these intellectual labyrinths is always by instantiation – by turning to a real example and the one which we have lived ourselves is that of carboniferous capitalism, particularly in the form of coal-based capitalism.[10] As Clark and Jacks put it: 'the switch from a self-sustaining organic economy to a mineral resource depleting inorganic economy was central to the British Industrial Revolution' (2007 2) and hence central to the emergence in full destructive power of the Capitalocene. The Great Northern Coalfield of North East England was one of the earliest zones of industrial capitalism in the world, and coal transport was the origin of railway technology on a global scale. Coal mining in the UK which had enormous political and cultural saliency – in 1911 one in every 13 occupied adult males in the UK was a coalminer and many more worked in the rail and marine transport associated with coal – has gone altogether, in the aftermath of a massive defeat of industrial unionism in the 1984 miners' strike. Coal is absolutely material and was experienced materially both in production and historically in much of consumption. This was all of a social-economic-cultural *and* natural system of production, reproduction, relationship with the natural world: hewing coal put the pitman in direct contact with the material through his own body as he lay on his side with his hipbone in a cracket slot holding his body out of what was mostly a damp underlay and kived – undercut the coal face – before the shotfirer brought it down and he shovelled it into tubs for the hand putter to move to the pony haulage putter. The life of a colliery village revolved around the pit, and even in large towns with mines and coal shipping, coal production and consumption shaped lives.

However, in what way did coal have agency? Sure it made possible coal-based industry in extraction and though electric power generation – coal by wire. Its production shaped the nature of daily lives in production and reproduction. It was constitutive of the radical reformism which created welfare capitalism. Its production and the social relations established in its production shaped the possibility space for the development of the social world. But it did not act itself. It was acted on and around and shaped, constrained and made possible but was not in any way a direct actor. It had a relationship with complexity theory because it was coal-fired steam engines which led Carnot to develop an understanding of thermo-dynamics which brought complex systems into physics. And burning coal is the major factor generating climate crisis and threatening the continued future of human civilization on this planet. So in a way, coal has agency but an agency in terms of potential, not in terms of direct action. We are all for materialism but a human-mediated *and* enacted materialism. With much of the new materialism, we tend to agree (although very much not with laws' ontology of unknowable flux) even if we say so what else is new?

Unger asserted that

> a practice of social and historical explanation, sensitive to structure but aware of contingency, is not yet at hand. We must build it as we go along, by reconstructing the available tools of social science and social theory. Its absence denies us a credible account of how transformation happens.
>
> (1998 24)

For us the complexity frame of reference is precisely the 'practice of social and historical explanation' necessary not only to explain transformation but as a basis for making transformation happen. As Marx said, the point is to act in the world to change it. Why do we need a credible account of: how transformation happens? Because without such an account, we cannot inform our actions in order to achieve the transformations we want. All the variants of the post-modern programme deploy a radical pessimism *against* agency. Structuralism denied the reality of agency. Post-structuralism in whatever guise denies the possibility of informed agency to ends. There is a way forward. It depends on agency. Social science conducted in the complexity frame of reference can inform that agency. Of course, this makes social science ideological in the very real sense that programmes of knowledge can be used for particular ends. That commits us to a class-based conception of universalism. Under those colours we are happy to run.

Notes

1 It might be considered that the issue is not ontological but epistemological. In other words, micro-emergentists might admit the complex character of the emergent social but argue nonetheless that we can explain/know macro-social phenomena in terms of simple rules. We disagree. For us the micro-emergent tradition endorses a fundamentally reductionist account of how the world works and that is an ontological position.

2 We might more properly say a socio-ecological world since the inter-relationship between the natural and the social is particularly evident in relation to ecological disaster and the potential for catastrophic global warming.

3 Latour is one of the founders of ANT, although he has asserted that he is not an ANT and has a critical, one might even say flirtatious, relationship with the current.

4 Leach's account of political systems amongst hill peoples in Burma fits very well with a complexity-founded understanding of such change and its drivers.

5 That said, it seems to retain an understanding of human beings as the determined product of structures rather than in any sense as autonomous selves.

6 Their impact has not been trivial in other fields of science and practice but only as discourses. They have had minimal impact on the actual practice of investigation and none whatsoever on social actions informed by investigation. Foucault's emphasis on power is an exception here, but for us he says nothing that Lukes (2021) did not say better and in a more intelligible fashion.

7 That is, they see an inherent indeterminacy in chaos. First, chaotic systems are not inherently indeterminant. They are deterministic if we can measure precisely enough. Second, complex systems are not chaotic systems. Rather they are relatively stable over long periods.

8 Foucault is certainly the big baboon in the troop of les singes de France. For a coherent and radical differentiation of his approach from that of complexity theory, see Price 1997.

9 Although to be fair, they do deploy possibly the only useful idea Althusser ever had, that of the ideological state apparatus.

10 Both of us had grandfathers who were seamen in the coal trade, both killed in consequence of doing that job. GC had a grandfather who was a coalman – delivered coal to houses. DSB had a grandfather who was a coalminer alongside two great grandfathers and, albeit briefly and incompetently, his father. He is the first in four generations in a direct line since his family came out of famine struck Ireland not to have worked down a County Durham Pit.

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4 Evolutionary social theory

In this chapter, we focus on evolutionary social theories as theories that have sought explanation in terms of systems as units of social change. Even when not explicitly modelled, the language of evolutionary change runs through a broad swathe of social theory. Parsons's conception of social systems theory as an evolutionary process, for example, is based in a convergence of influences from a range of academic fields (1970). He identified his initial study of biology and his subsequent engagement with economic theory via Alfred Marshall and Pareto and sociological theory, particularly the work of Durkheim and Weber, as informing an evolving understanding of the social system.

Two points need to be made at the outset in defining the scope of a short chapter on the subject. First, we set aside all arguments of the biological determinist type as representing inadequate understanding of the social and base our examination closer to Williams's (1983) discussion of potential rather than simple determination. Second, and related to this, we are not attempting an exhaustive review of evolutionary theories but rather taking a route through some major accounts to consider how systems concepts can illuminate the study of complex social systems. We start from the point made very straightforwardly by Hirst that 'Any definition of the term "evolution" will entail a definite concept of evolution, a specific theoretical system, and a specific evolutionary process constituted by that theory' (1976 17). We will look at the explicit systems theorizing of Spencer, Parsons, and Luhmann to discover consistencies with our exploration of complex social systems. We will consider the resonances between complexity theory and historical materialism in relation to emergence. We will trace some of the principal debates to consider how complexity treatment of systems both resonates and differs in its concepts of cause, adaptation, mechanisms, and prediction that we have identified as core to complex systems.

Evolutionary social theory builds on a tradition in social thought since Aristotle has been framed more or less explicitly in terms of 'biologism' (Bock, 1955 129). Spencer, for instance, directly compared 'the kind of thing' a society is to an organism that grows to maturity, reaches limits, and divides or is overwhelmed. Given this heritage, organic evolutionary ideas have, not surprisingly, been prominent in social theory. Social and cultural evolutionary theory, however, has frequently used an analogy of gradual organic growth rather than the more dramatic shifts associated with natural selection (which even Darwin was initially reluctant to acknowledge). In the 19th century, debates on evolution in social theory were conducted in a dialogue with natural sciences. These founding ideas are apparent in Comte's study of social dynamics, but most explicitly in the work of Spencer who published an account of evolutionary progress in 1857. His theory related more closely to Lamarck than to Darwin, but he read and drew insights from Darwin's

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Origin of the Species and indeed coined the term 'survival of the fittest'. Evidence of this productive dialogue is also apparent in letters from Darwin to Spencer (Carneiro, 1973). Darwin attributed the inspiration for his theory of natural selection to reading Malthus's essay on population (Hirst, 1976). This cross-fertilization of ideas continued through the 19th century with concepts borrowed in both directions. Spencer, for instance, identifies the source of the concept of division of labour as originating in political economy and subsequently adopted in biological science (1967).

These discussions gradually ceased in response to the growing dominance of Newtonian science from which the distinction between the subject matter and methods of natural and social science emerged. Dilthey's distinction underpinned a shift from ideas of necessity and natural science models of causality that is reflected in the division between approaches in social theory of Comte and Weber. Differences in the character of theory-evidence links in Newtonian science that separated physics and chemistry from evolutionary biology have subsequently been challenged. Debates within evolutionary social theory reflect these differences about foundational questions of ontology and epistemology, and we can see their consequences in disputes about causation as 'historical' versus 'law like' as well as in the discussion of macro–micro relations.

System theoretic explanations

As systems theories, explanations with an evolutionary character have emerged alongside, rather than deriving from, natural science and continue to play an important role in social theory today. Rather than seeking a definitive narrative of theoretical development, we will focus on aspects of the work of Spencer, Parsons, and Luhmann to consider how they can contribute to a complexity framed explanation of continuity and change.

Defining evolution

In focusing on evolutionary theory we find that some useful axes for discussion have been identified in the commentaries by Hirst (1976) and by Sanderson (1990), each seeking to typologize a vast range of work in this domain. Evolutionary theories attempt to explain, over long historical periods, the nature and meaning of social change. In one sense, they are archetypal modernist attempts to create order out of a narrative of history, an ambitious project that inevitably gives rise to a range of interpretations whose categorization exposes fundamental ontological differences. Sanderson uses a distinction identified by Toulmin between evolutionary and evolutionist theory. Here, 'evolutionary' theories describe change in terms of successive historical periods (process) while 'evolutionist' theories interpret history as the outcome of transcendent patterns with a progressive nature and are therefore teleological. The distinction lies between a mechanism type account (that could be consistent with natural selection), identifying the causes of change but implying no ultimate resolution; and a history that suggests a master direction in social change. Confusion over this fundamental distinction is one of the major misunderstandings evident in discussions of specific evolutionary theories which Sanderson particularly identifies with Giddens's critique.

A notable attempt to combine the two processes was made by Sahlins and Service (1960, see Hirst 1976). They describe a specific adaptive process as embedded within a more general evolutionary movement towards successively higher forms of development. In this approach, 'specific evolution' identifies the creation of evolutionary stages by adaptive responses leading to new forms of differentiation. 'General evolution' describes a grand movement in a particular direction and reflects a tendency for differentiation to create successively higher evolutionary forms. The relationship between the two is inverse because potential is based on the achieved level of adaptedness of a system. The more perfectly adapted a society is to its environment, the less likely it is to be able to change implying a more linear universe than the one we have described. For complexity theory, the notion of inevitability denies emergence and is therefore untenable, and the incorporation of a teleological component is problematic. Both Toulmin's mechanisms type explanation and Sahlins and Service's discussion of specific evolution permit an interpretation in which causal relations describe contingent, system-environment interactions that are emergent and adaptive without prefigured outcome. Hirst's review of approaches that have used an evolutionary narrative is helpful. He suggests they can be categorized as three types:

1. Progress teleology explanations, identified with Spencer and Lamarck.
2. Histories of successive development which have no meta-theoretical principle underlying them such as that proposed by L.H. Morgan.
3. Darwinian, abstract general theory of natural selection which applies universally.

These distinctions differentiate a range of evolutionary explanations in terms of their explanatory principles, and while the categorization may be neither complete nor mutually exclusive, the identification of dominant type is useful. It is suggested that this failure to distinguish between the motor forces of change in these types reflects misunderstandings by theorists such as Giddens, whose response has been to reject evolutionary approaches per se (Sanderson, 1990). Particularly valuable for us is the distinction between cultural, developmentalist accounts and a mechanism of change explanation. We will explore the consequences for social theory below, but first we want to deal with the problems raised by directionality, usually conceived in the form of progress, that is firmly embedded in popular as well as some scientific discussion of social evolution.

Progress, teleology, and directionality

The first notion that we want to despatch is the idea of progress which raises a number of problems, not least, of definition. The concept of progress has been formulated in modernization theories as an inevitable process, involving improved adaptation of organizational and institutional structures to better control the physical and social environment. Historically, anthropologists and sociologists have formulated accounts of human history describing a general progressive development from lower to higher forms of social being. In the 20th century, this has been read as productive efficiency creating surplus based on technological advance. Such accounts expose two fundamental sets of assumptions. First, they present unilinear renderings of history as an onward march with specific direction, and second, they present their ethnocentric value judgements as universal (see Granovetter 1979). We can read both problems in the works of Spencer and Parsons, alongside an awareness of causation as complex. Despite criticism of the teleological dimension of Spencer's theory, for instance, his account of structural differentiation, leading to ever greater heterogeneity, leaves open the way for contingency. 'Every active force produces more than one change-every cause produces more than one effect' (1868, p37). Parsons was aware of it and claimed to avoid this pitfall by achieving an objective basis for his theory in terms of adaptive capacity:

I have tried to make my basic criterion congruent with that used in biological theory, calling more 'advanced' the systems that display greater generalized adaptive capacity (1966 110)

Yet there is clear ethnocentrism in Parsons's judgement of what constitutes an advanced society/profession. Parsons saw American society as having reached the furthest point in evolutionary progress, based on weakening ascriptive status in industrial and democratic change, diversity of ethnic and religious base, and its manifestation in the evolutionary concept of citizenship developed by T.H. Marshall (1950). Parsons's judgements represented the dominant ideas of their time whose ideological status, as Williams (1983) pointed out, was difficult to recognize. Not only are such theories frequently unknowingly ethnocentric and gendered, they pose insuperable problems for any attempt to categorize evolutionary stages. Rank-ordering societies in terms of any notion of efficiency is clearly questionable, and (setting aside the problem of defining a society's problems objectively rather than in terms of its own priorities) the notion of adaptive capacity remains impossible to empirically document or measure (Granovetter, 1979).

Not all teleological explanations involve ideas of ultimate progress. Equal weight is given in some theories, to darker prospects for humankind. Weber's evolutionary account predicts progression towards a more negative future in rationalization/bureaucracy, while for Spencer, society continues to develop in a progressive way until it reaches its own system limits and hence its ultimate demise.

So, our first distinction lies between progress and other accounts of directional change, inherent in the concept of evolution of which progress explanations are more appropriately seen as one type. Sanderson (1990) reminds us that although critical of the evolutionary frame on these grounds, Giddens developed a directional theory of time–space distanciation. Perhaps we can then formulate the issue more precisely, in Hirst's terms: 'There is a vast difference … between recognising teleological elements in social practice as an object to be problematised and explained, and constructing explanations of social relations in terms of teleology' (1976 125).

If, instead of regarding history as a narrative of single direction, we recognize direction as produced through iteration of systems processes, we can retain the notion of direction while dispensing with linearity. Anthropologists such as Marvin Harris have treated history as a 'mixed bag' raising the question of whether there is any discernible directionality in the aggregate of all changes. In this view, history is appropriately represented as multiple adaptations made in response to multiple stimuli, both originating in the environment and, internally, in the reflexivity of agency. Change with direction is an aspect of the irreversibility of systems over time which suggests a non-teleological directionality of evolution, paralleling the approach we identified above as Darwinian. It implies no necessary tendency and no master causal process. The notion of a hierarchy of social forms disappears, as do ideas of purpose. This provides us with a notion of directionality, inherent in any explanation which considers each historical period to be emergent from the conditions of the one before.

Working with evolutionary social theory in dealing with how change can be understood as ordered presents us with an equally fundamental question of what constitutes the unit of change. Thus far we have used the language of system/environment as taken for granted concepts to allow us to focus on the kinds of change that evolutionary theory seeks to explain. We need to consider to what in the social world the concept of system corresponds. We can draw on the work of Spencer as the most explicitly system-focused theorist, writing in terms of history, structure, and function. Parsons took the discussion forward in explaining system reproduction as the work of actors, albeit in accordance with functional norms. We will consider Luhmann's account of the system because it is constituted in a very different form in terms of both its components and its temporality. In each, system theoretic ideas are produced that provide insights and challenges for accounts based in complex dissipative systems.

Society as a system: the concept of system

Spencer established evolution as a process involving society as a system. Indeed it was Spencer, rather than Darwin, who introduced this way of conceiving of evolutionary change. The order-seeking property of Spencer's concept is immediately apparent in defining systems as having an internal unity that distinguishes system from environment: 'a whole whose parts are held together by complex forces that are ever re-balancing themselves – a whole whose moving equilibrium is continually disturbed and continually rectified' (Spencer 1868 p544, quoted by Carneiro 1973 P84). Spencer has been criticized for concentrating on equilibrium, but such notions of system and of euquilibria being broken and re-established imply inherent dynamic character as systems operate by 'seeking through adaptation to establish and maintain an equilibrium in the face of a changing environment' (Carneiro 1973 p80). Moreover, systems and their subsystems operate in the context of an environment, which is itself characterized by change. Social change can be explained by the social system developing through a progressive process of differentiation to create increasingly complex structures. These structures become stratified in response to specific functional imperatives. Differentiation leads to increasing specialization, but their interdependence is essential because the framework maintains the coherence of the whole. Spencer describes this by analogy with biological processes:

> Evolution establishes in them both [social and organic structures] not difference simply but definitely connected difference – differences such that each makes the other possible.
>
> (p5, 1876)

In discussing systems through analogy with the biological organism, Spencer's concept of system can be challenged for accepting the constraints that organicist theorizing imposes (Bourdieu et al, 1991). However, the analogy is not held to be perfect, and Spencer clearly differentiates social and biological systems, most importantly in terms of the function–structure relationship: 'In the one, consciousness is concentrated in a small part of the aggregate, In the other it is diffused throughout the aggregate; all units possess the capacities for happiness and misery' (1883. P479). This qualification has significant implications for the kind of functionalism that can be developed. The aggregate can be treated as a 'whole' in terms of cultural norms or collective conscience, but there is no organizing principle, no simple hierarchical set of rules to guide its development. The internal development of the whole is a subject that must still be explained. In itself, of course, this raises one of the most vexed questions in social theory, the relationship between structure and agency.

Founded in a rejection of history analysed in terms of individual agency alone, Spencer sought social explanation in terms of regularities across time and place that could nevertheless be abstracted to provide law-like statements. His conception of evolution was concerned with mechanisms of adaptation, and the degree to which this was conceived of as unilinear, necessary, or endogenous has been the subject of ongoing debate. Social evolution advances through successive forms from primitivism to civilization, and this is associated with growth in intelligence, language, and hence social progress (1883). The relationship between structure and function sees the two as intimately interrelated, and his discussion of evolution is particularly concerned with these relations (Carneiro, 1973). More interesting for us is that we can detect, albeit not a particularly developed, sense that the whole is emergent: 'Hence arises in the social organism, as in the individual organism, as life of the whole quite unlike the lives of the units, though it is a life produced by them' (1883, p479).

Systems are constituted by relatively permanent structures, existing in a relationship with their constituent units but not delimited in time by them:

> By a catastrophe the life of the aggregate may be destroyed without immediately destroying the lives of all its units: while, on the other hand, if no catastrophe abridges it, the life of the aggregate immensely exceeds in length the lives of its units.
>
> (ibid.)

Spencer introduces ideas of criticality, again, through comparison with biological organisms. Teleological criticism is supported in the inevitable process of complexification and differentiation, which eventually leads systems to exceed their own limits and break down, returning to a simple condition. Dissolution occurs when environmental change deprives the social system of the necessary energy to maintain established structures and functions, a process initiated when the environment pushes the system beyond the 'impassable limit' of evolution (Carneiro, 1973 94).

The linearity apparent in Spencer's work arises here in the proposition of reversibility, as Spencer clearly saw in the evolutionary process an inevitable return to simpler structural forms. Translated into the language of dissipative social systems, increasing system complexity represents the ability of the system to import energy from its environment. What is absent from Spencer's conceptualization is a concomitant system ability to export internal disorder to its environment. This absorption rather than exchange of energy explains the basis of Spencer's system as having inevitable limits. It is a major departure from complex dissipative systems, and in this respect the charge of teleology against Spencer seems justified. Carneiro has defended this charge, however, arguing that 'only when evolution is regarded as a unique series of historical events can it be said to be irreversible. And then the argument is probabilistic, not deterministic' (1973, p92). If evolution is defined, as Spencer does, as 'a succession of general states of a system', the notion of moving to simpler forms of organization, although manifest very differently empirically, could be consistent within complex dissipative systems. The qualification would then be that dissolution would take place only along this dimension of complexity/simplicity. It would not be inevitable, neither would it re-establish prior forms in any sense empirically recognizable as a 'return'.

However, we interpret Spencer's work, his contribution to theory of an evolutionary form was significant for its discussion of the nature of systems, subsystems, and environments in processes of interaction which involve critical axes of transformational change as well as adaptation. He drew on the historical record to develop an empirically related discussion of systems change in the real world that inspire a useful dialogue with complexity concepts.

Parsons took up the explicit theorization of systems in the early to mid-20th century to move the discussion in an idealist direction. He began by rejecting Spencer but returned to his work as his own systems theory developed in line with a gradualist neo-Darwinian synthesis (Harvey and Reed 1994). He attempted to delineate the basis of relationships between subsystems, framed as the symbolic systems constitutive of culture. Like Spencer, his early studies in biology are apparent in his work, and he distinguished his interest in the physiological form of the concept and in homeostasis from Schumpeter's conception of physico-chemical systems. Parsons directly addressed social and cultural evolution as exhibiting 'continuities with organic evolution' (p851, 1970).

> developments in biological theory and in the social sciences have created firm grounds for accepting the fundamental continuity of society and culture as part of a more general theory of the evolution of living systems. One aspect of this continuity is the parallel between the emergence of man as a biological species and the emergence of modern societies.
>
> (Parsons 1971, p. 2)

Parsons (1951) used organic analogy quite extensively to model relationships between roles and functions within organizations. As in Spencer, differentiation plays a major explanatory role as structures progressively divide functionally to create qualitatively different units within the system. Each unit evolves greater internal complexity, and the process of increasing structural differentiation brings about greater independence within subsystems over time. It is here that Parsons discerns the increasing individualism apparent in modern society and the concomitant need for integrative mechanisms to solve problems of anomie.

While Spencer identified structure as developing functionally, giving rise to new systems with new problems of adaptation, Parsons explained system functioning through an account of cultural systems, incorporating the dimension of action. He drew on both Durkheim's ideas of collective conscience and Weber's focus on the social action of individuals in communication of information, symbols, and meaning to achieve an account of the action of individuals fulfilling roles. He described social systems as the outcome of interactions between the individuals that comprise them. Although analytically distinct, systems interact in a hierarchically ordered form in which the individual's role is defined and governed by shared rules of a normative culture. The cultural system sits at the top of a hierarchy and feeds down through community, polity, economy, from society to personality. Individuals are then conceived as performing multiple roles across subsystems which are complex and interacting. These roles are institutionalized through a process of allocation which, if successful, results in system integration. The social system is internalized in the individual through normative culture, which is highly functional, and it is this form of explanation that has resulted in criticism of Parsons's over-socialized agents. Subsystems must be both interpenetrating and interdependent. Distinguishing analysis of the structure from process, Parsons used structural forms to identify specific points in evolutionary progress, based in four universals of technology, language as a basis for communication, kinship rules, and religion.

Society is conceived as a system comprising subsytems, each of which is orientated to achieving defined goals. Structure and function are related through differentiation creating subsystems in his formulation: AGIL (adaptation, goal attainment, integration, and latent tension management), which together represent the functional prerequisites for the systems' survival. Economic, political, social, and legal subsystems interact for economic adaptation to sustain political goals, achieve social integration, and supported by formal legal process facilitate the maintenance of the system as a whole.

Parsons's early training in biological theory is evident in his approach: 'action process involves combinations of factors that have different functions for the system in which they are combined … one major aspect of these functions is control of the cybernetic sense' (1970 p850).

Systems exchange symbolic information with each other to remain in equilibrium. The assumption of a homeostatic loop in which stability is restored is a major point of departure between Parsons's conception of system and a dissipative conception. Parsons used his 'analytical realism' (1970, p830) to formulate concepts of system and subsystem, drawn from Pareto. His work primarily focused on internal environment and system stability and showed awareness of, but did not develop, a dynamic analysis of social process. It is here that Parsons identifies the genesis of his concern with the relationship of structure and function (ibid, p831), although he rejected the structural-functional label arguing that they operate at different levels.

> It became evident that function was a major general concept defining certain exigencies of a system maintaining independent existence within an environment, while cognate of structure, as a general aspect of such a system, was process.
>
> (1970, p849)

Taking an approach through cybernetics, Parsons incorporated stability and change in a theory that both maintained the distinction between the processes of pattern maintenance essential to system continuity and those of structural change 'as broadly analogous to the basic biological distinction between physiological processes by which the state of the individual organism is maintained or changed and evolutionary processes involving change in the genetic constitution of the species' (ibid., p851).

This highly abbreviated description has sought to highlight how relevant concepts of systems and subsystems are deployed in Parsons's theory. Parsons's claim to have made an original contribution is focused on two endeavours: the specification of the nature of systems and their relationships, and the systematic development of a theoretical framework for their analysis. His over-emphasis on equilibrium has been widely criticized for underplaying the importance of conflict, while his focus on the ideal level seems to have led him to underestimate the significance of the material to social systems. The problem of the intersectionality of systems is not addressed (Walby, 2007). These criticisms are significant, but perhaps more important for our discussion is the charge of reification that has also been levelled at Parsons's theory.

> For Parsons what adapts is a social system, or at least one of its physical subsystems … it presumes the existence of something that cannot logically have existence. It presumes the actual existence of a social system as a thing apart with 'needs' and 'imperatives' of its own.
>
> (Sanderson, 1990, p20)

This difference over the ontological status of systems is fundamental and is one we will return to later. Parsons's rejection of it, however, resonates with complexity in arguing that his intention was to develop a relational theory. This

> very central concept depended heavily on the conception of the abstractness of the subsystem under consideration. Thus a 'society' is not a concrete article but a way of

> establishing certain relations among components of 'action' that are distinctive relative to the manifold of concrete reality.
>
> (1970, p839)

The success or otherwise of this attempt does not concern us here. The relational basis of systems thinking is a clear thread that can be drawn through social theory, however, and we will return to it in the chapters that follow.

We now turn briefly to draw on Luhmann's approach to systems because, while his debt to Parsons is clear, Luhmann takes the discussion of systems in a radical constructivist direction. Following Spencer and Parsons in seeing evolution as inherently a concept dealing with systems change and reproduction, Luhmann (1995)proposes a more complex system conception. He develops his account of autopoietic social systems as complex self-referential systems that reproduce themselves through communication. Everything that can be defined as social must be explained in terms of interaction and communication rather than individual agents. Rejecting dualist explanations of structure and agency in which the individual is the ultimate system unit, systems become systems of communication and (following Maturana and Varela) they are autopoietic, operatively closed in the sense that environmental pressures create internal perturbations within the system but do not in any way determine the system's response. The system adapts to its environment in a way consistent with its own internal conditions. The relationship between system and environment, therefore, is one in which the environment might 'trigger' a response but cannot determine its nature.

The 'system makes the difference between system and environment' (Rasch and Wolfe 2000, p36) so that 'reality' is effectively constituted by the system's self-description. 'The question whether it is the world as it is or the world as observed by the system remains for the system undecidable. Reality may be an illusion but the illusion itself is real' (ibid. p37).

The system, then, is defined by the network of its production processes rather than by its components. System operations are dependent on structures which are, themselves, produced by its operations. The core element of the social system, then, is communication, and each subsystem determines what communication is relevant to it:

> A social system comes into being whenever an autopoietic connection of communication occurs and distinguishes itself against an environment by restricting the appropriate communications. Accordingly, social systems are not comprised of persons and action but of communications.
>
> (Luhmann 1995, quoted in Mingers 1996 p288)

Communicative events are constituted in tripartite information, utterance, and understanding relationship, and it is in requiring the third element that the process can be described as social. This defines interaction in that each communicative event involves a decision to accept or reject the communication. These communicative events are the means by which systems reproduce themselves. They are immediate and have no temporal duration so that there is no possibility of reified structure. Autopoiesis takes place entirely at the level of communication, and the 'agent' in conventional terms does not constitute a relevant unit.

Although the genesis of Luhmann's system can be discerned in Parsons's work on communication, people, their consciousness, and their actions alongside the physical environment constitute the environment of the social system rather than being components of the system itself. He respecifies the system–environment relationship and develops the account of systems responding to triggers rather than being determined by external change. It is a theory that does not explicitly deal with power relations and one that operates at a deeply abstract level.

Luhmann's theory is based on several challenges to traditional systems theory that are usefully summarized by Knodt in her foreword to *Social Systems* (1995) as:

> 1. the principle of a unified, autonomous subject 2. The idea of the social as a derivative sphere of intersubjectivity, 3. The corollary of communication as an interaction between subjects, 4. The notion of communication as a transmission of mental contents between separate consciousnesses and 5. The corresponding idea of language as a representation of such contents.
>
> (Knodt,1995 pXXV)

Society is, functionally, world society, structurally differentiated, existing within a complex environment. As with Parsons, but in very different form, complexity increases over time through segmentation and stratification so that subsystems become autonomous and self-referential. Society is then 'the indeterminate outcome of the interaction between these independent but interdependent domains' (Mingers, 1996 p285).

The understanding that society is no longer a controlling centre but becomes this indeterminate outcome addresses the difficulty of society as a system that Parsons and other traditional systems theorists have encountered. The representation of a single, all-embracing, system with component subsystems creates difficulties for both conception and empirical work in global society (Mingers ibid.).

Luhmann's approach has been criticized as suppressing the time dimension, collapsing synchronic and diachronic processes (Reschke, 2005). This is a significant issue for complex dissipative systems which have history and are hence not engaged in circular processes. If this time dimension is explicitly added, the circles of self-referencing elements in systems theory diagrams become 'spirals' that move through time and 'characteristics space' of (more or less changing) actors and objects. The more iterative characterization of system change that Reschke proposes is of course more consistent with complexity, and it is one that we will develop further.

Thus far we have identified thinking about systems in terms of their constitution and relationship to environment, and this has brought structure and function to the fore as core ideas of reproduction. The mechanisms of reproduction that occur as adaptive responses within an overarching process of evolutionary change have been dealt with rather differently in these theories revealing core ontological differences.

Adaptation, adaptedness, and emergence

A central distinction needs to be made between adaptation and adaptedness as frequently confusing two very different ideas (Sanderson, 1990). While adaptation describes a process within which the system changes to respond to its environing conditions, seeking equilibrium; adaptedness describes how far any social pattern is optimal for the individuals within it. The distinction between seeking and gaining equilibrium that we raised in Spencer's work is relevant here. It is a fundamental distinction for complex dissipative systems because to conceive such a condition as possible would imply stasis. It is difficult to imagine an empirical social system that could conform to such a description, and, of course, even Parsons, in seeking to explain reproduction in terms of order, recognized no ultimate evolutionary state could be achieved. We see its more recent manifestation, however, in the 'end of history' account proposed by Fukuyama (1989).

Adaptation is central to both Spencer's and Parsons's work. Spencer describes it as a process for 'bringing a system into equilibrium with its environing conditions' (Carneiro, 1973, p84). Non-linearity is evident in the distinction between processes of direct and indirect equilibration. Direct equilibration is proportional and gradual, while indirect equilibration is the system's more dramatic response to initial conditions. In the latter, when system change is impeded by structural rigidities, the whole breaks down and is replaced by more flexible systems. System flexibility is therefore critical to ensure survival, and the relationship between evolution and equilibration is conceived as contingent. It is an implied acknowledgement of the importance of adaptive capacity rather than adaptedness as a characteristic of the system.

We identified the discussion of differentiation and increasing complexity in Parsons as the adaptive response of goal-oriented systems. Parsons seems to suggest that systems can, in the process of evolution, develop this adaptive capacity. A system's 'adaptive upgrading', is core, constituting its longer-term flexibility and adaptability. These forms of structural differentiation constitute the basis of the typologies he then constructed to define the place of societies in the evolutionary hierarchy. Parsons was primarily concerned with adaptedness, and he focused less on the system dynamic explanation that could explain how societies passed from one stage to the next. Adaptive capacity is a hallmark of advancement that indicates not only the ability of the systems to shape itself, as Luhmann's closed system implies, but also 'includes an active concern with mastery, or the ability to change the environment to meet the needs of the system, as well as the ability to survive in the face of its unalterable features' (see Granovetter 1979 p499). So although evolutionary progress is not inevitable but depends on the resolution of problems posed by differentiation, an assumption is made that short- and long-term efficiencies are entirely consistent. As Granovetter has argued, the consequence of this kind of thinking in empirical research was to encourage some in political theory to try to develop scales to measure adaptability, while in economics it further supported modelling on the basis of rationality.

The concept of emergence, a fundamental dimension of realist-based complexity, is a familiar notion in social theory. It is evident in Durkheim and we find signs of it in Spencer and Parsons. While Parsons initially rejected emergence at a social level because of his commitment to explanation based in individual action, the social re-emerged in his later work. A more developed notion of emergence is evident, however, in historical materialism, where we find it in Engels's discussion in *Anti Duhring* (1877), as the transformation of quantity into quality, one of two laws of evolutionary change. The concept of differentiation in increasing social complexity is the basis of the example of population pressure in which structural complexity increases to match the increasing demands made on the system by a growing population. Population pressure gives rise to fission, social structures can be suddenly complexified by reaching critical size, and indeed this is the mechanism Spencer identified as the basis of reversion to simpler forms when critical limits have been exceeded. We suggested that the notion of ultimate breakdown results from the absence of a dissipative relationship of exchange between system and environment. Carneiro (2000) pinpoints this process in terms of systems, working within 'elastic limits' which, when overstretched, give rise to major structural change. Based in supervenience and irreversibility, historical materialism has considerable potential for complexity theory. The importance of an emergentist explanation becomes clear.

Historical materialism

Our critique of evolutionary theories so far has focused on the work undertaken specifically on systems concepts either as a counterpoint or as a basis for analysing social systems with complexity theory. An important strand neglected, in consequence, has been explanation that constructs history in terms of mechanisms based in the material impetus of productive forces and relations that frame complex realism. Change is modelled, not in the inevitable, teleological account that critics suggest is evident in Marx, but rather as historical, the contingent result of processes in which structural inequalities are practices operating in an interpretation of determination offered by Williams (1980). Systems are reconceived as multiple and intersecting based on a reading of Marx that rejects accusations of technological determinism and developmentalism. The evolutionary dimension of Marx's writing involved both material and relational elements as the motor of change. The position is very clearly argued by Holland, and we quote extensively from his debate with DeLanda on this aspect of the nature of Marx's theory:

> The capitalist system need not – indeed should not – be conceived as a social totality (of the kind DeLanda assumes Marxism must entail). There are other important systems or abstract machines besides capitalism informing social relations, including systems of racialisation, genderisation, and so forth. And it must be an empirical question whether – or the extent to which – one system becomes dominant with respect to others (and not a foregone conclusion, as in even Althusser's invocation of the 'lonely hour of the last instance' [1969: 113] in which the economic system would always determine which system is dominant). As DeLanda proclaims, From the perspective of a bottom-up methodology, it is incorrect to characterize contemporary societies as 'disciplinary', or as 'capitalist', or, for that matter, 'patriarchal' (or any other label that reduces a complex mixture of processes to a single factor), unless one can give the details of a structure-generating process that results in a society wide system.
>
> (1998: 271)

Systems are historically emergent from the interactions of relations within and between them. Moreover, Holland argues, no single descriptor is adequate when systems are interpenetrating. What Marx and, in following him, Deleuze and Guattari provide is

> *a bottom-up account of the capitalist system as a self-sustaining autocatalytic loop growing by positive feedback and probe-head drift.*
>
> (Holland 2006 195)

While challenging DeLanda's interpretation, Holland argues that its value lies in reminding us of the dangers of allowing a structuralist interpretation of Marx,

> so out of step with the best understanding of material processes in history available today, thanks to important developments in nonlinear mathematics and science. Through lenses offered by De Landa, Deleuze and Guattari and complexity theory, we are now able to read Marx's accounts of capitalism and primitive accumulation in a new light, and see Marxism as both benefiting from and contributing to a novel, nonlinear version of historical materialism.
>
> (Holland, 2006 p195–6)

The considerable promise of historical materialism has been discussed in Reed and Harvey's work. Its utility in reconceptualizing systems intersectionality in terms of social inequalities of gender, ethnicity, and class is further developed by Walby (2007).

An attempt to incorporate the insights of historical materialism in an explicitly evolutionary systems theory has been made in the theory of evolutionary materialism developed by Sanderson (1994). This has sought to respond to criticisms of evolutionary systems theory using a systems theoretic account that recognizes the motor forces of change in material relations and activities. Sanderson (1994) identifies evolutionary materialism as a theoretical strategy:

> a highly abstract set of assumptions, concepts and principles designed to serve as a broad theoretical guide to explaining empirical reality. Its purpose is to function as an orienting device for the formulation and empirical assessment of theories in much the same way that we have identified complexity theory as a frame. As such it contains numerous theories which are linked and specific propositions (or sets of propositions) designed to explain specific phenomena.
>
> (p48, 1994)

Based in an agreement with Harris's materialist evolutionary theory, Sanderson develops a theoretical approach which he claims can explain the most significant events in world history. In a move consistent with Holland's discussion above, he emphasizes that, rather than a single evolutionary process, we are seeking to document a considerable range of evolutionary processes underway at any point in time:

> Social Evolution involves processes that occur within social systems of all levels, for example dyads, age sets, kinship groups, social classes, complex organisations, societies, any of the institutional sectors of societies, and various types of inter-societal networks.
>
> (1994, p50)

This seems a promising frame in seeking to identify mechanisms of both stability and change in systems in relation to empirical evidence, which incorporates power. The theory is based in an understanding of systems as non-linear and non-teleological, in which the principal causal factors at work are the material conditions of existence; the theory proposes that modern evolutionary change is driven by the accumulation of capital. Notions of progress or regression become empirical questions based in historical period and particular dimensions of social life. Challenging Giddens's claim there is a lack of concept of agency in evolutionary theory, Sanderson points to materialism as considering the course of social evolution to be driven by individuals acting to further their own interests but as bearers of inequalities in power which means that the consequences of their actions are often the unintended (1997 p104). Social structure, then, is a 'product of human intention but it is not an intended project' (p399, 1999). Although not explicitly stated, we can read this as an explanation of systems interacting in ways that have both mutually advantageous and contradictory effects giving rise to inevitable non-linearity.

Sanderson has used this theory to frame an extensive empirical review of historical change; however, two related points of departure from a complexity-conceived system are apparent: the significance of emergence, and the methodological individualist approach that Sanderson adopts.

The absence of a concept of emergence is apparent in its starting position that social evolution is only the response of particular individuals located at a particular point in time and space to the conditions that they face. It leads to a reductionist formulation with a major claim, that grounded in different motors of change (individual agency versus random variation), the social has a predictive quality that biological systems do not. What is problematic in Sanderson's account of systems, then, is the lack of conception of a relationship between contingency, history, and therefore, emergence, which leads him to predictions based on Kondratieff economic prosperity. While much in Sanderson's account accords with an ontology of complex social systems, the denial of emergence and consequent commitment to the predictive capacity of the social is a major point of departure for us.

The theory serves a second purpose for us in drawing into focus a further and increasingly significant body of work in social theory on co-evolution. Returning to the evolutionary account of systems as interpenetrating that Holland identified in historical materialism, we can incorporate this dimension of systems theory in our review. This work has developed both within the social sciences as well as across fields of biological and social science in the discussion of co-evolution of genetic structures with cultural forms (Boyd and Richerson, 2005). In the latter, the concept of co-evolution seeks to incorporate biological and social fitness in terms of an interaction between genes and culture. Dual inheritance theory describes how the intertwining of cultural and genetic characteristics can bring about effects that contradict genetic fitness (2005). Cultural explanations have also been drawn on to explain individual behaviour that is orientated towards group/other survival rather than individual genetic survival through conformist transmission of group values (Dietz et al, 1990). In policy terms, it has underpinned research into decision-making processes in co-evolving physical and social systems (Gerrits, 2008). How these interactions are formulated has consequences for the way that we think about structure and agency as well as macro- and micro-level processes that we will develop in the next chapter. Based in accepting Carneiro's discussion of reversibility alongside Sanderson's argument that genetic evolution in humans is not a relevant temporal scale, we nevertheless have to acknowledge the co-evolution of ecological and social systems means that it would be impossible to replay history in the way that Sanderson suggests.

We have foregrounded the problems raised in Sanderson's methodological individualist approach here not only because of its effects in his particular theory but because it is reflected more widely across the social sciences. The problem is replicated in Guala's discussion of the foundation of the evolutionary programme in social philosophy (2010) in developing a rational action account based on knavery/game theory, and of course in evolutionary economics founded in Campbell's random variation epistemology. While this model of change, in contrast to mainstream economics, incorporated cultural change and institutional contexts, it relied on a neo-Darwinian random selection mechanism to suggest best outcomes, thereby failing to recognize the significance of power and conflict or the possibility of other forms of action. These approaches have been challenged from within institutional economics where the dominant rational actor approach has been questioned (Hodgson, 2006) while economic historians such as David (1985) have long identified the significance of path dependence as disrupting rational development. David argues that it is only through the study of historical dynamic processes that we can adequately understand economic change. Across social theory these models of human action and interaction have been deeply contested, yet they continue to appear unqualified in new fields of inquiry such as agent-based modelling.

Conclusion

The purpose of this chapter has been to look at how the character of explanations in social theory can contribute an evolutionary account of social change in a complexity-relevant way. We have specifically identified some divisions in social evolutionary theory because of their consequences for the way that systems concepts have been deployed. The discussion of reproduction and system stability in Spencer and Parsons and Luhmann's focus on systems as autopoietic have brought different ways of thinking about system-environment relationships. There is a debate about the degree to which these theorists are developmentalist, teleological, or employ ordinary causal explanation as well as the particular relationships envisaged between cultural/ideal and material factors. We are not seeking definitive position on these debates, rather in the dialogical mode of theory development, we hope to identify some grounds for a conversation with complexity here.

While Spencer, Parsons, and Luhmann dealt with systems differently, the functionalism inherent within their approaches appears to offer lesser explanatory potential than complexity theory's open systems approach. Evolution can be heuristically described as a single process, but the models of evolution developed in social theory recognize interactions of short-, medium-, and long-term processes of adaptation in which co-evolution brings change, which is not usefully defined either as 'progress' or 'regress'. We used Carneiro's analysis of Spencer to support an interpretation of social systems as dynamic wholes interacting with their environments, including other systems. In such explanations, the notion that there could be any completion to evolution is clearly inappropriate in assuming a point of ultimate fitness which implies an unchanging environment. It highlights the significance of the temporality of systems that we will develop in Chapter 6.

In this chapter, we have identified the genesis of evolutionary ideas in biological and social theory, originating in a shared discourse in which they dealt with common concerns about causal processes. This is important because it demonstrates processes of reasoning that applied in both settings, in which the role of theory in relation to the empirical was dialogical rather deductive. In the 20th century, Hirst pointed out that social evolutionary theories became different enterprises: 'they have been philosophies of history or systems of historical generalisations' (p16, 1976). Hirst used this distinction to argue that there can be no genuine common ground between biology and social because of their different nature.

> Natural selection specifies a set of relations between the things it designates by its concepts (mutation of characteristics, selection pressures etc.) and a range of possible conditions under which they apply. It is a general theory of the mechanism by which species are transformed and unlike an historical generalisation it is not confined to a given set of transformations.
>
> (1976, p26)

Rereading this work in complexity terms, however, we can identify random mutation in biological evolutionary theory as a mechanism-based explanation analogous to a social conception of conditions of variation in which agency must be incorporated. In both, mechanisms are effective as causes in some circumstances and not in others as we saw at the beginning of the chapter.

We want to suggest that evolutionary theories can, with amendment, include retroductive explanation of systems change which parallels Darwin's approach to scientific reasoning. Far from the social Darwinist substantive theory that the selection process yields 'best' outcomes in a normative analysis, we have based adaptive processes within the context of Gould's (1977) reading of evolutionary processes that recognizes contingency. We can go on to suggest that complexity theory can perform the role of abstract general theory that Hirst regards as a missing foundation for evolutionary social theory.

In complex dissipative systems, the capacity to import energy from environment gives rise to increased structural complexity, while the system can export internal disorder to its environment. This explanation seeks to account for adaptation while recognizing that this is ultimately adaptation to an ever-changing environment. The nature of environment as dynamic also brings forth the discussion of co-evolution of social and physical systems and the unintended as well as intended consequences of action. Conceived as process that is ongoing, the notion of improvement is unnecessary.

Teleological explanation sets progress explanations according to a principle of organization, a basic identifiable cause. While there is a debate about the degree to which Spencer and subsequently Parsons employed teleological ideas, as we have seen, there is no necessary teleology in evolutionary theory of the biological or social kind. Ultimate functionalism is not inherent in evolutionary theory, and we can use mechanisms of adaptation without drawing any conclusion about their ultimate success. While the rationale of functionalism, in as far as it expresses an adaptive response, may shape system action, there are no simple unilinear cause–effect outcomes to detect.

We have dealt with evolutionary theory as a systems theoretic explanation, of the same kind as complexity theory, and have favoured explanation based in ordinary cause over time, and in contexts, replacing notions of teleology and progress with those of change. Evolutionary social theory is inherently systems theory but, in Toulmin's terms, theories of complex systems are not inherently evolutionist. The social world, in dealing with agency, has to cope with considerably greater complexity than that of random variation. It requires a view of the human agent orientated to the future as well as the present, to consider the combination of purposeful and other actions, unintended consequences, and individual and collective agency as they interact across history.

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5 Structure and agency

Introduction

We have identified that a persuasive causal account can be developed retroductively, at an aggregate level through traces, but also that we need to understand long-term outcomes of processes of change, through an understanding of how that change is produced. In doing so we need a science that takes us beyond the limitations of positivism without falling into the trap of ultimate relativism. One theme of this book has been that, often in different terms, this reality has been recognized by social theorists whose insights have much to offer for the complexity frame.

This chapter will take us into discussions in social theory that seek ways of dealing explicitly with the epistemological consequences of a world conceived as real yet constituted by open systems giving rise to outcomes that can be explained but not predicted. The interrelationship between ontological and epistemological position is fundamental, and in rejecting positivism we can embrace a recursive relationship between the reality we perceive and the way we perceive it. Our focus is on how social science can deal with the nature of the world in terms of the action of people within it and the effects of structures as shaping and shaped by action. This was formulated as an explicit problem by Lockwood (1964) in terms of system integration and social integration and later by Dawe (1970) describing the historical split into 'two sociologies', between those who interpreted the world in terms of the structures that created regularities, in a normative functionalist account, and those who saw the world as springing from the action of individual agents. While the former challenged the action account for its inability to explain continuity, the latter charged their opponents with reifying the practices of human beings over time. The distinction has been extensively discussed by social theorists who have recognized the need in some way to incorporate both for adequate explanation, struggling to provide a convincing account of how that can be achieved. In this chapter, we focus on the work of Archer and Bourdieu. We do so in the spirit of Bourdieu's approach to theory as a dialogue, a building, and working process. To think about their theories and their relevance for complexity, we will briefly outline the core tenets of their work and build from their commonalities as well as their divergences. As with the previous chapter, our aim is to raise the questions and identify themes relevant for complexity theory rather than to compare the work of two theorists. We begin with Bourdieu's framework, which develops an explicit relationship between epistemological questions and practical social theory as the basis of useful explanatory work. We will then go on to Archer's relationship between ontology, and explanatory methodology developed through critical realism, which involves explicit reflection on how agency is produced and is causally effective.

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Reading Bourdieu as a complexity theorist

We begin by challenging a misconception that is apparent in literature about Bourdieu's work, suggesting that he deals with reproduction but cannot understand change. This misconception comes from the tendency to discuss habitus as an isolated concept and to equate habitus with Bourdieu's account of agency. Bourdieu's habitus can only be understood in articulation with field, a relationship that provides fruitful discussion for complexity. It recognizes reflexivity in agency and posits a complex conception of interaction with field in the use of several forms of capital. It sets this discussion in terms of a clearly developed relationship between ontology, epistemology, and empirical methods which grounds the conceptual articulation of his theory. Bourdieu's discussion of the relationship between epistemological and empirical is important here because it will underpin our subsequent attempt at synthesis in a complexity frame. His starting point is that concepts have value only in so far as they can be 'put to work'. Before going on to consider how Bourdieu frames these concepts, we will briefly outline his approach to the status and purpose of theory, because it has been the treatment of his concepts of habitus field and capital absent this discussion that has resulted in claims that Bourdieu is structuralist.

Ontology, epistemology – theory and evidence

To understand the status claims of Bourdieu's theory, we must appreciate that he rejects grand theorizing in the way of Parsons, rather seeking a formulation developed in dialogue with empirical reality. His approach is consistent, in this sense, with Bhaskar's question: what must the world be like for science to be possible (2008 13)? As we pointed out in the previous chapter, all theory is based on, often implicit, ontological assumptions. A discussion of structure and agency raises inevitable questions about those aspects of the social world that are real, having causal power, but not visible to the senses. Bourdieu acknowledges this reality and identifies the need to develop theory that can account for what we observe while recognizing that this means that deep structures are implicated:

> my entire scientific enterprise is indeed based on the belief that the logic of the social world can be grasped only if one plunges into the particularity of an empirical reality, historically located and dated, but with the objective of constructing it as our "special case of what is possible" action as Bachelard puts it, that is as an exemplary case in the finite world of possible configurations.
>
> (1998 2)

Theory is important for seeking understanding in and through application. The researcher's role in uncovering this reality is to develop a theoretical explanation that can grasp the operation of underpinning structures and processes as they are manifest in the real history of the social sphere.

Bourdieu's first principle for theorizing is that no explanation can provide an account to cover all actions and relations, system structures, and practices within a system of relations. The social scientist is, rather, to treat theory as a somewhat plastic tool of explanation. It must be applied in empirical settings to have meaning or validity but is not imposed upon the empirical so much as developed in relation to it. Here he points to C. Wright Mills's distinction between theory as 'splitting and polishing concepts' and theory as 'generative'. His work on the logic of practice focuses on this gap between theory as logic and theory as practice arguing that to develop theory of the social world we need to understand it as practical logic:

> based on a system of objectively coherent generative and organising schemes, functioning in their practical state as an often imprecise but systematic principle of selection, has neither the rigour nor the constancy that characterize logical logic, which can deduce rational action from the explicit, explicitly controlled and systematized principles of an axiomatics (qualities which would also be those of practical logic if it were deduced from the model constructed to account for it).
>
> (1990 102)

Theory develops 'models' of systems of relations which stand, as models always do as metaphors of reality. It is this relationship between epistemology and empirical investigation that makes Bourdieu's work compelling for complexity theory, in which the aim is to achieve scientific laws, expressed as normic statements that can 'give reasoned scientific accounts of reality' (Reed and Harvey 1992 357). It highlights the need for explanation that rejects the application of both a master functionalist logic and methodological individualism. As with Bourdieu's objections to Althusserian Marxism, insights from theory are a starting point for understanding rather than an eternally applicable description to impose on the world. It is in positing abstract social theory that the danger of reification lies.

We encounter here the recursive relationship between theory and empirical work which returns us to our opening concern with the relationship between the nature of the world and the nature of social science. An example from the development of social science itself helps to illustrate. Bourdieu points to Kuhn's structure of scientific revolutions as making a sociology of science possible. While Kuhn's insight into the nature of scientific development was radical, he explained punctuated periods of normal and revolutionary science as internal to science itself. Bourdieu argues that if we see science as a field in which conflicts occuring between 'orthodoxy' and 'heresy' at points of crisis that bring about paradigm change (ibid.), we can, then, break with both the logicism that posits the discovery of a priori rules and immutable laws and the relativism that treats science entirely as something that is 'done'.

What is to be developed then is not simply an account that relates structure to agency, but one that acknowledges this interaction as much within the scientific enterprise as within other fields. Structure and agency are to be understood in this recursive relationship, and it is through material inclusions, social structures as dispositional structures, objective chances as expectations that, Bourdieu argues, we develop practical knowledge.

Structure and agency

Based on this understanding of how we use theory, we can go on to look at the relationships proposed between structure and agency in Bourdieu. It is useful to begin by establishing a distinction between agency and action. Although it is not a distinction made explicit in his writing, it does pinpoint a difference he makes between what people do and the component of that that describes its conscious, reflexive, creative dimension. Action comprehends both the reflexivity of agency and the non-reflexive, reproductive elements that are consistent with the structural context. What people do is a contingent outcome of those structural and contextual elements working in interaction with conscious, rational, and affective interpretations of meaning. Structure–agency relationships involve both current and past actors, whose actions have become sedimented into structures. Bourdieu adds an important dimension in explaining the actions of current agents, that structures do not exist as external to, and imposed on, actors but are embodied within them. What we perceive and how we interpret the world is shaped by our experience of it, not simply cognitively, but as we interact with it bodily. Our relationship to the social is therefore embodied. The world is a different place for women and men, for rich and poor, and so on, leading to tendencies based in our position within it that are not necessarily reflexively appreciated. The interaction of these facets of our experience means that it is not possible to identify structural factors in any simple way as contributing to a way of being as aggregate groups.

The debate about class and patriarchy that was particularly strong in the 1990s illustrates the point clearly. Moving from a battle between advocates of the primacy of class versus the primacy of patriarchy, a recognition developed that for both women and men these structures interact in different ways to produce an experience of reality that is internalized. Structure is treated as real in the sense that it has causal powers which are contingently effective, based on interactions between structures and agents' orientations to them. Historical evidence is abundant. Actors dealing with structures of class and of patriarchy do so from different positions and with different interests, even within the same family. These structures are both externally imposed and internalized. If this were not the case, it would be difficult to explain people reproducing structures that are based on their own subordination.

In his engagement with the nature of theorizing, Bourdieu lays the basis for a relationship between structure and agency that sees them in interaction, as generative mechanisms, denying simple reduction to containers of structure and containers of agency in which one occurs prior to the other. Structural determination does not present us with simple master causes translated in correspondence, as Potter highlights, 'given an ontology of structured depth and levels whereby different causal processes are at work within different temporal frameworks, determination becomes a much more complex notion' (Potter 2000 239). We have noted this interpretation in Williams's notion of base and superstructure in Marx's theory in which determination means setting limits and boundaries rather than direct and unmediated cause. This brings forth a more layered and contingent understanding of the role of structure that works well with our complexity frame.

While attempts at analysis are inevitably reductionist in some sense, Bourdieu is pointing to models of recursive relations between structures and embodied positions in interaction with cognitive, reflexive responses. This accesses enduring explanatory principles consistent with the notion of the world as emergent, in which causes operate contingently so that apparently minor local variation can bring change in some contexts and not in others. If empirical reality is seen as 'a particular case of the possible', then Bourdieu is identifying the significance of contingency resulting from causal mechanisms that become efficient in some social spaces and not in others. Far from a world in which agents could be conceived as the 'cultural dopes' that Garfinkel identified as the actors of structural functionalist theory, the significance of agency as strategy and reflexivity is clear:

> nevertheless if the social world with its divisions is something that social agents have to do, to construct, individually and especially collectively, in cooperation and conflict, these constructions still do not take place in a social void, as certain ethnomethodologists seem to believe.
>
> (1998 12)

Bourdieu's work shows some clear common threads with critical realism. There is no doubt that Bourdieu attributes causal powers to structure and to agency. He starts from the position that objectivist approaches describing reproduction ignore the medium of reproduction which is human actors, albeit in reproductive phase acting as the bearers of structures. Bourdieu's language use seeks to avoid common sense definition, so that the concept of 'strategy' is used to acknowledge creative action in pursuit of goals while disavowing its conventional basis in rational action.

Strategy is introduced as a 'feel for the game' which sees actors encountering a field in which external structures have ongoing force. Indeed it is one in which agents have been equipped by their inheritance and accumulated experience with more or less ability (or resources of capital) to handle those structures and play the game, but one in which they nevertheless engage as creative, reflexive individuals and members of groups developing, defining, and redefining their interests. We will outline the role of the field below, but here we want to point out Bourdieu's identification of the field as the focus of social explanation is a clear indicator of the primacy of relations within which the outcomes are the result of an interaction between regularities of structures (seen as relatively permanent) and the reflexive agency of collectives and individuals. At each turn, Bourdieu qualifies action as neither simply reproduction nor as wholly rational and voluntaristic. Indeed we must understand continuity and change in terms of causal powers in configuration. He refers to Elias's figurational approach in 'The Court Society' which saw actors reproducing 'burdensome ceremonies' because the risk of initiating reform was to upset the precarious balance of power among factions within the ruling class. Bourdieu points to similarities with Japanese and French education systems that we might also recognize in the UK:

> worn out parents, exhausted young people, employers disappointed by the products of an education which they find ill suited to their needs, are all the helpless victims of a mechanism which is nothing but the cumulative effect of their own strategies, engendered and amplified by the logic of competition of everyone against everyone.
> (1998 27)

In describing this aspect of reproduction his point is that we are not dealing with structures that have been imposed from above against active resistance, but rather that the outcome is one which actors have produced, albeit in the context of structures that they have not.

Structure and agency are analytically separate but actually embodied. Embodiment does not imply an elisionist approach, indeed discussed in terms of dialectics and relations. Bourdieu implies a clear analytical distinction. His core concepts of habitus and field are established far from direct approximations for agency and structure, as comprehending a relationship in which habitus is part of action rather than agency, and fields comprise both the structural relations sedimented from the past (actors long dead both individual and collective) and those current. This relational approach is the foundation for elaborating Bourdieu's central concepts of habitus, field, and capital. What is frequently not recognized is that, for Bourdieu, the dynamics of this relationship between interests, positions, expectations, and context mean that change is the only constant.

Based on a relational/dialectical approach to structure and agency, habitus and field then take on quite different connotations, making way for a much more dynamic understanding than is conventionally allowed by critics. It has provided us with a basis for complexity-consistent appreciation of structure–agency relationships which can help us in thinking about complexity framing in empirical research as well as for evaluating agent-based modelling and the appropriateness of prediction.

Habitus explains the reproductive dimension of human action, being the pre-conscious orientations to action with which actors are equipped to reproduce the world as it is 'without either knowing they are doing so or wanting to do so' (1998 26). When this is achieved, habitus and field exist in 'ontological complicity'. Individuals are carriers of habitus, but this is not synonymous with agency. Bourdieu consciously uses the language of mechanisms to describe the process but actively rejects the 'conspiratorial fantasy' that can result from simplistic application of mechanistic metaphors. Expressed as a system of dispositions, habitus is Bourdieu's attempt to 'escape from structuralist objectivism without relapsing into subjectivism' (Bourdieu and Wacquant 1992 61).

Bourdieu's rejection of the notion of 'rules' is another point at which we have to be precise about his definition of terms. He uses rules, not as regulating behaviour but as a shorthand for describing regularities. Actors are strategic rather than rule following. Habitus as a feel for the game cannot be separated from the game that takes place in a field of relations in which agents at times rely on those dispositions but are always liable to reject, innovate, and create.

So what we have is not simply a description of structures that in some way shape or determine agency and therefore action. We have an explanation of how structure enters into action. The analysis is not focused at the individual level but identifies individuals as the units of data. Bourdieu is looking at what is social, expressed in terms of relations. This is neither an argument against the creative power of agency nor one for methodological individualism.

So we have developed a conceptual reading in Bourdieu that gives habitus a role in explaining action that must be incorporated with reflexive agency expressed in relation to specific fields. Elder Vass (2007b) has captured Bourdieu's meaning in his discussion of habitus as only part of behaviour, in Elder Vass's terms the metaphorical rather than the literal sense. What we are interested in here is the dimension of Bourdieu's work, consistent with his relational ideas, that has promise for complexity framing. In complexity terms we learn about the mechanisms through which actors are constituted to reproduce those structures through their own practices. So we interpret habitus as a non-reductionist concept whose embodied nature encapsulates 'practical sense emergent from experience' (Lau 2004 382). We hope to unite this with Elder Vass's step in co-determining, consistent with Archer's emergentist approach later.

Collective action in Bourdieu

The further significant dimension of Bourdieu's work that needs to be addressed lies in the role of collective agency in reproduction in which his particular concern is to understand class reproduction. Bourdieu sets the discussion of social class ontologically in relational terms: 'what exists is a social space, as space of difference, in which classes exist in some sense in a state of virtuality, not as something given but as something to be done' (1998 12).

We can describe the relations of class in capitalist society which are inherent in the structures of that form. Class is not determining but always enacted. Bourdieu recognizes two uses of the concept of class: as a sociological category that is a basis for description and as a collective actor. The significance of empirical engagement with particular understandings of structure becomes immediately apparent in the nature of explanation being produced and the agency of change. Bourdieu's response reflects generally his reaction to attempts in the 1960s and 70s to draw substantialist lines of demarcation around groups such as social classes and to develop the reified explanations that Parsonian functionalism was accused of producing. His own work on class expressly rejects the substantialist approach in favour of a relational interpretation. He cites E.P. Thompson's work on *The Making of the English Working Class* as an approach that sees class as something that is 'done', and further, as discernible over time. Consistent with his general approach, Bourdieu does not set out propositions about class that apply always and everywhere but rather discusses it through his empirical work, most explicitly in *Distinction: A Social Critique of the Judgement of Taste* (1987). In this study, he investigates relations of class, referring to Marx, but paying particular attention to Weber's distinctions between class and status, to recognize the significance of the symbolic as well as the material for the reproduction of dominant class positions. So while location in the division of labour broadly approximates to dominant, petit bourgeois and working classes, a much more complex set of relations can be discerned in reality in the distribution of capital (particularly economic and cultural) as a 'set of actually useable resources and powers' (1987 14). Class reproduction comprehends, therefore, not only control over material resources and inheritance, but also the distribution of cultural capital which is carried by individuals and reinforced in institutionalization in organizations. Cultural capital is important because it is a highly effective, yet disguised, form of transmission of dominance. So a model emerges of three broad hierarchical categories internally differentiated according to the differing distributions of capital and by trajectory over time. These locations have existence in social space as a 'quasi reality' (1992 27) but display neither the gradations of empiricist categories nor the sharp cleavages of Marxist analysis. It is a model of three interacting dimensions with inherent dynamism that occurs through struggles for domination and control. Class does not necessitate a unity or consciousness but rather a potential for that unity exists in the social space.

He goes on to remind us of the nature and use of theory in qualifying this point:

> Things are, of course, much more complicated, but I think that this is a general proposition that applies to social space as a whole, although it does not imply that all small capital holders are necessarily revolutionaries and all big capital holders automatically conservative.
>
> (1992 108–109)

Field

The discussion of embodiment and the need to understand agents as both reflexive and embodying structure is only part of Bourdieu's theory, incomplete without appreciation of the operation of these complexes at individual and collective levels within a field. The field takes on considerable explanatory importance not as a structure, but rather as a 'social space' produced by actors rather than being self-producing. Fields are 'systems of relations which are independent of the population these relations define'. So while there are times when habitus and field seem coherent, the field is one of relations of force which precisely, because it is the site of transformational struggles, is always dynamic. It has common features with systems but, Bourdieu argues, rejects the conceptual framing of systems theory in terms of equilibric, order-seeking systems that are associated with Parsons's functional reproduction: 'While a field could be clearly said to exist before the individual enters because of actions of past actors, it is shaped in reproduction or change by current actors' (1998 32).

Field expresses a relationship between structure and agency which reflects interaction not simply between habitus and field but reflexivity, habitus, and field. It has history and is emergent in a way that resonates with complexity:

> The effects engendered within fields are neither the purely additive sum of anarchical action, nor the integrated outcome of a concerted plan … It is the structure of the game, and not a simple effect of mechanical aggregation.
>
> (1992 17)

In case we are tempted to treat fields in the 'literal' application identified by Elder Vass (2007b), in discussion with Wacquant, Bourdieu gives us this reminder:

> (Fields) major virtue, at least in my eyes, is that it promotes a mode of construction that has to be rethought anew every time … It forces us to raise questions about the limits of the universe under investigation, how it is 'articulated', to what extent and to what degree etc. It offers a coherent system of recurrent questions that saves us from the theoretical vacuum of positivist empiricism and from the empirical void of theoreticist discourse.
>
> (1998 110)

The key to Bourdieu's emergentist approach lies here. It is in the very acknowledgement of the impossibility of creating final theory to explain everything that he demonstrates his recognition of the actual world as the emergent result of configurations of causes, one 'case of the possible', which in the real world involves interests in intersecting fields that may, at different times, cohere or be in conflict. Indeed in answering a question from Wacquant about the interrelations of fields his main concern is to avoid 'theoreticist' theory and to reinforce the significance of dialogue between theoretical models and empirical work that is practical social theory.

While drawing on Elias's figurational approach, Bourdieu's development suggests that rather than a structure of relations between people, we are dealing with a structure of relations between positions. What emerges from this discussion is a theory that exists to engage with practice. The promise of Bourdieu lies in his methodological focus, and we have a clear approach to empirical analysis. There are lines of continuity here with Walby's (2007) reconceptualization of complex systems in terms of intersectionality. Sharing Bourdieu's evident dissatisfaction with the conventional conception of systems for its flattened and reductionist representation of class identity, Walby points to class and gender as non-nested intersecting systems in a world in which sets of social relations never fully saturate. Such a concept 'allows the possibility of analysis of multiple simultaneous complex inequalities while retaining concepts of social structure and system' (ibid., p460) seeking to address the difficulties that Bourdieu tried to overcome using fields.

Bourdieu emphasizes that the focus of social explanation lies 'not with individual actors but with the field in which both individual and collective agency is exercised'. The subject of social scientific study is the level of interaction, and in this sense we see echoes of agreement with Luhmann in trying to identify what is specifically social.

The centrality of ontological and epistemological frames for situating concepts of habitus, field, and capital in terms of a relational theory is frequently neglected by critics.

Reminding ourselves of this foundation, we find three key interrelated principles of Bourdieu's theory that make his work particularly relevant for complexity theory; it is intended always to be discursive rather than didactic, to incite and invite a dialogue; it is empirically based and sees the value of theory in how it helps us to understand the world rather than for schematic application; and as a result it seeks explanations that are local, recognizing the significance of interactions, emergence, and the non-predictability of agency. It is expressly about developing models and informing questions that help us to look with a particular focus on the network of social relationships.

Understanding these relationships as complex, based in nonequilibric systems that are emergent continually reproducing but also as changing, is the foundation of an approach that we would want to deploy. In addition, in acknowledging the significance of transitive knowledge of an intransitive world, identified by Bhaskar, we have considered the issues involved in describing a world constituted by relatively permanent structures but at the same time constructed through the experiences and interpretations of actors within it. To understand not only system regularities but their contingent manifestation shaped through meaningful action, we need to incorporate these sensitivities into our methods and approach. We can now go on to Archer's work to develop our complexity framing further in terms of agency. We begin with Archer's central concept of morphogenesis.

Archer's morphogenesis

Archer's commitment to realist philosophy gives her work immediate resonance as a complexity-consistent theory. Based on her formula social ontology – explanatory methodology – practical social theory (1998), she proposes an ontology of structures and of agency in a stratified reality in which the transfactuality of mechanisms rests on historical rather than pure contingency. This implies that explanatory methods must involve relational concepts. Archer then sets out to develop a theory of the relationship between structure and agency that avoids the sins of the 'downward conflationists' (1996) such as Parsons's model of centralized direction and goal-seeking behaviour of interdependent subgroups with reproduction occurring through simple feedback mechanisms. Equally problematic for Archer are 'upward conflationists', those interactionists for whom the world is the product of the action of current agents, but who draw no explanatory strength from the actions of those 'long dead'. All of this seems reminiscent of the discussion above. Archer goes on to contend that the 'third way' taken by central conflationists represents an ill-conceived attempt to overcome these oppositions by the 'transcending' structure and agency dualism. Here Archer takes particular issue with Giddens for his discussion of duality of structure and agency (11995) but also with Bourdieu, on similar grounds (2003).

Archer developed her morphogenetic approach, following Walter Buckley, in relation to the principle of emergence in which structure and agency, distinct and irreducible, interact to bring forth emergence over time. For Archer, human beings, social structures, and cultural entities each have own distinctive existences and influences on social outcomes. Morphogenetic social theory identifies roles for structure and agency in which both intended and unintended action give the social world its shape. Society emerges from the agency of its actors, their collective goals, their conflicts, and their negotiations. These are not all current actions but include those laid down in structural forms over time; it is on this basis that she reminds us of Comte's observation that 'the majority of actors are dead' (1996 696). The first principle, then, is that structure and agency exist as ontologically separate, and explanation lies in mapping the interplay between them (Archer 1998). It is a significant premise for Archer that these are not two sides of the same coin, as Giddens proposed, a duality, but (following Bhaskar's discussion of individuals and society) different kinds of things.

Morphogenesis describes a process of system change that emerges from interaction (as compared with the reproduction of morphostasis) to produce a world that is neither planned nor a result of determinate master structures. The morphogenetic cycle instead comprises three analytical phases: structural conditioning which Archer identifies as prior to agency; social interaction which takes place in the context of the structural condition; and structural elaboration, the outcome of the interaction between the two over time. This elaboration in turn becomes T1 for the next phase in structural conditioning and is the basis for activity dependence. Current structures arise from the actions of people in previous cycles so that there is a 'temporal escape' of structure from past actions: 'we are all born into a structural and cultural context which, far from being our making, is the unintended resultant of past actions among the long dead' (Archer 1995 253).

The ontological separation of structure and agency is fundamental as we have identified above so that methodologically we are dealing with an analytic dualism, which, Archer tells us, offers a better resource for practical research. Archer contends, contra Giddens, that morphogenesis is not only a process that is unending but also one that has outcomes. The elaborated structure has properties that, because it is emergent, cannot be reduced to the practices of the agents producing them. While all social systems are the results of the action of individuals, their manifestation cannot be reduced to the individual. It requires that structure is prior in time because agents must occupy roles before they act upon or change them. Archer uses the example of a bagatelle board in which 'each subject is involuntarily launched' (2003 345). Moreover, time scales for structural change are longer term so that there is an endless cycle of 'structural conditioning/social interaction/structural elaboration – thus unravelling the dialectical interplay between structure and action' (2003 458). This involves a simpler and more discrete set of relationships than that proposed by Bourdieu and runs the risk of presenting structure and agency as containers that can be separated in empirical analysis. It is a dimension of Archer's explanation that has been criticized as presenting a mechanical version of structure that under-recognizes the significance of the relational (Sibeon 2004 142). Sibeon further challenges Archer's focus on present-day structures as the result of past actors for neglecting the degree to which current social contexts are dependent on current actors.

Morphogenesis of agency

Archer's later work focuses on the agency side of the relationship, developing a complex stratified conception of the actor to distinguish its collective and individual social moments from persons, the personal, and social self. Agency is discussed in terms of a morphogenetic cycle based in a stratified realist conception of human beings. Archer distinguishes 'persons with their strict numerical identity, who are also actors as incumbents of roles, and agents through their determinate relationship to the distributions of society's scarce resources' (2003 118). Agents are collectivities that can be powerful corporate actors as well as less powerful primary agents sharing the same life chances, so that we are all primary agents by virtue of being born into social positions. Morphogenesis takes place at the collective level as groups organize to claim power from existing elites, while at the individual level the elaboration process means roles are redefined. Both agents and actors exist within persons, and here we come to the centrality of reflexivity as, for Archer, human beings possess reflexivity as a power. Individuals develop personal identity through the reflexive process of an internal conversation which denotes our ability to monitor ourselves in relation to our circumstances.

So we start from a range of causally contingent structural and cultural properties that in the morphogenetic process are prior to agency. Archer seeks then to explain how such cultural and structural properties can be understood in relation to human projects: 'structural and cultural factors do not exert causal powers that become efficient in relation to human beings, but rather in relation to our emergent powers to formulate social objectives' (2003 132–133). Emergent powers arise out of the interaction of ontologically distinct but interacting structure and agency so that people's emergent powers (PEPs) are established alongside structural emergent powers (SEPs) and cultural emergent powers (CEPs). The internal conversation, through which the morphogenesis of agency is achieved, is a process that comprises 'thought as internal speech; a movement from inchoate premonitions to articulate utterance' (2003 63) combined with listening to ourselves 'as we phrase our thoughts'. The first two stages draw on James's work on 'inner dialogue' while Archer adds the final stage to complete the 'internal conversation'. The structure is reminiscent of Luhmann's communicative event although here the communication is with oneself.

This inner conversation is 'a ceaseless discussion about the satisfaction of our ultimate concerns and a monitoring of the self and its commitments' (quoted in Elder Vass 2007b 331). Archer describes three conversational phases in the inner dialogue: 'discrimination, deliberation and dedication':

> We survey constraints and enablement under our own descriptions (which is the only way we can know anything); we consult our projects which are deliberately defined to realise our concerns; and we strategically adjust them into those practices which we conclude internally (and always fallibly) will enable us to do (and be) what we care about, most in society.
>
> (2003 133)

The major motor force of reflexivity is a mature ability, which has causal power in relating personal and social identity: 'reflexive deliberations constitute the mediatory process between "structure and agency", they represent the subjective element which is always at interplay with the causal powers of objective forms' (2003 130). Those forms, nevertheless, 'rebound(s) upon us, affecting the persons we become, but also and more forcefully influencing the identities which we can achieve' (2000 10). Society enters the self, following Pierce, through the 'me' of habitual action, a voice that is differentiated from the 'I', and from the 'you' of our projected future wishes. The distinction between the three voices is an analytic one that permits the interplay necessary to a morphogenetic process across the lifespan of the individual.

In the morphogenesis of agency, the role of structure is mediated through the internal conversation. We are now dealing with multiple time spans for structural change, lifespans in terms of social identity and more immediate morphogenetic processes that are ongoing in the internal conversation itself. These cycles must themselves intersect. Some questions arise about the role of current agents and current parts mentioned above, and the synchronous relations that both Sibeon (2004) and Elder Vass (2008) have identified as under-theorized. Archer's integration of founding ideas in analytic dualism and morphogenesis has been questioned by Zeuner (1999) challenging the coherence of analytic dualism and systems thinking, arguing that Archer has taken morphogenetic thinking to dominate so that agency itself becomes a morphogenetic cycle which in turn becomes part of the social system. The model conjures up an interaction between voluntarism and constraint which takes place in conscious deliberation but does not deal with the numerous aspects of daily life in which we act without awareness of projects, reproducing a world through predispositions in terms of tastes and beliefs about appropriate behaviour that Bourdieu was concerned with. This significant source of reproduction explains the regularities, that actors produce, must be incorporated into the explanation.

Collective agency

A final note in the discussion of Archer's account relates to collective agency, a highly significant feature of reproduction and change to be incorporated in any theory of agency. We quote extensively from Archer below because she identifies both argument and counter-argument in her discussion of groups as collective actors over time and specifically relates to class. The departure from Bourdieu's approach to the definition of social space according to relations is evident and this pinpoints the issue of analytic dualism. Archer challenges the notion of the durability of the group:

> The real force of the objection thus comes to rest not upon critics maintaining that groups can show a greater durability than structures because they can point to 'teachers' as a 'group' retaining continuity before and after the emergence of state educational systems, or the same for 'doctors' as far as the inception of the national health service was concerned). Rather, what I am criticizing is their (implicit) notion that the 'group' remains fundamentally the same, that is, they are pointing to the same entity. If this were the case, as seems quite persuasive at first glance, then it would indeed prevent one from ever talking about a pre-existent structure and would also effectively demolish 'analytical dualism' by removing its temporal mainstay which is what makes events tractable to explanation. Thus, we would be back to the simultaneity model of central conflation. However, this critical viewpoint is fatally flawed by the naive nominalism with which it treats 'the group'. It supposes that just because we can use the label 'working class' over three centuries of structural changes in Britain, that we are talking about the same 'group'. We are not, any more than this is the case for 'teachers' or 'doctors' above.
>
> (1995 74)

The excerpt raises the question for us of why Archer describes social class in these terms. While it seems reasonable to suggest that particular occupational groups change in form and nature so cannot be seen as the same thing over time, it is more difficult to apply these ideas to a changing working class because of different material conditions. Given that the definition of class under capitalism is precisely a relational definition comprehending the producers and exploiters of labour power, it is fundamental to the realist frame (and to the relational account we are developing) to argue that these relations define the group, which consequently does have continuity. Indeed, elsewhere she accepts that it is consistent to say certain social relations are implied by material conditions. Quoting Lockwood, 'there is nothing metaphysical about the general notion of social relationships being somehow implicit in a given set of material conditions' (Archer, 1996 684). It is an important point because it is on this basis that Archer contends that structure is prior and opens the way to understanding structure as immanent in relations.

Structures that are inherent in the class relationship under capitalism and structures (understood as relatively permanent but perhaps less enduring) that contain Archer's examples of change in control and professionalism in groups can be differentiated, although we might see class position of the profession as a significant dimension of its level of control. Here, then, we would want to bring forward the position we have already identified in Bourdieu as explaining how our class position is both pre-reflexively within us and acted upon reflexively; it is a conception based on the experience of class in which 'each member of the same class is more likely than any member of another class to have been confronted with the situations most frequent for the members of that class' (Bourdieu 1977 85). This implies neither determined behaviour nor full agentic freedom but rather that the outcomes of interactions are uncertain, acted out in every social situation.

Archer's discussion of structure and agency offers an explicit systems theoretic account that coheres with non-linearity, emergence, and therefore of course the significance of history. We have developed an account of Archer's work that has much to say about intended action and of course its unintended consequences, but which, perhaps, neglects the degree to which our own subjectivity is an outcome of a social world. We need to explain satisfactorily how entities possess emergent properties at a structural level, and this has been taken forward by Elder Vass (2007b, 2010) in developing a reconciliation between Archer and Bourdieu.

Resonances for complexity theory

This discussion of structure and agency has looked at two approaches to highlight their potential for working in a complexity frame, paying adequate attention to the dimension of agency. Both theorists stress consistent relationships between ontology, methodology, and engagement with the empirical. Both in fact deal with structure and agency as irreducible to each other. For Archer, these foundations lead to morphogenesis which seems to lead to a sequential, if cyclical, relationship. For Bourdieu, we have another level in the recognition that current actors come to fields as both carriers of structural relations, and as reflexive actors embodied.

In some respects, their two approaches reflect differences of focus. Archer develops a greater emphasis on the voluntaristic and reflexive in her account of how structure enters into agency. Her focus is on the projects, the reflexive element of human action identifying the dynamic nature of subjectivity, defining and redefining meaning and our goals in relation to it, perhaps at the cost of the myriad ways in which we unthinkingly reproduce social contexts. This emphasis on reflexivity and identity raises questions about how far structure can be seen as constitutive. Bourdieu's main attention is focused on why and how reproduction occurs through habitus, acknowledging the significance of strategy in relation to reflexive action in fields.

In reading this work for its complexity insights, we started from the position taken by Bourdieu and Archer that theories are tools for developing explanations in relation to the empirical. The value of theory lies not in its elegance or comprehensiveness, but rather in its ability to account for situated and embodied experience, here formulated as consistent with a loosely nested and reflexive reality. It is dialogical rather than law like, and this has consequences for the way we treat its insights. We began, for example, by pointing to Bourdieu's distinction between 'rules' as regularities rather than regulating devices determining behaviour, because regularities are always the achievement of actors, within contexts, including those of time and place. This is a founding principle of a complex realist ontology, yet it is also the basis of the split within complexity that we discussed in Chapter 2. The combination of these insights has made us question the possibilities of ABM, modelling structures through rules that can learn in a way that might approximate to a habitus, but cannot contain that reflexive, creative 'doing' of individual nor the at least equally important dimension of collective agency.

We have interpreted Bourdieu's discussion of the internalization of externality in Elder Vass's 'metaphorical sense' on the basis that habitus is a set of dispositions that tend to causally influence us. We are not suggesting that structures exist within the individual, but rather, agreeing with Elder Vass, that there is no contradiction between this and the ontological work of Archer.

Both Bourdieu and Archer would subscribe to a social sphere as a not necessarily planned, or even wished for, outcome of processes of struggle which include conflict and negotiation between individuals and groups in conditions of differential levels of power. Both would see structure as having emergent properties, while at the same time emergence is also evident, for Bourdieu particularly, in collective as well as individual action. In these respects, both theories seem consistent with a flexible realist ontology.

What we have in consequence is a way of understanding the significance of process and relations in terms of the interaction between structure and agency in which habitus is part of action and only an element of interaction so that causal powers are historically contingent and the actualization of those powers is shaped by context. Fields of relations encompass relatively permanent social structures but ones that are open to change because of the constant nature of struggle within the field. If the world is situated and reality is embodied, we may model a reflexive reality, but we cannot use it as a strong frame for predicting future regularities. It can give us information about tendencies without being able to predict their actualization.

Emergence at all levels

We will draw briefly on Elder Vass's (2007b, 2010) analysis because while his acceptance of Archer's description of Bourdieu as structurationist seems to us to miss the emergence and dialectical relations we have found in Bourdieu, the potential he suggests for marrying insights from Archer and Bourdieu seems promising for complexity theory. In concluding, we consider the common and overlapping insights from these theorists can help us to think about how social theory has actually engaged with the world in terms of the roles of class, gender, ethnicity and through which it has investigated institutions of educational and family life.

We need to be able to explain individual and collective agency and how the latter particularly, sedimented in social structures over time, exists in a continuing recursive relationship with structure. The elaboration/refinement of Archer that Elder Vass offers seems consistent with Bourdieu's approach, handling emergent relationships that arise from the actions of actors long dead but continuing to manifest in structures that do not simply exist but rather must 'be done'. We can then go on to agree that most actions are co-determined by habitus and reflexivity. His clarification that different aspects of the same behaviour can be attributed to habitus or reflexivity, rather than different aspects of different behaviours, is a useful one. He suggests that Archer's internal conversation is a substantial contribution, a theory of reflexivity to complement Bourdieu's theory of habitus. We can, then, think of action as emergent from the interaction of habitus and the internal conversation. It may be that we can, taking the relational work of Bourdieu in articulation with Archer's work on agency, develop an approach to empirical work that is consistent with complexity framing.

Elder Vass suggests a further step in his reconciliation, developing Archer's emergentist account from embodiment in its social structural moment to pay attention to its physical relation to the world.

> Like all emergent entities, human individuals are part of a hierarchy of such entities, each with emergent causal powers of their own, including in this case both the biological parts of human beings and the higher level social entities composed (at least in part) of human beings).
>
> (2007 326)

Presenting a theory of action based on the emergence relation between dispositions stored in neural networks (which he identifies with habitus) and the mental, reflexive decision-making process by Archer, Elder Vass suggests we can see co-determination that implies a further dialogue is necessary to accommodate the co-evolutionary processes we alluded to in the last chapter.

So we have ways of thinking about formulating change, the role of structures and the role of agents in terms of systems that are somewhat different from the order-seeking closed systems that were proposed in some evolutionary theories. In common with complexity theorists, the focus of both Bourdieu and Archer is to develop a scientific approach to explanation based in disciplined observation and the reflexive stance of the scientist in relation to their objects of investigation. Both theorists apply notions of emergence and concomitant irreducibility of layered entities. The resonances with a social ontology of complexity theory based on dissipative systems is clear. The aim of theory is explicitly addressed as generating models that can be useful for describing the world rather than predicting outcomes from complex configurations of causes in interaction with conditions. Time is crucial, both synchronic and diachronic explanations, underlining the significance of system history, its unpredictability, and its irreversibility. Its policy implications for the shift towards modelling for prediction is considerable.

This chapter has applied a different lens of explanation than the previous chapter which reviewed long-term tendential explanations of change in terms of the historical evolution of systems, often but by no means necessarily, as an expression of master teleological processes. We have focused on understanding processes of reproduction at the level of the agent, and this has been useful in identifying why teleological accounts are unhelpful. Having identified co-evolution at macro-level in Chapter 4, we have briefly pointed to its processes in closer view to consider the body as social in Bourdieu and identify the significance of its constitution as organic, as Elder Vass does, to identify processes analogous to co-evolution at the level of the agent.

In far from equilibrium systems language, we have considered the processes involved in system evolution in terms of sensitivity to initial conditions (Reed and Harvey 1992) as actually the unending process of collective cooperation and conflict in pursuit of interests that reproduces or redefines its nature at any point in time. Change can be radical but it is not random; its antecedents can be clearly identified. The court society and the education system may continue to be organized in ways that suit almost no one for substantial periods of time, but their change, when it comes, will result from interaction of forces and interests that we can identify and whose causes we can explain. When it comes however, the unintended consequences of these interactions may yet serve to frustrate the intentions of its movers, a condition we see constantly replicated in top-down policymaking processes. Modernist notions of progress disappear and we have an explanation based in complex social processes, founded in relations of power that can leave behind the rational actor.

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6 Time and place

Introduction

The Newtonian conception of time and space treats them as a frame, a grid of coordinates within which social action takes place. Positioning time and place as invariant and unilinear, however, removes agency as constitutive of both and so presents difficulties for understanding social systems in complex terms. Rethinking their role involves moving from concepts that have dealt with an absolute world, external to and shaping the social world, to developing a view of how it is recursively shaped. It is worth reminding ourselves of the nature of theory, defined in relation to critical realism, by Sayer as rejecting the 'Humean view of theory as tied to the search for regularities among events, rather than to the (realist) search for necessity in the form of causal mechanisms, regardless of whether the events they produce are unique or common' (2000, 120). We are not seeking to generalize outcomes precisely because systems are open, and interactions can bring about divergent outcomes which are, nevertheless, amenable to causal explanation. We have already encountered this understanding of theory in Bourdieu, who, saw its role as to understand the empirical as 'a particular case of the possible'.

This chapter focuses on challenges to accepted Newtonian spatial and temporal concepts that seek to respond to observed changing social relations in globalization and to accommodate the insights of post-structuralism. In doing so our purpose is to engage in a discussion with some key complexity relevant ideas and we begin by raising questions about how we can deal with time and space in causal terms.

We will focus on the nature of the theoretical ideas underpinning conceptual development, setting aside discussion of empirical work with those concepts. While this seems to run counter to our overarching principle that theory is about its application, what we are seeking to identify here are the ways in which time and place/space concepts have been naturalized in application. The distinction is identified by Gilbert Ryle 'many people can talk sense with concepts but cannot talk sense about them' (2009, ix). Ryle uses a cartographic metaphor describing, 'people who know their way around their own parish but cannot construct of read a map of it much less a map of the region or continent in which their parish lies' (ibid., ix).

Theorizing space and time

To consider the ontological status of space/time in realist-based complexity we can draw on Sayer's (2000) discussion of space/place as in a contingent rather than necessary relation. Based on explanation in terms of configurations of necessary and contingent

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relations, and the structures and powers that express those relations, temporal and spatial relations express conjunctures of contingently related phenomena:

> there is no contradiction in saying retrospectively how I got here and what was necessary for me to get here, while acknowledging that beforehand it was contingent whether I would. The open nature of social systems and the contingent nature of their reproduction means that the historical and geographical course of development is always an open question.
>
> (2000 127)

Human capacity for accommodating different spatial conditions is characteristic of this contingent status. This is the crux of difference in the conception of systems and the role of agency that we saw as convincing criticism of functionalism in the previous chapters. We have drawn on Bourdieu's argument against objectification of the empirical world from the stance of the social scientist who, using retrospective analysis, implies that ritual or repetitive acts, that are regular in time and space, are inevitable. Yet as we saw, such acts are not predetermined, their outcome cannot be known because they are the continual achievements of actors producing them. In his rejection of functionalist accounts, Bourdieu identified a world in which time and place are crucially involved in those achievements.

An empirical observation of the regularity and therefore, the very high probability of such an event occurring is of a very different character from a prediction, making the notion of social laws problematic. Moreover, Bourdieu reminds us, to reintroduce uncertainty is to reintroduce time, and with it, agency, 'with its rhythm and its irreversibility, substituting the dialectic of strategies for the mechanics of the model, but without falling over into the imaginary anthropology of "rational actor" theories' (1992 99).

This contingent expression of relationships of time and place does not, however, imply rejecting a role for necessary relations, as Sayer emphasizes:

> Just because the coexistence of A and B in a particular form is contingent, it does not follow that there is nothing necessary about how, once configured in this way, they interact. On the contrary their interactions follow necessarily from their constitutions and relative dispositions.
>
> (2000 138)

The discussion of place and time rejects the battle between idiographic and nomothetic explanations, the one focusing on the need to contextualize while the other seeks abstract general laws. The critical realist solution establishes a rather different status and objective for social scientific knowledge. It is summed up for us by Sayer, quoting Bhaskar, 'Contextualising and law seeking approaches should … be seen not as competing but as extremes of a continuum ranging across different kinds of objects' (2000 137).

In this chapter we want to pay attention to the role of space and time in constituting meaning while acknowledging contingency, based on our recognition that it is fundamental to human communication to share meaning across different space-time conditions. Sayer argues the enterprise of 'theorizing space' does not advance because it cannot as space is contingent, and impossible to theorize beyond context in the way that, for example, class relations can be (2000 126–8). This view has been widely supported in more recent spatial thinking, expressed by Thrift et al. 'Contexts have their own dynamics and what comes out of them is often unpredictable, excessive and certainly is only partly open to what we call theory' (2008, 10). Yi Fu Tuan (2008) similarly comments on the lack of progress with a theory of spatiality, suggesting that as a cultural construct the appropriate academic work on space is taxonomy, comparison, and tracing the evolutionary course of concepts rather than theoretical development per se.

The relationship between time and space and their interaction is central to empirical observation and theoretical work. It must be dialogical because these concepts are necessarily inspired by observation and refined in the process of theoretical exploration. We begin with an understanding of space and time that focuses on how resources, in the very widest sense of the term, are distributed. As becomes increasingly obvious in conceptual development, space and time are difficult to separate, however, we begin here by, focusing on each in turn before reuniting to think about where time-space concepts take us.

Space

Our first distinction lies between the topographical view of space and later recognition of its social constitution. Those working with spatial concepts have rejected the agentless form of empty topographical spaces. Soja criticized previous treatment of space as 'the domain of the dead, the fixed, the undialectic, the immobile – a world of passivity and measurement rather than action and meaning' (1989 37). This is fundamental for establishing the subsequent nature of the field because it shifts the study of the spatial world from one looking at what is going on within defined boundaries to understanding how it is produced and how it produces. It reflects, in Massey's (1992) terms, that not only is space socially constructed, but that in its turn the social is spatially constructed. The implications of this core proposition are summed up by Lee (2002):

> geography cannot just be added on. It is intrinsically a part of social practice and it is intrinsically disruptive as it necessarily brings context and instance properly to bear. It is there all the time (albeit constantly changing in form and influence from place to place and from time to time). However, the claim is not that geography matters because of some intrinsic qualities of place, space or the geographical imagination which define a separate and defensible field of study. It is, rather, that these influences cannot be ignored and that perhaps the most significant formative influence of geography on social life is that it cannot but stress instance, particularity and practice. (2002 352)

The shift to understanding the spatial as socially constructed was the basis of renewed interest in the nature and importance of space and place for social outcomes and processes, and hence of interest in geography in the social. Foucault and Miskowiec argued that space was replacing history in pre-eminence social explanation (1986). This highlighted the active role of space in capitalist production (Massey, 1984) and related structural questions about how space is expressed and experienced in terms of class (Bourdieu, 1999), gender (Massey, 1994), and race (Wilson 1987).

These debates about the nature of space and place were prompted by major structural and social change in late 20th-century globalization, from the critique of capital industrialization and its necessary relations to space, to arguments about the freeing of capital from space in the globalized post-industrial era (Lash and Urry, 1994).

In responding to global change, space has also come to be understood in terms of networks and flows. Such changes are characterized as the replacement of modernist order with more free floating network societies (Castells 1996) understood as complex and interconnected, constantly bombarded by information (Urry, 2003). Moving a long way from the original conceptualization as a container for action, the term space has been used to cover them all, with consequences in confusion and unacknowledged disagreements (Massey, 1992).

It would be wrong, to give the impression that conceptual development has been linear. The identification of space as socially produced, has a long history. It was core to the work of sociologists such as Simmel who focused on social space and its effects on social interaction, as well as to Durkheim's evolutionary perspective on the development from mechanical to organic solidarity through a changing division of labour. Despite these insights, the topographical view continued to dominate until Lefebvre's (1991) work on its social construction. Lefebvre suggested that space is produced in interpenetrating networks with different temporalities which become imposed on present space in three dimensions of material interaction, communication, and meaning. These insights became a major resource for spatial theory in the latter decades of the 20th century with a focus on examining the role of spatial practices, particularly those concerning property and capital (Harvey 2001), and the need to understand both individual and collective/institutional practices over time, as influential in forming the spatial landscape. In delineating its role, Harvey drew a distinction between space conceived of as absolute, space as relative, and space as relational that seems highly relevant for our discussion, echoing the relational focus of structure and agency that was identified earlier.

> If we regard space as absolute it becomes a 'thing in itself' with an existence independent of matter. It then possesses a structure which we can use to pigeon-hole or individuate phenomena. The view of relative space proposes that it be understood as a relationship between objects which exists only because objects exist and relate to each other. There is another sense in which space can be viewed as relative and I choose to call this relational space – space regarded in the manner of Leibniz, as being contained in objects in the sense that an object can be said to exist only insofar as it contains and represents within itself relationships to other objects.
>
> (1973 13)

This tripartite scheme established important distinctions in the conceptualization of space because it identifies three ways in which the relationship between necessary and contingent relations has subsequently been elaborated. It finds echoes in the work on rethinking time that we will look at later.

These developments have given rise to both reconceptualization and changes in dominant focus that have been summarised by Jessop et al. (2008) in terms of four 'distinct and largely sequential spatial lexicons of territory, place, scale, and network'. Each of these has competitively dominated attempts to understand spatial processes at particular points in time. The categories are useful for framing the debates about spatial concepts because each indicates something of the relationships that we have been elaborating on between structure and agency, particularly the degree to which space can be seen as shaped or shaping and the possibilities for creative human agency.

Noting the one dimensionality of each of these 'turns', they identify a changing dominant interest in the field. Territory, denotes the empty space that we have characterized as the dominant Newtonian view against which all others have been formulated as challenges. The concept of 'place' was developed to acknowledge the relative dimension of spatiality and deal with identity, uneven development, and differentiation within places. Locality studies developed in the context of social polarization to learn how actors were dealing with the constraints of structural change. The reality of shared physical location, implying shared social locations in terms of constraints and oppressions was understood as the social mapping onto the physical (Bourdieu, 1999), being placed alongside a recognition that the physical also maps onto the social, (Dorling, 2001). Place was conceived not merely as a structure in the sense of structure regarded as a 'thing': it was understood as created by and constraining agency in the context of relations of power. The objectification of agentless places remained, although some of the literature began to consider the significance of locality in relation to localism, linking community to place (Cohen 1985). Dramatic changes brought about by globalization also inspired scalar thinking to consider the relations between local and global, national and regional, sometimes thought of as nested levels. Recognition of the interpenetration between them, particularly in the context of new technologies inspired the final turn to networks completing a relational shift in rethinking the role of place, boundaries, and spatial relations. Some influential postmodern and post-structuralist critiques went so far as to reject structure entirely in favour of a geography of networks and flows.

Our focus is less on the substance of these debates than on the kind of world they describe, moving from solid, bounded territory to indefinable spaces of networks and relations. Within them however are ideas of scale that could be conceived as hermetically sealed, in which physical boundaries serve to demarcate the arena of interest from powerful cross-cutting interests that were no longer physically grounded. This new reality was encountered in dealing with the British 'regional problem', which was defined in terms of industrial restructuring as a result of the disorganization of capitalism and the spatial separation of capital from the workforce (Lash and Urry, 1987). Regional change brought about internal polarization but clearly responded to global restructuring. Interregional relations constituted by economic, social, cultural, and political forces undermined simple notions of regional economies and regional identities. As Sayer (2000) points out, attempts to provide nested accounts can 'scramble' relations in reducing and reconstituting them. The new regional geography sought to work with scales arguing that space and time were constitutive rather than passive/objective. It involved rethinking the very conceptualization of regions and 'the regional question' in an era of increasingly geographically extended spatial flows and an intellectual context where space is frequently being imagined as a product of networks and relations, in contrast to an older topography in which territoriality was dominant (Amin et al. 2003).

Scalar thinking also faced challenges in focusing on the role of bounded places as the basis for actions or identities, again confronting the impossibility of hermetically sealed regions given their particular structural location in capitalism (Hudson 2007 p1158).

In questioning the simple coherence of territories as a basis for organization and operation of social processes, attention turned to cross-boundary forces and to reconceptualizing scalar notions of national, regional, and local while retaining some of their core assumptions. This was not reflected in the politics of regional planning and the continuing importance of both constructs is noted by Allen and Cochrane (2007) in their attempt both to challenge scalar ordering and to highlight the significance of a politics of scale:

> The argument was … that … any region, is made and remade by political processes that stretch beyond it and impact unevenly. To assert that regions are political

> constructs therefore, is not to suggest that such constructs 'contain' in some way the very governance relationship that 'invented' them.

Two geographies are identified, one that deals with the outcomes of political interventions and reflects the fact that such definitions are 'real in their consequences' (Thomas, 1928), while the other considers relationships and interactions that have no clear boundary consistent with region. This latter sensitivity led Allen and Cochrane to suggest it would be more appropriate to think in terms of a 'more diffuse and fragmented form of governance as a "regional" assemblage, rather than as a series of regional institutions that are territorially fixed in some way' (2007, 1172).

Recent development in scalar thinking, then, appears to incorporate relative and relational dimensions to focus on reconceiving boundaries beyond their dividing roles, consistent with Cilliers's (2001) argument that boundaries must be seen not only as dividing but as connecting, as sites of interpenetration that make it difficult to separate inside from outside in any simple way. While relational thinking has mostly been associated with post-structuralist thinkers, Harvey, Amin, Massey, and others have developed their work along these lines. Rather than the flat ontology of ungrounded networks, their approaches draw on all three levels. In regional geography, in particular the relational turn challenges an exclusively territorial focus. Its consequences for policy and application arise in reframing reality to include the relevance of multiple perspectives and scales away from simple nested hierarchies and towards more fluid and interpenetrating networks.

Thinking about space in relational terms has become a key recent development in human geography, but one which raises its own problems of definition and theory (Jones 2009). We can return to Lee to identify the project, suggesting a fundamentally different role from taxonomy:

> So the critical point about geography as a catalogue or project of classification is not that it is hardly the basis of a worthwhile intellectual project but that, without relational notions of difference and the formative power of spatial scale, it stands in danger of essentializing people and places as sets merely of internal relations rather than as complex and never-completed products of difference.
>
> (2002 342)

The notions of flows and networks then, that represented such a crucial departure from territorial thinking, are now incorporated in interaction with a material world to acknowledge that multiple concepts must be used simultaneously. Space must be seen in terms of networks and flows, not as alternatives but in addition to understanding the role of boundaries as real in spatial configurations (Amin et al., 2003).

This move away from nested thinking is one that seeks to incorporate agency. We started by using Jessop et al.'s (2008) framework to address the tendency towards one-dimensionalism apparent in separate exploration of territory, place, scale, and networks. They argued that, rather than alternatives, each concept contributes to a field of operation in which structuring principles can be seen in specific historical conjunctions and power relations. Their approach seeks to overcome the competitive focus on what are closely intertwined theoretical and empirical issues. Drawing on Geertz's notion of thick description in ethnographic work, they propose an analysis that includes more than one dimension in historically located and spatially specified and contextualized description, a tendency they suggest is beginning to be evident in the use of concepts such as glocalization. These dimensions are to be subject to a polymorphic analysis and Jessop et al. produce a Territory, Scale, Place, and Network grid (TPSN) whose aim is to respond to the challenges of developing complex-concrete analyses that are systematically, reflexively attuned to the polymorphy of sociospatial relations. (2008 392).

The TPSN framework necessarily carries the dangers of any theoretical scheme (as is recognized by Jessop et al) that frameworks can be treated as abstractions imposed on the world. Mayer (2008) has argued that the framework presents the danger of reifying while Paasi (2008) expresses a related concern in its ability to recognize the importance of agency. These criticisms highlight a relation between theory and empirical research as more appropriately one of continual reconceptualization to take account of the changing nature and meaning of these concepts in use. For our purposes it sums up a focus on theoretical work on interactions in contingent relations which pays attention to the ways that space is both constituted and constitutive in multiple dimensions.

These developments in relational thinking are equally evident in an approach by Jones (2009) working with Poincare's 'phase space' to identify four dimensional space-time in which dynamical systems perform, that 'acknowledges the relational meaning of space but insists on the confined, connected, internal and always context specific nature of existence and emergence' (Jones 2009 489). He argues, 'the spatial project for relational thinkers is to replace topography and structure-agency dichotomies with a topological theory of space, place and politics as encountered, performed and fluid' (2009 492). This emphasis on both the relational and material in the 'space of the possible' (2009 499) parallels Bourdieu's discussion of relations. It is contrasted by Jones with the relational thinking that, 'insufficiently problematizes boundedness, inertia, power, and time – and internalizes spatiality into our cultural constructions to such an extent that it perhaps loses senses of space as something external or given' (2009 499).

This challenges the flat ontological networks proposed in postmodern relational accounts. Networks are not seen as free floating but are grounded in existing and previous historical/spatial material contexts. The approach has obvious resonance with Massey's analysis of the sedimentation of rounds of accumulation in capitalist development. Consistent with the focus developed in earlier chapters, the project becomes one of understanding both the roles of objects and their relations as transitory or relatively permanent. Territory and its governance are now reconceived as 'plastic achievements' (Jones, 2009 p495) just as we have described the structure–agency relationship in the last chapter. These perspectives are evident in describing relations between structure and flow in Hudson (2007), and reflected in the discussion of process and permanence in Harvey (2004).

Complexity and space

So we have a concept of space that can work with complexity framing. The reality of place as a container continues to have effect but this is no longer the empty container of Newton but is reconceived as interacting levels and networks in a loosely nested open system in which the outcomes of the processes are never, in a mechanical sense, determined, but always actively achieved. Further, the reflexive relation between 'observed reality' and the production and adoption of concepts in both academic and policy literatures itself reflects a recursive relationship that continues to shape the field.

These conceptualizations of social processes and structures in spatial relation have responded to changes identified in globalization and in technologies that have had significant consequences for the organization, use, and meaning of space. While the relational move has been identified by geographers primarily with post-structuralist interpretations and flat ontology there is a significant challenge to that view in recent work bringing spatial processes into an active relation, consistent with a complexity understanding of causal configurations.

Thrift (1999) commented on a surprising lack of interest in complexity theory among geographers, despite its obvious resonances in dealing with distribution and the 'messy' nature of things. As a theory which is, arguably, 'about, precisely, the spatial ordering that arises from injections of energy. … Its whole structure depends upon emergent properties arising out of excitable spatial orders over time' (1999 32). Thrift found explanation for that lack of enthusiasm in its preliminary adoption by quantitatively focused academics mostly inspired by CA and ABM, (approaches, that we have described as restricted complexity) (O'Sullivan 2008). More general complexity work with modelling has been developed in challenging aspatial economic theory. This approach models spatial assumptions in a heuristic way to produce multiple 'cases of the possible' but does not seek to predict which will occur.

There are significant sympathies between the approach to ontology of the spatial theorists identified here and a theory of complex social systems as open and interactive. Our brief discussion has sought to highlight the effects of conceptions of place, among academics, addressing policymakers as well as those of actors responding to the material and cultural realities of a particular social world.

We should note finally that while the conceptual development (talk about concepts) that we have discussed here has significantly undermined the dominance of the traditional notion of local as a container it nevertheless reappears in much empirical literature (talk with them) such as that describing a nested relationship between local, regional and national levels as was noted by Allen and Cochrane (2007):

> in many respects the vocabulary has remained trapped within a framework that attempts to identify new territorial settlements, even if the size and nature of the territories has changed.
>
> (2007 1162)

Thinking of space as a setting continues to underpin objectified approaches to empirical work while the container concept of space shapes policy and planning, as Sayer argues in the 'place matters' literature there has remained a tendency to conduct 'political economy without ethnography' (2000 134) and yielding explanations of restructuring as the economics of the local.

We have looked at work favouring emergence based on relationships between structures and agents in particular material conditions and relations. While Massey's notion of sedimentation has strong structural connotations, its purpose was to emphasize that the histories of places, laid over time in experience of place matters, and at least in some degree frames the orientations of actors within them.

Time

Recent attention has focused on the role of time in illuminating social processes, particularly as developed in the work of Adam (1990, 1995). Adam challenges social theory structured according to Newtonian time and associated dualist thinking, suggesting ways of reconceiving it to allow social theory to engage with the greater complexity of social reality. While we have reflected theorists' rejection of dualisms in previous chapters, they nevertheless pervade social scientific thought and, beyond that, policy and practice. They thereby shape the social world because, as we identified above, they are real in their consequences.

In the remainder of this chapter, we will review issues raised for complexity theory by reflecting on approaches developed by Durkheim, Elias, and others, but will focus primarily on Adam's work pointing to the consequences of ontological and epistemological framing of temporal knowledge. Just as space was treated as a container, Euclidean time has been taken for granted and it is only in recent decades that time has received significant attention in social theory, leading to a wide ranging but largely un-integrated field of study (Adam 1990).

Historically, the Newtonian concept of time framed research in an unreflexive way because its operationalization as clock time became so dominant in western industrial societies that it excluded other forms. The linear perspective of enlightenment rationalization had significant consequences for epistemology, separating observers from their subject matter and developing reductionist science: 'Abstraction from context, objective observation, quantification of sense data, the single fixed focus, preference for space over time and the association of the "real" with visibility are its innovative features' (Adam, 2003 p62).

Clock time, both a consequence of and an instrument for the development of capitalism in conditions of increased structural complexity, has become naturalized. So although multiple concepts of time can be detected in the literature they largely remained implicit (Elchardus, 1988). Most frequently, where time is problematized, the precision and context independence of clock time is opposed to natural time of the seasons, although as Elchardus points out, the selection of relevant measures of 'natural' time is, itself, social. Seasons are significant to agrarian societies, while for many in industrialized and post-industrial societies more abstract clock time has become an intuitive basis for social life. As the creature of industrialization, clock time is accompanied by four other temporal strategies; commodification; compression of time; control of time; colonization of time (Adam, 1990). For Adam it is the unintended consequences of reflexive modernization that challenge this naturalization and have thus drawn the attention of social scientists to the sources of time construction and the power relations inherent within them. While awareness that, used alone, clock time undermines understandings of social process was evident in the work of Durkheim and Elias, constraints on its success spring from this Newtonian framing.

Social time/natural time dualism

The primary distinction in social theory between natural and social time was developed by Durkheim to enable research into the collective rhythms of social life. He defined social time as a construct, changing across cultures and over history. While introducing radically new thinking this established a dualism in which nature as non-reflexive was separated from social as reflexive systems (Nowotny, 1992). The distinction between social time of interaction and individual time was particularly concerned with the organisation of collective life in highly differentiated societies in which time becomes a normative basis for interaction and control. It led to a view that all time is social time, because only humans live an 'in time' existence. This has subsequently been challenged by research in natural sciences, where organizational aspects of organic and inorganic existence are also found (Adam 1990).

While Durkheim opened the way to an analysis of contingent social time there was little subsequent attention in social theory to time alone. Sorokin and Merton (1937) developed the theory along functionalist lines and time continued to be treated by most theorists as external to actors, flowing inevitably onwards, producing the second dimension of the grid upon which social life was played out. In Elias's work (1992) placing time at the centre implied discussion of time and meaning as a significant focus explaining social reproduction. It played a prominent role in examining relationships of structure and agency in his figurational sociology.

Time as continuity/discontinuity and recurrence

Elias considered two forms of time continuity/discontinuity and recurrence. He argued that the purpose of the concept of social time was to allow narrative and meaning to be endowed within those significant individual and collective processes that constitute history. Social time was conceived as a way of developing this meaning as fundamental to both individual and collective identity. Continuity is important as the core of identity in which people use shared symbols and memory at both individual and organizational levels, enabling reasonable expectations of, and orientations to, the future yet without the certainty of prediction. Its meaning for any account of structure–agency relationships is then fundamental, and its complexity relevance lies in the emphasis it places on agency. Structures can be discerned in interaction both in individual and collective history, as not simply existing, but as interpreted and reconstructed. Time is defined as

> a symbol of a relationship that a human group of beings biologically endowed with the capacity for memory and synthesis, establishes between two or more continua of changes, one of which is used by it as a frame of reference or standard of measurement for the other or others.
>
> (Elias 1992 46)

Timing is redefined away from its conventional understanding to denote its essential role, to mark people's capacity for connecting relations between meaningful objects, thereby creating a narrative. These core conditions of relative change and sequential ordering are the crucial basis of human agency because they allow actors to impose meaning and order and to develop the collective coordination that we have identified as essential to differentiated societies.

This view of time as relational, expressing relationships between continua, highlights the selection of criteria for measurement and hence its social construction. For Elias, time has three dimensions, being understood as social, individual, and natural. Social time has meaning in the coordination of organizational, family, and informal activities; as the individual biological time it is associated with health and well-being and as natural, the time of the seasons. Its role in creating order was evident in comparative studies. In modern society self-regulation is pervasive. Neither biologically nor metaphysically inspired, it responds to demands on the self that originate in the social world, which Elias described as the civilizing process. Time becomes a focus of constraints on freedom and the exercise of self-limitation that permeate the actions of individuals and organizations.

To explain the mechanism by which this self-control operates, Elias used the concept of habitus to denote the internalization of temporal norms that allow the organization of social life. These internalized norms ensure continuity in reproduction and make discontinuity manageable;

> Temporal norms would seem to play the eminently social role of guaranteeing the organisation of work, the systematic satisfaction of reciprocal expectations in people's behaviour towards each other, at the same time as they express evaluations and moral positions in the face of the fundamental experience of change and the awareness of death.
>
> (1992 9)

An unreflexive habitus allows reproduction to occur without conscious choice but as we discussed in the last chapter, is nevertheless a process in which actors must work to maintain identity at individual and collective levels. Referring to Bourdieu's image of habitus as operating as a 'fish in water' a metaphor used to describe harmony between habitus and field, we can see the active work the fish does in filtering the water, experiencing events, and remaining in harmony. In Elias's view, this internalization is so pervasive that the inability to break the social rule of punctuality is not an individual trait but rather an internalized habitus, socially acquired, which becomes part of the personality structure. Internalized self-restraint, as a necessary part of a civilizing process, enables a smoother fit between individual and social. We could be in structural functionalist territory were it not for attention to time, which Elias reminds us, means that the future is open.

In a move that is highly resonant with complexity, Elias rejected dichotomous thinking about natural and social impulses, arguing that they represent a figuration of interdependence, which exist in a tension that is constantly managed. He challenged the opposition of control within work and freedom outside of it as imaginary extremes that are never realized. Instead, he proposed that social actions need to be understood as multiple, interconnecting, and interdependent. In figuration, rather than a simple oppression/freedom dichotomy, people subordinate their impulses to what they consider to be overriding necessities. Individuals construct their own time in relation to these social and natural constraints. The study of time offers a better way to understand structure and agency relationships as they are manifest in experience, allowing the researcher to understand the unfolding action, the processes of choice making, and the significance and effect of condition and contexts (Tabboni, 2001). This has been developed in Adam's research into the use of time as creative action and meaning in the context of a complex of social conditions (1995).

Elias offers an interactive real that challenges Durkheim's opposition between individual, social, or natural times, pointing to levels of human experience of change and choice more consistent with our complexity frame. Individual experience is conceived as built on choices made out of existing and interpenetrating social and natural times, we would term it loosely nested and interpenetrating structures. This view has strong resonance with our discussion of Archer's internal conversation, dealing creatively with personal projects in relation to expectations and previous experience. For Elias time is not just collective rhythm, as Durkheim conceived it, but also a social construction which varies with the development of civilization. It is not simply a measure but a means of control whose ultimate expression is clock time.

These approaches to time significantly advanced their use in social theory but in the end neither overcomes the problems of Cartesian dualism (Adam, 1990). Durkheim's social time was established in relation to it and, despite his recognition of the problems posed by dualist thinking, Elias is unable to escape it. His attempt as synthesis ultimately fails because his analysis is caught in an epistemological framework that views reality as something that can be abstracted and reduced.

Significant challenges have been mounted to the Newtonian conception of time as absolute objective and quantifiable. Adam's work brings forward developments in natural science that include alternative conceptions of time; the relative time of Einstein in which the significance of the frame of observation becomes relevant, and quantum time with its flow and its non-reductionist oneness in which the irreversibility of time becomes apparent. These concepts alongside the rhythmicity of life cycles, processes, and structures that can be described as iterative (albeit not goal directed) rather than repetitive, suggest a more multiple and mutually implicating analysis of time. Human social life forms part of a whole time rather than being, dualistically, in opposition to natural time. By undermining structuring in terms of dualisms the possibilities for multiple interacting relations arise that are consistent with a complexity frame.

Time and systems reconceived

We briefly highlight the work of Luhmann here because he has paid explicit attention to the forms of time identified by Adam above, and in doing so has proposed an explicitly temporalized system. In evolutionary theories we discussed systems operating through time conceived as closed and equilibric, modelling stasis rather than dynamism through feedback loops with a longer-term directionality. Luhmann reworked these closed systems with a radical conception of their essential temporal nature, incorporating linear and cyclical time, constituted at every level of existence. A temporalized system is one that deals with the irreversible passage of time by developing structures that link acts that are by their nature ephemeral. For Luhmann structure is an inherently temporal concept because it encapsulates the relationship between elements 'across temporal distance' (Luhmann, 1995 p282). This rethinking proposes systems in terms of events, occurring in the present. The autopoietic social system comprises communicative actions that are meaningful as part of a recursive network of information, communication, and comprehension. Time is understood as difference between past and future and distinguished from chronology and motion. His focus on the present emphasizes that past and future are selections. The past is reinterpreted so its meaning is present specific, while the anticipated future is similarly selected in the context of past/present interpretations. A core distinction is established between this and abstract, objectified, context independent, translatable/comparable and reversible 'world time' in which he points out the connection between what lies in past and future is, in principle, contingent. These insights are taken forward by Adam, who develops the discussion of time in explicit complexity terms. Defining the problem for theory in the challenge posed by dualisms, she considers those attempts to overcome in terms of dualities, transcendence, and synthesis made by Giddens, Luhmann, and Elias have been unsuccessful.

To study time in its complexity, Adam tells us we must seek the relations between

> time temporality, tempo and timing, between clock time, chronology, social time and time consciousness, between motions, process, change, continuity and the temporal modalities of past present and future, between time as a resource, as an ordering principle and as becoming of the possible. Or between any combination of these.
>
> (1990 13)

Adam's critique of work on social time describes a field of 'conceptual chaos'. Challenging the still-dominant position that all time is social time, she argues that human time and social time must be understood as part of all time to enable us to think of time beyond Newtonian/Cartesian structure that has constrained social theory in the past. Elias's relational time was undermined by disciplinary science to be overcome through a postdisciplinary approach in which time is the common denominator. Adam proposes the alternative of a holographic metaphor in which the relationship between part and whole is not reductive but in which the whole is encoded within its parts. The holographic principles of

> simultaneity, mutual implication and complexity, the time aspects that pose such insoluble difficulty for traditional social theory appear manageable for a theory based on holographic principles.
>
> (1990 160)

This reconceptualization has consequences for time understood in terms of mutual implication and as fundamentally interdisciplinary, consistent with the view on the prospects for social science we identified in the Gulbenkian Commission report. It implies some 'points of departure' for social theory. Adam suggests a 'levels' approach, inspired by thermodynamic theory, replacing the Newtonian frame, accepting multiplicity and mutual implication. Identifying that we are, create and experience time simultaneously, Adam draws on Mead's fluid conception of consciousness as a continuum grounded in the emergence and the principle of sociality (1990 163) and on Giddens's discussion of the time spans of Duree (daily life), Dasein (lifetime), and Longue durée (history) acting simultaneously on the present. Time's irreversibility is based on the impossibility of undoing the past, while the impossibility of knowing the past lies in the mediating role of present knowledge. This complex understanding of time takes us a considerable distance from the Euclidean conception we started with.

Rethinking temporal order: the future

This focus on processes of construction and authorship alerts us to the significance of the future to social theory, an interest that is gaining momentum in postdisciplinary studies (Adam, 2008). Social theory dealing with agency has viewed the future as a time open to influence, important in shaping the present. In distinguishing action from behaviour, Weber identified actors as future oriented, acting in response to imagined, perhaps never realised futures. The reflexivity of actors makes time a core element of explanation so in explaining reproduction we are dealing with space-time that is materially instantiated and space-time that may never happen. As with Luhmann, dealing with the impact of the future on the present requires rethinking time beyond its naturalized linear form. Adam has developed this, in the context of the under-determination of the past, to emphasize the importance the causal directions future to present and past in temporality and to allow, then, that the future to present dimension is materially constituted, as Adam expresses it the 'for-ness of things' (Adam, 2008):

> future oriented and future-created knowledge practices have material effects that reverberate through the entire systems of physical, biological and cultural relations and processes. (2008 6)

We come to a conception of time that brings agency to the fore while recognizing that time is constructed out of and in relation to the real, reflecting the essence of complexity thinking.

Space and time

Having considered the scope of the concepts of space and time in isolation, we can respond to an emphasis in each field on the importance of the other in a non-reductionist explanation. While Jessop et al (2008) discussed four spatial concepts that required polymorphic treatment in a historical context, we can now extend this analysis to the temporal nature of systems, beyond the naturalized historical context implied. The need to complete this shift is increasingly acknowledged, expressed here by Jones (2009) quoting LePoidevin:

> Combining these two, we have a picture of space and time as an ordered series of states of affairs concerning the properties of and relations between, concrete objects and their fields of force. The extent to which we think of space and time [and the practices and politics therein] as independent of their contents will affect our view of their boundedness (or unboundedness).
>
> (Le Poidevin, 2003: 238, emphasis added by Jones)

Recent work on space has argued the need to reconfigure its relationship to time in order to develop concepts that move on from Newtonian framing, conceptualizing instead a field of space-time (Massey, 1992). Einstein and Poincare are drawn upon for their discussion of four dimensional space-time, in which, rather than abstraction the focus is on how they are inextricably interwoven. As Amin et al. point out, 'space is not static nor time spaceless' (2003 80). Neither can be understood in the absence of the other.

Spatial theorists are working with both interdependent and discrete sets of networks and relations at different spatial scales, existing simultaneously but involved in processes that must be understood over time. Notable attempts to develop space-time concepts in the work of Giddens (time-space distanciation) Harvey (time-space compression) have sought to incorporate this mutual implication and much empirical attention has focused on the analysis of processes of copresence and stretching over time and place made possible by money economies and modern technologies. Returning to a distinction identified in Chapter 5, this can also be represented as a development in which system integration is progressively replacing social integration as the relevant social relations become founded in expert systems.

Similarly, Adam argues that in developing complex accounts of causality each dimension needs to recognize the ways in which others are implicated. Geographers, Adam says, use space-time trajectories and Giddens time-space distanciation, but both have treated their interaction as a grid rather than curved in the relative way that Einstein identified. To illustrate the significance of the reworking she proposes, we conclude with a lengthy quote from Suteneau (2005) that highlights the implications of processes of interaction in time and space, analogous to one we identified in terms of transformation of quantity into quality and which pinpoints complexity:

> What I would like to emphasize here is that the acceleration of interactions and their predominantly informational nature do not simply mean that things occur faster. If we modelled this process, what we would see is not simply a more rapid dynamics,

as if the model were running on a faster computer. The pace of communication will always interact with the rhythm of various parts of our system. Some of these rhythms are biologically conditioned while others are the consequence of psychological or social mechanisms. Many, if not all, of these rhythms may also change when interacting with each other. Depending on their phasing, these rhythms alone may generate outcomes that could be very different from each other, encompassing surprising scenarios. However, the situation is not simple and many other factors can be expected to change too, which adds to the difficulty of making meaningful forecasts. We should be prepared, therefore, to face unforeseen implications, such as changes in the quality of human communication and forms of creativity.

(2005 133)

Complexity relevance

Recent work on time and place has moved from a focus on order/disorder dichotomies to recognize that the simple images of continuous time and bounded places are an insufficient basis for understanding social production and reproduction. Developments in the study of both time and place acknowledge multiple relations between causes and effect and their frequent disproportionality. Immediately we can see resonance with complexity theory in which both spatial and temporal dimensions are core and their interaction is the basis of emergent social relations.

In earlier chapters, we identified the importance of space and time in complex social systems in social theory that deals with system history as punctuated, rather than as smooth unilinear development of societies bounded, even nested in space. In this chapter, we have developed that account further in recognizing the importance of time and place in the dynamics of systems.

We used time in earlier chapters to emphasize the irreversibility of open systems. This supports an understanding of the practical and precarious achievement of actors working with structures in recursive relationships that are iteratively rather than repetitively framed. The exclusion of time in structural functionalist thinking provided an illustration of the consequences of working with closed equilibric systems and we recognized that patterns and regularities can only be detected over time. Building on this appreciation comes an understanding of time as active in the processes of reproduction, contributing to their emergent character. Emergence is the expression of structure–agency interactions in systems, over history, in which these crucial contextual elements of space and time operate in contingent ways.

In discussing work done within the disciplines of geography and sociology we have seen challenges to dichotomous thinking and the objective stance of the observer, leading theorists from each discipline to identify their implications for politics, ethics, and responsibility. For Massey, this is evident in the ways that space is represented by powerful agents including the state as well as the ways that people experience space collectively and assign symbolic meanings to it. Adam similarly identifies the implications of a view of complexity of times as the context for ethics and for theory development:

> For today's' social theories this means globalized times and social relations; technologies based on both Newtonian and non-Newtonian principles; environmental hazards created by the scientific and industrial way of life outside the control of science, economics and politics; satellites transmitting virtual realities, pictures of wars being

> fought across the globe, while real bombs still kill and maim people and destroy their lives, to name just a few of these contradictory complexities.
>
> (Adam, 1995 p153)

In complexity theory, Adam finds recognition of the mutual constitution of the material-spatial-temporal. To this triple basis of analysis is added a fourth based in Capra's discussion of the importance of meaning, 'the evolutionary emergent property of the social', which returns us to the significance of stance, because as Adam notes, 'social investigators interpret a pre-interpreted world', so the meaning must be understood as knowledge practices to emphasize the active construction of knowledge and the authorship of those constructing it, in comparison with the distance between subject and object in Newtonian science. This locates responsibility because a complexity perspective divests us of the 'view for nowhere'. Such practice

> transforms recognition of complexity from a rather non-committal and circumspect pluralism into a critical and radical social science criterion whose points of departure are both continuous with and distinct from allied postmodern and feminist critiques of that tradition of thought.
>
> (1995 p151)

Adam has welcomed complexity theory for time analysis – its focus on emergent properties, non-linear dynamics, and feedback loops. It points to the need to acknowledge both the incommensurability and contingency recognized by postmodernists and the importance of 'unified measure and totalising tendency of quantified spatialized time' (1995 156). To do otherwise, she suggests, is to return to the binary thinking that both postmodernists and complexity theorists reject.

So we can identify some clear resonances between developments in theories of time and place that have significant promise for complexity theory. Both take a holistic and systemic approach that can be best served by postdisciplinary study. In both, the notion of developmentalism, that was identified as central to some evolutionary theories, is replaced by notions of openness, emergence in which abrupt change is possible, and finally, both have developed in rejection of functionalist equilibric ideas.

This work has substantial implication for theory focusing on emergent and co-evolving processes. Structures are no longer defined in terms of stability across time, rather systems are continually produced and reproduced in the work that actors do in the world. Social laws are abandoned in favour of an ontology of mutable systems composed of individuals and collectives in the relationships they are establishing with past and future. Luhmann's insights point to reflexivity in temporalized systems in which structure is conceived as enabling reproduction across time. The critical realist basis of complexity theory allows a search for the contingent in open systems because it expresses the nature of a social world that can explain both unique and regular events in terms of configurations embedded in time and space. In dispensing with bounded, dualist time and space we have approaches that can explore interactions, networks, and emergence in recognition that the study of time and space is the study of social reproduction.

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7 Social theory and macro system transformation

There is a tendency for students to learn the canon of theory in their undergraduate careers and to leave it behind on graduating, particularly as they enter the applied professions. Social science cannot exist without theory, yet in these days of 'choice' this fundamental element is being marginalized in favour of 'pick and mix' degree programmes. Thus far we have considered theory relating to complexity-relevant processes and relations. In this chapter we want to seek out resonances between the epistemological and methodological concerns of some major macro-social theorists and the complexity frame. Maintaining an active role for theory has value for practical social science for two reasons: it problematizes the everyday language we use to allow us to achieve a critical relationship to it and, related to that, it enables a dialogue with evidence. It is a tool for thinking with rather than a formula to follow and it is in this spirit we want to use it here.

Whatever approach is taken in the social sciences, the language of the everyday is the language of our concepts. Without definition and theoretical examination, importing those 'common sense' concepts becomes inevitable and leads to what Bourdieu et al describe as 'spontaneous sociology', because ordinary language carries with it a 'legacy of ideas … a petrified philosophy of the social' (1991 21). The difficulty in achieving a break between the two was as apparent to early social theorists as it is to us today. It can result in reducing social relations to personal relationships and writing out structural realities. The tendency for social science to adopt physical science metaphors brings further problems, again importing ideas and the content of concepts along with its language. Social theory seeks to achieve those breaks from common sense usage, to enable a critical approach that can draw out causes beyond the level of individual action, without losing the significance of action. In this way, we can identify relatively permanent structures as embedded and embodied in individuals and groups and as having causal power. Whether the research be at the level of world systems or local community, breaking from the preconstructed objects that we come to our studies armed with represents a necessary first step.

In our complexity frame, we adopt an approach to theory which engages with it as dialogue. We are not interested in Marx or Weber studies for their own sake but rather in how we can use them to understand the social world in research today. It is a discursive rather than cumulative process which develops explicit and systematic concepts and models in articulation with data. In this way social science takes on the active and always evolving character that echoes the world it studies. We have chosen Marx, Weber, Polanyi, Wallerstein as our figures in this chapter to access important theoretical approaches that cast light on the world in a way that we can read through complexity. It is, of course, a small selection from a much larger body of theory and represents a process of exemplification rather than an attempt to provide a complexity canon. Our aim

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is twofold. First, to argue that theory is essential because without it empirical research necessarily draws on the unexamined constructs of the everyday, and second, to resist the equally problematic blind adoption of any theoretical frame. A complexity frame requires robust engagement because, as Bourdieu et al quoting Bachelard reminds us, 'The knowing has to evolve with the known' (1991, 9).

Approaching our aim through particular theorists serves to limit our ambition rather than to define the field of interest. In keeping with our discursive approach we want to identify themes from our chosen theorists that are fundamental to complexity analysis. None restrict their work within disciplinary boundaries and all have been significant figures in disciplines across the social sciences. To identify the themes that work most effectively with a complexity frame we will review ontological and epistemological positions and go on to consider how these theorists talked about causal processes, the nature of systems, and the dynamics of their operation.

Marx

The discussion and interpretation of Marx's work has seen the commitment of lifetimes of scholarship which cannot be mirrored in this short review. Our purpose here is to understand type and structure and to look for signs that can lead to a complexity-consistent analysis across a range of social theory. We will briefly sketch out the basis of Marx's materialism and dialectics to provide a context for examining his approach to the study of social change.

We start from his ontology which is a materialist one formed in response to both Hegel's idealism and Feuerbach's humanism, leading him to propose that an adequate understanding of social change must comprehend the role of the forces and relations of production in the formation of social class. In the ontological account that develops across Marx's works emerges a conception that we can frame in terms of systems and dynamics, manifest in history. Hegel adopted the dialectical approach to solve Kant's dualism of freedom and necessity, concluding that consciousness determines the conditions of social existence (Veltmeyer,1978). While privileging the ideal, Hegel did not reject the material in this formulation, rather Marx's criticism in *Grundrisse* is that,

> Hegel fell into the illusion of conceiving the real as the product of thought concentrating itself, probing its own depth, and unfolding itself out of itself, by itself, whereas the method of rising from the abstract to the concrete is only the way in which though appropriates the concrete, reproduces it as the concrete in the mind.
>
> (quoted in Rioux 2009 p599)

Although early Marx drew on idealist philosophy, he developed a view of class in terms of a historical dialectic that is the product of material determination and conscious action. Feuerbach moved man to the centre and made materialism the core principle. What Marx rejected in Feuerbach was his failure to comprehend the significance of practice. The young Marx, writing his sixth thesis on Feuerbach made this explicit:

'Feuerbach resolves the religious essence into human essence. But the essence of man is no abstraction inherent in each single individual. In reality it is the ensemble of social relationships' (1970 122).

Starting from the material base Marx applied Hegel's dialectic to the economy so that the relation of wage labour to capital becomes central and capital and its oppressions must be opposed in praxis. For Marx as a young Hegelian then, the crucial shift was to adopt Feuerbach's humanism and to correct what he saw as Hegel's inversion of the relationship between material and ideal allowing him to start from the material world and from man's production of that world.

> The first historical act is thus the production of … material life itself. And indeed this is an historical act, a fundamental condition of all history.
>
> (1970 48)

The nature and quality of this production have been at the core of debate on Marx's analysis and there is evidence in Marx's writings to support both strong economic determinism and a more nuanced reading which includes political and other social factors such as ethnicity and leadership. While various readings of Marx are permissible, we are most interested in that showing Marx as involved in a more complex undertaking than outlining a simple linear cause–effect relation (Needleman and Needleman 1969). Marx appreciated that understanding social forms must include the actions of individuals, groups, and institutions in history. The implication of this anti-reductionist reading of Marx is to reject methodological individualism and to pay attention to micro-foundations in articulation with macro-social theory (Levine et al 1987, Wright et al. 1992). It suggests that the heavily structural/determinist reading of Marx's theory turns what Marx intended as epistemological distinction for the purposes of analysis into an ontological difference.

Abstraction, relations, dynamics

We have noted from Sayer (1992), countering abstraction, that everything has space/time dimensions. We now want to look at the vital role abstraction plays in model/theory building as a first step in explanation. Abstraction is a method widely used across social science research because it allows causal explanation to be built but is never intended to stand as a representation of reality (Oliviero 2020). It is a heuristic device that allows the researcher to focus on constitutive structures from which their cultural manifestation in time and place can be understood. Marx used abstraction consciously as an analytic device and this has been the cause of some heavily deterministic readings of his work. The nature of his abstraction, however, points to the way he handled a world he understood in relational and dynamic terms. Ollman sums it us well for us;

> The view held by most people, both in and out of the academy, is that there are things and there are relations, and that neither can be subsumed in the other. On this view capital is a thing that has relations with other things, and it makes no sense treat these relations as parts of capital. Marx rejects this logical dichotomy and views capital itself as a complex Relation composed of its ties to other … relations … Moreover, since these relations extend backward and forward in time, this also makes what capital was as well as what it is likely to become essential parts of what it now is. This view, which Marx took over from Hegel, is known as the philosophy of internal relations.
>
> (Ollman 2003 176)

This, the foundation of dialectics, has significant consequence for his approach to analysis in that elements exist only in relation to each other. The constitutive role of a philosophy of internal relations becomes clear beyond Marx and has given rise to reflection on process-relational ontology in response to recent focus on network theory (Prey 2012). Network theory is a conception of 'things with relations to each other' that suggests a flat ontology of relations that are structured in terms of inclusion/exclusion (Prey 2012), Marx, by contrast, materialized relations so that the concept of exploitation, for example, is fundamentally internal. Treating the mutual interaction that takes place in relations of production as analogous to those within 'an organic body', Marx rejects their treatment as 'things' to emphasize their dynamic quality. This process ontology is materialized in production in Marx in a way comparable with Whitehead's process-relational approach (see Prey 2012). Production is defined in Marx in terms of activity and process and it is in engaging in productive activity that the actor produces himself. Unlike Castells' account therefore we have a materialization of a network rather than a transformation in terms of knowledge. In fact this relational character is necessarily prior to the network.

This ontological position necessarily implies that abstraction cannot work in a reductionist way. The purpose of abstraction, rather, is to explore phenomena, identifying relevant structures, mechanisms, and processes within their larger dynamic, temporally ordered context (Joseph 1998). Although frequently written about as proposing capitalism in abstract terms then, we can certainly read Marx's process of abstraction as creating models which could be realized in their specific conditions in history. The capitalist system appears as a result of the interaction of forces and relations of production but specific tendencies, such as the falling rate of profit, can be discerned as structuring those interactions. While no simple general account can be generated, we can discern the effects of those forces and relations empirically and historically (Tabb 2010).

The notion of an economic base and political, legal, and ideological superstructure established in *Grundrisse* (1973) and developed in *Capital* (1996) acknowledges their inseparability. In focusing on the material, Marx recognized it as including both sides of an interaction whose outcomes in human practice provide evidence of structures that persist over time, that are relatively permanent (Thompson 1980). It is an approach to analysis that, as we will see, has much in common with Weber's concern with both generality and specific manifestation of phenomena. Marx and Engels's discussion in *The German Ideology* points to an approach that

> is not devoid of premises. It starts from the real premises and does not abandon them for a moment. Its premises are men, not in any fantastic isolation and fixity, but in their actually empirically perceptible process of development under definite conditions. As soon as this active life process of development is described history ceases to be a collection of dead facts.
>
> (quoted in Roiux 2009, p595)

What emerges from this view of materialism is that it involves a dynamic and relational ontology. Transposed to a systems account the dynamics of class struggle, social relations of production, and falling rate of profit mean that it is impossible to see systems as closed. In open systems, the outcomes of such dynamics are contingent on interaction. In looking at relations rather than static entities we can begin to see non-linearities emerge. The structures that underpin regularities are abstracted for heuristic purposes but do not exist beyond their reproduction in practice. This applies at all levels and grounds his evolutionary account. Although Marx's work has been characterized as predictive in a strongly determinist sense, what we find are descriptions of forces, clearly identified as providing the dynamics of system change.

His theory then is based in contextual discussion of causality as it becomes manifest in history rather than as a set of law-like statements. It is, moreover, based in multiple causality as well as in cause as configurational in which historical events can trigger change. Adopting Williams's (1973) discussion of determinism in terms of setting limits and boundaries as opposed to a direct correspondence, brings coherence to this ontological view of a complex dynamic process and to an epistemological approach through the study of history.

So we can ask, how does Marx's theory provide an account of change and the rise of capitalism that shows consistency with complexity theory? In outlining the transition from feudalism to capitalism *The German Ideology* provides a systems account in which structural conditions are central. Marx's dialectic brings the role of class in history to the fore, the moving force being a class seeking to overcome its alienation through struggle within capitalism. A second radical shift appears in the method advocated, with a historical approach bringing a new perspective. Instead of a conception of a doctrine of man he proposes studying forces and processes in history, based in the material conditions of production. Forces become the actions of 'men' produced in response to a material base which is, itself realized through the actions of men, sometimes in conscious action but also carried through routinely. This is very clear in *The Eighteenth Brumaire*:

> Men make their own history, but not of their own free will; not under circumstances they themselves have chosen but under the given and inherited circumstances with which they are directly confronted.
>
> (1973 146)

His purpose was to understand the world in order to change it and that focus on action brings us forward to the overcoming of Newtonian separation and to the relevance of the participatory nature of knowledge.

One of the most debated issues in Marx's work is the problem of historical causation and we have argued that he does not solve this in terms of extreme determinism. The explanation of social change, rather, takes an evolutionary form in a discussion of emergent social forms that we can recognize in complexity terms. We have discussed historical materialism in Chapter 4 but it is useful here to identify the significance of direction and some characteristics of the explanatory type that we would want to embrace.

A Darwinian evolutionary process relies on theories of variation and natural selection, which need to be distinguished. Given existing variation in local form, selection processes can begin to operate. Variation is necessary for processes of selection to work, but there is no necessity for hard determination, indeed a deterministic reading is based on confusion of the two (Carling,1993). We will see this distinction worked through in world systems theory in relation to relative positions of core, semi-periphery, and periphery.

The German Ideology (1973) sets the development from feudalism to capitalism in an evolutionary frame, demonstrating the open system character of the world Marx was describing. System dynamism is founded in class struggle because, as Marx famously stated in the *Communist Manifesto*, 'the history of all hitherto existing … society is the history of class struggle' (1992). This renders it inherently unstable and vulnerable to periodic crises.

The system is constituted by forces and relations of production which interact in complex ways so that any actual manifestation is a historically specific one. Marx was working in terms of a 'real' that was underpinning but not necessarily directly empirically accessible, a relationship core to critical realism and one that also becomes apparent in qualitative debates (Maxwell 2012). Empirical work was essential to develop an analysis of actual historical development and the possibility of revolution. This non-determinist approach to change offers an account combining both complementary and contradictory forces with system boundaries in historical trajectories that can, but does not inevitably, lead to revolution. It yields a narrative of dynamic open systems with unpredictable but explicable outcomes. The significance of micro-foundations is clear and consistent with this understanding of complex causation. 'Evolutionary theory allows for a retrospective explanation for the transitions that in fact took place. But the specific sequence of the transitions is a consequence of countless exogenous events' (Wright et al. p79 1992). Marx made this explicit in his discussion of historical study (Needleman and Needleman 1969).

Weber

Weber's own life and work exemplified the rejection of strict disciplinary boundaries and while he was formative in the development of the academic study of sociology, his work is read widely across the social sciences. It was fundamental to Weber's theory of social action that it was situated historically and empirically so that subsequent efforts by Parsons and Schutz to strip out the historical dimension in service of a general theory of social action was inimical to Weber's writing (Zaret 1980). Weber is known for his stance on the relationship between facts and values and his rejection of a value-free social science. Value-laden enquiry is inevitable because, Weber argued, the construction of the empirical object is an interpretive process, so that not only the choice of problems, but the very objects of inquiry are value based. It implies that the researcher is involved in a conscious practice of examining how data is constructed in relation to the constructed object of study.

He proposed an opposition between social and natural science in recognition of the inappropriateness of Newtonian principles for the study of the social. For Weber, the distinction lies in the nature of those interests in that social science must deal with meaning. In natural (Newtonian) science the use of quantity and abstraction makes for general laws without historical dimension. The concern of social science with action and meaning presents other problems which require qualitative understanding and for which the rich context of historical dimension is necessary. In discussing the construction of theory, Zaret (1980) argues that Weber was making an explicit epistemological rather than an ontological distinction. If facts are constructed on the basis of theory then adequate methods and the criteria for 'fact' status are equally theory based. For Weber, the combination of historical and sociological analysis was necessary to understand a particular form of social organization in terms of the conditions of its emergence. His discussion of comparative design and ideal types as models significantly contributed to methodological development in the social sciences, an understanding that fits well with the complexity frame.

Explanation in social science must be adequate therefore, not only at the level of cause but of meaning and Weber constructed a method to understand on the one hand, 'individual events in their contemporary manifestation and on the other the causes of that being historically so and not otherwise' (Weber 1949, quoted in Zaret 1980 p1184). Clearly, social science must deal with complex relations of cultural and historical dimensions and this can only be achieved, as in Marx in the first instance, through a process of abstraction. Weber developed the concept of the Ideal Type as a way of systematically dealing with two related features of social change, to understand both how a specific phenomenon becomes manifest and is at the same time characteristic of a wider class.

Ideal types

The rationale for ideal type lies in the need to incorporate meaning in social science. In dealing with an endlessly complex reality, abstraction provides the basis for conceptual tools and modes of analysis to grasp theoretically significant phenomena and relationships. In doing so it departs from the general law-seeking approach of positivist science. Weber describes the process in relation to the spirit of capitalism as beginning with a 'provisional description' that can be documented 'in almost classical purity' (Weber 1992 p14). This ideal type is later related to religion. The initial independence of the two concepts allows clarity in building their relations in history.

The construction of an ideal type is based on understanding action and meaning in combination, using variation from rational action as a pointer to research questions. Swedberg (2018) provides a useful table setting out the procedure for constructing an ideal type in Weber's terms, although as he points out, Weber's specification ends at the point from which confrontation between abstraction and reality takes place.

What Weber was doing is highly recognizable as a complexity approach, constructing a model of interactive elements whose indicators can be both quantitative and qualitative. Instead of seeking a single dominant cause it looks for causes in combination springing from both material and cultural realms. It can thus acknowledge the significance of human agency in causal configurations. This focus on configurations has been related to the more recent work of Charles Ragin (1987, 2000) in the development of Qualitative Comparative Analysis (QCA) that we will look at in Chapter 10. As with the ideal type, both variables and cases are necessary to develop explanations that have depth and generality. The method is attractive in being 'combinatorial by design' (Agevall 2005, p9). In this way, Weber was offering tools that could be used to understand any historical period while rejecting attempts at prophecy (Albrow 2020). The 'historical individuals' that are identified by Weber are built in terms of 'multiple and conjectural causation' which as Agevall identifies, was foreign to his critics and the source of much misinterpretation (Agevall 2005, p15).

Elective affinities

A further complexity consistent strand of Weber's work is his employment of the concept of elective affinities. This is particularly interesting because it so obviously fits with our recognition of multiple and configurational causation. Affinity suggests a similarity in terms of compatibility of some kind 'a sympathy of one thing for another' (McKinnon 2010, p17), while elective implies 'choice' although not in the rational sense but rather in terms of tendency to fit. The attraction is mutual and therefore can be understood from either side.

Weber used elective affinities in a somewhat ill-defined way and the secondary literature seeks further meaning in terms of Goethe's novel of that title. The term originated with Bergman, a chemist, who used it in describing how the combination of elements could bring about the emergence of a new substance. Goethe adopted this idea in his novel seeking to show combinations and attractions among his characters. It is a concept that does important work, being used by Weber as a way of relating two phenomena, whether ideal or material without having to imply that one is simply caused by the other. This allowed him to acknowledge connections without presuming any determinate relationship (Thomas 1985). The causal relationship is one we can easily recognize:

> Elective affinities cannot readily be understood in the standard terms of cause and effect. Rather, pursuing the chemical metaphor, the elements form bonds and together produce a new substance because of the characteristics of each element and this is better understood as a kind of 'emergence'.
>
> (McKinnon 2010 123)

While this account is hardly consistent with Weber's professed methodological individualism it clearly provides a complexity consistent way of dealing with social research. It rejects the positivist linear cause–effect relation, allowing explanation to be focused while at the same time acknowledging that much more is present to be explored. It acknowledges configurations, for example of the protestant ethic and of the spirit of capitalism and points, contrary to Parsons, to a dialectic concept that 'becomes realized through mediation, interaction, merger, metamorphosis' (de Paula 2005 4).

Polanyi

One theme of this chapter has been that no theory can be drawn upon unreflectively and we have posited theory as discursive rather than as a set of universal laws. This was a clear basis for Weber's explanatory method of ideal type and is equally evident in Polanyi's work. Polanyi's project in The Great Transformation (1944) was to explain the transition from traditional to market society and, like Marx and Weber, he approached explanations through mechanisms that were not necessarily visible in an empiricist way. His interest was in the interaction between social, political, and economic spheres and, like them, he rejected ideas of disciplinary boundaries as a basis for studying them.

Polanyi developed his explanation of social change in terms of theories, class, and institutions. His focus on theories was central, dealing with them as not only ideas but as rationales for organizing human action that therefore have shaping power. Theories can become the 'truths' we live by and his concern was that the theory of market economy was inimical human goals (Polanyi 1944, p45). In consequence of free market theory becoming hegemonic it appears as an expression of 'human nature' rather than behaviour required for living within the system. In doing so it shapes social relations. A core concern for Polanyi was this naturalization, making the idea a basis for social organization. His historical work demonstrated the theory status of market society. The process which turned separate markets into a self-regulating market economy was seen as a natural phenomenon but was clearly a social creation whose existence must be explained.

> the control of the economic system by the market is of overwhelming consequence to the whole organisation of society: it means no less than the running of Society as an adjunct to the market. Instead of the economy being embedded in social relations, social relations are embedded in the economic system.
>
> (60 1944)

The role of such theories is to legitimate a particular order. This explains the dominance of classical economics that regards the market as a self-regulatory organ in which actors act rationally in pursuit of gain. For Polanyi, by contrast, the economy is always embedded in the social and reciprocity and redistribution are fundamental concepts for which there is abundant historical evidence.

The creation of a 'science' of economics fulfils that legitimation by establishing 'laws' governing the market economy that facilitate natural attribution.

> The law of diminishing returns was a law of plant physiology. The Malthusian law of population reflected the relationship between the fertility of man and that of the soil. In both cases the forces in play were the forces of Nature, the animal instinct of sex and the growth of vegetation in a given soil.
>
> (130, 1944)

Such naturalization permitted problems such as poverty to be reinterpreted as natural outcomes rather than human constructions. Free market theory both represented the economy as autonomous and its unfettered operation as the basis of human freedom (Block and Somers 2014). As a result a 'double movement' occurred in a dialectical process of materialization and social protection. The commodification of everything, including 'false' commodities of land, labour, and money, was responded to through laws and institutions seeking to counter this disembodying of the economy. In Polanyi's terms it represented a 'stark utopia', ignoring the reality that economies are everywhere embedded in the social and that change is possible through the political system.

Polanyi developed his account by studying societies across time to demonstrate how, as economies evolved, ideologies evolved to legitimate them, arguing that historical knowledge is essential to an understanding of economic systems. As with Weber, reformulating the significant elements of the historical account changes what counts as relevant evidence.

> Broadly … All economic systems known to us up to the end of feudalism in Western Europe were organised either on the principles of reciprocity or redistribution or householding, or some combination of the three. These principles were institutionalised with the help of a social organisation which, inter alia made use of the patterns of symmetry, centricity, and autarchy. In this framework the orderly production and distribution of goods was secured through a great for variety of individual motives disciplined by general principles of behaviour. Among these motives gain was not prominent. Custom and law, magic and religion cooperated in inducing the individual to comply with rules of behaviour which, eventually, ensured its functioning in the economic system.
>
> (Polanyi 1944 57)

Replacing social standing as the primary motivation while economic gain becomes secondary, suggests a very different way of looking at a market society. It is an explanation that allows for much greater internal diversity, conflict, and adaptation than the natural law style of the free market suggests. Explanation can be found in terms of the interests of different groups and the institutions created to embody interests which are not wholly or even predominantly economic. Polanyi questions the exclusive economic basis of class formation. Theory must be situated and historical and Polanyi formed his explanations in the long run. He recognized the destructive effect of commodification of nature as posing longer-term threats such as climate catastrophe.

By bringing reciprocity and the distribution of social resources into the account, he proposed a world of greater complexity. The evolution of socio-economic and political systems is an interactive process in which individuals both shape and are shaped. The significance of institutions is as an expression of collective will, producing mechanisms which we can see as actors in the system, not wholly reflective of that will but having some level of autonomy once created. Nonetheless, there is no simple correspondence because putting it in systems terms, parts do not act in a simply functional way and are replete with unintended consequences, an insight clear from Weber's work. In this more complex configuration, we do not need to seek ultimate causes. Rather than a basis for causal explanation, institutions are part of the evolutionary process. The collective will remains unrealized until an institution is created to provide the mechanism for it to act in the world. The double movement sees a system in which contradiction and accommodation required of capital can serve to sustain market resilience (Carton 2014). Polanyi sees institutions as limiting the nature of action and therefore institutional analysis as central. In just this way we will see Wallerstein describes state action in the world system as shaping and limiting.

Echoing Weber, Polanyi rejected the separation of observer from observed phenomena as inappropriate for the social sciences, precisely because theories shape understanding and responses to experience and therefore affect history. Polanyi was interested in the relation of theory to practice and the role of theory in shaping the world based on an understanding that observations are not possible absent a theoretical frame. In common with Marx we have the rejection of methodological individualism and with both Marx and Weber, a search for an explanation in terms of social and historical factors.

We can see a further theme also underpinning Wallerstein's (1996a) plea to open the social sciences in Polanyi's concern at the impact of the transformation from traditional to market on the ecology of the planet. The commodification of natural and human substances means that natural habitat is threatened with annihilation. This recognition takes on enormous significance in our current context.

World systems and Capitalocene

Drawing on insights from Marx, Weber, and Polanyi, among others, Wallerstein came to conceive of world systems theory in response to dominant theories of development in the 1960s and 1970s. While sharing common roots with complexity theory in Bertalanffy's general systems theory the two were 'ideologically separated at birth' (Grimes 2016). However, their common foundation makes complexity consistent work a rational next step and was an approach Wallerstein later developed.

Wallerstein's contribution brings systems framing to the level of analysis that includes the whole world economy in a system based on a single division of labour. This was a radical shift from comparative country analysis, a way of making sense of relative positions in an interactive form. Rather than comparing levels or phases of development as favoured by the modernization school, the explanation of individual country positions lies in their histories and their relations to others in the whole system. Although Wallerstein (1974) originally proposed world systems without reference to complex systems theory, he developed that dimension particularly in relation to Prigogine's work in the 1980s.

We should begin by noting that Wallerstein rejects the notion of world systems 'theory' in exactly the way that we have rejected it for complexity. In discussing his leadership of the Gulbenkian Commission we identified that his aim was to escape the disciplinary boundaries placed upon social scientific inquiry in the 19th century. World systems analysis seeks to reject the nomothetic/idiographic divide and opposition between point in time and the longue durée to develop an analytical frame that can work in the 'unexcluded middle'. Rather than a specific theory he argues, what is necessary is world systems analysis. Theory must be treated as a basis for contextual analysis rather than as universal law. It finds common ground with complexity in recognizing its historically situated position.

The analysis of the world systems involves a set of dynamic processes that take place within and between states defined as core, periphery, and semi-periphery. Wallerstein (1996b) emphasizes that it is a perspective and an intellectual movement that includes relations and processes across the whole world that structure it politically, economically, and culturally. World systems are distinguished from world empires, characterized by a single political system. Based on a single division of labour, world systems exhibit many different political and social-cultural systems. So we have an explanation of an inherently dynamic set of forces acting structurally and across time. In the 19th and 20th centuries, the capitalist world economy has become the only world system.

As in Weber's ideal type, working with the world systems model requires both historical specificity and analytical generality. The approach explores the whole system over time and, as with Marx and with Weber's ideal types, relies on a process of abstraction. Acknowledging the danger of importing meaning, the term 'society' is explicitly rejected in favour of 'historical system to avoid the common use of society and state as interchangeable' (Wallerstein 2002). Wallerstein's reframing has consequences for the mode of research and implies the theoretical shift which rejects single disciplines. He describes this process underway in the Ferdinand Braudel Centre for the study of historical systems and civilizations, where analysis not only focuses on la longue durée but also seeks to develop a more appropriate epistemology for social science. It brings together researchers with a wide range of skills and knowledge to facilitate exploration. Wallerstein cites two crucial, interrelated elements in Prigogine's work, 'the fundamental indeterminacy of all reality' (2002, P369) and that 'time irreversibility was absurd' (2002, p370), underpinning an understanding or order as relatively permanent but subject to bifurcations that may be understood in retrospect but not predicted.

The analytic work undertaken in this frame seeks to understand both periods of stability and of crisis in terms of order and far from equilibrium positions. It seeks to chart cycles over time through the complex interdependencies that allow for relative positions to change so that states can move between core, periphery, and semi-periphery status. Explanations for particular states and their relative positions must be through an understanding of the whole system, not from the standpoint of any individual entity. In this it reflects Marx's (Hegel's) philosophy of internal relations. The process is an evolutionary one. Incorporation into the world system shifts states out of their pre-existing political and economic trajectory to form a new set of core-periphery relations which brings with it a changing set of class relations. At the same time incorporation brings changes in relations within the world system itself. Wallerstein explicitly frames this in terms of complexity ontology, commenting on methodological individualism;

> once we recognise that reality is irreducibly complex the very notion of a monad is meaningless. To say that society is composed of individuals tells us no more than to say that molecules are composed of atoms … the reality of structures as determining

> outcomes no more negate the reality of biographical actions than asserting the reality of psychological processes negates the reality of physiological processes.
>
> (1996b 20)

This departure from dominant analysis that positions some nations as 'advanced' while others require 'development' demonstrates that peripheral countries exist as they do precisely because of the needs of core countries. At the same time this division of labour requires neither shared political structure nor culture. Through 'a series of accidents – historical, ecological, geographic–' (2000, P86), North West Europe has become core while Eastern Europe became periphery and Mediterranean Europe a semi-periphery. From their different starting points, the convergence of powerful interests gave rise to strong state mechanisms in some areas while their divergence in peripheral areas brought weak ones. Theories of the globalization of capital have significantly misunderstood the nature of this dynamic: 'Thus capitalism involves not only appropriation of surplus value by an owner from labourer, but an appropriation of surplus of the whole world economy by core areas' (2000 P86).

World systems analysis regards the role of institutions as central. National boundaries were created by capital in defence of its particular interests in its particular location but as with Polanyi, once institutions of the state are created, they become in some degree autonomous. They are elements in a system which is dynamic and in which removal of trade barriers is not to be read as growing internationalization through an evolutionary lens but rather as profit-maximizing local responses to shifting international ground.

This in turn makes a very real difference to the nature of analysis because classes, ethnic groups, status groups, are phenomena of world economies whose nature has been confused by analysis within nation states. These principles set the scene for a further development in terms of Capitalocene that we will discuss below. World systems analysis, then, works in an inextricable relationship with the epistemological foundation that we have already considered. Wallerstein sums it up as

> a concept of world systems that have structural parts and evolving stages. It is within such a framework, I am arguing, that we can fruitfully make comparative analyses-of the wholes and of parts of the whole. Conceptions precede and govern measurements. I am all for minute and sophisticated quantitative indicators. I am all for minute and diligent archival work that will trace the historical series of events in terms of all its immediate complexities. But the point of either is to enable us to see better what has happened and what is happening. For that we need glasses to discern the dimensions of difference, we need models with which to weigh significance, we need summarising concepts with which to create the knowledge that we then seek to communicate to each other.
>
> (1974 102)

The relationship between Wallerstein's epistemology and his conception of a dynamic world system characterized by history, emergence, and non-linearity denies any attempt to view the world through a single disciplinary lens. In many ways, this resolves difficulties faced in the work of Marx, Weber, and Polanyi who recognized the need to incorporate cause and meaning, general and specific. It also involves rethinking time. The resonances between social theory and complexity theory here are considerable. Braudel tried to reinstate la longue durée in response to a focus on specific time. Prigogine, reacting to specific duration in physics places emphasis on the 'arrow of time' as irreversible. Wallerstein's response is to work in the 'unexcluded middle':

> We must stand rather on the ground of what I should call the unexcluded middle both time and duration, a particular and a universal that are simultaneously both and neither – if we are to arrive at a meaningful understanding of reality.
>
> (2000 165)

We want to turn now to a major reconceptualization that has moved us further in thinking about capitalist world systems in complexity terms. We focus on the Capitalocene because the prior conceptualization of an Anthropocene, while recognizing human impact, continued to conceive of nature and society as separable categories. This cartesian dualism, as we've seen from Ollman, gives substance ontological status but fails to recognize the double internality of relations. Capitalocene, in opposing the undifferentiated attribution of a geological period to human domination, argues that an examination of the processes and effects of that domination exposes its source in the system of capitalism rather than simply industrialization. It recognizes differentiation in terms of power and responsibility and links to the work on co-evolution that we noted earlier, takes us beyond social theory that treated 'nature' and society as separable. This involves radical rethinking but as Moore (2016) points out, attempts to overcome this dualism have remained largely philosophical while 'all our concepts of "big history" – imperialism, capitalism, patriarchy and racial formations – remained *social* processes' (Moore, P4 2016).

Capitalocene

The point that emerges from our exploration of the work of these theorists is that capitalism is not simply a mode of production but shapes and is shaped by cultural regimes. The proponents of Capitalocene take this analysis further in exploring how capital has shaped environments since the time of Columbus.

We have identified the common roots of world systems analysis and complexity analysis in Bertalanffy's general systems theory and with Wallerstein's discussion of the work of Prigogine. Taken together with the overcoming of a historical dualism the continuities in theorizing between the natural and social world are clear while the human exceptionalism contained in much previous social theorizing becomes highly questionable. In much the same way that we have discussed the complexity consistency of structure in agency and agency in structure in Bourdieu, we see the logical consistency of humans in nature and nature in humans (Moore 2016).

Wallerstein (1974) identified a 500-hundred-year-old phenomenon, characterized by the basic division of labour and focused on relative positions within the capitalist world system. Capitalocene works from the system level to examine its power in environment making, creating climate change in which ecological catastrophe and pandemic result. It focuses on the history of this making through putting nature to work, recognizing the significance of both ideas as shaping and the role of states in configurations.

The core of the Capitalocene argument is this shift from an additive relationship of humans and nature to that of humans in nature and nature in humans. It starts from a position that we have seen with our other theorists in this chapter of recognizing that capitalism is not simply an economic system. With Polanyi, it explores its embeddedness but in a much more explicit way, considers how capitalism has organized nature.

This conceptualization incorporates dialectical relations within the 'web of life' (Moore 2018) as a totality. From the starting point that capitalism as a system relies on continued growth, Moore identifies the role of 'cheap natures' in a number of forms that can be grouped in terms of unpaid work, existing beyond the zone of commodification, that has been drawn upon to allow growth. This has long been a theme of feminist literature recognizing the unpaid work of women in the domestic sphere, the low paid work of women in the formally productive sphere, and Moore extends it to other exploited labour to describe 'islands of capital' among 'oceans of cheap Labour'. In defining some labour as 'natural', free and cheap labour is created for the reproduction of capital.

The conception of Capitalocene takes us on from Marx in examining how capitalism relies on society in nature, the premise being that the nature/society dualism is wholly socially constructed. It invokes a philosophy of internal relations that gives ontological status to both substance and relations. It relates to Weber's discussion of rationality and ideas shaping the material world either directly or indirectly (Kalberg 1980). Moore (2018) cites both Marx and Maturana and Varela in building this point:

> If nature *includes* humans, if humans are a 'natural force'(Marx 1973 p612) if human thought is embodied in an unbroken circle of being, knowing and doing' (Maturana & Vareal 1987, p25) we are presented with a challenge and an opportunity. Both turn on a historical method that moves from humanity *and nature* toward a double internality: humanity- in- nature and nature-in-humanity.
>
> (2018 242)

This, as Moore points out, takes us from a fragmented understanding of capitalism to one that recognizes capitalism's necessary conditions as 'oceans of potentially cheap natures' (2018, p242). Moving from Marx' argument that abstract labour is the 'substance of value' which is 'an economic phenomenon with systemic implications', he asks, 'how value relations are a systemic phenomenon with a pivotal economic moment' (2018, p242).

Conclusion

Our aim has been to open a discussion with some of the rich theoretical insights available to us from two centuries of social science. In a search for complexity relevance we have moved from Hegel's consciousness creates social being to social being determines consciousness without removing agency, uncertainty, or historical process. We have looked at major theorists who have sought to understand relations of structure, agency, time, and space in history. Each of these accounts is necessarily an interpretation from the perspective of complexity theory. In replacing the linear and reductive framing, abstraction that has previously been used to identify determinism and laws has been replaced with abstraction prior to investing nuance and context. A non-determinist Marx and a configurational discussion of cause in Weber emerges.

We have focused on a reading of Marx's theory which describes feudal society in terms of the development of trade, class struggle, and rural/urban division but does not read from this an inevitable transition to capitalism. We have pointed to the importance of micro-foundations and the possibilities in doing so that do not lead us into methodological individualism. We have used readings that engage in a dialogue with ongoing empirical work. Recognition of a marked absence in Marx's theory emerging from his focus on wage labour and the resulting naturalization of women's productive labour becomes a valuable insight for the conceptualization of Capitalocene.

If we take the route suggested here we see a theory which deals with systems that are open and dynamic, that therefore have history. For both Marx and Weber the search for a definitive account is inevitably frustrated because of their awareness of the nature of complex causation that needs to be situated in history.

Polanyi's focus on the role of theory as real in its consequences and in the complex relationships between markets and institutions emphasized multiple causality and the non-linearity of the double movement. The salience of world systems analysis as explicitly based in complexity theory and concepts of Anthropocene, and most convincingly Capitalocene emerge from there. These post-humanist approaches take us beyond earlier theorists who put humanity at the centre, building on theorizations that make sense of a world in complexity terms. Viewing through a complexity lens implies setting aside deeply engrained habits of thought and in doing so we can recognize that these theorists were putting theory to work to offer both a body of knowledge and tools for exploring the world today.

Wallerstein quotes Alfred North Whitehead:

> Modern science has imposed on humanity the necessity for wandering. Its progressive thought and its progressive technology make the transition through time, from generation to generation, a true migration into uncharted seas of adventure. The very benefit of wandering is that it is dangerous and needs skill to avert evils. We must expect, therefore, that the future will disclose dangers. It is the business of the future to be dangerous, and it is among the merits of science that it equips the future for its duties. ([1948: 125] 2000: 168)

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8 Complexity theory and the qualitative research programme

We have deployed the notion of complexity as a frame and we return to this way of thinking in relation to qualitative approaches to inquiry. Qualitative researchers have not shared a single taken-for-granted paradigm and those who have not adopted simple realism have had to consciously consider the roots of their inquiry. They have developed theoretically grounded bodies of work based on internal paradigmatic coherence while often disagreeing about the nature or existence of a 'real' or the possibility of reflecting it. The breadth and depth of qualitative research cannot be covered in a single chapter so our aim here is to focus on areas of qualitative work that find resonance in realist inquiry and the complexity frame of reference. In doing so we will question the foundational opposition between natural and social science as expressed in post-modern work that has turned to relativism.

For qualitative researchers, the need to respond to dominant positivist standards of evidence, validity, and reliability, has generated both attempts at compliance and wholesale rejection of the real. In either case, positivism has framed the debate. Qualitative research has necessarily been distinguished from quantitative because, in searching for meaning, it has worked with human 'subjects' rather than objects. There is a substantial literature on the implications of this difference for negotiating access and maintaining field relations that acknowledges not only a continuum of involvement (Gold 1958) but the ethical and personal costs in negotiation and renegotiation that are explored in Whyte's entry to *Street Corner society* (1943). It is hardly surprising that the second half of the 20th century saw sociologists examining the conduct of normal science as a social activity. Alongside this exploration, undermining the hegemony of positivist unity, the technological ability to deal with non-linearity, has enabled a different understanding of the natural world, leading to a new dialogue. We will look at these converging strands before moving on to develop the case for complex realism in relation to established qualitative approaches.

In the space created by the demolition of the idealized description of positivist science and a new ontology of a world that is interactive, non-linear, and emergent, it is possible to reach an accommodation that sees the constructed as real. We can identify several strands of knowledge development, culminating in the late 20th-century reaction against positivism, which have contributed to freeing qualitative work from attempts to impose inappropriate methods and standards of evidence. Qualitative researchers, for whom no explanation is adequate that fails to understand meaning, have been at the forefront of questioning positivist science as a basis for understanding human agency. Whether from the hermeneutic emphasis on meaning, phenomenological concern with lifeworlds or the critical theorists' concern with power and inequality, the focus lies in learning about the experience from the perspective of research 'subjects'. This has been informed by work

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in the ethnographic tradition, notably Weber's discussion of *verstehen* and Chicago school sociology. Despite its relegation to the lower levels of evidence hierarchies qualitative inquiry has continued to provide a rich vein of valued evidence.

The quantitative/qualitative divide within the social sciences was founded on the view that behaviours of social groups could be studied by seeking positivist regularities but that meanings and interpretations required different methods. Qualitative researchers argued that quantification resulted in abstraction and hence a failure to comprehend a whole case, setting, or system, expressed succinctly by Strauss : 'Qualitative researchers tend to lay considerable emphasis on situational and often structural contexts, in contrast to quantitative research is whose work is multivariate but often weak on context' (1987 2). They have consequently resisted the mid-20th century tendency for social scientists to take on the role of objective observer and recorder. Nonetheless, the dominance of positivist thinking and the power of its methods in conferring authority put qualitative research in a defensive position when making claims to evidence.

Complexity entered the social sciences initially through developments in natural science but for many of us it came as a recognition, perhaps in Denzin's (1989) terms an epiphany. It deals with issues in knowledge that we have actively struggled with throughout our lives in theory, methodology, and empirical research. Being 'outsiders' to the world of normal science, qualitative social scientists have actively considered the basis of their practice. The dubious basis for separation of natural from social has drawn attention to the intersubjective nature of science, as it is practised, as a social construction. 'Epistemology concerned itself with the "truths" of established science but failed to understand its variation from those truths in practice' (Bourdieu 2004 3), permitting an idealized, Mertonian, structural functionalist characterization of positivist science. Examined in these terms the struggles over capital and power that are exhibited in every social field become visible, while processes of control and exclusion undermine the claimed objectivist basis. The very contextualization of scientific inquiry opened up the 'Pandora's box' of the laboratory and became one basis for the demise of its objectivist status in bringing

> to light a whole set of facts which strongly call into question the scientific epistemology of the logicist type … and reduced science to a social life with its rules, constraints, ruses, effects of domination, cheating, theft of ideas etc. The realistic and often disenchanted vision that sociologists have thus formed of the realities of the scientific world has led them to put forward relativistic, even nihilistic, theories which are the very opposite of the official representation. There is nothing inevitable about this conclusion and one can, in my view, combine a realistic vision of the scientific world with a realist theory of knowledge.
>
> (2004 3)

Science is a social activity and within it there are struggles over 'what is at stake'. Bourdieu reflects on Poincare's use of the term 'convention' as letting:

> the sociological wolf into the mathematical sheepfold and into the always somewhat pastoral vision it encourages, when he uses the word 'convention', a term who social implications he does not at all spell out but which calls into question the idea of universal validity and invites one to consider the social conditions of this conventional validity.
>
> (2004 79)

This dis-embedding of a simple objectivist view of science has been a powerful spur to relativist thinking. We are with Bourdieu in seeking a basis for knowledge that can recognize the fundamental truth of science as an activity conducted by humans, while arguing that it is only through clinging to the 'dogmatic ideal' of absolute and individualized knowledge that relativist arguments can prevail. To get beyond this we need a 'dialogic and argumentative knowledge of the scientific field' (2004 3), which encapsulates Bourdieu's reformist reflexivity and echoes Campbell's approach to realist inquiry.

The significance of convention in framing approaches is substantial. In its origins Newtonian science was unnuanced because the linear positivist frame was itself a device for making applied science practical in a more complex world. Science developed its conventions in framing the world as linear to manage calculations that could be done with pen and paper:

> [This] habit became so ingrained that many equations were linearized while they were being set up, so that the science textbooks did not even include the full nonlinear versions. Consequently most scientists and engineers came to believe that virtually all natural phenomena could be described by linear equations. As the world was a clockwork for the 18th century, it was a linear world for the 19th and most of the 20th century.
>
> (Stewart 1989, quoted in Davis and Sumara 2005 312)

The negative consequence of adopting this positivist rationale in social science was identified by Schon (1983) noting that the focus on 'problem solving' capacities of positivism through technical rationality ignored the more difficult undertaking of 'problem setting'. Schon describes the former as the 'firm high ground' and the latter as the 'swampy lowlands' where 'situations are confusing messes incapable of technical solution and usually involve problems of greatest human concern' (Schon 1983 42).

Qualitative research has been described as going through 'four moments of consciousness of such significance they can be regarded as revolutions' (Lincoln 2005). Starting with the separation of natural from social, Lincoln identified, 'the crises of representation and authority/legitimacy', 'the performative turn' and lastly 'the recommitment of social science to social purpose and social justice'. She argued that a new consciousness emerges from these processes leading to a view of the world as socially constructed and dispensing with the concept of a real. The problem of an external reality with its own causal powers is rejected. Within qualitative research there have been attempts to systematize sets of beliefs in terms of paradigms, notably Guba and Lincoln's (1994) categorization. Their categories initially established as positivist, post-positivist, interpretivist, and critical paradigms were later joined by the participatory paradigm (Guba and Lincoln 2003). The differences have significant consequences for framing research and the authors identified potential for 'interbreeding' using the example of the influence of critical theory and action research on their own work: 'Consequently to argue that it is paradigms that are in contention is probably less useful than to probe where and how paradigms exhibit confluence and where and how they exhibit differences, controversies and contradictions'(Lincoln and Guba 2003, 254).

We find much common ground between this and Cilliers' (1998) discussion of the relationship between complexity and post-modernism. We acknowledge valuable concepts such as bricolage and the image of 'interweaving'. We recognize 'the fluidity of categories in research, the denial of absolutist criteria for judging reality'. We are wholly at one with the argument that, 'a goodly proportion of social phenomena consists of the meaning making activities of groups and individuals around those phenomena'. Subsequently Guba and Lincoln (2008) rejected cross paradigm working

> the assumption that there is no single 'truth' – that all truths are but partial truths; that the slippage between signifier and signified in linguistic and textual terms creates re-presentations that are only and always shadows of the actual people, events, and places; that identities are fluid rather than fixed-leads us ineluctably toward the insight that there will be no single 'conventional' paradigm to which all social scientists might ascribe in some common terms and with mutual understanding. Rather, we stand at the threshold of a history marked by multi-vocality, contested meanings, paradigmatic controversies, and new textual forms.
>
> (2008 281)

We draw a different conclusion. The tendency to ally the real with 'things' has been adopted not only by positivists but also by constructivists but if we reject this and accept that the real can be constructed the problem disappears. Hence we can recognize the 'gap between reality and its representation' that produces an impetus towards 'literary, narrative as well as poetic forms of representation' (Lincoln 2005 29–30). Our purpose here is to argue that the complexity frame of reference allows us another way of thinking about this relationship. We assert in agreement with qualitative critical realists such as Maxwell (2012), that those meanings are real.

This is a restatement of points we have made in earlier chapters but we think it important to return to it here because it has been the source of rejection of science and evidence in significant areas of qualitative research where the argument led many to radical constructivism. Starting from a disavowal of the separation of natural and social that is implied in both Guba and Lincoln's constructivist and Williams' (2020) complex realist approaches we argue that the basis for complexity explanation lies in recognition of a socially constructed 'real'. This reality underpins the use of quantitative and mixed methods as much as it does qualitative methods. It is consistent with our review of structure and agency as embodied and embedded within each other. Analytical separation must not be taken to stand for a belief in lived separation.

The move towards acknowledging complexity has found support in a critical appreciation of the practice of traditional science while the complexity frame of reference itself emerges from developments in technology that have allowed science to handle a world not framed in linear ways. Stewart (1989) sums up our differences with both Williams and Guba and Lincoln:

> the notion of inter-objectivity does not lend itself to brief description. It is difficult to define in part because the idea relies on a rejection of some of the most deeply engrained assumptions about the nature of the universe, such as the tendency to separate phenomena from descriptions of phenomena. (This separation is maintained both in classical science, with its emphasis on objective fact, and within many post-modern discourses, with their emphases on intersubjective accord.) Importantly, the point here is not that things change by virtue of how they are described, but that the actions of the describer are affected by the descriptions. As actions shift, the physical texture of the world is altered—a point that has been dramatically demonstrated by many of the environmental and medical concerns that have arisen over the past

> century. These are, as Capra (1996) describes them, 'crises of perception', difficulties born of particular habits of action and interpretation.
>
> (quoted in Davis and Sumara 2005, 315)

Representations may be shadows, but they are real representations with consequences and when history is written it will also be both real and contested. It is an ontological commitment that is not seeking the reality 'behind' the representations but recognizing the constructed nature of the representation and acknowledging that such representations are also real in their consequences. Establishing a basis for qualitative research in critical realism we can draw on Maxwell who points to the total compatibility of ontological realism with epistemological constructivism, arguing that epistemological and ontological perspectives can be used 'not as a set of "foundational" premises that govern or justify qualitative research, but as resources for doing qualitative research' (2012 13).

This position allows us to deal with the issue of cause that has been such a significant source of division within qualitative research. While some researchers eschew the notion of cause (Lofland and Lofland, 1984), others have dealt with it in quasi positivist terms (see Maxwell, 2004, 2012 for discussion). We want simply to restate the point that, taking a process and mechanism approach, we can see that meanings and interpretations are causes of action. This is fundamental to qualitative work that is interested in 'thick description', takes a narrative form, seeks to understand the case in its full context, to understand the meanings and beliefs of those involved, and to describe in order to explain. Qualitative researchers have studied the how and why questions that are at base causal questions through deep knowledge of single and small n cases to achieve an understanding of interaction because causality has been understood as non-reductive.

We agree with Maxwell: 'Concepts, meanings, and intentions are as real as rocks–they are just not as accessible to direct observation and description as rocks' (2012 18). Maxwell's discussion of the concept of culture draws on a rich body of qualitative work, showing that culture can be regarded as real, not directly observable but comprised of mental and symbolic entities. Our view of the social as embedded and embodied suggests that we can regard culture as also physical and behavioural (Bourdieu, 1999). A further important insight from Maxwell is that culture is not shared but *participated* in and this is a foundational position in the complexity frame which becomes manifest in our approach to researching the social world. Shared conceptions and symbols are real because they are real in their consequences but participation does not imply a conventional view of culture as a uniform system but rather as a distributive system. This view of culture allows for the complexity that is characteristic of a social world which recognizes its manifestation in time and space and the unequal nature of participation arising in those interactions.

This is consistent with our understanding of communal phenomena. We have already identified these attributes in the notion of community, mistakenly treated in governance and policy in the UK as both uniform and wholly positive. To recognize that 'community' is differentiated and has a 'dark side' (Bourdieu, 1986,1999) is to recognize struggles for power and the ability to appropriate and place value on knowledge. These aspects of culture and community are real and it is only the simplifying categorization of natural and social that ignores that reality.

What is clear from the foregoing is that freeing ourselves from positivist claims to evidence is difficult because it involves rejecting bases of knowledge that have been hegemonic and therefore formed unquestioned foundations for the development of methods. To maintain a stance that claims an ontological real alongside an epistemological constructivism it is necessary to perform the epistemological breaks that Bourdieu identifies (2000). One of the themes of our work has been that while the complexity frame of reference is a 'new science', it grounds ways of looking and thinking about the social world that social scientists are familiar with. This is because these approaches have revolved around something real which has been the anchor that has held social research and theory together. It is not surprising then, that many approaches and methods in qualitative research can deal with the world in complexity terms. We want to look at examples of ways of doing qualitative research that are coherent with complexity's fundamental positions and in doing this we have argued that we need to set aside the antinomy of positivist versus interpretivist.

Grounded theory

Grounded theory was developed by Glaser and Strauss (1967) in response to dominant social scientific theory building without systematic engagement with empirical data. This focus on learning from the field came from their work on death and dying and their desire to develop a set of methodological principles that could bring 'rigour' to ethnographic work in its broadest definition. By building theory through concepts derived directly from data and using constant comparison the aim was to develop those substantive, middle range theories that could become the basis for formal theories at higher levels of abstraction. Analysis begins with exploratory open coding in which data segments are compared and categorized and hence the initial process is fragmenting. Axial coding, developing relationships between categories, is then a process of making connections within the data that can comprehend causal conditions, through linking context, action, and interaction. Categories and their properties are conceived of at different levels of abstraction, the main aim being to build an emergent set, rather than a set adopted from existing research. 'In short, our focus on the emergence of categories solves problems of fit, relevance, forcing, and richness' (1967 37). Hence it seeks both patterns and the connections within the data that make sense of those patterns.

While its core process is one of constant comparison, the two original authors subsequently developed the approach along different lines that suggest rather different logics of theory building. *The Discovery of Grounded theory* (1967) argued that the researcher entering the field should be as free of preconceptions as possible: 'An effective strategy is, at first, literally to ignore the literature of theory and fact in the area under study' (1967 37). Glaser continued to hold this position (1978,1992) in seeking to avoid researcher bias shaping theory through selection, emphasizing the tight connection between data and theory. A student of Merton, he sought primarily to keep an approach of innocence from either responding to impressions or to existing knowledge by repeated rounds of relating theory to data. In this way, he argued, researchers can combat personal prejudices or favoured theories.

While Strauss and Corbin (1990) also wanted a strong relationship between data and theory, they argued that it is of primary importance to develop transparent methods of data handling. They emphasized keeping a research journal detailing the process of the research, reflections, and decisions made at each step. This was necessary because bracketing is not possible and transparency lays decisions and interpretations open to challenge. In engaging with the presence of the researcher in the research they sought to use procedures of reflexivity to bring 'rigour'. So while theory continues to be emergent, analysis is an interaction between researcher and data rather than a simple 'reading'. Qualitative research uses 'natural analysis' but in responding to scientific problems, does so in a more self-conscious and 'scientifically rigorous way' that echoes the basis of reflexivity and critical analysis that we identified in Bourdieu and in Campbell's critical realism. The constructed nature of theory is acknowledged in Strauss and Corbin's writings and relating data to concepts by advocating micro- analysis of data line by line, Strauss and Corbin are trying to avoid bias in order to hold on to the real.

The debates have continued and seen significant divergence. Glaser (1992) contended that two variants of grounded theory had emerged, identifying the major differences between himself and Strauss and Corbin in these terms and arguing that Strauss and Corbin's modified Grounded theory engaged in preconception, undermining the emergent nature of theory. The major split between Glaser and Strauss centred on these issues but this brief outline has sought to recognize the stress both laid on emergence in understanding data and building theory. Their different beliefs about the potential for discovery from the data alone and the degree to which the researcher constructs the research appear consistent with a positivist/interpretivist split.

This debate can also be characterized as a debate about the relative merits of induction and abduction (sometimes called retroduction). In its original form, the grounded theory process could not be described as wholly inductive, indeed Glaser and Strauss (1967) described analytic induction as useful for testing theory, the purpose of grounded theory being to develop categories. The objective is saturation of data rather than any claims to proof. Hammersly's (2010) conclusion that Glaser and Strauss were 'urging we should not be so pre-occupied with testing ideas that we dismiss anything that does not seem to be immediately verified or verifiable' seems reasonable. Nonetheless, the relationship between theory and data in the grounded theory has a strong inductive quality. While induction builds from specific observations to develop its conclusions, abduction, by contrast, is a process of making sense emerging not only from the data but from other knowledge and influences so that the analytic framework focuses on the nature of the system within which those cases are manifest. As an abductive process of collecting and connecting (Richardson and Kramer 2006) it chimes with Maxwell's (2012) approach.

The debates in grounded theory reflect a series of tensions, framed in positivist/interpretivist terms that have structured qualitative discussion of meaning, action, and cause. The approach to handling data follows from the positivist or constructivist frame in which it is situated and the claimed relationship to, or rejection of, objectivity. In arguing that there is no need for a justifying ontology or epistemology because grounded theory works in all paradigms, Glaser aims to gather data without first having it 'filtered through … pre-existing hypotheses and biases' (1998 3). The question becomes whether theory can 'emerge' from data innocent of 'forcing' in Glaser's terms or whether there is an inevitability in science being done by agents, the direction taken by Strauss and Corbin and constructivists such as Charmaz (2006). Acknowledging the realist position in which the researcher is active throughout in determining questions, directions, and analysis while reflexively maintaining an explicit relationship between theory and data seems promising.

A major feature of grounded theory that resonates with complexity is the explicit intention to integrate both qualitative and quantitative data systematically to develop theory. We can agree with Glaser that, 'all is data' and hence that 'grounded theory is a general method that can be used on all data in whatever combination' (Glaser 1998 42). The aim of 'ensuring conceptual development and density' (Strauss 1987 8) through a methodological approach founded in American pragmatism involves searching for patterns, observing interactions, and understanding cause in terms of mechanisms, a way that is coherent with the complexity frame of reference. Dey (2007) argues that the ease of translation to computer-based analysis has made it the 'mode of normal science' for qualitative researchers (p494). Castellani et al. (2003) recognized the potential of grounded theory to underpin grounded neural networking and the use of the self-organizing map. This exploratory analysis technique can incorporate quantitative data in searching for connections and patterns which result in open-ended models rather than verifiable results. Identifying five affinities between self-organizing maps and grounded theory, they suggest the approach can incorporate both quantitative and qualitative data in an integrated rather than sequential way. In doing so it can leave behind the division that has structured ways of knowing in social science for over a century.

Case study

We have discussed case-based methods elsewhere in the book in looking at qualitative comparative analysis which uses this frame to develop configurative accounts of cause. Here we want to look at the rationale for the case study that makes it so persuasive as a complexity approach. The method has a long history in sociology and anthropology. Although downgraded as not generalizable during the years of positivist hegemony, the case study's ability to capture context and interaction ensured its continuing use in qualitative research. Discussion of the case as a method was brought to prominence towards the end of the 20th century by Ragin and Becker (1992) and by Yin (1984) who sought to systematize it in terms of reliability and validity. In the work of Park and colleagues in establishing the Chicago School of sociology, we find resonances with our discussion of evolution and ecology (Helmes-Hayes 1987). Park was concerned to explore a 'human ecology' comprised of systems and subsystems interacting in time and space. The focus on space led to a substantial body of work in mapping human action and movement exemplified in models of successive waves of occupation in urban areas.

A realist take on case-based research sees the case as a constructed unit, defined and selected by the researcher, and to this degree would be in complete agreement with the constructivist view. As realists however we would go further and say that it is constructed from something we can observe, inevitably filtered, selected, inevitably a subject of negotiation. Any form of ethnographic work that seeks to comprehend and convey a world, to describe experience beyond our own, and to communicate that to others falls within this description. Case studies seek to do that and hence can cover a much wider range of qualitative investigation than we would necessarily recognize under that label.

The underlying rationale for case-based research is its recognition that variable-based analysis undermines any understanding of complex social processes. It is an exploratory approach in which processes of selection frame research questions. The case is the unit of study because, like narrative, it maintains focus on the significance of context throughout the research. It seeks, in complexity terms, to understand a system working and to do so it needs to comprehend the core features and dynamics of the system. One of the primary insights of complexity framing for social research lies in its recognition that social systems are not inherently stable or simply functional. Looking for the dynamic, the interactive, and the unpredictable guides research questions towards the intended and unintended consequences of action in systems (Stacey 2010), features recognizable in Park's human ecology approach. They allow observation of not only intended and unintended outcomes within but also the effects of co-evolution of related systems. Processes of incorporating, adjusting to, subverting new rules and resulting changes in practice can be observed so that the system is understood through its patterns of activity. While understanding rules, interventions, and drivers of change at a formal level is important, our interest also lies in what frustrates as well as what facilitates processes and change. Within this frame we can see the possibilities of negotiated order (Strauss 1978) that we have identified in discussing evaluation research.

Byrne (2011) describes a 'strong programme' in case-based research founded on these ontological premises, recognizing, with Glaser, that 'all is data' rejecting qualitative/quantitative divide and embracing multiple methods. In essence, all complexity studies are case studies. Context is never simply a background but is fully interactive. Foreground and background become changing research foci because their significance is subject to the current and potentially changing choices of the researcher in responding to emerging findings. Case studies are also stories and hence we can identify the role of process tracing (Bennett and Elman, 2006). The phenomenal range of their application to settings is also mirrored in the breadth of purposes to describe, to generate, and to test theory (Eisenhardt 1989).

The role for comparison is clear because what matters is not only in finding similar outcomes across cases but in learning how different outcomes can emerge. Thus, it is possible to detect the non-linearities and their emergent nature. The complexity frame of reference helps us further here in addressing the question of what constitutes the case in determining how and where we can look to draw the boundaries around our system of interest (Byrne, 2009). The participation of researchers and local actors in mapping the system is an essential element of developing this understanding. Causal relations can be discerned through immediate and longer-term historical accounts of states of the system and the occurrence of significant events. We do this by focusing on features of the social world that qualitative researchers have long prioritized, agents and their interactions within and across settings.

Narrative

Narrative is a core feature of qualitative research, sympathetic to our nature as 'storytelling creatures' Lincoln (2005). The elicitation of a narrative is a joint creation of the researcher and the researched. In social research, the narrative is analyzed, not simply as an individual story, but as one situated in a social context. Chase's (2005) history of narrative research identified the early studies we have cited as case studies as using a narrative approach and comments that they were, to limited degree, interested in subjective experience but 'primarily interested in explaining the individual's behaviour as an interactive process between the individual and his or her sociocultural environment' (20005 653). Our realist approach rejects dichotomizing subjective experience and behaviour but the focus is clear. Further development of narrative methods by feminist researchers (Woodiwiss et al 2017) not only placed emphasis on individuals as social actors but also brought in the meanings and power relations that are present within the narrative. The process of narrative inquiry can be varied but essentially involves developing storylines, usually through interviews and documents such as journals. In evaluation, it draws on interactions with an intervention set in its particular community and cultural context.

The narrative approach can relate history, context, and cause in terms of meaning without committing the violence of ultimate abstraction. It places engagement between researcher and researched at the core and recognizes an inherently dynamic field as stories emerge through the process of construction. This involves the elicitation/co-creation of stories whose meaning is made initially between interviewer and interviewee but then 'enacted' between the researcher as narrator and the research audience as another listener (Savin-Baden and Van Niekerk 2007, p464). While narrative inquiry has been identified with constructivist approaches there is no necessary basis for this. Bourdieu (1999), for example, used the method in dealing with the concept of habitus as embodied relations of structure and agency (Bourdieu, 1999).

Narrative provides us with an organizing frame that has clear affinities with complexity in terms of its historical structure and we can identify its use in process tracing as a 'a continuous and theoretically based historical explanation of a case' (George sand Bennett 2005 30). Case studies use narrative as a way of maintaining the historical character of the field. In single cases, narratives can provide information about interactions in causal relationships in a 'fluid and powerful way' (Abbott, 1992, 72) which suggests its value lies not only in doing social science but in communicating to policymakers. Using AT&T as an example, Abbott suggests, 'A social science expressed in terms of typical stories would provide far better access for policy intervention than the present social science of variable' (1992 72).

Drawing the distinction between narrative as presenting a picture and narrative analysis, Maxwell (2012) points out that while many research accounts use narrative to retain context, the use of narrative as a frame has analytic potential. As a connecting rather than a categorizing device, narrative analysis seeks the connections within the data that provide causality. As in grounded theory, this involves coding and to this degree initial abstraction is part of the analytic process.

Narrative analysis has given rise to the significant concept of epiphany (Denzin,1989**)** capturing transformational experiences that represent breaks in linearity. Epiphanies express life events or experiences that significantly reshape or disrupt a continuity to reframe thinking, those that give the narrative a before and after quality. Time becomes an active feature of the analysis conceived in terms of continuity, cycling, and rhythm. This recognition of construction demonstrates a clear alliance with participatory inquiry in the collaboration between researcher and researched (Connelly and Clandidnin 1990).

Making sense through storying and re-storying, description and interpretation, narrative structures have been used to understand a range of processes, including the development of identities, life course events such as illness and dying, as well as implementation processes involved in programme initiatives (Riley and Hawe, 2005) and community development research (Dixon and Martin 2017). Clanindin and Connelly (1989) describe its use in understanding organizational process to discover the 'underlying unity' in apparently inconsistent practices. In programme evaluation we can see explicit use in opening up the 'black box' of the intervention that we have discussed in Chapter 12. The stories of the implementation process, the experience of institutions, interactions and the developing history of a policy intervention entering an existing policy space all become pertinent to the picture being developed.

Participatory inquiry

The recognition of participation as the basis of knowledge creation represents a significant departure from positivism, requiring rethinking the status and purpose of evidence. It deserves the status of a revolution in proposing not simply another scientific paradigm but a thoroughgoing revaluation of the production of authority and knowledge (Heron and Reason, 1997). At the forefront of this rethinking is the recognition that construction and use of knowledge are shot through with power relations. Hence our interest in participatory enquiry is not in specific methods employed (often both mixed and innovative), nor in the challenges their use presents, but rather in how a participatory paradigm can illuminate research in a complexity frame.

Participatory inquiry begins from the premise that we are all participating actors and arguing therefore that the purpose of enquiry is 'to forge a more direct link between intellectual knowledge and moment-to-moment personal and social action, so that inquiry contributes *directly* to the flourishing of human persons, their communities and ecosystems of which they are part' (Reason and Torbert, 2001 p6). The four paradigmatic assumptions at the base of participatory inquiry involve seeking practical rather than abstract knowledge and recognizing that all human knowing is participatory, experiential, and normative.

As participatory research involves an iterative process through dialogue between co-researchers, it is an evolving process, methodologically pluralist and constantly subject to reflexivity. Reason and Torbert (2001) have pointed to the need, in contrast to the positivist 'out there' world, to study the 'in-here' subjective and 'among-us' interactive worlds. The distinctions of first-, second-, and third-person research are used to discuss levels of engagement and distance, their main object being to question the traditional 'scientific' basis of research. Advocating the interweaving of first-, second-, and third-person research, Reason and Torbert argue: 'that a full-fledged social science after the action turn will not just describe an external reality, but will support personal, social, and epistemological inquiry and transformation' (Reason and Torbert, 2001 p30). We can draw on participatory work to open up new questions about purpose and approach when simple objectivity is no longer the gold standard.

A range of types of participatory forms of inquiry have developed overlapping approaches including action research (Argyris and Schon 1974), cooperative inquiry (Heron, 1998), collaborative, appreciative, and Collaborative Developmental Action Inquiry (Reason and Torbert 2001, Torbert 2021). Approaches vary in terms of degree of co-researcher involvement in research design and analysis but are fundamentally framed around the rejection of the notion of research 'subject' and the distancing of researcher from researched. Once it is recognized that the dichotomy of thought and action is a false one the focus on methods and rigour as expressed externally through objectivity is not valid. It gives rise to a re-evaluation of proposed method in terms of experiential, presentational, propositional, and practical knowing and emphasizes 'quality of knowing' over method (p9). Moreover, Torbert (2021) argues, participatory approaches capture a greater breadth of domains of human experience through their focus on the present rather than the positivist past, in iterative feedback loops, and in first- and second- as well as the positivist third-person voice.

While most participatory action research has been undertaken in community interventions and local policy implementation, appreciation of these paradigmatic and power issues has brought wider application, for example in extending to impact assessment (Burns, 2015). Often when working with powerful actors in systems we refer to co-production rather than participation but in practice the terms are synonymous. The affinity between systems thinking and action research is clear (Midgely 2015) as action researchers seek to understand complex interaction contexts in which several agencies and individuals are acting in response to organizational pressures, ethics, and values. Midgely argues action research emphasis on intervention does not undermine scientific observation but rather uses it to inform intervention goals. The process of defining the system is one of drawing boundaries based in ethical judgements through a rational and dialogical process. The very purpose of scientific inquiry, Whyte (1989) reminds us, is primarily to solve practical problems and secondarily to contribute to theory. Whyte argues it can do both.

Participatory research recognizes the impact of power, structural and political context on the production and authority of knowledge. Freire's (2005) discussion of the 'banking' concept of education illustrates the point well. He contrasts the notion of active educator and passive learner who 'banks' given knowledge, compared with 'conscientization' in which an interactive relationship occurs. Transferred to research done on or to people the parallel is clear. Participatory inquiry not only foregrounds power in research findings but in the iterative nature of research processes of sampling, data gathering, and analysis. Midgeley (2015) describes how power can be embedded in research using an example of engaging homeless young people in the design of new services through a 'playful' approach in contrast to the more authoritative, rational engagement of professionals. The emancipatory potential of both engaging in shared research and foregrounding issues of inequality and power lies in enabling participants to plan action at multiple levels and a very different research sensibility emerges from looking, thinking, acting in cycles. Goals, questions, and methods must be developed in cooperation and are by definition emergent.

In consequence of these reformulations of subject, object, and process in inquiry, the approach taken by Shaw (2002) to challenge the concept of organizations as having existence separate from our own activity pertains. Shaw writes rather about 'conversing as organizing' (2002 11). This works well with our discussion of culture as participated in and with negotiated order that sees organizational knowledge as embedded in the discussions that go on on a daily basis. Group norms emerge that set the context for new rounds of negotiation that create the pathways for action. The outcomes of this emergence are both patterns of interaction and tangible projects and both are real Moreover, Lichtenstein argues, these two types of outcomes represent a continuum that may be described as weak or strong emergence: 'What appears to be a single outcome is more likely to be a tangible expression of an "equipped ecology"' (449 2015).

The shift from managing information and knowledge in organizations as a thing to managing them as flow is wholly consistent with the complexity understanding of process and in keeping with our attempt to shift away from a conception of real that results in dualism. When we think in these terms the significance of narrative becomes very clear as communicative interaction (Stacey 2010).

Applying our earlier discussion of critical realism based on the complexity frame and action research we can see how direct implications can be drawn. The emergent is fundamental to participatory inquiry because it starts from the position that power to direct the course of inquiry does not lie with the researcher, but rather is a dialogical and evolving process of group identification of goals and decision-making.

The relevance of participatory inquiry to complexity framing has been related by Reason and Goodwin (1999) to post-modern biology. In proposing a 'science of qualities' their six principles of rich interconnections, emergence, iteration, holism, fluctuations, and edge of chaos are familiar. The insights of complexity make emergent processes comprehensible and helps us to understand why linear-based methods of both understanding and managing them are likely to fail. Beyond that, the participatory nature of knowledge points to questions about the status and purpose of inquiry because, in the absence of claims to objective knowledge, ethical questions cannot be subordinated to the needs of science.

Conclusion

The complexity frame suggests an approach to defining questions for qualitative research. Starting from a world that we acknowledge as dynamic, emergent, unpredictable it posits those manifestations as real, explicitly foregrounding conceptions that are often implicit, even in post-modern constructivism. We can embrace the recognition that the world cannot be objectively known, as qualitative social scientists have sought to do, and we can acknowledge intersubjectivity without losing the ability to explore it in a range of ways that yield valuable knowledge. We can transcend the quantity/quality divide as a convention that dominated only while social scientists were working with a model of science that used linear accounts. The consciousness of epistemological issues that has been necessary to qualitative work in responding to positivism, could usefully be reflected across science, be the methods quantitative or qualitative. The critical reflexivity involved in that process is crucial, while the knowledge derived must be both theoretical and practical. The primary test of science, as Whyte reminded us, is one of usefulness.

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9 Quantitative methods for exploring, understanding, and possibly predicting complex social systems and their intersection with natural systems

Measuring, tracing, and sorting things out

Numbers are measures. Numbers measuring complex systems are attributes of the systems, not things with causal powers external to them. To say that is to follow Abbott's (1998) challenge to the notion that what we measure are variables which are understood as things possessing real causal powers external to the cases being measured. Measurements are attributes describing traces of the character of the system, ideally at multiple time points so we can see the attribute describing a trace. Attributes belong to cases and are not disembodied things outwith the cases. Traces describe the trajectories of cases, the paths through time which the systems take with a focus on when those trajectories involve radical changes of kind, phase shifts in complexity terminology. There are two orders of changes of kind. One is that our cases, always for us considered as complex systems themselves, can change the kind of thing they are. The other is that the set of categories of things to be within the possibility space, the real as opposed to abstract mathematical attractors, can also change. The central purpose of measurement is to generate descriptive attributes which allow us to say something about the whole character of our cases as particular complex systems *and* to allow us to sort them into categories of kind.

Sorting systems into kinds when we are dealing with multiple systems of the same order is a crucial step in 'tracing'. In the social world sets of cases which belong in the same population can be thought of as ensembles of systems. The sets represent attractors in the possibility space at any time point. There is a range of techniques for sorting. The primary use of the learning algorithms, usually based on emulated neural nets, is sorting things out. One of the older computer-based methods for sorting is cluster analysis.[1] Hierarchical methods are good for exploring the whole range of differences. Clusters generated depend on what variate trace data is used to construct them and then on the method deployed. Tracing is a process of classifying across a range of time points. These may be calendar time based or may derive from significant stages in a process. Any longitudinally ordered data set measuring cases at different time points can be used in this way, for example, the large cohort panel surveys which Leisering and Walker (1998) consider make the study of social dynamism so much easier and demonstrate the necessity for taking dynamism, change, as crucial in social research.

Tracing is of great value in exploring any case-based administrative data which has a longitudinal basis. Dyer (2001) used it to map the careers of offenders moving through a custody diversion system. She classified offenders on the basis of their personal, psychiatric, and offending attributes on admission to the system and then reclassified them at

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key points during the processes through which they passed. This mapped their trajectories through the system. These were related to outcomes defined as any reoffending and violent reoffending. Many administrative systems in health, social work, and education generate data of this form and the potential for application is enormous. Tracing makes narratives from quantitative data. We can do it with single cases as well as with ensembles whenever we have time ordered data in longitudinal form. When we have multiple cases it is always useful to classify as a basis for differentiation but the story of single cases is often very useful at any level – micro for individuals and households, which is a way of tracing life courses, meso, e.g., for localities and city regions, and macro at all levels from the nation state to the world system itself. Life history quantitative narratives have value both for sociological history and in practice particularly in relation to epidemiology.

Comparing by exploring difference

The comparative method is an essential tool of social science and, even if often implicitly, of biological science. Darwin certainly used it. Ragin notes:

> an idea widely shared by social scientists – that it is possible to derive useful empirical generalizations from the examination of multiple instances of social phenomena [i.e. from comparable cases].
>
> (2000 332)

What we are looking for is differences. Difference is how we engage with cause in comparisons: 'Where there is a difference in the effect, there is a difference in the cause' (MacIver 1942 26).

Systematic comparison across cases allows us to use difference as a way of guiding action to achieve outcomes. We do this by looking retroductively across sets of similar cases and identifying the factor(s) which produce different outcomes for cases which are otherwise very much the same in terms of character. This is particularly useful in searching for accounts of causality at the micro and meso levels (see Byrne 2009; Blackman et al. 2011). Systematic comparison gives us the information we need to move systems of a given kind towards a desired outcome state, provided we have adequate information about systems of the same kind which have achieved or not achieved that outcome state.

Any adequate method for exploring causality in complex systems has to be able to handle first complex causality, causality in which there is a reflexive relationship between the system and both its own components in the assemblage and other systems with which it intersects. Then we have to be able to deal with equifinality, with multiple paths towards the same outcome. The key idea which makes Ragin's method of Qualitative Comparative Analysis (QCA) useful here is that of configuration. A configuration is a set of attributes, whether regarded as binaries in crisp set QCA, multiple attributes in MQCA, or fuzzy (partial membership being possible) where the attributes stand for membership/degree of membership of the set. Configurations are sets where the members of the set possess the same set of attributes. The set theoretic logic of QCA is radically different from the variable logic of regression-based methods. Since causal power is not allocated to disembodied variables the method does not allocate 'partial causal power' to discrete variables or even to combinations of variables. Rather it identifies the multiple associations of sets of attributes, configurations, with outcomes. The truth table, the first output of any

QCA, is a contingency table with the final column of cells being the outcome. Gerrits and Verweij explain the congruence of QCA and complex realism:

> QCA … accepts that, while reality is real, we perceive and reconstruct it. To study this complex reality and find causal patterns that may explain the emergence of events, processes or other kinds of social phenomena, we need to make choices, as with any study. We need to choose the descriptive and explanatory conditions to include in the analysis, and how to define and operationalize (i.e. perceive and reconstruct) these conditions. QCA provides a dialogical research framework in which the researcher moves iteratively – based on actualized events (i.e. cases) – between observations and the discovery of mechanisms, thereby covering Bhaskar's three dimensions. In this way, the operations of QCA articulate the iterative nature of complexity research as a critical realist cycle of scientific discovery.
>
> (2013 178)

The method was developed for small to medium N-sized data sets which often describe all the cases in a population. This is different from conventional statistical methods which are dependent on a logic of inference which assigns statistical significance to models on the basis of the assumption that the cases in the data set have been randomly sampled from a large population. Often in social scientific research we have attribute data about all the cases. The qualitative component of QCA requires detailed qualitative engagement with the cases in order to develop attribute sets. This is possible with small to medium N examples, for example, and of particular significance in applied social science with all the instances of a particular complex social intervention. When we engage in the co-production of data sets, as Blackman and Wistow did in their study of health target achievements by English Local Strategic Partnerships, the medium N can be large since crucial attributes are both identified by practitioners/policymakers, and constructed by them in data collection.

QCA has been used as an alternative to conventional statistical methods with very large data sets (Cooper and Glaesser 2008; Byrne 2012; Ragin and Fiss 2017). It is more difficult, but not impossible, to deploy QCA for large sets of relevant cases where there is not a pre-existing, and ideally time ordered, data set. Inspection regimes for public organizations generate reports, usually repeated across time, which describe cases. It is straightforward to construct a QCA for 3,000+ English secondary schools using the quantitative data sets. However, it would be impossible to read through all OFSTED (the inspection agency) reports for all schools at all inspection points in order to construct attributes. There are two approaches possible here. One is to use the quantitative data to construct typologies of the cases, using cluster analysis or a data mining tool, and then sample within the emergent categories to select examples for detailed qualitative investigation and attribute construction – to reorder the norm of engagement with medium N QCA to be quantitative, then qualitative, then quantitative again. The other is to construct a truth table and then explore the qualitative information for 'contradictory' configurations and seek to explain the differences in outcome found. Contradictory configurations are configurations where the outcome is not identical for all or even for a large majority of cases possessing that attribute set.

Gerrits (2012) noted that QCA has difficulty in dealing with time as a dimension along which causality unfolds. Hino's (2009) approach to Time Series QCA (TS/QCA) has considerable potential for dealing with this. Both sequence, a focus of political science after Tilly (1984), and duration have causal potential. The inclusion of sequence in a QCA analysis leads to a massive expansion in the number of possible configurations. However, in exploratory work, the number of potential configurations is not really an issue. Instead, we should, using the complexity frame of reference, be prepared to ignore configurations which have no cases in the population of interest. In possibility space terms these configurations are located in the empty quarters and can be eliminated by using the 'Don't care' specification in Boolean reduction as suggested by Hiro. This is a practical recognition of the insignificance of the potentially causal sets which are simply not there in the possibility space.

Pagliarin and Gerrits (2020) lay out four strategies for the deployment of the approach:

1. They remind us that the word 'qualitative' is in the title of the method and assert that we must engage with a qualitative – one might say hermeneutic – approach to all the materials we have about our cases.

    > Rich, qualitative data are essential for understanding the conditions under which changes in system states (transitions or shifts) occur. Qualitative data achieve this not through reductionism, but through progressive contextualization because it is in past and present contexts that information about the conditions behind the emergent whole appears.
    >
    > (2020 8)

    They cite the approaches of thick description and Riain's approach to extending the ethnographic case study (2009) as appropriate modes for engagement with qualitative materials.
2. They assert that we need to deal not in variables external to the cases but in 'conditions' understood as representing emergent processes rather than abstract aspects of cases. While Byrne et al. (2009) deployed NVIVO to generate new differentiating attributes for cases, Gerrits and Pagliarin are going much deeper. That said, and in agreement with Gerrits and Pagliarin's assertion that a grounded theory approach can be used to open up 'the black box of conditions' (9 2020), NVIVO is a useful tool for developing a thematic analysis through which this can be done.
3. They want to be able to account for relative stasis and change and see how both relate to what they call 'conditions'. Time ordering by some mechanism is essential. For complex systems an arrow of time matters.

Haynes (2018)[2] developed the approach of Dynamic Pattern Synthesis which combines cluster analyses with QCA.

> The sequence of this combined method is to undertake the CA [Cluster Analysis] first, to make informed judgements about the most "real" clusters to test. These clusters and then considered as an outcome variable against variable definitions using QCA, to see how well this validates the clusters, to consider adding additional explanatory variables to the QCA, and to consider whether an additional outcome variable should be added to the final model.
>
> (2018 58)

This approach has particular value in policy contexts.

Social network analysis and neural nets – useful to explore but not perhaps to explain mapping connections kinship connections – are a staple of social anthropology

The development of computer packages which incorporate the logic of mathematical graph theory has led to a considerable extension of the approach. Social network analysts endorse relational social science. They focus on the social connections among entities (individuals, institutions, organizations) as opposed to a focus on the character of the entities themselves. Plainly connections are important for the social world but we have reservations about assigning complete importance to them. The vocabulary of graph theory and the way in which it can be operationalized in computer packages are useful for generating descriptive accounts of connections. When social network analysis is deployed as part of a multi-method approach to researching complexity it can be very useful. On its own, it can generate important descriptions but cannot get beyond this.

Neural net techniques have an obvious connection with complexity. Garson (1998) gives a full account of their use and potential in social science. While single-layer neural nets can be understood in terms of probit regression models, multi-layer nets cannot be represented in formal mathematical terms. They work by learning on the basis of algorithms which guide that learning. Neural nets are trained on a set of data and then are 'let loose' on new data and work through that data in a black box fashion to generate results. Related approaches include self-organizing maps and genetic/evolutionary algorithms.[3] These approaches have massive commercial importance and considerable potential for use in social science. The commonest commercial use is targeting through classification. Neural nets generate output through data mining which has the same character as the output of numerical taxonomic techniques like cluster analysis but they do this in a different way. Neural nets generate weightings as they proceed but no formal status can be attached to these weightings. One area where neural net-based data mining and related methods have potential is in health data. Castellani and Castellani (2003) present an account of using neural nets in health informatics which is also an excellent introduction to the general ideas behind the approach. Neural nets do not enable us to develop a grounded science of complexity. Rather they help us to predict in relation to particular complex systems on the basis of information about lots of other complex systems. In a way, they are idiot savant versions of experienced practitioners and are able to draw systematically on enormous amounts of data about enormous amounts of cases in the way a clinician draws on that experience in the traditional method of medical education – the demonstration of a case.

Modelling and simulation

A model is a tool which is intended to represent the essential aspects of something without reproducing everything about the thing being modelled. Although the contemporary emphasis is on models represented in code form in digital computers, designers have used analogue models for a long time. Modelling in engineering and science is always for a purpose. The form of modelling which has developed over the last 40 years and deploys the resources of digital computers as a basis is simulation. The Oxford English Dictionary gives three meanings for this word, the first two of which are all about being false. The third is

> The technique of imitating the behaviour of some situation or process (whether economic, military, mechanical, etc.) by means of a suitably analogous situation or apparatus, esp. for the purpose of study or personnel training.

Simulations are working metaphors because they are intended to be like the systems which they simulate and thereby provide a basis for exploring how those systems actually work. Much of the complexity modelling in the social sciences is in the tradition described by Morin (2008) as restricted complexity. That is not to dismiss it. There are useful things to be learned from restricted complexity approaches if their limitations are always taken into account.

Macy and Willer (2002) in an influential contribution argued for a turn from 'Factors to actors' in computational sociology. They followed Gilbert and Troitzsch's (1999) account of three phases in the development of sociological work which relied on the capacity of digital computing.

1. Macrosimulations based on differential equation models that predicted population distributions as the outcome of sets of factors (variables) considered at least implicitly to be causal to them. The original models used linear equations but physicists have developed models using non-linear differential equations.
2. Microsimulations – Here a set of rules are applied to a large sample of cases drawn from the population of interest. An important example of their use is in predicting the impact of changes in taxation systems. In macrosimulation, the rules/equations/algorithms are applied to whole system parameters. In microsimulations they apply to the cases making up the population of the system as a whole. Microsimulations do not permit the cases to interact or adapt.
3. Agent-based modelling (ABM) – here the agents are not passive cases to which rules are applied without any interaction with other cases or capacity to adapt. Instead, agents are autonomous, interact, follow basic rules (although developments in coding have allowed for agents to change behaviour), and from their interactions complex emergent whole systems develop.

Macy and Willer saw ABMs as permitting theoretical exploration although they did acknowledge that the tool generates micro-emergence only and is not well equipped to cope with pre-existing social structures or agentic actors other than individuals. In the early years of ABM little attention was paid to the relationship between ABM modelling and data descriptions from reality. The models tended to operate without calibration against the social real, although this was much less the case for models in ecology and related areas. In many discussions of the ontological/epistemological status of models there is a turn to the notion of mechanism as distinct from that of covering law. In general the notion of law has been either explicitly or implicitly abandoned in most modelling.

The use of real data as the basis for specifying an agent-based model is certainly a great improvement on the purely abstract approaches which used to dominate agent-based work but still remains wholly micro-driven. In contrast, the understanding of mechanism which informs the critical realist programme as developed on the basis of Bhaskar's original specification does provide, as Reed and Harvey assert, a basis for understanding of the complex social which allows for structure and regards generative mechanism as the basis of that structure. The problem of fictiveness applies in general to any simulation which does not in some way engage with real descriptions of the social or indeed any other world through the use of real data. Any model is a metaphor, a point which we will reiterate continually, almost indeed continuously, in this text. Models which engage with data do have a connection to reality against which their isomorphism with reality can be assessed. Those that don't have data have no way of checking out what is actually happening in the world. To use Hedstrom and Aberg's (2005) terminology, they have no mode of calibration. Agent-based modelling was originally predominantly fictive. That has changed. Now there is a clear recognition that models of any form need to be calibrated against real data.

Crutchfield reflected on the nature of investigation into causality in complex systems and the role of computer simulation thus:

> Very few nonlinear problems can be solved in closed form; that is, the underlying mechanisms and solutions cannot be expressed in mathematically tractable ways. We need a new approach. People today say computers can fill the mathematical void. I would agree with this only partially since powerful computers can be programmed to simulate models that are as complicated as many natural systems. Beyond the conceptual hygiene that comes from porting the ideas of the mathematics to a computer language, often little is gained. More to the point, having petabytes of simulation data is not the same thing as understanding emergent mechanisms and structures. We need to understand mechanisms and structures to build predictive theories – theories specific enough to be wrong.
>
> (2008 42)

We very much agree but many engaged with simulation used to believe that the specification of a set of rules is a valid mode of specifying causality. For example, Gilbert and Troitzsch argue that it enables us

> to take theories that have conventionally been expressed in textual form and formalize them in a specification which can be programmed into a computer. This process of formalization which involves being precise about what the theory means and making sure it is complete and coherent, is a very valuable discipline in its own right. In this respect computer simulation has a similar role in the social sciences to that of mathematics in the physical sciences.
>
> (2005 5)

This is Hayles's platonic forehand at a merely abstract level. The key issue is establishing any correspondence between the elements of the programme, which in agent-based modelling translate into rules for agent behaviour, and reality itself – the problem of isomorphism. Holland (1998) explicitly confines his discussion to systems for which rules can be established, allows the possibility of developing such rules for other systems in the future, and suggests that there may be a set of systems for which such rules cannot be developed. For us he is letting in the distinction between restricted and general complexity. Simulation models can be developed to deal with restricted complexity. Calibration of them against real data is a valid and proper way of establishing isomorphism with reality. For any modelling of whatever form of a real complex system, calibration against descriptions of the world, which descriptions can include both quantitative data and qualitative accounts, is the only mode of establishing validity available to us.

Castellani and Rajaram (2012) extend the range of modelling in a most interesting way. They identify three modes of modelling which have been deployed so far in complexity studies:

Age-based modelling.
Dynamical modelling which is equation based.

Statistical modelling based on aggregate data to which we return in the last section of this chapter.
And add a fourth – case-based modelling.

This last has three components which comprise a theoretical blueprint based on the complexity frame of reference, a set of instructions based on the case method approach which deploys the idea of assemblage, and a mathematical toolkit. This last is founded on the representation of cases as n dimensional vectors: 'Each case is a k dimensional row vector where each x_{ij} represents a measurement on one of the variables being used to model a complex system. (2012)

Castellani and Rajaram deploy a process-based understanding of the social world but develop this through an account of coupling, (which reflects the interests of relational sociology and the methods of social network analysis), and of social knowing which brings in agency. They also deal in collective terms in a way which corresponds closely to Cilliers (2001) specification of boundaries among systems as being open and allowing causal influences precisely based on spatial propinquity. There are a lot of important ideas in this paper including the understanding that ties are not just relationships but also interactions in that connected entities at whatever level, and indeed across levels. That means cases have causal powers on each other through these relations. Their approach which deploys cluster analysis as a typological device, social network analysis to describe relationships, and a simulation exercise based on the products of the first two approaches, and does so within a carefully established qualitative description of the social areas they are examining in relation to public health issues, has very considerable value. Schimpf et al.'s 'Cased-based modelling and scenario simulation for ex-post evaluation' (2021) provides a demonstration of how this enables us to use modelling to get to grips with working out what works, when and where, and how different things work in combination to generate outcomes, even within the same programme.

The modelling community's response to the challenge represented by the COVID-19 pandemic has provided us with material of absolute relevance to a discussion of the role of modelling in a research programme which accords with the complexity frame of reference and demonstrates encouraging signs of an engagement with that programme in its general rather than restricted form. In March 2020 the *Journal of Artificial Societies and Social Simulation* published a special article 'Computational models that matter during a global pandemic outbreak: a call to action' (Squazazzoni et al. 2020). The authors stated that:

> Understanding the pandemic requires fine-grained data representing specific local conditions and the social reactions of individuals. While experts have built simulation models to estimate disease trajectories that may be enough to guide decision-makers to formulate policy measures to limit the epidemic, they do not cover the full behavioural and social complexity of societies under pandemic crisis. Modelling that has such a large potential impact upon people's lives is a great responsibility.
>
> (2020 abstract)

Data is vital. Squazzoni et al. made a particular point that in a context of crisis the normal mode of scientific publication of findings as this has developed since the 19th century is not appropriate. The rationale for this departure from the norm (of which we completely approve) was that:

> When policy decisions and people's reactions depend on perceptions of the future, and scenarios are probabilistic and largely unpredictable, computer simulation models

> are seen as a viable method to project future states of a system from past ones in a non-trivial manner. What we see today in many media are predictions of the exponential growth of the number of infected persons based on equations that capture stylized populations and the distributions of their different states. However, any social or behavioural scholar can spot that these projections do not consider relevant factors of social complexity, which are intrinsically crucial to the modelled dynamics and a negligible exogenous force. Not recognizing social complexity can undermine the credibility of findings, and thus we call for urgent initiatives to: (1) improve the transparency and rigour of models to understand theoretical premises and details and (2) promote data access to help contextualize and validate models across various levels of analysis (i.e., micro, meso, and macro). This call is even more urgent when simulation findings can rapidly affect public policy decisions (e.g. on possible consequences of certain policy scenarios) and/or motivate individual actions (e.g. impact upon decisions to stay at home to 'flatten the curve').[4]
>
> (2020 1.8)

Brian Castellani concluded:

> that many of these models do not think about disease transmissions from a case-based configurational perspective. Regardless of the method used, a case-based configurational perspective is anchored in four core arguments that deeply resonate with the majority of computational methods used today. First, the case (which in this case is someone with COVID-19) and its trajectory across time/space are the focus of study, not the individual variables or attributes of which it is comprised. Second, cases and their trajectories are treated as composites (profiles), comprised of an interdependent, interconnected sets of variables, factors or attributes. Third, the relationships and social interactions amongst cases are also important, as are the hierarchical social contexts/systems in which these relationships take place. And, finally, cases and their relationships and trajectories are the methodological equivalent of complex systems – that is, they are emergent, self-organizing, nonlinear, dynamic, network-like, etc. – and therefore should be studied as such. … Given that many public health models do not embrace this approach, they often struggle to demonstrate the differential impact that the spread of COVID-19 will have on different populations and subgroups.
>
> (2020)

Castellani was part of a team developing a model to help policymakers and practitioners at local levels across the North of England. The abstract to a paper published in JASSS as a response to that journal's call to action sums up what was done:

> This paper presents JuSt-Social, an agent-based model of the COVID-19 epidemic with range of potential social policy interventions. It was developed to support local authorities in North East England who are making decisions in a fast moving crisis with limited access to data. The proximate purpose of JuSt-Social is description, as the model represents knowledge about both COVID-19 transmission and intervention effects. Its ultimate purpose is to generate stories that respond to the questions and concerns of local planners and policy makers and are justified by the quality of the representation. These justified stories organise the knowledge in way that is accessible, timely and useful at the local level, assisting the decision makers to

> better understand both their current situation and the plausible outcomes of policy alternatives. JuSt-Social and the concept of justified stories apply to the modelling of infectious disease in general and, even more broadly, modelling in public health, particularly for policy interventions in complex systems.
>
> Badham et al. (2021 Abstract)

JuSt-Social is a descriptive model made to be used to help local planners in Local Resilience Fora or their equivalents – cross governance local bodies drawing on all relevant services and agencies to confront crises and developed in part in response to previous flooding emergencies – consider policy options in response to the current state of the pandemic in real time in their locality. It focused on the two distinct processes of infection of non-infected people and subsequent progression through a series of epidemic states because it was these processes which drove the translation of infection into demands for hospital beds and deaths. This allowed local planners to think about alternative futures and was greatly valued by them as a method of doing something which addressed local contexts and the necessarily local character of the pandemic as it progressed.

In the first edition of this book, we were to say the least critical of much of the thinking behind and forms of models in practice which existed at the time of writing. Things have changed and the response to COVID-19 demonstrates that clearly. We are much more enthusiastic about how modelling is now being done and about the prospects for social modelling in the future. A key aspect is always co-production. JuSt-Social was developed in dialogue with its users. Those who thought such dialogue unnecessary were flat wrong.

Big data – backwards, now, and the future

Kitchin's (2021) stylistically innovative book *Data Lives* has the subtitle: 'How data are made and shape our world'. Since the development of data systems by states, data always has shaped our world. Statistics have never been simply described. As descriptions of complex social systems they have had causal power in shaping the future trajectories of those very systems. Kitchin focuses on the enormous extension of the datasphere in consequence of computing power and the generalization of digitally based technology. Data has always been big when we look at the statistical constructs of censuses and large-scale social surveys but it has become much, much bigger. In part, this is because the filing systems of the state and other organizations now exist as interrogatable electronic records. Now, records exist not as handwritten or typed entries in a physical ledger but as textual entries in an electronic record and could be extracted with far greater ease. There is nothing new about such records. They are just so much easier to get at.

Kitchin while recognizing the importance of filed records pays far more attention to the way electronic devices generate data about their users and other devices survey not only human users but also produce real-time records of so many other elements of the social and natural world. He described (2013) big data as:

- Huge in volume, consisting of terabytes or petabytes of data.
- High in velocity, being created in or near real time.
- Diverse in variety, being structured and unstructured in nature.
- Exhaustive in scope, striving to capture entire populations or systems (n = all).
- Fine-grained in resolution and huge uniquely indexical in identification.

- Relational in nature, containing common fields that enable the conjoining of different data sets.
- Flexible, holding the traits of extensionality (can add new fields easily) and scaleability (can expand in size rapidly).

So big data is big – lots of it. It comes at users at great speed. There is no such thing as continuous measurement but there is very rapid continual measurement – that is repeated measurements often at very short time intervals. Structured data has some sort of organization in relation often to specific cases. Unstructured does not. Much of it takes the form of text documents or records Estimates suggest that about 80% of all data in the 'datasphere' is unstructured. Unlike sample-based data, big data often tries to measure all the cases in the relevant population. This is particularly true of big data generated by administrative processes. Fine-grained means go down far towards and often includes the micro level of the smallest identifiable relevant case set. Indexical means that data is linked to specific cases and to specific measurements. Relational means that one data set can be linked to others, usually through some kind of common case identifier. Flexible means that more data can be imported in the form both of new measurements and new cases.

Anderson (2008) argued that the ready availability of machine extractable and machine-readable data made a scientific method based on covering laws or even on mechanisms redundant. Correlation replaces cause. Look for the patterns and all will become clear. We agree with Kitchin that, to put it more forcefully than he does, this is nonsense. To illustrate let us take one of the most important uses of data (misnamed as artificial intelligences[5]) analytical tools – to mine the enormous data sets of health records which record life histories of diagnoses, treatments, and outcomes. These are ways of replicating the individual and collective experiences of clinicians in developing more effective combinations (complex causation) of interventions or treatments to get better outcomes. Randomized Controlled Trials (RCTs) deal with whole populations of cases rather than particular categories of similar cases, everybody instead of the near neighbours, and are poor at dealing with treatment protocols which involve more than one intervention – involve complex and interactive attempts to effect change. Patterning allows the development of more effective intervention which is necessarily related to the basic biological science established in other ways. What clinicians do on the basis of data mining is still in the domain of the art of medicine.[6]

The COVID-19 epidemic illustrates how data is generated and what can be done with it, both as science and as governance practice. COVID-19 death and infection data not only had administrative value but also became hot news topics across the media. There was nothing new about the death data since recording death by cause has been one of the core components of vital statistics since the development of registration processes in the first half of the 19th century. However, there was still a counting issue in relation to operationalization. Was a death attributed to COVID-19 if that condition was the sole recorded cause or was it so attributed if there were other conditions which were present and contributed to the death? Recording infections was a chancy business since many infected people were asymptomatic or had such mild systems that they did not present for testing. Although great claims were made for data 'scraped' from various communication devices and computer usage this has not proved to be significant in patterning COVID through the pandemic.

At the level of science, there have been interesting developments in relating COVID-19 mortality data to other data about the people who have died. Nafilyan et al. (2021)

used data from the ONS Public Health Data Asset to pattern COVID deaths differentiating by ethnicity.[7] The Public Health Data Asset contains data on 29 million adults aged 30–100 living in private households in England derived from various sources including the 2011 census. It is certainly big data. Tying COVID mortality data to it demonstrates both its relational character and the ability to extend it easily. Data sets on this scale can be used for multiple purposes including examinations of inequality in all dimensions.

It is data sets like the Public Health Data Asset – Big because it ties together information from a whole range of other sources – that have the most potential for retroductive analyses of life-course outcomes for individuals. What is new is the ability to set everything together by indexing to cases. We are tempted to say that is administrative data which generates useful big data and really not much else matters but we await developments.

Statistical modelling

Breiman (2001) identified two cultures in statistical modelling – the stochastic and the algorithmic:

> There are two cultures in the use of statistical modeling to reach conclusions from data. One assumes that the data are generated by a given stochastic data model. The other uses algorithmic models and treats the data mechanism as unknown. The statistical community has been committed to the almost exclusive use of data models. This commitment has led to irrelevant theory, questionable conclusions, and has kept statisticians from working on a large range of interesting current problems. Algorithmic modeling, both in theory and practice, has developed rapidly in fields outside statistics. It can be used both on large complex data sets and as a more accurate and informative alternative to data modeling on smaller data sets. If our goal as a field is to use data to solve problems, then we need to move away from exclusive dependence on data models and adopt a more diverse set of tools.
>
> (2001 199)

Cobb (2015) called for a rethinking of the undergraduate statistics curriculum in a direction very much in sympathy with Breiman's argument noting that if Tukey (1977) had had access to current computing technology he would have used it as the basis of exploratory tools which let us see what the data are telling us. While neither Breiman nor Cobb are calling for the abandonment of statistical models, they are saying there are other, and probably more useful, ways of doing things. And yet regression-based models persist and inform a great deal of quantitative work across the social sciences. There have been sophisticated developments of them, notably multi-level modelling, which do at least begin to address the reality that causal force operates at different levels in the world. Williams (2020 169) asserts that while a statistical model – most such models are regressions – has to be moderated by a theoretical account of a mechanism, it can serve a useful purpose. We agree with one enormously important proviso.

Statistical modellers almost always talk about finding THE model that fits the data. What is missing from this approach is any grasp of the significance of equifinality. Causation has to be understood as multiple. To put this in the language of statistical modelling there is not one model which fits the data but possible many models. The initial truth table in QCA and the reductions the Boolean element of that technique achieve are ways of dealing with equifinality. In practice, the total amount of variation in a data set 'explained' by for example a logistic regression model is usually not all that much. The set of configurations in a truth table can often account for almost all the variation in relation to an outcome precisely because they are multiple.

In regression models, the emphasis is almost always on slope coefficients which has always seemed to us to derive from a desire to replicate the equation-based representation of physical laws so that what is described looks like proper science. There is also a desire to see just how much of the variation each contributing variable explains in relation to an outcome. This is the ultimate in reification of the causal powers of variables, either as directly measured or as derived from some factor analysis. Variates almost never operate alone. Social and much of natural/ecological causation is always complex. There is considerable disquiet in the philosophy of science in relation to the value of covering laws themselves but looking like 'science' has led to higher status in the world of the contemporary academy.

Conclusion

Critical realists often reject the value of quantitative modes of exploration, description, and causal reasoning and see only descriptive value in statistical measurements.[8] We do not agree but we very much do not privilege quantitative work over qualitative modes. Abductive/retroductive reasoning can be deployed in relation to all forms of description of the social and natural worlds and of the crucial interfaces between them. Despite the general public's equation of controlled experiment with *the* scientific method, a great deal of contemporary science, particularly in physics, is done by simulation based on data collected. Simulations are not closed-off and abstracted experiments. All science is changing in the era of the complex. For us the important science for application in confronting the enormous issues that face us as a species in the only world we have is most likely to be done by a variety of methods and deploy the full toolkit of empirical exploration and causal reasoning. So we now turn to mixing methods in the final chapter in this section.

Notes

1 Pang-Ning et al. (2019) in their *Introduction to Data Mining* go through these procedures and include a discussion of cluster analysis techniques. Henig (2015) has a useful technical discussion of how to establish good clusters, i.e., clusters which do correspond to a real set of differences.

2 Chapter 3 of this book provides a first-rate account of the logic and methods of both cluster analysis and QCA.

3 Castellani and Rajaram (2021) provide an excellent account of these approaches.

4 There is much we agree with in Squazzoni et al.'s general argument but we absolutely do not agree with their endorsement of Popper's rejection of planned grand social interventions.

5 Misnamed because they do not analyse in the sense of explain by breaking up into components. Rather they do indeed look for patterns in terms of categories of courses of disease in relation to life histories. They describe rather than analyze.

6 Castellani and Rajaram (2021) present not only a clear and informed discussion of the techniques and mathematical foundations of big data methods, but also engage with the ontological and epistemological debates from an explicitly complexity frame of reference.

7 No analyses have yet been published which analyse COVID-19 death by any measure of social class although there was a very clear occupational hierarchy evident in analyses done on deaths during the first wave.

8 Dyer and Williams (2021) confront this position with an example which illustrates the value of case-based accounts of trajectories in relation to outcomes.

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10 Complexity and mixed methods

The complexity frame of reference attaches no privilege to any mode of scientific inquiry, other than regarding experimental methods, whether bench or randomized controlled trial, as inappropriate when dealing with complex systems and emergent properties. The paradigm wars in the social sciences in the late 20th century were, to put it bluntly, fatuous. Bryman's *Quantity and Quality in Social Research* (1989) should have put a stop to them. For most of UK sociology, it did. However, in the US assertion of the primacy of quantitative reasoning persisted, not least in King et al.'s (1994) *Designing Social Inquiry* which deployed the linear logic of regression analysis to inform the conduct of qualitative research. By the late 1980s, Bhaskar's critical realism was increasingly deployed as the ontological basis of social investigation. Teaching in the 1990s, Byrne treated Bryman's book as implicitly realist because it identified the positivist logic which underlay many causal accounts in qualitative work. Likewise, Marsh's *The Survey Method* (1982) based its understanding of what surveys were and what they could tell us about the social world on a realist framing. With the increasing influence of the realist programme, the assertion of the incommensurability of quantitative and qualitative modes of investigation and modes of reasoning has ceased to dominate.

'Mixed Methods'[1] are increasingly employed in social research particularly in applied fields of study and in work done for governance especially evaluation.[2] There are now journals, for example, *The International Journal of Mixed Methods Research*, which promote the programme. There is debate about what constitutes mixed methods as opposed to research which simply employs a plurality of approaches. The distinction between multiple and mixed methods is usually articulated in terms of when in the process of research integration of the outputs of the different modes happens, if it happens at all (Bazeley 2009, 2016). A good deal of the criticism of mixed work points out that in reporting findings there is no integration across modes. Quantitative statistical findings are often reported quite separately from qualitative text, even when this is done in the same report, article, or book.

Integration or synthesis?

The key word is 'integration'. Poth (2018) is representative in asserting the need for integrative thinking throughout the research process. We approach the relationship between the complexity frame of reference and mixed methods through thinking about a close but not quite identical synonym – synthesis. Abbot's (1991) discussion of 'the lost synthesis of history and sociology' helps here. For him what was required was

> the synthetic revolution that Abrams and others wished to see – the merging of determinate and contingent explanations in a fully historical social science. (1992

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> 202) …the historical sociologists accepted from history the positive value of limiting generalizations and mastering details but reinterpreted social science's belief in causality in qualitative (original emphasis) terms.
>
> (1992 211)

It helps to look carefully at the language of social research as a basis for understanding what research is doing and comparing that with what the complexity frame of reference suggests it should be doing. The 'key words' include data, models, effects/outcomes. To start with the last paired couple – effects/outcomes. We deal in the social sciences with cases which have to be understood as complex systems (see Byrne 2009). Cases have relations which describe their interwoven character with other systems. The world is constituted – note the world, not just the social world – from systems at every level and their inter-relationships. The world and all that is in it changes, including changes in system relationships. What matters is the state of the systems of interest to us at any point in time – along their interwoven trajectories. Byrne (2011) argued the outcome/effect is the actual state of the system at any point in time. We need to explore sections of the trajectory where things stay much the same – remain in the same attractor – for a relatively long period, and those where the trajectory describes a period of qualitative change, a phase shift, a move to a new attractor in the possibility space.

What does the word 'data' mean? Dictionary definitions usually begin with mentions of quantities as descriptions which are generated by processes of measurement before going on to mention data as also having a possible qualitative character. Both quantitative and qualitative data can be descriptive but only numbers at some level of measurement can be entered into mathematical modelling processes of whatever kind. As descriptions both should be understood in the terms suggested by Byrne (2002) as traces of the state of the system. The word trace has an inherently dynamic character. It implies seeing not just where a system and its relations are now but also where they have been and where they might go in the possibility space for them depending on social actions

Scientific realisms, including critical realism, understand our empirical descriptions of the world as made by us out of our investigations of the actual world, itself as made by the underlying generative mechanisms of the real. Empirical accounts are made things but made from something, not nothing. Reality has a voice alongside the voices of scientists, our procedures, and our instruments. From a complex realist position all our findings – a word which we much prefer to data despite their etymological identity because findings allow us to incorporate relations into our account in an explicit fashion – are made things but made from the actual and in interpretation of the underlying real itself. Our models are necessarily abstracted, i.e., reduced in complexity, ways of describing causal relations in and among the systems of interest to us. They are articulations of cause which depend ultimately on narratives, even when expressed in quantitative form and are always buttressed by qualitative materials since those are the foundation of our interpretations – interpretation is everything.

We generally express our causal accounts in terms of models – abstractions from the complex which reproduce the essential patterns of reality in the terms expressed by Cilliers. One interesting, if contestable take on all of this is given by Goertz (2017) in terms of the research triad of causal mechanisms, cross-case analyses, and case studies. The title of his book includes the term 'multi-methods research' and the subtitle argues for 'an integrated approach'. As with the development of process tracing (George and Bennett 2005), Goertz continues to deploy a language of variables assumed to have causal powers which operate outwith the cases which form the objects of study. This is not commensurate with a complexity framed understanding of causal processes but his account of the relationships among the three modes is informative. We disagree radically with his assertion that statistical approaches' use of the idea of cofounding variables is to be understood as a recognition of the reality of equifinality. As he notes the different configurations, even after Boolean reduction, which are generated by Qualitative Comparative Analyses (QCA) do accord with Von Bertalanffy's understanding of equifinality not only in terms of complex causation but also possible different multiple forms of causation. The reference to multi-methods is telling. For Goertz integration happens not in the process of the conduct of the research but through recognizing that cross-case methods, including explicitly qualitative comparative work, are concerned with cross-case causal inference, and case studies, implicitly in ideographic form, are concerned with within-case causal inference. So Weber's account of the relationship between Protestantism and the emergence of capitalism is an example of cross-case inference, whereas Malinowski's account of the role of the Kula cycle in the Trobriands, which results in structural functional account of it, is a within-case causal inference.

What is missing from this is social theory and the role of generalization in buttressing theory beyond the middle range. That is precisely what the historical sociologists were calling for. Leach's account (1954) of change in the political systems of highland Burma is, as Firth said in his foreword to the book, explicitly organized around models. While Leach was always an opponent of easy transfer out of context, his adherence to a structuralist position meant that his work, or rather it is perhaps better to say his interpretations of his own fieldwork and others' fieldwork, was informed by an understanding which despite his use of the term equilibrium was in fact a version of what we would now call far from equilibric thinking. Shifting equilibria is a way of talking about dynamic systems in change. So here in the essentially ideographic interpretation of anthropological fieldwork, we find something which corresponds to the qualitative generalization of historical sociology, and at the same time in a much more explicit way deploys the idea of models as a basis for framing accounts.

We continue this chapter by outlining examples of work done in a complexity congruent fashion which deploy mixed methods. First, there are some important assertions to make in relation to the way social research as done can contribute to the real issues facing us on this planet as we go forward through the 21st century. Much social science research is done and presented in forms which reproduce both a Newtonian form of causal account, whether this is presented in quantitative, qualitative, or mixed form, and work as part of incremental normal science rather than contributing towards ways of understanding which can enable us to confront wicked issues in general and the key challenges of inequality and impending climate catastrophe in particular. There is a difference between applied research directed towards informing policy and practice and research in a disciplinary mode. These are not dichotomous sets – they are fuzzy and overlap – but the distinction has heuristic force. There are differences across disciplines but the tyranny of the journal article as the basis of academic careers has had a malign influence on most of them. Other than in monographs the findings are presented in discrete packages in the journal article. In some disciplines, notably US sociology and political science, what dominates is post facto modelling which purports to explain patterns of outcomes at some level. The examples presented in Goertz (2017) generally fall into this category. Even in applied fields work done to inform policy and practice tends to take this form. Our major objection is to the belief that these scraps of knowledge construction on their own should be the basis on which social scientific understanding is constituted. The fragmentation of academic disciplines and fields into ever more defined and siloed sub-elements with their own institutional and intellectual scaffolding exacerbates this. What is missing at every level is synthesis towards a purpose. Our discussion henceforward will be directed towards deploying the complexity frame of reference in helping us to do just that.

Who does research and why do they do it?

We need to think hard about the role of intellectuals in the 21st century, particularly in high- and high-middle-income countries which contain more than half the world's population and through both production and consumption generate nearly 90% of the greenhouse gases, the drivers of climate catastrophe. Gramsci's distinction between traditional and organic intellectuals helps here but does not really deal with the way in which contemporary capitalisms – the plural is important to indicate varieties of form – have much larger proportions of their population educated to third level. In these societies there now is a large group of technical intellectuals, not only in the traditional forms of engineers, etc. associated with production of material commodities, but also in roles associated with the operation of the state including the very large sectors of state activity concerned with health, education and related services, and – of crucial importance – planning processes in relation to the development of policy including at the interface of the social and the ecological. A lot of social research is done in these areas and by these people. Some have tenured University jobs, many are contract researchers in academic institutions, others are employed at various levels of governance for example in the analytic professions of the UK civil service. This research is done to be applied and some of the most interesting innovations, both in technique and thinking, have developed here rather than in the core disciplines of the social and related sciences.

Applied research is done as a basis for action. Some of it is done as action research. Action research is never a matter of researchers standing apart from a process of change to describe and evaluate it. Rather the research is part of the process of engendering change. It is important to make a distinction between participation and action. Participation means the engagement of the human actors present in the context of the research: at the fully developed level of participation, in elements of its design, execution, and interpretation of results. When fully developed participatory research is done it is more usually described as co-production because the human actors have existing power as gatekeepers and from their institutional roles. Action research is inherently and absolutely about research where the objective is change and the research forms part of the action towards change. Research can and, in terms of Cilliers's specification of the ethnical necessity for understanding social complex systems to be based on presence within them, should be both participatory and action oriented but the two aspects are different. In this sense, all research which has a policy/practice orientation is action oriented. However, in fully developed action researcher the researchers are directly engaged in the process of social transformation whereas much applied research is handed to others to interpret and act upon. Another distinction is important. That is between research as projects – single pieces of investigation – and research as programmes – larger-scale linked series of research projects for funding purposes each piece being described as a work programme. A lot of research is done in the latter format, sometimes in academic modes as with the locality studies funded by the UK Economic and Social Research Council in the 1980s or with a specific applied focus as with CECAN (Evaluating Complexity across the Nexus) and the

UK Research Councils' combined programme 'Rural Economy and Land Use' (RELU see Lowe and Philipson 2009).

And then there is synthesis of research as reflective scholarship. There is not enough of this because it does not fit well into the format of either journal articles or research monographs. Scholarship is different from systematic review. Systematic reviews are carried across large numbers of research investigations as reported in relation to a specific research question of the kind: is there a relationship between the number of surgical procedures carried out in relation to a specific condition and the outcomes of those procedures usually defined by short-term survival rates? Usually, they are associated with some notion of a hierarchy of evidence as presented inter alia by the Cochrane collaboration which would place observational data studies of the volume/outcome kind well below the metanalysis of data sets from multiple Randomized Controlled Trials (RCTs). Reflective scholarship in contrast is the process of reading through lots of examples of research of all sorts of kinds and seeking to see what we can draw from them in terms of understanding an issue of concern to us. Byrne's (2019) review of *Class After Industry* was an example of reflective interpretive scholarship drawing on quantitative work including time series of descriptive data and cluster analyses of microdata, and qualitative work based on ethnographic, oral, and documentary history and related modes. In contrast with systematic review, it is informed not by assessment of technique but by a focus on what the research is telling us about the issue. We do need to know that the research has been done competently but not by judging it against a rigid protocol based on a naïve ontological perspective. In a way we can think of reflective (and reflexive) scholarship as being rather like the hermeneutic circle. Reflexivity matters. Interpretation always has to be informed by a clear exposition of the position from which it is being conducted. This is even more important than the deployment of reflexivity in the research process itself. We will develop this argument in discussion of specific examples.

North Tyneside CDP – action research in a context of transformative social change

The North Tyneside Community Development Project (CDP) was part of a UK-wide programme of action research initiated by the UK Home Office in the late 1960s and inspired by the Johnson administration's 'Great Society' 'War on Poverty'. It was modelled on the Ford Foundation's 'Grey Area Projects' described in Marris and Rein's *Dilemmas of Social Reform* (1967). The projects combined action and research with two teams originally intended to operate separately, one engaged in innovatory action and the other describing the processes of innovation and evaluating outcomes. The intention was to run the programmes in order

> to discover and develop methods of helping the severely deprived to make personally constructive use of the social services; and to assess the implications for the future development of these services in their normal forms of organisation, including the establishment of more valid and reliable criteria for allocating resources to the greatest social benefit.
>
> (Working Party 1968)

The Home Office identified two factors as causal to deprivation. One was deficits in terms of skills and orientation in the population of deprived areas – very much on the lines of although not explicitly referencing the 'culture of poverty' arguments advanced in the US by Moynihan and Glazer (1969).[3] In the UK this was usually described in terms of a 'cycle of poverty' with an emphasis on intergenerational transmission of deprivation. The other, which reflected the contemporary interest at the national level of governance in organizational issues, was that the local government was too siloed in departmental empires and that services were delivered in a poorly coordinated and inefficient manner. This was a central principle of the Baines Report (1972) on the need for a new organizational structure in the local authorities created by the restructuring of English and Welsh local government in 1974. There was a clear intention that by operating cross-departmentally at the neighbourhood level the CDP teams would shake up ossified systems through functional conflict. Many but not all of the CDP teams rejected the deficit model of the populations of their areas, assigning their deprivation not to the people themselves but to a capitalist social order in general and in particular the social costs of the developing process of deindustrialization which transformed the UK from the most industrial society there ever has been into its present post-industrial form.[4] Bertrand (2017) provides one of the better summary accounts of the overall CDP experience.

Here we want to focus on the actual mode of research deployed in the North Tyneside CDP where Byrne was the Research Director. He had become interested in the methodological implications of action research through reading Marris and Rein's (1967) book as an undergraduate and had written an MSc dissertation on action research. There was a move in the radical CDPs to break down the distinction between action and research as processes and the action and research teams in work roles. In North Tyneside, there was an explicit commitment to research in the service of action which at least was based on a commitment to the notion that knowledge was achieved in the process of acting. The complexity frame of reference had not been formulated in its current form in the 1970s but what was going on with the North Tyneside CDP was a set of different but related complex interventions, in a set of neighbourhoods, also complex systems, embedded in a locality, again a complex system, in a conurbation, again a complex system – and so on up to the world system – which was at every level from that of the individual people upwards interwoven with other complex systems of governance, culture, and economic structure, and of not only capitalist but also patriarchal social relations.

The CDP joint team worked by drawing together every available form of existing evidence from historical documentation describing the development of North Shields and in particular the creation of the Meadowell social housing estate which comprised much of the CDP area through processes of slum clearance in the 1930s. This historical evidence included time series of data from decennial censuses and reports of the local Medical Officers of Health and administrative microdata describing the households allocated to social housing over the period since its construction. In collaboration with local residents, who did much of the interviewing, general surveys were conducted to establish household structures, residents' views as to important issues, and their desires for change and development. As the project progressed these were supplemented by specific focused surveys, e.g., addressing residents' wishes as to whether older housing areas in central North Shields should be redeveloped through clearance or improved as they were using available central government grants. Team members met regularly with local residents on an individual basis in relation to advice about welfare benefits and in 'action group' meetings as well as more informally. The team was present in the areas throughout the working day and beyond at evening meetings and subsequent socializing – there was a lot of going to the pub. The group meetings were minuted and team meetings regularly reviewed progress and documents of various kinds.

What was going on was a multi-method and mixed method set of research inquiries embedded in what was a participants as observers ongoing ethnographic programme. One important aspect of the team's work was that, with considerable input from local residents in oral historical contributions, they sought to locate the CDP area within the whole dynamic system of the locality of which it was a part.[5] So we paid attention not only to housing within the CDP area but also to how the whole housing system in the locality had developed from the mid-19th century onwards. We explored not just our own case of the CDP area but also in terms of its relations with the urban system as a whole. One example was the exploration of the role of local building societies and local government's implementation of national policies in developing owner occupation as a tenure system in the 1930s. Byrne (1989) revisited the whole North of Tyne urban system to explore changes subsequent to the closure of the project deploying that range of investigation. Understanding the dynamics of urban systems and their spatial subsystems is crucial for any intervention within them.

The research outputs from North Shields included contributions to a series of inter-project reports on the consequences of industrial change, housing policies, and the austerity imposed by the Labour government after its recourse to the IMF in the late 1970s. North Tyneside CDP paid particular attention to women's work at a time when some of the project workers and other women on Tyneside were developing an important second-wave feminist initiative as the 'socialist women's action group'. The output of the local project was a series of locally focused reports covering industrial change, housing, the action groups, play and recreation, and the experience of 'working for change in a working class area'. Looking backwards at these now informed, by the complexity frame of reference, we can see them as descriptions of system change in progress informed by systematic and careful process tracing, both of the system changes at micro, meso, and macro levels, and of the actual work done by team members in full participatory engagement with local people. This was action research done in a dialogical fashion from the bottom up. To put it bluntly and in the maritime vernacular of Shields,[6] this was pissing to windward against national policy and global trends, particularly globalization of production. This demonstrates unequivocally the need for action to happen at all social levels in order to achieve meaningful change, but as an experience it was great fun (a view shared by many local residents) and as something to reflect on from a mixed methods perspective it is important.

The Rural Economy and Land Use Programme – RELU

> Rural areas in the UK are experiencing a period of considerable change. The Rural Economy and Land Use Programme aimed to advance understanding of the challenges caused by this change today and in the future. Interdisciplinary research was funded between 2004 and 2013 in order to inform policy and practice with choices on how to manage the countryside and rural economies.
>
> The Rural Economy and Land Use Programme enabled researchers to work together to investigate the social, economic, environmental and technological challenges faced by rural areas. It was an unprecedented collaboration between the Economic and Social Research Council (ESRC), the Biotechnology and Biological Sciences Research Council (BBSRC) and the Natural Environment Research

Council (NERC). Additional funding was provided by the Scottish Government and the Department for Environment, Food and Rural Affairs.

RELU was a collaboration across the social and natural sciences and there is an informative and particularly useful literature on the implications of this kind of cross-disciplinary research. Here the mixing of the methods was not just within the programme of the social sciences but involved a mixing of natural science approaches with social scientific modes. Lowe et al. in an important reflection on the experience of this programme argue:

> that effective engagement depends upon overcoming basic assumptions that have structured past interactions: particularly, the casting of social science in an end-of-pipe role in relation to scientific and technological developments. These structurings arise from epistemological assumptions about the underlying permanence of the natural world and the role of science in uncovering its fundamental order and properties. While the impermanence of the social world has always put the social sciences on shakier foundations, twenty-first century concerns about the instability of the natural world pose different epistemological assumptions that summon a more equal, immediate and intense interaction between field and intervention oriented social and natural scientists. The paper examines a major research programme that has exemplified these alternative epistemological assumptions.
>
> (2013 207)

One notable characteristic of RELU was the mode of assessment of project proposals. Lowe et al. describe it thus:

> RELU required that each and every project creatively combine social and natural scientists, but not specifying how this should be done. Project proposals had to show how they would effectively combine research staff and perspectives to maximise their synergy. This called for strategic leadership and innovation in project design, while project management had to make space for inter-disciplinary exchange and synthesis. A thorough approach to the assessment and selection of research proposals was essential. It was found to be vital to have two separate elements. First, a rigorous peer review by relevant experts of the strengths of the scientific components of a project proposal, and second an overall assessment of its quality of integration and the strategic importance of its interdisciplinary collaboration, done by assessors with a breadth of understanding and experience of interdisciplinary research.
>
> (2013 217)

There was no in advance specification of how this combination across disciplines was to be achieved but specification of the ways in which synergy was to be achieved was a key criterion in awarding funding. The significance of inserting interdisciplinarity and an acceptance of the validity of mixed methods into grant award decisions is especially relevant as Lowe et al. note in relation to the development of programmes of research which offer insight into how to confront climate catastrophe.[7]

A central feature of some of the empirical studies in RELU was the incorporation of the knowledge of stakeholders accessed through the classic qualitative methods of interview and focus groups into the development of models. The models were often sophisticated and incorporated simulations but the experiential knowledge of stakeholders and eliciting their tacit knowledge through dialogue is a key part of co-production/participation in mixed methods. The RELU studies are especially important given the need to develop a research programme which contributes to confronting climate crisis and stand not only as exemplary illustrations of research practice but also provide important reflections on the interface of the ecological and social sciences. Berkes (2004) identified

> paradigm shifts in ecology and applied ecology. I identified three conceptual shifts—toward a systems view, toward the inclusion of humans in the ecosystem, and toward participatory approaches to ecosystem management—that are interrelated and pertain to an understanding of ecosystems as complex adaptive systems in which humans are an integral part.
>
> (2002 621)

This is absolutely correct and the RELU programme provided multiple examples of how to work in a way which must be the form of ecological interventions going forward.

CECAN – Centre for Evaluating Complexity Across the NEXUS

We will describe CECAN in our discussion of the deployment of the complexity frame of reference in evaluation. Here we want to discuss an example of how the work of that programme, an innovative exercise based on an explicitly complexity frame of reference, and the deployment of methods to be applied rather than pre-specified work packages, demonstrated the utility of mixed methods. This was an approach to using the existing database of the UK Environment Agency in relation to waste crime.

The Environment Agency is the regulatory and enforcement body dealing with waste crime across the UK. Waste crime is a big criminal business with a social cost valued at around £800 million per annum. The agency investigates instances of waste crime with the case being defined as the site on which the crime has happened. The frontline investigators, mostly experienced former police detectives, investigate, report on their investigation in qualitative and quantitative terms, and then the agency acts to enforce using appropriate powers although often on the basis of negotiation before formal enforcement and records the outcome of its intervention in relation to resolution at whatever level, including failure to resolve. The agency's own research staff were interested in using Qualitative Comparative Analysis as a basis for interrogating their own large database as it would allow them to incorporate both what was recorded in numerical terms and enable them to use the rich qualitative data to develop additional quantitative attributes to input into a QCA analysis. The advantage of deploying QCA is that it allows for the demonstration of both multiple and complex causation in the configuration sets.

It was the actual process of developing towards the deployment of a QCA approach which introduced other methods into the practice. This piece of work was absolutely co-production in the form of its development. It began with discussions with agency research, evaluation and policy staff and proceeded through a focus group where key participants were the investigators. This generated engagement by the investigators who were both keen to see their work be used to develop effective interventions and were familiar with the practice of research informing this from their own experience of intelligence-led policing. The output of that discussion was a key qualitative element in shaping up thinking towards developing a QCA protocol but this was reinforced by a follow-up survey of investigators and enforcement staff to develop the format for the QCA exploration. Co-production will always involve multiple methods, even if this is done in an implicit fashion. In this instance, the synthesis takes the form of the construction of the QCA interrogation although it is anticipated that the results of that will be interpreted in discussion with all involved in the co-production process (Proctor 2017)

Scenario making – looking to the future

The *Cambridge English Dictionary* definitions of the word 'scenario' include a description of possible events or actions in the future. Three elements in this are important. A scenario is something that is possible. The complexity frame of reference take on this is that we are dealing with states of a system which exist in the possibility space for that system. We describe them – we lay out what they will be using evidence of what is now, of drivers which will shape the possible future trajectories of what is now, and by noting the implications of and among those drivers including purposeful actions towards creating a specific system state among those which are possible. Bell's version of scenario sums it up:

> Scenarios are the representations of alternative futures (by which) analysts sketch a paradigm (an explicitly structured set of assumptions, definitions, typologies, conjectures, analyses, and questions) and then construct a number of explicitly alternative futures which might come into being under the stated conditions.
>
> (1967 865–66)

Note that by implication, following Juarro (2011), we have admitted teleology into our philosophical framing.

The future is not yet. We cannot access it by conventional abductive/retroductive reasoning which seeks to deploy evidence to explain how the current state of a system came to be as it is. We have to project. Of course, we do use abductive reasoning about the current state of the system of interest to inform our projections for the future, but they cannot be used to specify for complex systems in the sense that what will be is a simple development of what is. We do not have laws in the sense of laws of physics which will prescribe. Brewer, in an explicitly complexity informed discussion of the issues posed by climate change, puts it thus:

> There are no data about the future on which to rely. We are challenged to imagine many different and possible 'futures' as humankind seeks to exert its mastery and control. … No one can predict the future but we can invent and make the future.
>
> (2007 159, 160)

As O'Connor (1987) said in his classic discussion of 'the meaning of crisis', in moving beyond crisis, that state of a system which has to change, it is not a matter of what will happen but rather of what will be made to happen.

Scenario building is a conventional tool in policy development which draws on imagination of what might be informed by knowledge of what is and seems to identify a set of possible outcomes depending on the interaction of causal processes, usually described as drivers, outwith our control, with our own actions directed towards creating specific instances of the range of possible futures. This is almost invariably done on the basis of mixing methods and typically includes crucial stages of co-production in the generation and interpretation of the scenario set. Brewer explains why this is necessary:

> Obviously many disciplines and methods can contribute to the analysis of a problem. The problem, embodied in one's evolving appreciation of it, points out, perhaps demands, which disciplines and what methods should be brought to bear. Calling attention to multiple methods lessens a prevalent tendency to celebrate methodology at the expense of substance. Methods have blind spots that focus attention on highly selected aspects of a problem while blocking others. … One must counteract this by viewing problems with different methods or approaches and working to assemble their partial insights into something approximating a composite whole.
>
> (2007 163)

The UK Government Office of Science's *Futures Toolkit* (2017) provides an outline of methods of constructing scenarios as ways of imagining possible futures, although it identifies scenarios as just one style of doing that work alongside visioning and SWOT analyses. However, that useful document is intended to inform discrete policy interventions. In the 21st century, we are faced with the combination of increasing inequality, including pressure on the middle mass in the social order, and impending climate catastrophe. What is needed is a way of formulating scenarios of possible futures on a whole social order scale alongside a recognition that it is differential power which determines which will come to pass. As to methods it is a matter of all hands to the pumps – the extreme of mustering the crew to action in a foundering ship. Brewer, while arguing for all possible methods, gives a good account of the potential of simulation and gaming approaches as part of the repertoire. Dilaver (2015) argues for a synthesis of grounded theory and simulation approaches as a mixed methods strategy and this has clear potential as part of the scenario construction toolkit. Byrne (2021) drew on fictional scenarios from near future science fiction and argued for their value as an element in social science thinking about the future.

Scenario construction is a key part of climate science although most elaborated scenarios have been constructed on the basis of atmospheric physics and chemistry with biological responses taken account of to some extent. Physics and Chemistry models, even when simulations, can appear to generate predictive accuracy. To be fair those making such models always identify their limitations first in relation to biological ecology, although some attempt quantitative incorporation of that, and then with an honest recognition of the agentic character of social drivers towards the future. This is a particular problem in relation to the interaction of climate change and inequality. Economists' attempts to model this have taken the form of General Equilibrium Models at the macro level with all the limitations that equilbric and usually linear approaches imply. If we are going to understand how climate change and inequality will work out together, we need more than that. We need a serious understanding of how power works.

Power is not a given. Power can be created by coherent action. And scenarios have to be understood as a source of power. Allen (1997) noted that his models of development of the urban system would have an influence in shaping the way in which things were done in the future. They had causal powers. Foucault's language of discourse is relevant here but discourses are embedded in relation to material resources and other forms of capital in Bourdieu's sense. So scenario construction is more than a matter of expert technical work. In discussions there is always reference, entirely properly, to the need to engage stakeholders in co-production. We will conclude this chapter with a proposal for a mixed methods action research form of scenario development which draws on the history of confrontation with capitalism as a basis for addressing the reality of the capitalocene.

A proposal for a city region level of developing policy and practice towards confronting both climate crisis and inequality

In the *Futures Toolkit* (2017) there is an example of scenario construction organized around a scenario – Trading Places – which 'describes a future where economic power has shifted to the eastern economies and where markets and cultures are open to each other' (2017 88). The description of Britain[8] is remarkably optimistic:

> Many people thought that the UK was in deep trouble when the financial services sector relocated large chunks to Dublin, Paris and Frankfurt after the UK's 'passport' was revoked – but they were wrong. The last decade has seen the former UK replace financial services with environmental services, effectively swapping out one wealth creation process with another – albeit with different measures of success. 'Old' economy industries such as gaming and medical technologies remain strong in Britain, but the real opportunities are in circular economic development and environmental protection. It has been a fortunate transition – socially as well as economically – and one which the UK government would like to claim credit for. But they can't. Credit goes instead to the innovative partnership formed by environmental businesses, by the UK's young, talented and compassionate workforce and by an education system that has nurtured them and provided the skills they need to create a sustainable future. A partnership, of course, that reflects the new reality of Britain and who really runs it. Perhaps that is why people in Britain smile so much – because they have taken control and are now working hard to deliver what they value and care about. No-one seems that bothered that the economy is still flat rather than growing or that taxation is relatively high and people are less well off financially than a decade ago. Perhaps that's because anything is better than the drawn out and deep recession that cost Britain so much pain post Brexit. … Some wonder if cities are about to change fundamentally. Certainly, distributed networks and remote working mean that concentrated population is no longer absolutely required for success and, many would now say, has gone way past what is sustainable. Local communities are strong. Government reforms have led to decentralised decision making as far as possible. Public services still have some way to go to achieve the level of integration and efficiency that citizens demand, but the new crop of people coming into local politics have a high sense of responsibility and are making good progress.
>
> (2017 89)

All will be well, and all will be well, and all things will be well.[9] There are some interesting things in this optimistic outline of the future. We will present an outline of how action research, modelled on the development and expansion in a UK region including two city regions of something like WISERD – the Welsh Institute of Social and Economic Research – and even more closely integrated into governance, might proceed towards something like this desirable outcome.

Here Patrick Geddes specification of how to plan is worth remembering – Survey, Plan, Implement. Survey means to establish what is and for us that means not just the present pattern of land use, crucial though that is, but the whole structure of relevant interwoven complex systems of governance, services, economy, and people. We will discuss how this might be done in the area of the North East Local Economic Partnership (LEP) covering Greater Tyneside, the northern two-thirds of the North East Region of

England. Historically the region was the significant level for integrated planning, particularly during World War II, but there has been an administrative shift in England towards geographies which correspond with the city regions that Senior recommended in his Minority Report to the Redcliffe Maude Commission.

Greater Tyneside is a depressed and relatively poor post-industrial city region. It was one of the most industrial places in the world until the 1960s and before World War I was one of the most prosperous. It is now deindustrialized and is experiencing all 'the costs of industrial change'. An action research unit dealing with transition to a sustainable future would begin by collating all available documentary and statistical evidence describing the current condition of the city region and its trajectory towards that condition. They would reinforce this with oral historical testimony drawing both on the conventional stakeholder groups usually admitted to the political arena but also on the wider public in a mode exemplified by Warren's (2018) work on Teesside. It would necessary to make participation part of an assertion of the crucial significance of the exercise in confronting both climate crisis and inequality. One idea would be to establish a youth parliament with representatives elected from secondary schools and FE colleges to be one of the audiences for the results of the survey and to have a role in commenting on policy developments.

What is required in policy development is the preparation of an extended version of the Structure Plans of the 1970s which took the form of a 'written statement, illustrated diagrammatically, of the local planning authority's policies and main proposals for change on a large scale' (Ministry of Housing and Local Government, 1970, para 1.10). The key phrase is 'change on a large scale'.

> 'Structure Plans were to be essentially written statements, accompanied by a key diagram, that set out broad issues and policies concerning all aspects of the development of a local area over the next 15 years or so. The subjects to be covered by this integrated plan included population, employment, housing, shopping, leisure, transport and the environment. The scope of the plan was intended to include the relevant activities of all agencies in the area, including private companies, nationalised industries, central government and households, as well as those of the local authority itself. The general proposals in the plan had to be supported by analysis and reasoned justification, and had to take account of the likely future availability of resources.'
>
> (Barras and Broadbent 1979 1)

Then local government in England was a much more powerful state actor and there were institutional arrangements which reinforced the capacity for indicative if not directive planning, but the process was very much top-down and while there was no real controversy about structure plans, the lower local planning element often was controversial particularly in relation to the clearance of working-class housing.[10]

An action research programme organized around and in support of something like a Structure Plan directed towards the goals of increasing equality (and equity) and confronting climate crisis could draw on the whole repertoire of appropriate research procedures including all forms of modelling and co-produced simulations. A good starting point would be participatory systems mapping as outlined in the CECAN work by Barbrook-Johnson and Penn (2017, 2022 forthcoming) to establish just what components there are in the system and to focus on deploying resources in a particular direction.

All this would depend on a reconfiguration of power relations in and around governance but without something like it then things will go very badly. It requires a radical rethinking of the purpose of science towards action, something interestingly asserted as the role of operational research in the Structure Plan era. A lot would have to change but then this is a time of change and change is possible. In the end, it requires a very different politics. We can but hope.

Notes

1 Not just in social research, there is a very strong move towards mixed methods in ecological science – see the discussion in Lowe and Phillipson (2009).
2 Poth's *Innovation in Mixed Methods Research* (2018), which is subtitled 'A Practical Guide to Integrative Thinking with Complexity', is an explicitly complexity-informed practical guide in the applied tradition.
3 Harvey and Reed (1996) outline the ideological character of this use of the notion of 'culture of poverty' and the radical distortion it imposed on Oscar Lewis's original formulation of the term.
4 The CDP teams were pretty much the first to draw attention to this process as described in their joint publication *The Costs of Industrial Change* (1977).
5 This was not really done in the later 'Imagined Communities' examination of the area under the ESRC which albeit on a much smaller scale was conducted in the period 2012–7 as part of the ESRC/AHRC Connected Communities programme. That focused on the original CDP area itself in relation to change since the 1970s but did not set those changes in the context of changes within the wider locality/city region/national context.
6 Byrne is a native of South Shields from on his mother's side from a merchant navy family and aal together like the folks of Shields – there is a big estuarine river separating the towns but that was a mode of communication as much as a barrier.
7 In their valuable study of the experiences of natural scientists working with social scientists reported in this article Lowe et al. note that the natural scientists often found economists least open to moving across boundaries despite initial expectations that common quantitative programmes would make this easier than with those who work in qualitative ways. Experience seemed to be that the reverse was the case.
8 So perhaps Northern Ireland has been dumped on the Republic of Ireland and good luck with that.
9 With apologies to both Julian of Norwich and T.S. Elliot or the band played believe it if you like.
10 Something which was repeated in the 2000s in the appalling pathfinder programme.

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11 Modes of governance

The academic and grey literature engaging with governance deploying the language and perspectives of the complexity frame of reference has grown exponentially in the 21st century. The need to write of 'academic and grey literature' explains why this is so. The complexity frame of reference has been deployed in both theoretical writing and empirical investigation across the whole set of disciplines and fields of study – political science, public administration, social policy, health, housing and urban studies, *and* ecological policy – which have as their field of interest the public sphere defined as both the operations of government at all levels from local to global and the networks of other actors/interests which engage in governance as a process. At the same time, it has begun to inform the actual operations of governance at all levels. Complexity has entered into all aspects of governance thinking at the highest level of understanding, that is, in relation to what governance systems are and what they are for. It is beginning to inform management across the public sector, providing a radically different set of principles from new public sector management which is far better suited to confronting issues of inequality, pandemics, and climate crisis, and restores principles of democratic engagement and participation.

Crisis is everywhere in contemporary discussion of governance. Another word widely deployed in governance which requires consideration is 'resilience'. The *Oxford English Dictionary* defines this as 'The quality or fact of being able to recover quickly or easily from, or resist being affected by, a misfortune, shock, illness, etc.; robustness; adaptability'. What does it mean to think about the socio-economic–cultural–natural: systems (all those words are necessary) with which governance is concerned as 'resilient'? In clinical usage, recovery and restoration of health is one meaning but a great deal of clinical work now is not about restoration of previous health state but restoration of enough functionality for an elderly person to maintain a satisfactory way of living. That implies diminishment. It involves continued existence but in a changed condition. So resilience is an ambiguous term. It can mean systems being pushed back to what they were, which was the intention of governance at global and national levels in their interventions in response to the financial crash of 2008 but it can also mean the system continued to exist but in a transformed/changed/phase shifted form. In the literature these two meanings are seldom distinguished. In the context of oncoming climate crisis, in the era of the Capitalocene, it is the latter which should concern us. Özkaynak et al. explain why, deploying the complexity congruent notion of coevolution:

> The coevolutionary perspective recognises that all the different aspects of a holistic system are interdependent and evolve together; a change in one affects and brings about changes in all the others. … to assume that the existing form of modernity,

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> including the technology that coevolved with industrialisation, is the *only* possible form of modernity would be a mistake, since in the nineteenth and twentieth centuries carbon-based industrialisation also coevolved with a historically specific socio-economic system—capitalism. Although different varieties of capitalism currently exist, which place more or less emphasis on market regulation, particularly with regard to income and wealth distribution and environmental protection, experience so far indicates that all have a structural requirement for growth.
>
> (2012 1130)

We are in the Capitalocene and capitalism requires growth which in its present form is carbon deploying and climate crisis inducing. Things have to change but the question is 'change to what?'

The penetration of complexity thinking into disciplines and governance

Morçöl (2014) identifies three streams in the use of complexity in academic studies of governance, namely the identification of governance as a domain of practice, network studies, and complexity theory itself, arguing that they 'should be joined together under a *complex governance networks* theoretical framework to make better sense of policy and administrative processes' (2014 6). Governments, both in policy formation and implementation, have been obliged by the increasing complexity of the problems they face to engage with other actors in order to function effectively. Unlike Sowels (2021) he does not identify the ideological and interest-driven basis of this increasing complexity. In many areas, far from making governments more effective at dealing with issues, this increasing complexity has made the delivery of services far less efficient, notably in relation to the introduction of private providers into the UK NHSs.[1]

If there are multiple actors then there are connections among them through networks. Morçöl argues for a complex governance networks approach to political science. At the academic level this makes sense, but things have moved on very fast beyond this because of the way complexity has entered into the vocabulary and practices of governance itself. There are important suggestions in a disciplinary sense in Morçöl's review. He notes that a variety of methods have been developed to engage with complex governance and points towards the work of Ostrom (1990, 2005).[2] It is necessary to distinguish between networks in the sense deployed by Morçöl and the idea of nexus. The two are related but while network is used to define specific connections, the idea of nexus as deployed in governance describes the character of interwoven systems. The important complexity-informed research programme CECAN (Centre for Evaluating Complexity Across the Nexus) was concerned with policy and programme initiatives which operated in the interconnections among the food, energy, water, and environmental domains. The case studies in this programme included explorations of the future of farming, a data exploration exercise using QCA to review what is effective in confronting environmental crime, working with the UK Food Standards Agency on regulations for the future, exploration of energy initiative programmes including renewable heat programmes, and work on bovine tuberculosis. What characterized all of these was that the work involved a mix of academics, consultants, and civil servants from the analytical professions. This for us is the way things should be done although there are problems. Kirsop-Taylor and Hejnowicz

(2020) reviewed the experience of *Natural Resources Wales* – an example of a multi-scalar, hybrid, adhocratic organization designed to meet nexus challenges in the interwoven domains of water, energy, and food. For them:

> The 'nexus' is not just a physical description of a 'system'; however, but also a critical conceptual approach (or lens) to appraise challenges and problems from an integrated and holistic perspective. … As such, the nexus represents a suite of multi-layered biophysical, socio-economic and governance systems and ways of thinking that demonstrate core properties of complex systems such as non-linear dynamics, feedbacks, emergence, self-organization and uncertainty.
>
> (2020 2)

They state: 'public agencies are not naturally disposed towards the interdisciplinarity and complexity posed by nexus thinking' (2020 3).

Johnson et al. (2017) edited a series of essays titled *Non-Equilibrium Social Science and Policy*. The clue is in the phrase 'non-equilibrium'. The emphasis on solutions based on equilibria is wholly wrong when dealing with complex systems which in their very nature are non-equilibric and where trajectories do not lead towards equilibria. Herbert Simon made important arguments in this direction but his radical and appropriate understanding was distorted in academic economics to fit with the neoclassical/neoliberal hegemonic frame. Rosewell in a chapter on *Complexity Science and the Art of Policy Making* works through examples of her own practice showing how difficult it was to get policy decisions made which broke with traditional methods, particularly in relation to transport planning and economic development. Her emphasis is on narrative as a basis for developing the art of policymaking:

> Many years in practical policy making show that it is part science and part art. Complex systems science, as a form of story telling, can create links between them.
>
> (Rosewell 2017 159)

We agree!

The issue of appropriate methods of research for investigating complex governance is considered by Teisman and Gerrits (2014).[3] A key starting point for their review is expressed thus:

> We follow Byrne and take his concept of 'situated complexity' as our point of departure in understanding social reality. Byrne suggests that social reality is a compound of generic or recurring patterns in conjunction with idiosyncratic events, i.e. events that are local in place and temporal in time. As such, social reality is not exclusively governed by universal laws, but each event is not unique either. Causal patterns are comparable across cases, yet at the same time, they differ with unique elements generated by unique contexts or circumstances.[4]
>
> (2014)

This leads them to a case-based focus in developing a complexity-informed research programme for appropriate disciplines and fields. They discuss three complexity congruent approaches: Qualitative Comparative Analysis, Dynamic Network Analysis, and Group Model Building. These are approaches which intrinsically recognize the complexity of the systems they are dealing with and work in relation to it. There is often overlap with other approaches. Group model building has lots of elements in common with dynamic system mapping and with co-production approaches to the development of simulations. Dynamic Network Analysis has many common features with systems engineering approaches. QCA is a refinement of the comparative method which has always been a crucial social scientific tool in moving beyond the ideographic account of the individual case in search of causation and is particularly well suited to addressing equifinality and complex causal sets.

Marks and Gerrits (2017) make explicit reference to Schumpeter on creative destruction in relation to innovation. In the era of the Capitalocene is innovation as creative destruction really creative? Some is for example in the development of technologies of alternative energy production and transport. Others are not – for example, the enormous energy consumption by bot farms engaged in the making of bitcoins. There is good innovation – vaccines against COVID are a current example, and there is bad innovation – bitcoin generators; it depends on what the innovation is. In this last enormous category, it is very much a matter of what policies are adopted to resolve the social issues created by the creative destruction.

One valuable contribution here is Gerrits and Marks's *Understanding Collective Decision Making* (2017). This is case study informed – 'the dream of high speed rail in the Netherlands'; the effects of the Gotthard Base Tunnel on the Gotthard region of Switzerland; Sports in the City of Rotterdam as part of a revitalization programme for South Rotterdam; and airport and airport accessing transport development in Bangkok. The complexity frame of reference is central to the book's arguments and demonstration. There is an innovative use of the idea of fitness landscape and a central focus on time and the dynamics of decision-making in complex systems. Decision-making is crucial to governance and this original and fascinating work explores it through the lens of complexity. The studies are retrospective but they are valuable as a basis for informing the actions of decision-making in governance. In this respect, they stand firmly in the case study tradition in social science in general and organization theory in particular but set that approach firmly in complexity terms.

Morçöl's constructive critique of Ostrom demonstrates the value of her work on self-organization while raising concerns about the role of Rational Action Theory in her approach. Her work is interesting and has a political role in relation to what Marks and Gerrits (2017) note about the resonance between complexity thinking and much of the work in heterodox economics. One writer who stands on the bridge between the complexity frame of reference and heterodox, especially institutional, economics is Room (2011, 2016). An excellent summary discussion of a complexity take informed by institutional economics is provided by Room's commentary on the work of Dopfer and Potts (2008). Dopfer and Potts are concerned, to quote the title of their book, with *The General Theory of Economic Evolution*. This is an explicitly Darwinian take where technological innovations stand as an equivalent to genetic mutation as a source of transformation. Room develops his critique by passing their argument through the lens of Kauffman's NK(C) model which deals with:

> the adaptive interaction of diverse phenotypes within a shared ecosystem, modifying and warping their respective fitness landscapes. For Dopfer and Potts, this shared eco-system is the macro-terrain of the economy as a whole. … This is where multiple meso-trajectories of origination, adoption and retention interact, abrade and

> co-evolve. At this macro-level, co-evolutionary dynamics 'warp' the fitness landscape, not least for the first mover.
>
> (2011 55–6)

Potts states that 'accounts of economies as complex systems have provide a plausible new [i.e., not neoclassical] basis for the political creed of economic liberalism. Claims abound that "free markets are clearly self-organizing systems"' (Room 2011). Perhaps the most articulate summary of this position is that of Von Hayek in his Nobel Prize in Economics lecture. Room is bluntly dismissive of this turn and his discussion both of a Keynesian tradition which argued that without the state 'continually tuning the system parameters' (2011 63) through a range of measures including direct intervention in the system, crises were inevitable, and of recent events which demonstrate the system destructive effects of flocking behaviour in financial markets provides a convincing refutation of it.

> 'The complexity of the problem is the thing that first hit me': cabinet secretary Simon Case on Covid, his return to No.10, and working with Dominic Cummings.
>
> (Case 2021)

There are two interesting things in this comment by currently (June 2021) the most senior of the UK's civil servants. The first is that he identified COVID as a complex problem. The other is the reference to Dominic Cummings, who while not the first to try to set complexity at the heart of UK (or at any rate English) government did assert that it was complexity which confronted government, and by implication governance, and that the most important skill set in confronting it is synthesis. Cummings's ideas are expressed in a long running blog. Their origins in Hayek are evident. At the same time they are informed solely by a crude version of scientistic complexity and a belief that mathematics is really the language of reality itself.

Cooke and Muir (2012) argued for a relational take on governance, i.e., placing human relationships at the centre of thinking and practice. Stears extended this approach arguing against the way in which outcomes became the measurement of achievement in new public management (NPM) and part of an elitist and undemocratic approach to politics as a whole. There needs to be a change in the forms of politics and the distribution of power. Relationships are central and should not be a means to other ends.

> This critique suggests a politics that does not aim towards a known, identifiable end state. It rejects utopianism and embraces uncertainty. Taken to its logical conclusion, such an approach would be a radical departure for the centre-left, which has long defined itself in terms of a 'vision' for how society should be (in words such as 'equality'). In its purest form, Stears' argument is a call to abandon the pursuit of objective outcomes with politics coming instead to focus on the design of processes – especially ones that enable relationships. The specific 'ends' that people make of these 'means' – both individually and collectively – is then a matter for their own determination. This offers citizens, he argues, the prospect of both liberty and responsibility.
>
> (Cooke and Muir 2012 12)

This dismissal of politics towards ends is a common trope in many discussions of politics addressed in complexity terms. It is inane in a context of impending climate crisis where the survival of human civilization and even of humanity is at risk and where a central objective of governance should be doing something to confront this issue. However, while the dismissal of ends is absurd, the discussion of how governance should be done is not. An author who has developed this theme with specific reference to complexity is Chandler (2014a, b). He starts from a concern with resilience, for him an interactive process of relational adaptation. If the notion of relations is very broadly considered to include not only interpersonal but inter-institutional relations and the relations of the social and the natural, this is fair enough. However, it can be taken too far. To say:

> a complex problem such as that of social and economic equality, cannot be addressed through macro-level government economic and social policy, attempting to steer or direct the economy. Rather such a problem can only be tackled by working on the inner mechanisms and social dynamics which are seen to reproduce poverty and exclusion [We don't think he means capitalism] … governments cannot effectively 'touch' the problem through top down policy mechanisms but need to understand the real processes at work at the level of society and the individual to enable these inner mechanisms.
>
> (2014 9)

And proceeds to an assertion of the need to reveal mechanisms at the level of individual government of the self with: 'the relationship between the government and society and between the subject and the external world [to be] reconceived in a post-modern or more relational ontology' (2014 9), then a note not so much of caution as of 'what the ****' is appropriate.

In the fortunate third quarter of the 20th century in high-income countries, the era of what regime theory calls Fordism, it was precisely macroeconomic and legislative top-down policy mechanisms in relation to full employment, trade union rights, progressive taxation, and tax funded services which promoted the highest degree of social equality ever in human history since the agricultural revolution. Stears and Chandler seem to have, at very least implicitly, taken up the micro-emergent style of complexity thinking, and therefore move towards an emphasis on actors operating as discrete agents. For us all levels of governance and in particular institutions are necessary parts of the governance towards objectives. There is an absolute necessity for a radical redistribution of power in the direction of institutions, publics, and all levels of governance but we reject the assertion that ends remain unknown. Jessop has developed the idea of 'meta-governance', most recently deployed in his book *Putting Civil Society in Its Place: Governance, Meta-governance and Subjectivity* (2022). This makes a fairly superficial reference to complexity and complexity theory and draws for empirical illustration on various civil society focused research projects carried out by WISERD – the Welsh Institute of Social and Economic Research and Data.[5] The term is a useful on in so far as it describes the way in which higher governance elements, and particularly nation-states or federal/devolved states, have oversight and regulatory powers in relation to levels of governance below them and we certainly need to pay attention to civil society as a political domain, remembering always that it exists framed by the limitations of a capitalist social order.[6]

Getting it done – how institutions, structures and processes of governance should work in a complex world which is already in crisis

Whatever is the right way for governance to go in the current era of crisis, it is not the way of new public sector management and the audit culture (Strathearn 2000).[7] What follows draws heavily upon Haynes (2015) and his building on a tradition of the application of complexity thinking to management in general developed by Stacey (2000, 2007, 2010). By the 1990s, the problems of attempts under the rubric of NPM to manage by command, control, audit, and targets were not working in terms either of the general sensibility of public sector workers or the actual achievement of meaningful outcomes as opposed to box ticking in relation to procedures followed.

We need to set the complexity frame of reference in relation to governance in the wider context of how it has been deployed in organization theory and general management literature. Thrift identified what he called 'the cultural circuit of capitalism' (1999 42) as one of the three networks through which the metaphorical apparatus of complexity was being deployed in the production and dissemination of knowledge.[8] His description of this circuit is useful in thinking about why and how complexity is deployed in the management/business literature. There are two interrelated sets here. The first is work which emerges from the academy's engagement with business/management, an engagement embedded in 'business schools'.

Thrift pointed out systems theories with their engineering connections, the engineer being the technologist who worked with theoretically related and empirically validated knowledge in relation to projects and processes, are related to complexity and thereby much of the language was, we might say, 'comfortable' for the business academy. The deployment of information technology across business also aided the acceptance of a complexity frame of reference. IT draws on systems conceptions in engineering and on communication theory, both of which have informed complexity thinking. We can go beyond Thrift who does not, at least in that article, deploy a realist frame of reference, and note that management is an applied field in which both academics and practitioners are confronted with messy social reality itself. The turn to complexity was by no means merely linguistic/conceptual. A realist ontology allows us to note that reality itself had a voice and that one reason for the development of the complexity frame of reference was precisely that isomorphism with the world in which management was obliged to work. This gelled with the necessarily interdisciplinary character of the business education curriculum and the knowledge production undertaken by business academics.

The second 'business' set described by Thrift, that of the business gurus who sell their worldview to managers through books and events, does intersect with the business economy and even with his 'New Age' network, but again one of the reasons we can identify for the utility of complexity as a language for that sort of practice is exactly that the frame of reference resonates with the lived experience of the audience for it. Likewise the growth of management consultancy, which intersects both with the business academy and with the world of the management guru, provides another set of practices in which complexity can be deployed. Historically management consultancy tended to assert a black box account of processes and to emphasize only outcomes. New public management is absolutely a derivative of that practice mode. However, we find much more engagement now of consultancy with complexity and consultants who want to delve into the black box and see what is happening there in relation to outcomes from processes and practices.

There are two good collections deploying complexity in relation to management, with an overlapping set of authors: Mitleton-Kelly (ed) (2003) and Allen et al. (eds) (2011). Both have sections which explicate the ideas, sections which deal with the philosophical/epistemological issues involved in deploying them in relation to management, and sections which apply the perspectives. The Mitleton-Kelly connection distinguishes between complexity perspectives on organizational processes and the implications of complexity theory for management processes – we might say, between how academics see complexity in relation to organizations and how managers might deploy complexity in organizations. Allen et al. have a slightly different take distinguishing between 'complexity and organising' and 'complexity and managing' although the latter set corresponds closely to Mitleton-Kelly's set dealing with management processes. Usefully Allen et al. also include a section on 'interfaces' – that is, a set of chapters confronting the business/management take on complexity with takes from related fields, especially economics. Although it has a wider field of reference, Boulton et al. (2015) who argue for 'embracing complexity to provide strategic perspectives in an age of turbulence' sits within this approach.

One important writer in relation to the application of complexity ideas to the fields of organization and management is Stacey (2005, 2007, 2010). Stacey's work is coloured by an engagement with the set of social theory formulations reviewed in Part II of this book and particularly with the ideas of Elias and Bourdieu. However, he is also reacting against the dominant command and control/scientific management agenda which formed the classic curriculum of business schools since their development in the pre-war United States. The subtitle of his book *Complexity and Organizational Reality* (2010) is 'Uncertainty and the need to rethink management after the collapse of investment capitalism'. In it he challenges the notion that managers are able to make strategic choices and that the culture of maximizing shareholder value[9] has had massively deleterious effects on the social systems dominated by it. His review of the impact of mergers and acquisitions is particularly persuasive here.

There is a developing literature in which complexity approaches have been deployed in relation to empirical projects in the field of business and management. Such studies really address the questions: can the process(es) I am examining be understood and interpreted better if I think about them as complex adaptive systems (CASs) and does thinking about them in that way help me to understand their nature and inform managers and other decision-makers how to act effectively in relation to them? A key word used in these studies is 'examining'. The researchers have looked at their cases, sometimes comparatively although not deploying methods like QCA which would enable systematic causal comparison, and constructed narratives of them. There is relatively little work which uses quantitative methods in a quasi-traditional way, although there is some which uses simulation approaches, for example, Nair et al. (2010), which examined supply networks as CASs using cellular automata and a game theory framing. The simulation work stands on one side since it attempts to develop a predictive account. There is some work which deploys a complexity frame of reference in relation to time series of data. Derbyshire and Haywood's (2010) examination of the UK regional 'enterprise gap' is a good example with considerable policy implications. One book-length contribution which makes very good use of complexity is Gilpin and Murphy's (2008), where they bring the ideas out and use them by passing a series of case studies as it were through the complexity thinking mode of work. They use the term 'complexity based thinking' which is an exact synonym for 'the complexity frame of reference'.

Stacey made the point that in organizations which had to establish network relationships across not only their internal lines of command and control but also with other organizations and for the public sector, as Haynes emphasizes, with many elements of civil society, a different form of management based on negotiation and bargaining was required. Command and control did not work. Devine (1988) shows there was a very considerable degree of this kind of relational management on the Home Front in the UK during World War II. Byrne (2021) argues that that experience can serve as a model for the kind of governance necessary to confront inequality in a context of developing climate crisis. People have been thinking with and acting with the complexity frame of reference without deploying the term. That is to say, complexity is implicit in many of the emerging ways of thinking about management and governance, particularly when the focus is on whole systems and proper attempts are made to understand what the whole system is and what other systems of significance intersect with it. Haynes notes the significance of communication in relation to managing in these terms: 'Complexity theory implies that human communication (as the primary forum of system interaction) is at the core of understanding and managing a complex public service system' (2015 xv). We agree and agree that it is important to address the way IT systems operate in relation to the delivery of information and as means of communication. At the same time inter-personal direct communication matters.

Haynes draws on Luhmann's approach to the definition of subsystems in the social world in terms of how:

> specialist communication and language define these subsystems. While this gives them (and empowers) their specialist function, it also serves to separate them from the rest of society.
>
> (2015 29)

He picks up on Luhmann's arguments about structural coupling as the way in which systems are connected or not by communication.[10] This is an important insight although as Haynes points out (2015 30), Luhmann does not recognize the importance of micro action direct communication, particularly among elites, as took place in relation to the 2008 financial crisis across the political and financial subsystems. The same is true at many levels of governance and management.

Organization theory as applied to management has tended to emphasize the importance of the strategic level of management. Boulton et al. (2015 139) make the far from trivial point that while strategy is about the future all that seems to be to hand for analysis is the past. However, we do have some degree of foresight based on a reflection on past trajectories of systems in relation to their context understood as all the systems with which they are interwoven. The familiar SWOT (strengths, weaknesses, opportunities, threats) approach is a simple version of context setting in relation to both the organization itself and its wider context. The PESTE approach which seeks to identify trends in the political, economic, social, technological, and environmental domains, is focused on interwoven systems which constitute the broad context in which any organization operates. Those who follow Hayek in asserting that the future of complex systems is unknowable[11] turn markets based on the multiple tacit knowledge of entrepreneurs. But the reality is that future*s* – note the emphasized plural – are knowable in terms of the multiple options available in the possibility space and it is actions taken which will bring one of those futures about. Haynes (2015) offers an outstanding complexity-informed account of how public services should be managed in relation to their wider context and all the (dreadful word) stakeholders in that context. What about how futures should be created? For us in the current context of impending climate crisis, public sector planning for the creation of futures is the key issue.

The 1970s saw a radical reorganization of English local government, not only in spatial terms after Redcliffe-Maude (Senior 1969) but also organizationally on the basis of the Baines (1972) report which led to corporate organization in local authorities. This was meant to develop a power centre in LAs which could manage the system as a whole through a powerful Policy and Resources Committee and thereby overcome the problems posed by service-specific departmental silos. LA chief executives in association with directors of finance acquired locally the powers exercised in central government by the Treasury, although they often seemed to be better informed as to the purpose of what they were doing than the Treasury which seems not to think beyond resource allocation and placating finance capital. This explains why the UK Cabinet Office has taken on much of the strategic role and advisors have acquired so much power. There was a simultaneous change in planning practice associated at this time. Although the spatial reorganization failed to adopt the proposal of Senior's memorandum of dissent,[12] there remained a commitment to planning, not just in terms of land use, although that was a key feature because land use specification is a real power for shaping futures, but more broadly in structure planning. The 1970s were an era when with Wilson as prime minister[13] planning was in vogue. The Department of Economic Affairs never managed to overcome Treasury opposition and was abandoned but indicative planning on the French model was meant to be embedded in national, regional, and local plans. Community disorganization imposed from above despite the call by the Skeffington Report for participation played a part in discrediting planning as a mode of governance but planning is what is needed now.

A rhetoric of participation is prevalent across governance, although the logic of the new public management approach is deeply hostile to it in realpolitik terms. Participation has been an issue for planners since the 1960s, particularly in the aftermath of a series of planning disasters which resulted from attempts to impose a linear logic of control on complex socio-spatial systems. Skeffington (1969) even used a language of complexity to argue for the engagement of the wider public in planning processes. We can see a range of issues here. Participation is a mode of legitimation, although when it merely takes the form of consultation[14] it can have delegitimizing consequences. Skeffington in many ways prefigured Bang's (2002, 2004) conception of the need for cultural governance by noting that participation was necessary as a way of generating the information that planners needed in order to plan effectively. So participation is a source of information, information yielded by other actors in complex urban systems whose actions or potential actions have consequences for those systems and who possess information about those systems not available to the planning system personnel themselves. This resonates with Devine's (1988) call for democratic economic planning to incorporate the tacit knowledge of the public citizenry in general in contrast with Hayek's reliance solely on the tacit knowledge of entrepreneurs.

TIAA – Not TINA – there is an alternative

Haynes references Walker's (1984) argument for social planning as a strategy for socialist welfare. Walker was addressing how to organize the welfare state in all its aspects at the level of localities in order to redress the problems which arose from top-down management. There is still a lot of be learned from this book but the issues facing us after more than forty years of neoliberalism and in a context of impending climate crisis are different in quality because inequality has changed so much and climate crisis is an issue on the scale of a world war. The good thing about COVID-19 is that it demonstrated the capacity of governments and governance to act, even if the actions were often inept, interest serving,[15] and reliant on national strategies when the key levels for action were local.

There is an extensive literature on the difficulties of managing complex socio-ecological systems (SESs). Zia et al. (2014) argue for moving from the habit of control to institutional enablement deploying in particular the theory of Generalized Autocatalytic Sets (GACS):

> Within the GACS conceptualization, SESs cannot be 'optimally' managed due to the lack of determinism in predicting the evolution of their phase-spaces; instead, the policy and governance interventions can at best 'enable' the policymakers in SESs to nudge the system towards socially desirable, yet ecologically feasible phase-spaces, which in turn are continually revamped as new elements emerge in phase-spaces.
>
> (2014 82)

Although this is an explicitly complexity-informed approach, we are not convinced that 'tweaks in institutional rules' (2014 85) will actually have an autocatalytic effect of anything like sufficient magnitude to confront the issues that Byrne (2021) identifies in relation to necessary political action in the future. COVID-19 was an immediate crisis and provoked quick reactions. Climate crisis is slow and developing process although likely to have profound and dramatic effects at tipping points but then disasters will already have happened. Underdal (2010) identified the challenges facing long-term environmental governance in general terms but his specification can be applied to climate crisis itself:

> Some important processes of environmental change – including those of climate change and loss of biodiversity – share three characteristics that make them extremely demanding challenges of governance. First, time-lags between human action and environmental effect are very long, often extending beyond one human generation. Second, problems are embedded in highly complex systems that are not well understood. Third, these problems involve global collective goods of a type that links them to a wide range of human activities and leaves them beyond the scope of unilateral solutions. Social science research offers two essentially different models of collective response to severe challenges. One portrays effective response as collective action through central leadership and contraction of power. The other conceives of societal response as involving a variety of local activities undertaken by subunits of a complex and decentralised system. I argue that both models have considerable merit, but also that they respond to different types of challenges. Therefore, useful insights can be gained by specifying more precisely the circumstances under which each model applies.
>
> (2010 386)

We agree with all of this except the last sentence. For us it is not a matter of choosing between the central and the local but devising a system of governance in which they inter-relate with each other. Those like Colander and Kupers (2014) who assert solutions will be primarily bottom up are wrong. Scoones and Stirling's *The Politics of Uncertainty* (2020) arguments are similarly bottom up focused albeit with an emphasis on democratic rather than market solutions. They take an epistemological turn arguing that:

> uncertainties must be understood as complex constructions of knowledge, materiality, experience, embodiment and practice. What burgeoning uncertainties require lies less in escalating efforts at control, but more in a new – more collective, mutualistic

> and convivial – politics of responsibility and care. If hopes of much- needed progressive transformation are to be realised, then currently blinkered understandings of uncertainty need to be met with renewed democratic struggle.
>
> (2020 blurb)

Democratic struggle is required at all levels but the existing political structures impose a path dependency on what is possible. Byrne (2021) drew on Devine's (1988) discussion of the modes of organization of the UK in economic and social terms on the Home Front during World War II as providing a model of engagement across governance and civil society for this kind of process. Despite the use of the word 'complexity' in its subtitle, Crowley et al. (2021) do not really engage with complexity beyond some outlining of the frame. However, their emphasis on the significance of states and government as well as wider governance is entirely appropriate. There is much to agree with in Howelett's statement in his preface to this book when he reacts strongly against what he calls 'the governance turn' in contemporary policy literature:

> This turn is blamed (by Crowley et al.) for having led to an epistemological and practice oriented dead-end in policy studies by replacing more traditional and time-honoured ways of thinking about the state with abstract ideas about the merits of enhanced participatory processes. By continually promoting the advantages of collaboration and co-production over state-based goods and service delivery, this approach has de-politicised many aspects of contemporary policy studies and contributed as well to the failure of governments meeting many current policy challenges.
>
> (2021 ix)

For us participation has been monopolized in real terms by business and corporate elites and we are all for participation by the wider citizenry but states do have to act and politics matters. Lieven (2020), a former professor of war studies, states that it is nation-states that must act to avert climate catastrophe as that is the real threat facing them, not each other. What is preventing meaningful action against climate catastrophe is 'the lack of motivation and mobilization of elites all over the world and of voters in the West' (2020: xiv). Haynes locates the problem of the gap between conventional democratic politics in normal circumstances and what is required now:

> politicians are cautious in a pluralist democracy and reluctantly to suddenly change the fundamental direction of policy. Complexity theory questions the stable notion of this incremental adjustment and risk avoidance. … strategists and planners also need to be prepared for what happens in periods of sudden and unexpected change and instability.
>
> (2015 52)

Climate crisis is not unexpected but this holds for phase shift expected change even when that expected change has been treated like Ho's (2018) Black Elephant.[16] Politics has been ratcheted in a neo-liberal direction across democracies – China is another and vastly important but very different matter – since the 1970s. We have to knock the pawl out of that ratchet and do things in a very different way. In the UK we are fortunate to have a model which still has considerable cultural salience and can serve as a political basis for action.

Notes

1 There are examples of this in many areas including ultra-expensive PFI deals – one hospital for the price of three, mishandled design and build construction projects where the abandonment of public sector design and construction management expertise has led to cost overruns, defective construction and serious delays in delivery, the separation of housekeeping and clinical services as an aspect of PFI, and IT contracts which simply do not work. The last is general across UK public sector procurement.
2 Although Ostrom's work is founded in a rational action framework and however attractive this may be to economists, Mouzelis (1995) outlined how its analytical elegance does not stand up well in relation to complex social orders.
3 While we very much agree with their discussion of complex causality in this piece, we cannot accept the reality of theory transfer for the reasons outlined in Chapter 1. To be clear we think that scientific complexity after Prigogine served as a reminder of the complexity theme in post-Darwinian science as a whole but the deployment of the complexity frame of reference is not theory transfer.
4 This is certainly what David Byrne thinks but it is more elegantly expressed by Teisman and Gerrits than by him.
5 Although the empirical illustration consists of little more than descriptions of work programmes in these projects as designed without much in the way of interpretation of what came out of them.
6 Jessop remains a fan of Althusser and Poulantzas in addressing this domain. Thompson's magisterial denunciation of *The Poverty of Theory* should have put an end to that tradition but it survives alongside of course new variants of intellectual sterility.
7 Strathearn has also written on complexity.
8 The others were science itself and 'new ageism' which intersected with the business-/management-related cultural circuit of capitalism.
9 Although often this has in reality been a culture of maximizing monetary rewards of senior managers at the expense inter alia of shareholders!
10 This has considerable force and anyone with experience, like Byrne, of working in a project like the North Tyneside Community Development Project, which was specifically directed at disturbance in local government focus by operating in a cross-cutting mid-level fashion outside traditional departmental structures, will recognize the reality of mis- or non-communication.
11 Market-oriented economists and post-modernists share this pessimistic belief but the free marketeers present a solution to its implications, wrong though it is, whereas post-modernists seem to have nothing to offer but introverted epistemological maundering and existential dread.
12 Senior recommended City Regions located within provinces on a regional scale and containing second-tier authorities focusing on service delivery, alongside a small-scale community level. This excellent framework was not adopted.
13 Harold Wilson was an outstanding economist in the Keynesian tradition and had extensive experience as a war-time civil servant when he devised much of the statistical information system for governance. He knew what he was about and his early onset Alzheimer's was not just a personal but also a political tragedy.
14 Defined by Ambrose Bierce in his *Devils's Dictionary* as getting people to disagree with a decision that has already been made.
15 However strange Wolff's response to COVID measures has been, her term 'Disaster Capitalism' which refers to the ability of corporations to make profits from disasters is accurate in relation to much of the policy and procurement response in the UK and USA.
16 This is a combination of the elephant in the room and the black swan – we all know it is there, don't talk about it, and when it trumpets, say 'Oh, that was a black swan'.

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12 Evaluation in a complexity frame

The literature on complexity in relation to evaluation is relatively recent, yet the number of studies that deploy complexity theory ideas has grown rapidly. Since Sanderson (2000, 2003) focused on the significance of complexity for evaluation, there have developed organizations to provide guidance on strategies, approaches, methods, and tools for evaluating complex interventions. These include the American Evaluation Association's Systems in Evaluation Interest Group, 3ie (International Initiative for Impact Evaluation), Australasia Evaluation Society, and CECAN in the UK. The Magenta Book (2020) now includes UK government guidance on evaluating complex interventions. The literature reflects the search by academics and practitioners for better ways of understanding and describing complex interactions in practice.

In complexity science we assert the significance of context. Then a first point in examining evaluation must be to acknowledge, perhaps more than in any other area of social science, that the role and practice of evaluators is heavily circumscribed by the limits and expectations placed upon them by funders as well as by community stakeholders in a programme. For many commissioners evaluation is a practical piece of work undertaken with a set of political objectives, particularly since the advent of explicit 'evidence-based policymaking'. Evaluation has commonly been a retrospective exercise undertaken at a defined 'end' point and there has been little, if any, evidence of how it has informed change. Networked governance and devolution implies considerable variation from the idealized policy cycle that evaluation, in theory, informs (Hallsworth 2011). Downe et al. (2012) concluded this means commissioning should involve groups of studies across policy areas. Combined with the lack of transparency, accountability issues multiply with interaction between policies bringing further problems for commissioning and use of evaluation studies. The shift towards distributed governance has brought pressure to work across agency boundaries creating a fertile ground for mixed method approaches and appreciation of complexity (Walton 2016b). However, Walton identifies significant barriers to complexity evaluations in terms of resources, professional capacity, and existing dominant methods 'crowding out new theoretical perspectives and methods' (2016a, 416). The form of the traditional evaluation report, loss of context created by executive summaries, and word limits are challenges to complexity-informed evaluation.

The evaluation profession has historically been perceived as an atheoretical, technical arm, separate from academic social science. Most evaluation has been undertaken outside of the academia generating 'grey' literature in which structural context is treated as given and action is seen in voluntaristic terms. The role of monitoring has been poorly theorized, much of it being done 'in house' in small initiatives yet measures deployed in

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monitoring shape evaluation. This matters because it gives us some insight into the nature of the field and the issues that complexity framing faces in penetrating it.

Evaluation actors vary widely including private sector consultants, NGOs, in-house work, and evaluations undertaken in academic settings. In teaching evaluation to postgraduate students we used examples of evaluation reports conducted by private sector companies that failed to define the intervention and used cause–effect attributions of the post hoc ergo propter hoc type. The *Journal of Evaluation* (2021) special edition's focus on methods aimed to expand knowledge and use of complexity methods but there remains a persistent implicit positivism in much evaluation. Evaluators still need to start to engage with an explicit understanding of the relationships among ontology, epistemology, and methods in their work.

Professional evaluators are required to give account of programmes and initiatives that involve interaction between contexts and the actions of multiple stakeholders yet positivist accounts have dominated. There can be no more crucial place to be able to create evidence that is widely accepted and valued than in the application of social science to solving social problems in a direct and practical form. While it has to be acknowledged that such knowledge has less significance in policymaking than 'evidence-based policy' would suggest (Boaz et al. 2008; Sanderson 2009; Pawson 2006), there is now a general recognition of the need to find out what works and how it works. Discontent with the dominance of positivism led many evaluators to turn to a social constructionist frame. This approach recognized the importance of action and context and the need to understand more than can be grasped by simple measures of outcomes as with the products of RCTs. All interventions however simple are interventions in a complex world. So the complexity frame of reference is much more than just another approach. It is fundamental in framing the very field of interest, not just of programmes themselves but also the ways of thinking of those implementing and those receiving them. No programme is inserted into an otherwise static environment so we have to be cautious in using simple/complicated/complex distinctions in relation to evaluation (Rogers 2008). That distinction can be used to imply that complexity theory is relevant to only some interventions rather than providing the frame for all. We are currently familiar with outcome measures of a vaccination programme for those vaccinated in relation to disease progression but as we have seen in the pandemic, the uptake of vaccination is an explicitly complex issue. Moreover, it is important to make the distinction between output and outcome which may reflect the divide between simple and complex. Our concern is that this categorization of differing types remains intervention focused (Moore et al. 2019), neglecting interaction with other interventions and their wider context.

Room (2011) puts it succinctly in discussing Pawson's approach to realist evaluation:

> Policy interventions are not isolated: they are launched into a crowded world. Nor are these forerunners merely the detritus of policy enthusiasms long forgotten; in many cases their champions are still at work, seeking to broaden their scope and colonise the landscape onto which any new policy is launched. Nor is it just a matter of what combinations of policies co-habit a given landscape. What can also matter is the order in which they are introduced. Sequence and timing are important: change them, and the ways in which they shape each other, affecting the impact of each, will also change. There is, in other words, contingency in all of this. Not only must the evidence for a given policy intervention consider how it will work in combination with other policies; it must also differentiate according to the order in which these

> policies are introduced, in the various contexts which characterise the policy maker's domain, and having regard to the different time scales of their likely effects.
> (2011 83)

Room argues rather for an evolutionary form of realism that looks for generative mechanisms in the 'transformative synergies that develop among these interventions and their stakeholders and by which they co-evolve' (2011 83).

For complex systems any intervention can bring change to the whole system, either as minor perturbation or major phase shift. It involves interaction across but also within a nested hierarchy. The operation of specific interventions within a nested social order in post-industrial capitalism requires an understanding of the impact of neoliberal hegemony at the level of governance. The move toward complexity in evaluation poses questions of both evidence and ideology which become intertwined in the hegemonic status of the 'old' science. The claim by Blair's government that 'what matters is what works' sought to leave ideological and value issues behind and failed yet claims for ideology-free grounding continue (Sanderson 2009). The pretence of direct correspondence of intervention with simple measurable outcomes is based on the hidden nature of values in much modernist explanation. Science, allowed a narrative of objectivity despite evidence of its social construction, is overlain by ideology that emphasizes individualized approaches and explanations as opposed to communal/environmental/structural accounts. At a structural level every programme intended to address specific social problems sits within a much broader context of structured social inequalities. Unemployment initiatives have regularly sought to 'upskill' individuals in a competitive frame while ignoring the wider context of the labour market despite the fact that contextual labour market factors are likely to have a much more substantial effect on their outcomes (Byrne 2011). With agency in the frame we see the significance of actors' orientations and values in shaping implementation and take-up. The need to look at interaction at systems, organizational, and interpersonal levels in governance is clear and these relationships are exhibited in participation and involvement. Carried out within the bounds of intervention logic, and intensified by new public management, evaluation becomes a prop to hegemony rather than a critical voice. We can see, particularly in the field of health, an individualized conception of intervention emerging which strips out social and environmental context (Israel et al. 1998). Complexity framing can illuminate the nature of this process as Trenholm and Ferlie (2013) demonstrated in examining the effect of new public management on tuberculosis initiatives in London through a complexity lens.

Given this broader canvas, traditional evaluation by treating an intervention as a technical process and ignoring issues of power and how they affect commissioning, framing, and reporting is clearly ideological. This is reinforced by the aim for objectivity and the role of the expert and has given rise to the hierarchy of methods with RCT as gold standard. Under the dominant paradigm separating 'facts' from 'values' no critical voice is valid. While qualitative methods have long been acknowledged as providing valuable information, their scientific standing was considered poor with evidence denoted 'anecdotal' and its justification requiring a reframed understanding of validity and reliability. Timescales have often been arbitrarily imposed in terms of the objectives of policymakers interested in input–outcome, cause–effect stories to justify the allocation of funds or reorganization of service. In this framework the methods used are regarded as responses to technical questions that 'harvest' rather than create evidence.

The complexity frame implies a fundamental reframing of the nature of evidence, the role of evaluator as observer, the process of study, and the meaning of 'outcome'. The positivist objectivist stance that seeks to stand apart, gather evidence, and make an ultimate judgement of outcome remains attractive, and as Pawson (2013) points out this is not trivial because outcomes are important. Rather than an objective observer collecting fact like evidence, the evaluator acknowledges insights that have been core to qualitative work: the constructed nature of knowledge, the effect of standpoint, and the significance of meaning. Pawson and Tilley's work, laudably directed at making evaluation practical, has not embraced the full implications of this. It requires rethinking the potential for existing methods as well as the development of new ones.

All of this sits within a very different relationship to temporal span in evidence gathering and the expected appearance of outcomes. The relationship between the narrative of an intervention and evaluation has been characterized in ethnographic work as similar to Malinowski's 'long conversation taking place among the people with whom we live during fieldwork and in which we inevitably join' (quoted in Reynolds and Lewis 2019 p8). The notion of a dialogue over time is a powerful one. It reminds us that such reframing goes beyond judging only those dialogues that have resulted in agreement as successful (Connick and Innes 2003). Adopting a narrative approach, Haynes (2008) proposes a policy evaluation method starting with a 'storyline' considering 'timelines' and 'rates of change'. Looking at privatization in residential social care since 1980 he uses the time series data to identify three attractor states that explain system stability and instability. Time, then, rather than simply the arbitrary boundary of the evaluation, becomes an active element and we will see it later in the 'agile' approach recommended by Room (2013) and in metanarrative analysis developed by Greenhalgh et al. (2005).

Despite shifts in science objectivist ideas remain strong, witness the Audit Commission report of 2013 recommending rigour in those terms. While there is an accompanying recognition of complexity theory it reflects the 'niche' problem that we noted in approaches that seek to distinguish simple from complicated or complex interventions (Shiell et al. 2008). We see this in a number of spheres including UK Medical Research Council guidance where complexity ideas have been less well received (Craig et al. 2008). There is now some guidance on process evaluation which while not situated in a systems framework uses the language of mechanisms to represent dynamic field (Moore 2015).

Of course much evaluation takes place with an awareness of the poor fit between traditional approaches and the world they seek to describe. More nuanced strands in process-based and development-focused evaluations recognize that most of the valuable information for shaping programmes is lost in measuring only outcomes. These approaches seek to open the 'black box' of the intervention in a dialogue with programme and policymakers that considers how a programme is working and provides information for adaptation and informing future policy development. Qualitative research recognizes that people do not simply behave or respond, but act intentionally and in relation to future possibilities as well as current conditions. Evidence gathering is shaped according to the perceptions and intentions of those from whom it is gathered as well as those who gather it. It is not simply a matter of reporting experience but involves the orientations of actors to the intervention and their expectations of its effects.

The assumption of objectivity and the neutral standpoint of the evaluator has often been implicit even in qualitative work. Mowles (2014) argued that the attraction of complexity for evaluators is not in its radical reframing of science but, at least in part, in providing systemic abstractions that protect 'the discipline of evaluation by separating the evaluator from the object to be evaluated (p169). Questions of evidence and authority rely on a range of ontological stances held not only by evaluators but by commissioners and implementers. We can identify some key areas that underpin the reframing proposed through complexity and its claims to offer persuasive evidence for evaluation. Fundamentally this involves some rethinking in relation to theory, systems, mechanisms, and cause.

Theory

One of the first direct complexity discussions in the evaluation literature was Sanderson (2000) who identified the theoretical basis of policy intervention and the need for attention to that in policy evaluation. He argued not only that traditional evaluation approaches are 'scientized' (p458) but that the context of central authority and control within which they sit are, in the broadest sense, ideological, limiting the range of evaluation and undermining any attempt to understand how implementation actually works, or how it reflects on those wider institutional and political processes. The inappropriateness of an empiricist approach to policy analysis is widely recognized (Walker 2001) and increasingly understood by policymakers. The argument that all evaluation is theory based is more widely accepted and complexity approaches embrace that reality. This relationship to theory departs significantly from its use in positivist falsification, being dialogical and multi-layered. It requires explicit attention to the nature of the world being studied and how evidence can be gathered about it. Within this there is recognition of the significance of social theory for evaluation as the basis of explanation.

Within the complexity paradigm we can identify theory at lower levels of abstraction, including middle range and intervention theories. There has been much attention to middle range theory for evaluation to provide theory as guidance to empirical study. For Merton (1949) who initially proposed theories of the middle range, their purpose was to build toward the, as yet unfulfilled, objective of general theory. He defined theories of the middle range as

> theories that lie between the minor but necessary working hypotheses that evolve in abundance during day-to-day research and the all-inclusive systematic efforts to develop a unified theory that will explain all the observed uniformities of social behavior, social organization, and social change.
>
> (P448)

Merton's definition of middle range theories, removed from its modernist/positivist framing, offers us a valuable way of identifying abstractions that can be applied to empirical evidence. In evaluation we are dealing with collective actors such as organizations and communities, and emergent action in just the way we identified the 'micro-foundations' proposed by Wright et al. (1992) in Chapter 7. Theories of the middle range are designed to work at this level and to relate directly to empirical work.

While Pawson has focused on substantive middle range theories, their application is not straightforward (Jagosh et al. 2015) and the more fundamental question of their relation to 'grand theory' remains. If in using middle range theories in evaluation we employ constructs such as social cohesion, exclusion, or class, we are alluding to a more abstract theoretical level. Social capital, for example, is theorized in three different forms by Bourdieu, Coleman, and Putnam based on different understandings of structure–agency dynamics. We can consider both substantive and methodological theory. The methodological theories that can facilitate complexity evaluations include negotiated order and theory-based evaluation. Occupying the lowest level of abstraction are intervention theories, those that identify the specific mechanisms and paths required to create desired outcomes. Intervention theories often engage with substantive middle range theory, although this is not necessarily explicitly formulated.

Evaluation is a theory-driven process at all levels. We reiterate this point simply to reinforce its significance in reframing the dominant approach. Weber's recognition of the inherent nature of values in framing social research has real importance here. It alerts us, in addition, to implicit theories about how policies are developed and implemented as well as to theories about the effects and outcomes of specific programmes.

Callaghan (2008) called for attention to already existing complexity-consistent theories in the social sciences on the basis that these relationships and contexts have long been explored and we can learn a lot from them. Westhorp (2012) uses the idea of complexity consistency in terms of layering theories while Byrne (2011) sought complexity congruence. Theory plays into all levels, no longer for use as binary proof or rejection but as an active dimension of the process. Rather than seeking 'correct' theory, the process is one of engaging and reflecting because explicit theoretical underpinning matters at least as much as the evidence gathering tools. Social science is not a technical activity. If we are to overcome a dominant paradigm we need to consider ontology and epistemology as well as the whole range of actors' theories in implementation and evaluation. Acceptance of the 'new' science is a shared aim although some have expressed dissatisfaction with the continuing use of 'new' and reiteration of the complexity frame in articles and books seeking to apply it (Gerrits and Verweij 2015). We continue to use 'new' to remind ourselves that complexity framing needs to engage in a continual critical practice not only in relation to fields of inquiry but to itself. Becoming hegemonic might be comfortable but it is not conducive to a critical practice.

Systems

The complexity frame uses social systems theory and therefore chimes with an already substantial interest in the notion of system in evaluation. Patton (2016) discussing the work of Williams to delineate the use of systems concepts in evaluation notes that:

> as many as a thousand separate frameworks and methods fall under the systems banner because there is no single agreed-upon definition of a system. Nor is there ever likely to be. But having analyzed the great variety of systems frameworks, he asserts that there are three core ideas that characterize systems thinking: attention to the inevitable arbitrariness of boundaries, taking into account a variety of perspectives, and dynamic, entangled inter-relationships.
>
> (p252)

The use of systems concepts, particularly in open dynamic systems, has risen considerably as evidenced in a metanarrative review of 557 studies in public health using systems framework, which interestingly noted that half of them were published after 2010 (Chughtai and Blanchet 2017).

What we are *not* saying is that the definition of system doesn't matter. However, tolerance of uncertainty and awareness of the need to define and redefine are fundamental to a complexity approach. Definition must be achieved in response to a dynamic setting and a dynamic programme. It is this sensitivity that has drawn complexity researchers to alternative ways of framing empirical study using consistent approaches such as QCA and negotiated order. The very definition of the subject under study is multiple, plastic and needs to accommodate agency in the production and reproduction of settings. The recursive nature of the process should alert us to the significance of actors' beliefs and orientations in developing evidence. This also points to greater participation as a basis for definition of outcomes, evidence gathering, and interpretation (Israel et al. 1998). Being sensitive to context and action in system definition brings an awareness of social intervention as multiply framed, shaping how and where boundaries are drawn and to the need for 'a shift from describing social interventions and contexts to a more active and critical approach to selecting and justifying ways to frame particular social interventions' (Gates 2016, 68).

Causal mechanisms

The problems faced in evaluation are rarely simple technical ones of cause and effect. In a world that can be understood in terms of systems characterized by interaction and recursiveness, we need to think of the cause–effect relationship in terms of phenomena sometimes manifest and sometimes suppressed by contextual factors. It means that simple characterizations of outcome are unsatisfactory, while their definition becomes only part of the purpose of the evaluation. A primary question is whether and how the observed effects can be causally attributed to the intervention of interest. Evaluation design is based on how this question can be answered. As Byrne (2013) points out, in complex systems the cause is rarely the intervention alone, 'what matters is how the intervention works in relation to all existing components of the system and to other systems and their subsystems which intersect with the system of interest' (p219). This brings us to a discussion of causal mechanisms as structured entities. This has long been recognized in qualitative research where seeking to develop a deep understanding means gathering evidence in relation to 'what', 'when', 'how', and 'where' in order to build an account of 'why'. Identifying these elements allows relevant empirical evidence to be gathered in order to turn 'a possible mechanism to a plausible mechanism and may eventually lead to the identification of the actual mechanism' (Hedstrom and Yilkoski 2010, 6). Sanderson (2000) quotes Reed and Harvey (1992), 'we can at least reconstruct the particular constellation of structured choice and accident that led to the present reality' (Sanderson 2000, 442). The process is necessarily one of abstraction in foregrounding relevant matter while leaving that currently defined as 'irrelevant' as background, acknowledging the fluidity of relationships between mechanisms and the contexts in which they operate because causal configurations can change over time.

Distinguishing between correlation and cause is fundamental. The focus on mechanisms allows us to recognize complex causation in terms of context and mutual influence between causes and effects. Theory of change approaches to evaluation render complexity-consistent explanations based in methodological pluralism. Using quantitative statistical, modelling, and experimental methods provides surface analysis while an understanding of the institutional and structural context is required to go deeper and for this a qualitative approach is necessary (Harvey and Reed 1996). The main lesson for evaluation is the need to take an open, exploratory approach that focuses on explanation in contrast to prescribed methods that seek prediction and replication. Analysis cannot take place exclusively at either micro or macro levels. We are working in the theory–evidence domain in which structure–agency interactions are immanent. It is to qualitative methods that we turn to find the ways in which that world has been understood consistent with complexity framing. We will approach this in terms of a continuum that takes us from the complexity consistent to the complexity congruent.

Complexity consistent frames

Agreeing with Weiss (1995) that 'there's nothing as practical as good theory' allows us to bring to complexity a set of approaches based on insights achieved over years of social scientific practice. Sensitivity to relationships of structure and agency and to the importance of history makes us aware that abstracting from context does not provide generalizable knowledge. We start from negotiated order because it is neither directly evaluation oriented nor complexity informed but nonetheless has much to say about gathering evidence in relation to structure and agency in evaluation contexts. It is an approach that consciously works with collective and individual agency as well as structure in acknowledging interaction and presenting a way of understanding through observation. It brings the principal insights from complexity theory to the level of gathering data.

A complexity frame begins from the recognition of systems as interactive and historical in nature. It is tolerant of indeterminacy and recognizes that system boundaries may need to be redrawn in the dynamic process of implementation. Recognition of the system's evolving character draws attention to the inappropriateness of entirely 'top-down' approaches and the movement in that direction exhibited in direct performance management under the evidence-based policy agenda (Bevan and Hood 2006). We are looking at whole systems within which initiatives are embedded and the first advantage of the complexity frame is recognition of this wide-ranging lens as fundamental. It takes us out of the narrow focus on an initiative or programme and encompasses an appreciation of local context. We need a frame that comprehends relations of a range of actors in terms of levels of power and interests and can situate those within specific structural context.

Negotiated order

The concept of negotiated order is based on recognizing the recursiveness of structures as entities both shaping and shaped by the ways actors reproduce their settings on a daily basis. Like complexity it offers a frame for what we are observing and its interpretation. Strauss and his colleagues (1963) were concerned with how the 'order' that is achieved is the product of actors' negotiations and how organizational relations work to preserve, adapt, or change systems in a recursive process. Recognizing that framing action in wholly voluntaristic terms is inadequate, they sought a way to capture the patterned regularities created by structural positions. They proposed that these regularities are subject to maintenance or change through the ongoing negotiations of actors. Strauss later emphasized the structural dimension as core as he refined the concept to pay explicit attention to power in terms of the structural and negotiation contexts and to recognize the potential for 'coercion, persuasion, manipulation of contingencies and so on' (Strauss 1993: 249–50).

Negotiated order brings recognition of temporality, patterning and understanding systems, and renders it available to observation because we can develop empirical evidence of interactions between actors that exhibit both structural constraints and how actors work within and against them. Fine (1984), although criticizing Strauss's conception as a metaphor rather than a theory because it lacked testable propositions, usefully set out the four tenets of negotiated order which map onto complexity in terms of interaction, history, and contingency.

> First Strauss argued provocatively that all social order is negotiated order: ORDER is not possible without some form of negotiation. Second, he asserted that specific negotiations are contingent on the structural conditions of organization (a point occasionally deemphasized by his followers). Negotiations follow lines of communication, i.e. they are patterned, not random. Third, negotiations have temporal limits, and they are renewed, revised and reconstituted over time. Fourth structural changes in organization require a revision of the negotiated order. In other words, the structure of the organization and the micropolitics of the negotiated order are closely connected.
>
> (Fine 1984: 241)

Structural contexts create the conditions for action. This means that evaluations that focus purely on actors are likely to attribute failure to intransigence or incompetence while underlying structural cleavages are not addressed. Identifying systems boundaries in the problem of 'silo' working in health and social care, solutions to problems created by structures are sought in 'blurring boundaries' or the actions of 'boundary spanners' or 'reticulists' (Power1973). The focus turns to actors' skills in managing boundary issues and takes structures as given. Recognizing multiple perspectives as arising 'from differential statuses, experiences, and memberships in groups, organizations and local worlds' (Strauss 1993: 252) involves holding both structure and agency in dynamic relation. Social worlds emerge as the result of sedimentation of practice over time and represent the system's history. This is being recognized in accounts of health systems integration (Tsasis et al. 2012) as well as learning in complex adaptive systems frame (Jordan et al. 2010) and the conceptualization of mental health services internationally (Ellis et al. 2017). Here we have the complexity consistency that we have earlier discussed in the works of Marx and Bourdieu. New initiatives enter an already dense field of policy and must interact with existing structural conditions and pre-existing interventions. What results is a set of configurations of relations which have brought about their own patterned ways of communicating. 'Top-down' and target-driven approaches distort the process of reworking existing relations and processes and in turn bring both intended and unintended consequences.

We now have a way of approaching settings empirically with questions of how, when, and where, rather than the 'why' questions that we identified earlier. How has a particular order come about? What negotiations have been and are currently undertaken within what structural conditions? How have patterns of communication, both in form and in the content of accepted truths, developed over time? How have new policy interventions effected change in those patterns? This focus on the active and creative nature of social programmes alerts us to the problem of piloting mentioned earlier but also opens the way to a better understanding of processes in relation to changing contexts.

Negotiated order theory suggests reconceiving evaluation away from outcome towards understanding complex causal mechanisms manifest at local level. Transferable lessons are no longer direct but suggest, rather, creating space at local level for system objectives to be worked toward. It has the further merit of developing the kind of evidence useful for synthesis that we will consider later. This is a way of understanding and researching how organizations function that, while not specifically evaluation focused, casts light on interventions in context. From here we can move on to consider approaches developed directly within the evaluation field that are important for complexity framing.

Theory-based evaluation: theories of change

Theory of change evaluation starts from the premise that every intervention has a theory, or theories, of how change will be brought about. If frequently not explicit these theories can be rendered so. Generally it is the evaluation team that elaborates the theory of change. This involves clarifying all sets of assumptions upon which the programme is to be based. The evaluation design is then formed to develop evidence in relation to theories, both their success and failure. While theory of change has been criticized as linear it is clear that the approach lends itself to recognizing contextual complexity and interaction because it requires specification of causal mechanisms. Theory-based evaluation design takes the form of starting 'with the long term outcomes, work backward toward initial activities and then map required resources against existing resources' (Connell and Kubisch 1998 p5). This has a number of complexity-resonant advantages, not least the ability to incorporate information drawn from the broader theoretical knowledge that the complexity frame suggests. A primary complexity consistency lies in recognizing interaction between individuals, families, households, communities, and institutions as well as between theories and practice. The base requirement is that this complex configuration is discussed in terms of actors' expectations about how the programme will achieve its effects. Evidence gathering is then focused on questions of whether those effects are visible and, if not so, why the reality diverges from the theory. It incorporates a system-based understanding and research that can recognize the role of context, interaction, and emergence. The fundamental process of rendering underlying theories visible within the evaluation resolves the issues that can arise from differing theories of change being held implicitly within the group as a whole. It thus places front and centre an influential factor in the policy complex that has often been ignored in traditional 'black box' input–outcome models. We have a detailed account of programme theory which makes all elements of the process visible. It is about mechanisms and how evidence can be developed in detailing them. Some of the mechanisms assumed within the programme may have good supporting evidence from wider social research but each must be sustained within the theory of change being examined and can in turn provide evidence in refining those theories. In this way it establishes the relationship between social science knowledge and policy. Sanderson (2003) has argued that thus far theory-based evaluation has shown limited success in addressing theory–evidence link and this would undoubtedly be a focus for development in a complexity frame. Barbrook Johnson and Penn (2021) list discussions of theory of change in this vein. Nonetheless the potential for a theory-based approach to inform the process of implementation from the early stages by identifying what and how it is working is clear. Theories of change have been widely adopted and a range of evaluation approaches can be broadly subsumed within the approach: theory-based, theory-driven, theory-oriented, and theory-anchored intervention theories. This has become an accepted basis for complexity evaluation leaving the particular methods to be specified in relation to particular contexts and stakeholders.

The problem of complex causation is addressed directly in this approach. In terms reminiscent of Reed and Harvey (1992), Weiss (1997) pointed to its promise, 'or at least hinted promise', that 'If the evaluation can show the series of micro steps that lead from inputs to outcomes, then causal attribution *for all practical purposes* seems to be in reach'

(1997 70). While this is clearly open to the post hoc accusation it is precisely because it works in terms of mechanisms that it establishes the relationships within process between context, intervention, and outcome and makes them visible for confirmation, refinement, or refutation.

Critiques of theories of change are fleshing out the systems and mechanisms coherence with complexity framing. Critical systems heuristics, for example, offers a tool for bringing multiple perspectives to the fore by asking 'what is' and 'what ought to be' questions. Rather than resolving the difference this tool works by 'maintaining tensions between contrasting perspectives for critical purposes' (Ulrich and Reynolds 2020, p258). A further distinction within theories of change is worth noting here. Weiss distinguishes implementation theory from programmatic theory (1997 p72), implementation theory being concerned with the steps in how the programme is carried out while programmatic theory deals with the mechanisms that intervene between the delivery of program service and the occurrence of outcomes of interest. While both contribute to theory of change evaluation it is clear that the latter sits particularly well with complexity in focusing on both programme activities and how those activities give rise to responses from users and others, in other words, the interactions that give rise to non-linearity. It relies on clarifying the nature of the mechanisms developing within a narrative recognizing that they operate differently over time as context inevitably changes. The evaluation report that results is a 'systematic study of links between activities and outcomes' (Connell and Kubisch 1998, 10). As Connell and Kubisch have argued, theory of change is an approach rather than a set of methods, much as we have discussed in negotiated order.

Participation- and community-based evaluation

There has been significant interest in the role of participation in evaluation for obvious reasons. The notion of collecting data about social interventions without recourse to the constructions of users clearly fails. Based in an extension of Guba and Lincoln's constructivist paradigm, Heron and Reason (1997) define the participatory paradigm in terms of a mindset in a way that is familiar to us:

> The participatory worldview, with its emphasis on the person as an embodied experiencing subject among other subjects; its assertion of the living creative cosmos we co-inhabit; and its emphasis on the integration of action with knowing is more satisfying. To return to Ogilvy's terms it responds creatively to the emerging mood of our times, overturns the mechanical metaphor which underpins positivism, provides models for action inquiry and above all offers humanity a more satisfying myth by which to live.
>
> (P12)

One strong strand of evaluation practice has been community-based evaluation, developed alongside and often in concert with theories of change. Community involvement in new initiatives has come to be accepted practice in UK public policy. Israel et al. (2017) synthesize key principles for community-based research that fit well with a complexity frame. Its complexity consistency lies in not only in being context driven but in recognizing interaction, the constructed nature of knowledge, and the role of the researcher in power terms. This includes the power to decide how the evaluation happens, the power of commissioners and academics to develop and determine what is evidence, and the power to validate. Much evaluation research has already responded to this wider definition of its purposes and the need to incorporate more than commissioners' definitions and interests, placing values at the core and defining them in terms of community priorities. Problematizing knowledge in this vein exposes power relationships that lie hidden in a positivist approach.

These approaches, each developed outside the complexity frame, exhibit complexity consistency. There is familiarity with complexity ideas across evaluation studies, suggesting a potential for reaching back into those studies in establishing explicit complexity evaluations using frameworks such as that presented by Walton (2016a) which could inform a synthesizing analysis. This is a process of seeking to consolidate lessons for future complexity research. Our point is that the complexity consistency of these frames points us in directions that already form the basis of a strong body of evaluation practice. We consider that this more open approach can bring insights from a wide range of resources already available to us and, perhaps, begin to make the case for a cumulative dimension to evaluation in nascent form that can be approached more systematically through realist synthesis and meta narrative review. We turn to realist evaluation at this point in the chapter both to reflect the shift from complexity consistent toward complexity congruent work building upon insights drawn from both theories of change and community-based evaluations.

Realist evaluation

Realist evaluation (Pawson and Tilley 1997) as its name implies emerges from the same appreciation of the ontological level as complexity theory and the two become more intertwined in their application to evaluation practice. Bhaskar's critical realism provides a philosophical ontology which recognizes hierarchically nested world which is emergent and historically open. While paying attention to Bhaskar's critical realism Pawson (2006) is critical and contrasts it with Campbell's critical realism in terms of the difference between values and the dialogue of science: 'It is, in short, that between righteous indignation and organised scepticism' (2006, 20). Here we turn to Reed and Harvey to pinpoint the role of critical realism in relation to empirical study:

> Philosophical ontologies contribute to scientific activity by showing the scientist in broad outline what the world and his or her knowledge of it should look like. Philosophical ontologies, however, only describe the boundaries of an intellectual continent, not its surface details. The task of filling in the gaps on the philosophical map belongs to the scientist.
>
> (1992 358–9)

If Bhaskar's critical realism is limited by its recognition of constructivism, scientific realism as outlined by Reed and Harvey provides the possibility of developing persuasive evidence. Pawson and Tilley's (1997) context + mechanism = outcome formulation is now a widely recognized abstraction that focuses our attention on the need to draw boundaries so that the critical decisions lie in what we include in each component of the equation. This is a much harder thing to do in practice than it might appear and those working in this frame have found that in application it rather more difficult to achieve a separation between mechanism and context. At what point do we exclude and how do we draw the lines around an intervention's context? It is clearly not possible to frame any boundary once and for all at the beginning of the evaluation as contextual factors change both independently of and in interaction with the intervention. Indeed the CMO formulation has been criticized as implying a 'one way relationship'. Quoting from Pawson and Tilley, Barnes et al. (2003) suggest:

> Programs work by introducing new ideas and/or resources into an existing set of social relationships. A crucial task of evaluation is to include (via hypothesis making and research design) investigation of the extent to which these pre-existing structures 'enable' or 'disable' the internal mechanisms of change.
>
> (Pawson and Tilley 1997: 70)

> We suggest that an understanding of complex systems necessitates recognizing context as a part of the open system within which the programme is operating. As such it is part of the system and not external to it. The context itself is subject to change as a result of actions or activities beyond the scope of the programme, but also from the intended or unintended consequences of programme implementation.
>
> (2003, p269)

Replacing the + with the interactive & might solve this problem although it makes the equation more opaque for application. Realist and complexity evaluators use an approach that has not been prized by traditional evaluation, that is, the ability to remain open and critical and to be prepared to alter definitions and boundaries as the work progresses. Rather than trying to pretend the world is simple and apologize for being unable to produce a picture that accords, it relies on being tolerant of ambiguity and allowing it to play into the research as it progresses. Framed in this way it can draw on theory of change processes in a fluid and developmental form that allows internal feedback loops in multiple processes within the whole. Pawson's intent in rendering the equation to facilitate practice is a valuable approach but it results in an intervention focus that somewhat undermines the complexity we have identified as residing in the social system rather than one in the intervention. The context + mechanism = outcome formulation of Pawson and Tilley gives us a simple way to explain the practice of evaluation in complexity-congruent terms although we can recognize a number of challenges in practice in appraising evidence, using middle range theory and incorporating it into CMO configurations. In the next chapter we will consider methods that explicitly seek to retain all of the complexity in evaluation studies.

Conclusion

For evaluators, who must after all be alive to the problems of evidence, there has long been a recognition that programmes have their effect within a complex context and cannot simply be abstracted to judge degrees of success or failure. Outcomes may be defined in different ways on the basis of the different interests involved. This context involves social, political, economic factors and the potential for a dialogue between social science research and evaluation practice is clear. Stakeholders can have different expectations and objectives while agreement to participate in an evaluation study as a patient, service user, etc. is likely to be based on a range of beliefs about the purpose and value of the intervention, all of which shape responses. We have identified in Bhaskar's terms that science brings transitive knowledge of an intransitive world and the approaches to evaluation that we have discussed start from the point that every evaluation is a construction that builds a story of the intervention and its effects.

The essence of complexity framing does not lie in the new methods but rather, as Sanderson (2009) argues, in thinking differently. This has been the burden of this chapter. Evaluation is an important mode through which complexity thinking is entering what we might call the mindset of certainly the civil services of governance at every level, although there are not many signs of this in relation to elected representatives, at least in the English-speaking world. So getting evaluation understood in complexity terms is important and that is what this chapter has been doing. In the next chapter we present examples of work done, since instantiation is always crucial in showing how a way of thinking can be put to work.

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13 Explicit complexity evaluation

The complexity literature has grown so rapidly in social science because it has brought a moment of recognition, when things previously forced to fit into unaccommodating boxes gain their own shape and rationale. Offering ways of thinking about evident non-linearity and the role of history and context, it describes the world evaluators know and have sought to explain. Recognizing the troubled status of the concept of outcome as fragile and temporary, complexity evaluation places emphasis on learning and development.

We have described complexity framing in terms of a continuum from complexity-consistent approaches in which complexity thinking is implicit to realist evaluation at the complexity-congruent end based on a shared ontological frame. As a field develops, most publications focus on discussing appropriateness (Paina and Peters 2012, Khan et al. 2018; Morrow and Nkwake, 2016; McCool et al. 2015) while there is less evidence of methods in use. Adoption arises partly as a result of commissioners' beliefs about evidence leading to an understanding of the potential of complexity in the commissioning process (Cox and Barbrook Johnson, 2021). Now complexity is firmly present in the evaluation world, evidenced both by partnerships with policymakers such as CECAN in the UK and in attention in academic journals. It was a significant theme in the 20th anniversary edition of the journal *Evaluation*, followed by a special issue on complexity methods in 2021, and in edited collections of papers from a realist position in health services research (Greenhalgh and Papoutsi, 2018). The spread of complexity thinking in the professional evaluation community is also apparent in the *BetterEvaluation* website.

Retrospective complexity framing

If the major application has been in using complexity perspectives in framing studies, this does not suggest underdevelopment. We believe this is where the most significant impact on evaluation is likely to emerge, given that we have many existing methods that are appropriate for engaging with it, particularly from qualitative work. As Connick and Innes reflected in reviewing the effect of framing on collaborative policymaking:

> When we see our world as a complex system, in which learning, feedback and adaptations take place through linked, self-organizing networks, as opposed to a mechanistic model of inputs and outputs, we can understand better how collaborative dialogue processes function and produce the outcomes the authors have discussed here.
>
> (2003 196)

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This understanding is extending into a range of fields, including economic evaluation in health-care Lessard 2007), general practice (Sturmberg et al. 2014), and education (Hetherington 2013). Orton et `al. (2019) applied the frame to data from Big Local in 150 localities in England. Paina and Peters (2012) argued that scaling up health services in developing countries can be done more effectively when framed through complex adaptive systems perspectives.

Numerous studies have reflected on research previously undertaken outside the complexity frame. Cox (2012) reviews work undertaken co-researching with young people on an evaluation initiative within their communities. Woodside et al. (2015) revisited their research to apply fs/QCA (fuzzy set/Qualitative Comparative Analysis) to customer service evaluation data in casinos to discover a more 'nuanced' account of the relationship between casinos and problem gamblers. Chandler et al. (2016) returned to a process evaluation undertaken as part of a larger study evaluating strategies for reducing surgical fasting times. Dyson and Todd (2010) reflect on a theory of change evaluation of the full service extended schools initiative in the UK that aimed not only to introduce a new intervention but to integrate services from multiple interventions across child and family services. Walton (2014) describes an evaluation of the benefit of fruit in schools in New Zealand from a complexity-consistent perspective in an approach that allowed interaction effects and emergent outcomes to be noted. Caffrey et al. (2016) examined implementation experience in health service research systems.

Recent work gives more guidance on how to frame and undertake complexity evaluation as well as how to assess existing studies, considering how far implementers have responded to complexity awareness and suggesting how this knowledge could be used in design and implementation of new programmes (Larson 2018; Walton 2014; Gates 2016). New frameworks are being proposed to assist in the development of application such as Koleras et al.'s (2020) actor-based change framework, developed to apply to programme theory, and Bustamante et al.'s (2021) application in the sphere of agriculture and food. The use of complexity ideas in evaluation is emerging 'bottom-up' through adoption by those who see value in the frame. It may still have to overcome some reluctance from 'top-down' commissioners but again we are seeing change.

A developing literature about application and identifying the gaps and problems as well as advantages highlights the varied use of systems thinking and complexity science in evaluation practice in terms of framing, developing, application, and use. McGill et al. (2021), for example, synthesized evidence from 74 studies in the field of public health that have employed complexity frames and complexity-consistent methods. Now we want to look at some methods explicitly developed for complexity.

Developmental evaluation

Developmental evaluation is a complexity approach rather than method, framed in terms of the simple, complicated, complex grid that we have taken issue with. This matters not only because of the niche status it allocates to complexity, but more fundamentally because it ignores ontological complexity. Nonetheless developmental evaluation is framed in explicit complexity terms and has become part of the evaluation lexicon. Patton describes the emergence of this approach from an initial formative-summative evaluation design in a leadership programme in which participants:

> realized that they would need to be continuously adapting the leadership program as the world changed. To keep a leadership development program relevant and

> meaningful, they concluded, they would need, over time, to continuously update and adapt what they did; who and how they recruited people into the program; use of new technologies; and being attentive to and incorporating developments in public policy, economic changes, demographic transitions, and social–cultural shifts. They came to understand that they didn't want to improve a model or test a model or promulgate a model. … They wanted an approach that would support ongoing adaptation and timely decisions about what to change, expand, close out, or further develop. … And they concluded that they would never commission a summative evaluation because they wouldn't have a standardized model that could be summatively evaluated.
>
> (2016 253)

In keeping with its status as a 'mindset', Patton identifies eight principles to guide developmental evaluation design oriented to innovation in 'complex dynamic environments'. The approach responds to change, adaptation, and emergence in settings where uncertainty and turbulence are challenges. In such settings:

> premature specificity can do harm by constraining exploration, limiting adaptation, reducing experimental options, and forcing premature adoption of a rigid model, not because such a model is appropriate, but because evaluators demand it in order to do what they understand to be good evaluation.
>
> (2016, 254)

Patton's approach does not offer concrete guidance on doing a developmental evaluation although more practice guidance is appearing (Bamberger et al. 2015) and there are now several examples of its use. Tremblay et al. (2020) describe research using developmental evaluation in combination with community-based participatory research. They used the combined approach in relation to an affordable housing initiative in which developmental evaluation provided the framework for review and guidance of the developing programme. They identify shared elements of the mindset in that both involve collaboration in which those most directly concerned in the programme shape the evaluation process. The focus in both is on responsiveness and change. The methods deployed in this frame are generally not innovative in themselves but are well suited to taking account of complexity. Goertzen et al. (2021) discuss the use of developmental evaluation in an early stage study of the Complex Care Hub in Canada. They used a range of methods including an observation log and interviews to identify intended and emergent outcomes as basis for dynamic programme design and adaptation, facilitated by the interaction between evaluation team and programme.

Complexity framing with existing methods

Recent studies have used complexity framing directly across a wide range of fields including Wolf Branigan (2013) in social work; Bittencourt (2016) in relation to sports injuries; Rosas and Knight (2019), Matheson et al. (2018) in health promotion; and Ndhlovu, et al. (2017) in urban planning. The unifying frame allows reflection across settings and interventions, in a way reminiscent of the discussion of a shared language around complexity, and facilitates communication in interdisciplinary work. Eppel et al. (2011) describe three studies using complexity undertaken by each of the authors in the diverse fields of policy process, health promotion, and health inequalities. Roche et al. (2021) describe the use of complexity in two case studies in quite different programmes as a basis for synthesizing lessons for programme design, implementation, and evaluation. Many studies use well-established methods of qualitative interviews, documentary analysis, observation, and comparative case study in new conceptual frames There is substantial support in these articles for the argument that while many working in complexity frames seek to develop innovative approaches, the real need is to pay attention to the nature of the models and metaphors we use to represent reality.

The attractiveness of the language of complex adaptive systems has sometimes resulted in its use in relation to a less radical reorientation in thinking, reminding us of the need to maintain a critical appreciation of the ontological base of complexity studies. Critics point to approaches that fail to fully embrace the implications of complexity. Larson et al. (2018) note the increasing reference to complex adaptive systems in international development assistance alongside poor levels of application. Firstly, the focus tends to be on the intervention rather than context and the expectation that a simple transfer of successful programmes is possible. Secondly, there remains a tendency to see context as static thereby failing to incorporate its ever-changing nature, reflecting the continuing commitment to underlying positivist ontology. Rather than problematizing objectivity Goertzen et al. (2021) identify maintaining it as one of the challenges which suggests some difficulty in leaving behind positivist thinking. Their discussion of the conflict between timeliness and rigour is also an issue evaluators need to consider in complexity terms.

Developing new methods

Social scientists have sought to address the quantitative/qualitative divide through mixed methods in recognition of the constructed nature of our subjects of study and relationships of structure and agency but often this has involved parallel rather than integrated knowledge. Qualitative and theory-based methods are well embedded in evaluation and many can be used in a way that fits well with a complexity frame. At the same time a range of innovative quantitative methods have been incorporated into complexity approaches. The potential of computational methods is now being explored and appears to have promise, while remaining aware of the need for a critical approach to the social constructions involved in technology. The assumptions incorporated into models are the point at issue. Pawson (2021) has highlighted their significance in relation to model building for prediction in relation to the COVID pandemic, raising the question not of accuracy but of whether models can be 'good enough'. Through discussion of the assumptions incorporated into the models of disease transmission that have guided policy, he shows how the evidence base has been unable to comprehend the evolving nature of measures such as lockdown, while cross-country comparisons fail to recognize the different natures and order of interventions.

A primary organizing frame is the case and it is worthwhile reminding ourselves here of its character as requiring a rather more sophisticated analysis than a simple atheoretical definition might suggest. In Abbott's (1992) terms, it is the narrative that determines the case so the question becomes 'what do cases do?' The practice of using empirical data in a weak relation to theoretical construct is common. As Harvey tells us, 'Casing does not create the case-object, *but only a construct of the case-object*' (2009 16). Harvey outlines two consequences of a dissipative systems frame for case objects. Firstly, that they should be dealt with as 'complex, dissipative systems, as opposed to simple mechanical ensembles' and, secondly, that they are recognized as 'evolving sociohistorical entities … the joint product of the intentional activities of historical actors as well as of inertial institutions' (2009 29).

Case-based methods

Case-based methods have been widely adopted in complexity studies. Qualitative Comparative Analysis (QCA) can provide a comparative account of a number of cases in relation to the presence or absence of an outcome, describing causal mechanisms in terms of configurations of causes derived in terms of crisp sets, multi-value and fuzzy set comparisons (see Rihoux et al. 2009). Instead of reduction, all of the data relevant to outcome are retained within a truth table. Configurational approaches such as QCA are becoming widely used, particularly in relation to health but also in a wider range of applications including marketing and public infrastructure projects. Observing the lack of systematic research into a well-recognized problem of cost and time overruns in large-scale infrastructure projects and the ambivalent nature of evidence relating to management strategies, Gerritz and Verwej (2018) used multi-value QCA to understand how managers responded to unplanned events. The method becomes valuable for understanding cause in terms of interacting factors, not reducing but maintaining data in developing accounts of mechanisms.

New methods are both being developed and adapted from other fields, including models, maps, and simulation alongside qualitative and more conventional methods that embrace dynamism. Rosas and Knight (2019) describe the use of three systems methods, based in modelling and social network analysis, in relation to a complex health promotion intervention with young people. They use theories of change in articulation with other methods to achieve an explanation that can comprehend complexity. We can see this across a number of fields of practice as systems methods now have an established place in evaluation. The development of new methods as well as application in new fields necessarily raises questions about their transferability. McGill et al. identified a problem of definition common to a number of new approaches:

> a particular tension evident in efforts to apply complex systems perspectives to evaluation: namely, a fuzzy and contested sense of what constitutes and what does not constitute a complex system.
>
> (2021 272)

One of the major developments in innovative complexity methods has been in modelling, although used rather differently from the statistical causal model. Byrne (2011) describes the measurements used in evidence-based models as 'variate traces' that can help us to describe change in a non-linear system. The development of multi-level modelling and use of Bayesian techniques are among the approaches to evaluation in complexity emerging from the UK Centre for Evaluation of Complexity Across the Nexus (CECAN). The centre was established in 2016 to bring a multidisciplinary approach to the application of a range of methods in complexity science and evaluation. Its impact has already been substantial in establishing partnerships with government agencies and devising complexity methods for a wide range of policy areas (Barbrook Johnson et al. 2021, Bicket et al. 2021). The new methods address a number of purposes focusing on identifying, describing, and mapping systems (Simsekler et al. 2018), developing scenarios, configurations, and modelling agents' behaviour.

Pawson et al. (2005) remind us of the need to maintain grounded definitions that respond to stakeholders' views and dynamic contexts. This is embedded in approaches such as Participatory Systems Mapping (Barbrook et al. 2021). Modelling can offer opportunities to review alternative possible futures, as long as these retain the status of tools for looking rather than real descriptions. Scenario simulation is a new tool developed by Schimpf and colleagues (2021 117) aimed at facilitating the exploration of alternative scenarios to learn:

> in an accessible and reasonably simple manner, how policies take place in real-world open and dynamic systems, where (1) many intertwined factors about the cases being targeted affect outcomes in numerous ways, (2) many insights may only be available retrospectively and (3) controlling or eliminating all unpredictable events is unattainable.
>
> (Schimpf and Castellani 2020 or Schimpf et al. 2021 p17)

The case-based scenario simulation Complex-It tool combines ABM with case-based modelling (Castellani et al. 2019). Noting that scenario analysis is already widely used in evaluation research and case-based methods are becoming more popular, Schimpf et al. (2021) argue that case-based computational modelling can deploy the capacity of computational social science to facilitate a case-based approach to policy evaluation by exploring data through cluster analysis and then applying it to scenario simulation. They suggest a possible application to four studies to demonstrate its potential contribution.

> By generating and analysing 'what if' scenarios, evaluators, policymakers and researchers can consider past counterfactuals or anticipate future challenges, reduce uncertainty by condensing the envelope of plausible outcomes, make assumptions about affected systems or populations explicit, and conduct more thorough investigations of interventions into complex systems.
>
> (p118–9, 2021)

Befani et al. (2021) use simulated probabilities to assess the strength of evidence in relation to theories of change. They describe this as a 'third way' through diagnostic evaluation deploying Bayesian logic in relation to theory-based evaluation as well as system-based evaluation. Using simulation methods in ABM, they argue, allows them to not only validate causal relationships but to assess their strength. There are drawbacks to modelling but its proponents argue that it can help identifying what questions need to be asked and what data is needed. Schimpf et al. 'strongly emphasise' the learning environment that CBSS offers.

A recent innovation, JuSt-Social (Badham et al. 2021) is a descriptive model of the COVID-19 pandemic in the UK designed for planners to use at local level with primary emphasis on accessible knowledge and being timely. JuSt-Social is an agent-based model of the COVID-19 pandemic whose purpose is to use what is known about the pandemic and the effect of policy intervention in a 'coherent, internally consistent summary' by generating 'stories' that can connect events causally and theoretically.

> Descriptive models such as JuSt-Social are particularly suitable for local scenario planning, where plausible futures are required to understand the potential consequences of policy options for regional stakeholders and their communities. Further

> interrogation of the model can also reveal aspects of the encoded knowledge such as how specific scenarios arise, highlighting gaps and inconsistencies in the represented knowledge and facilitating estimates of uncertainty.
>
> (1.11)

The CECAN complexity evaluation toolkit (July 2021) provides guidance on commissioning, designing, managing, and using evaluation with a comprehensive reference list of appropriate methods.

These approaches are still developing and as yet there is little evidence of their application in practice. Stern, Saunders, and Stame in their editorial introduction to the 2015 anniversary issue of *Evaluation* commented that:

> the consequences and subsequent uses of systems and complexity thinking remain opaque. The difference between those who see complexity and systems as framing devices that highlight questions of boundaries, interactions, feedback and those who associate complexity with the need for new techniques such as modelling and simulations might well be one emerging divide.
>
> (385)

This is an important observation. Innovation needs to continue to offer valuable tools to assist in the fundamental project of framing complexity studies. It would be a retrograde step to develop a new set of techniques with the immediate attractiveness of alliance to traditional quantitative work while allowing the ontological underpinning to be forgotten. The burden of our argument has been that alongside technique and methods development it is necessary to maintain awareness of those foundational issues.

Application beyond immediate context: meta-evaluation/synthesis

While the primary purpose of evaluation is to assess performance and understand processes and outcomes of an intervention or policy, a second but significant question relates to the ability of its research findings to cast light beyond specific contexts and policy domains. Is it possible to develop transferable knowledge, founded on an understanding of the multiple relationships between context, its interaction with the initiative, and outcomes that can offer insights that can be reoriented to new tasks and settings? We have argued against generalization and identified the inappropriateness of piloting which might suggest abandoning the idea of learning beyond context. Yet this is such a core need for any policymaking system that thinking about what can usefully be learned is a high priority. The potential for evaluations to yield general findings has been limited by their very practical and applied (context-bound) nature giving them the technocratic air we identified earlier, while the funding and commissioning process has reinforced this usage. Reflection across a small number of cases can enable appreciation of mechanisms in different causal configurations. Westhorp (2013) describes a small-scale realist evaluation with a 'modified form of realist synthesis'. Roche et al. (2021) deployed a 'collaborative, reflexive, deliberation approach' to examine two case studies in very different contexts to reflect on the practice of applying complexity thinking. Transferability also arises in informing different levels of implementation. A number of writers have raised the question of the ability of local evaluations to inform national and higher levels (Stame 2004). Barnes et al. questioned the use of theories of change alone on this basis and turned to complexity for a more adequate theoretical grounding. The problem is one of making sense of evaluations in relation to 'policy mixes' based on different theories of change, policy domains, and governance levels (Magro and Wilson 2013).

Systematic review enjoys scientific respectability for its transparent and replicable method but does so by cutting out context and has been criticized for failing to encompass the full range of evidence. Boaz et al. (2008) sum up the issue:

> Attempts to transfer this methodology from its clinical research origins into policy research have revealed limitations – in the kind of (narrow) policy questions these studies are designed to address, in the limited range of research accepted by their restrictive quality criteria, and in the (only partial) usefulness of the outcome measures that they typically yield. Reviews are now being undertaken for quite diverse purposes. They do not just seek to answer the 'What works?' questions that have been considered to be appropriate to medicine. In public policy even that question must be reformulated as 'What works, for whom, in what circumstances?' (Pawson 2006). But even that may be too restricted. Sometimes prior questions such as 'What is happening in society?' and 'Why is it such a problem?' are the ones that policy makers and practitioners hope that research might answer. (p244)

Dixon-Woods et al. identified a range of approaches to synthesis of quantitative and qualitative findings using the distinction between integrative (primarily quantitative) and interpretive reviews which 'achieve synthesis through subsuming the concepts identified in primary studies into a higher order theoretical structure' (2005 46). Reflecting on the use of this approach as synthesis they point to its ability, rather than specifying concepts in advance, to develop concepts and the theories that they build. While grounded in data the process is not one of specification and checking.

Realist synthesis and meta-narrative reviews offer approaches which aim for 'enlightenment' (Weiss 1979) rather than direct transfer or specific approval of programmes. Rejecting traditional mapping of studies in systematic review style, Pawson et al. (2006) proposed realist synthesis as a way of dealing with the question of learning across studies by exploratory searching to allow relevant literature to be captured. This seeks to explore the causal mechanisms that operate within programmes to understand how they fulfil their purpose and what contributes to failure. It offers a model of research synthesis that is designed for complex interventions (Pawson 2013). Based on the CMO formula, it starts by identifying the nature of interventions as dynamic systems nested and interacting with other systems. The approach employs the structure of the Cochrane Review but elaborates it in terms not only of intervention theories but also by replacing a linear process with an overlapping and iterative one. Using the full range of available evidence, it seeks to examine the mechanisms by which programmes work, the context of their action and interaction, and what outcomes emerge. In this way it explains 'what works for whom, in what circumstances, in what respects and how'. The application of a realist frame by De Souza (2016) in contrast to Pawson's 'forward analysis' undertakes 'backward analysis' placing importance on 'investigating the history behind the pre-existing and prevailing structures and cultures operating in the context of interest' (p233). We will see appreciation of the significance of this understanding further developed in the discussion by Room below.

Greenhalgh et al. (2005) developed a meta-narrative approach to systematic review through 'storylines', the purpose being to understand each literature in its paradigmatic context, making sense of any apparent lack of coherence. The approach was developed in relation to Kuhn's description of the characteristic progress of paradigms: rise, dominance, and decline. The historical sequencing of the literature is clearly significant yet is ignored in systematic review. This approach not only retains rather than discarding any identified 'incommensurability', it renders it a matter to be explored and analyzed. Although criticized on the basis of lack of transparency in method (Faggion et al. 2017), the approach promises a more grounded understanding of a changing field Greenhalgh et al. (2018).

Room (2013) provides a very straightforward and persuasive critique of both linear and realist processes of review, arguing that while Pawson's realist approach brings consideration of time and contingencies, it remains intervention focused. Transformative realism, as proposed by Room, asks a different set of questions pertaining to 'synergistic dynamics' with other interventions, conditions and contingencies which give rise to 'substantially different directions', orders of introduction, effects of policy landscapes, and degrees of resilience in response to new interventions. Room is proposing a review that can comprehend a world of 'weaving' and 'reweaving' new with older policies to understand local policy landscape in terms of:

- The policy landscape or 'ecosystem' and the successive policy interventions by which it has been shaped.
- The historical development of – and antecedents to – the current social and political settlement.
- The principal stakeholders who are actively probing the policy landscape.
- The social, economic, and political connections among them.
- The capacities and resources they bring to the fray.
- The battle of ideas in which they are involved and the 'mental models' with which they variously operate. (p235)

This sits very well with our discussion of understanding of negotiated order being achieved on a daily basis. The 'agility' proposed by Room suggests a different kind of reporting. Rather than definitive findings it seeks to develop an 'on board guidance system' to 'navigate with the necessary agility a world that is too complex to fully anticipate' (235). He uses the process undertaken by the CDP projects in the 1970s as an early example of this approach. This notion of co-evolution and the recognition of interacting and developing histories has become an important theme, evident in Gerrits's discussion of the management of estuaries in Germany, Belgium, and the Netherlands (2018). In Marks and Gerrits (2019), the authors use a game theoretic approach to illustrate co-evolution, demonstrating the effects of history on interaction. Room's Contingent Historical Model seeks to apply this in integrating 'blind co-evolutionary dynamics with purposeful agency and the struggle for positional advantage' (p176).

Pursuing wider application and transferable knowledge there is a promise of evaluation rather than, at its worst, a perfunctory adjunct at the end of a single programme, being an accompanying voice in policy development that provides precisely 'enlightenment'. It abandons the accepted policy cycle of implementation–evidence–change. The significance of Room's transformative realist approach and the discussion by Greenhalgh and colleagues of 'storylines' lies in giving us a way of situating, appreciating what can genuinely be learned in a complexity frame and applied to contexts other than those in which the specific learning took place.

The possibilities for synthesis of evaluation studies in providing information for policy could be substantial (Melendez-Torres et al. 2015). We can draw on social science knowledge to understand structures, cultures, and power in operation and the potential for a form based in recognition of mechanisms rather than substantive cumulative knowledge arises. Successive evaluations that explore context, history, and new policy measures highlight the questions that need asking. Sheate et al. (2020) report on a meta-evaluation of environmental policy evaluations using a grounded theory approach to generating themes to suggest radically rethinking the policy cycle in a way that would be coherent with an 'agile' approach. They sought to examine studies in the field of environment to develop an understanding of barriers to learning in an ever-changing policy context. They describe a formal and structured review process in the EU policy cycle as making non-arbitrary and appropriate use of evaluation reports compared with lower instrumental use but higher strategic use in the UK.

> This means that in the UK policy making no longer (if it ever did) necessarily follows a typical policy cycle where evaluation is part of the cycle that leads to refinement of the policy. Instead, policy making and implementation are now seen as part of a more dynamic system, termed 'system stewardship' by Hallsworth (2011). This is a more flexible and adaptive model, but also means that evaluation now faces a 'triangle' of purpose (or policy goal), design (or policy direction) and implementation (or action, realization) simultaneously (rather than being part of a cycle) and needs to be responsive to changing circumstances. Both policy making and its consequences are more complex and so evaluation needs to be responsive to this complexity, as well as complexity intrinsic to the subject matter of environmental policy and the nexus.
>
> (2020, 5)

This context of multiplicity and interaction is discussed by Downe et al. (2012) in a meta-evaluation of a local government policy of 'experimental, pilot driven approach'. They conclude that groups of studies should be commissioned to recognize the interaction between policies with researchers cooperating and sharing information in relation to overarching themes.

There is as yet little evidence of cumulative substantive learning which appears appropriate for evaluations set in dynamic and multilayered contexts, but what we do find encouraging is the reframing taking place.

Conclusion

In this short chapter our purpose has not been to review the full range of evaluation taking place which in some way deploys complexity. This would require substantially more space and would not meet our purpose. What we hope to identify are directions being taken and the ways in which they cohere with the core paradigm of complexity. We have taken note of Merton's (1949) warning, however, that we should 'moderate our aspirations' in relation to theory and most certainly remain alive to the potential for new insights and challenges. In this we recognize that rather than simply defending it against critics, our purpose must be to use complexity to explore, adapt, and genuinely engage. Through relating to the world in a way that reflects a reality people know, the methods of complexity can allow social scientists, policymakers, programme implementers to develop a practice that no longer requires them to perform contortions with evidence to fulfil the positivist dream. The promise of reflecting reality is massive and disappointment with the usefulness of black box, outcome-focused methods of the past has brought a certain wariness about making such claims. Nonetheless the promise is huge and important and we must see where it can take us.

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14 Shaping research and policy

We described this text as an exercise in synthesis and the materials, arguments, and frames of reference being reviewed here are part of that process. The complexity frame of reference is being used and we need to say how. We do so because we live in an urban world facing a climate catastrophe. Fifty-six per cent of the world's population lived in urban areas in 2020. In 1900 that figure was 15% and 34% in 1960. In 1960 most of the world's urban population (55%) was in high-income developed countries. That proportion has now fallen to 23% with more than 70% of urban people living in high-middle and low-middle-income countries. In every inhabited continent except Africa where 44% of the population is urban, the majority of people are urban dwellers. China which contains more than half of the population of high-middle-income countries has gone from 20% urban in 1980 to 64% urban in 2020 with more people moving from the country to cities in the 1990s alone than moved from country to cities in absolute numbers in the whole of Europe in the 19th century. But, and this is a very big but, cities are very vulnerable to climate crisis/climate catastrophe. We pay attention to a developed complexity-informed literature which seeks to understand cities in terms of networks and processes rather than fixed structures but we also note the dependence of cities on networks which connect them with the rural and the maritime in relation to sources of food and energy. The vulnerability of cities when those connections are cut is illustrated by the experience of Leningrad during the Nazi blockade from September 1941 to January 1944. Of the original population of the city and its suburbs of 3.5 million, nearly 1.5 million died from starvation and cold. There was a large scale evacuation of women and children (1.4 million). Piskaryovskoye Memorial Cemetery in Leningrad/Saint Petersburg holds half a million civilian victims of the siege alone.

The shift from a rural, largely subsistence peasant global population to urban life has been very rapid. It is a phase shift, a qualitative shift in the form of the global human social system. Both of the present authors have ancestors who were subsistence peasants in Ireland who moved to the North East of England, one of the world's oldest industrial regions now a post-industrial region, in the 19th century. That region was based on carboniferous capitalism – on coal – and it is precisely the dependence of capitalism on carbon which is generating climate crisis. Much of global urbanization first in the West and then in the 20th century in China and the Global South was a consequence of industrialization, but what will happen in a world already present in much of the former advanced industrial countries and developing in China – now the workshop of the world – where a mix of increased productivity and global relocation to ever lower-income countries generates not only an economic process of deindustrialization but a profound cultural change?

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In the first edition we began with an examination of how ideas of complexity have been used in attempting to understand 'socio-ecological-technical' systems. This is where ecological science with its foundations in ecological biology has its necessary encounters both with the social sciences as disciplines and with processes of governance. If there is any absolutely clear demonstration of the fallacy of a distinction between the natural and social sciences and of the value of Khalil's argument that we should instead distinguish between understanding directed towards the real and the artificial (1996b) then it is in this field. The anthropomorphic consequences of the human social in relation to the whole of the rest of the eco-sphere are so profound that this reality is effectively right in our intellectual faces. We incorporate our review on this domain into our discussion of the urban and focus on global socio-ecological-technical systems through that lens. We deal with health, a domain in which the complexity frame of reference is assuming a dominant position in science, scholarship, policy, and increasingly, in practice, in the next chapter through a discussion of health systems as they reacted to the systemic crisis, not only for health but for the whole social order, of the COVID-19 pandemic. We have covered the extensive public policy literature, both academic and grey, on the complexity frame of reference when addressing both governance in general and the application of complexity thinking in evaluation, and we have presented our review of the complexity-deploying organization theory and management literature there because it seems best to relate the literature to the actual practice.

Complexity in disciplines

Some important areas and disciplines receive fairly minimal attention here. These include anthropology because most of the methodological issues relevant to that discipline overlap so substantially with sociology's concerns. History is absent because although there is no doubt that a complexity take on the narratives of history (with that term understood to cover both historical reality and history the discipline) is essential, most of the historical work of interest has been done at the intersection of historical sociology and social theory. The omission of psychology is a much more substantial issue. We do not deal with that discipline here primarily because most of the complexity-informed writing in it has been driven by a neuroscience agenda and tends very much towards an intrinsically reductionist conception of how to understand people and their lives. To a considerable extent, we are writing scientistic psychology out of the social sciences which might be just what many contemporary academic psychologists want anyhow. That said we must draw attention first to Castellani's (2019) *The Defiance of Global Commitment*, which develops a social psychology approach to the aspects of globalization which manifest themselves in terms of existential discontents. This book develops an approach based on an engagement with Freud's *Civilization and Its Discontents* to explore issues including Brexit and new populisms and demonstrates how the complexity frame of reference can reinvigorate social psychology as a discipline which engages with the other social sciences from which the scientistic turn has separated it. We must also note that the complexity frame of reference has had an influence on the practical areas where psychology engages with mental health policy and practice, domains in which scientistic psychology has had a minimal impact. Also, in this edition, in contrast with the first, we have not reviewed the role of the complexity frame of reference in relation to the field of education. There is a great deal of complexity-informed work in that field and for foundational examples we refer readers to the first edition. Education is a domain in which complexity has now influenced not only philosophical and methodological thinking about what education is and how it can be researched. Complexity has also deeply penetrated into discussions of educational policy and pedagogical practice. There is a great deal of good work out there and rather than attempt a brutal and superficial summary we have chosen to exclude education this time around.

Our treatment of economics, a discipline whose contemporary foundations in neoclassical microeconomic theory and dynamic stochastic general equilibrium models for macroeconomic theory are fundamentally challenged when the complexity frame of reference is brought to bear, focuses in this edition primarily on the socio-ecological domain in which conventional thinking in economics has been confronted by hard reality. Economists have engaged with the issues, often by turning back to a traditional conception of the discipline that problems can be solved by the allocation of prices as a means for managing markets in everything. Of course, this depends in turn on the (vain) glorious assumption that markets will result in optimal equilibria. The whole foundation of understanding of complex systems indicates that this is not the case for any far from equilibric complex system with a capacity for qualitative change. There are interesting examples of efforts to get beyond this in the 'ecological economics literature' and even a dedicated journal *Ecological Economics* which has published a number of articles informed to varying degrees and in different ways by the complexity frame of reference. There are two interrelated ways in which this literature has developed. One is by introducing ideas and metaphors from evolutionary theory, exemplified by Gual and Norgaard (2010) with an interesting development of the idea of co-evolution. The other is a turn towards micro-emergence and agent-based modelling/network methods represented by Wam (2010). Hodgson, one of the most important proponents of the institutional turn in heterodox economics, argues that the 'meta-theoretical' framing of Darwinian conceptions in relation to evolution

> are of particular relevance for ecological economics. Much of the policy agenda for ecological economics is a matter of appropriate institutional design to establish incentives that are consistent with environmental goals, to process flows of information to guide policy, and to deal with unforeseen disturbances and shocks.
>
> (2010 704)

As with Room's work (2011), we find the understanding of institutions being asserted as central to the understanding of complex systems with any kind of social component.

There has been a good deal of complexity referencing work since the publication of the first edition of this book, including not only academic papers and books but also an important contribution by the OECD (2017), much of which builds on Arthur's seminal work in association with the Santa Fe school[1] (2021). The significance of the OECD publication in their insight series Debate the Issues: Complexity and Policy Making is that the notion that equilibric approaches work at all in relation to the issues facing the world order in the 21st century had been pretty well abandoned. This text is related to the New Economic Challenges Initiative which involves leading Central Bank economists, one of whom, Haldane, from the Bank of England, is a contributor. In it Gurria, the Secretary General of OECD, says it is time to 'stop pretending that an economy can be controlled' by which assertion he means that the tools of conventional macroeconomic management do not work in times of radical phase shift and crisis. We agree although we do not agree that control is impossible, just not within a social order where market logics and corporate interests predominate over socio-ecological objectives. The ecological part of that is very important. Much of this writing (see, for example, Potts 2017, Kovacs 2019, Elsner 2017) is useful in an expository sense but generally it fails to set the economic sphere in the wider social sphere where other systems matter at least as much.

Heterodox economists do engage with complexity. Examples are Foster (2005), Brunner (2008), Fontana (2010), Koppl (2010), and Matunovic (2010). All these articles take a literary form in which equations, although not wholly absent, are essentially subordinate to textual development of argument. This is a move towards interpretation as a proper mode of scientific discourse and Koppl notes that in the history of economic thought even such a proponent and to some degree originator of the neoclassical paradigm as Robbins engaged with *verstehen*, interpretive understanding, and it was central to the thought of Hayek. Koppl endorses the re-emergence of 'literary economics' in language very similar to that deployed by Keynes's biographer Skidelsky (2012). Complexity treatments of economic systems have some way to go and certainly could fruitfully engage with the non-mechanistic elements in Marxist thought in developing an understanding which gels with the complexity frame of reference. However, things are moving and we may even soon see a dissolution of economics into an interdisciplinary complexity-informed social science which will actually have a good deal in common with classical economic thought as it was framed in the tradition of political economy.

Most of the complexity-informed writing which self-identifies as 'economics' remains at the level of theoretical and methodological exposition and argument. There are relatively few examples of actual empirical research which deploy economic concepts and arguments. Indeed the examples we have located are work done by authors who, while engaging with issues central to disciplinary economics, come from related disciplines or interdisciplinary fields. Room (2011) is one. Another is Haynes's *Public Policy Beyond the Financial Crisis* (2012) published as part of a series of 'critical studies in public management'. Haynes's work is fully informed by the complexity frame of reference but he also brings to bear the turn towards case-based methods proposed as a complexity-sensitive research strategy proposed by Byrne and Ragin (2009) and deploys all of numerical typologyzing (cluster analyses), Qualitative Comparative Analysis, and narrative historical examination in a cross-national comparative study exploring the trajectories of the crisis, causal factors in context, and different responses to it. This book is perhaps the most important complexity-informed empirically founded study of the economic and financial aspects of public policy published in recent years. Indeed the phrase 'complexity-informed' is redundant here. It represents a very powerful counterblast against the neoliberal arguments for globalized marketization as a solution to the crisis when it is plain that this ideological (certainly not scientific) paradigm was causal to the crisis in the first place. Haynes proposes a new form of public service provision based on contextually sensitive local management, public engagement, and mutual financial forms rather than 'public–private partnerships' engaging big state with big capital. And he has real calibrated evidence to support his arguments. Economists have much to learn from this kind of approach.

That gives us a good bridge forward to the implications of complexity for the discipline of economics itself. Balinta et al. (2017) present an extremely valuable survey of the literature in ecological economics. The abstract to their article explains what they are doing and why:

> Climate change is one of the most daunting challenges human kind has ever faced. In the paper, we provide a survey of the micro and macro economics of climate change

> from a complexity science perspective and we discuss the challenges ahead for this line of research. We identify four areas of the literature where complex system models have already produced valuable insights: (i) coalition formation and climate negotiations, (ii) macroeconomic impacts of climate-related events, (iii) energy markets and (iv) diffusion of climate friendly technologies. On each of these issues, accounting for heterogeneity, interactions and disequilibrium dynamics provides a complementary and novel perspective to the one of standard equilibrium models. Furthermore, it highlights the potential economic benefits of mitigation and adaptation policies and the risk of under-estimating systemic climate change-related risks.
>
> (2017 252)

The references to this article cover the domain of complexity-informed economic work on climate crisis up to the time of its writing. One very important point made by these authors is that the equilibrium models which are the foundation of most integrated assessment models (IAMs) convey a false impression of ability to control by macroeconomic manipulation taken alone. Balinta et al. are very keen on the development of agent-based models. We certainly agree that ABMs are a useful part of the toolkit of policy development but we must always remember that they are inherently micro-emergent and that all the criticisms expressed by Room (2011) which derive from the implications of the need to consider not only micro actors but institutions *and* the structural terrain on which all social action takes place do apply to the deployment of such approaches taken alone.

Understanding the urban and shaping it

In attempting to understand urban systems in the early 21st century after COVID-19 and in a context of impending climate crisis, it is helpful to begin by looking at what ecologists have to say about social systems. Norberg et al. make an explicit connection between ecological and social systems:

> Many systems, ecological, social and social-ecological, have multiple inherent feedback processes such that they experience several self-reinforcing states of attractions, called regimes. Shifts between such regimes can occur when the system is pushed across some threshold by external or internal dynamics. … Note that the definition of regimes differs from how it is used in the social sciences, where regime change frequently refers to the political actors or system in power. Resilience is defined as the capacity of a system to absorb disturbance and reorganize while undergoing change so as to retain essentially the same function, structure, identity and feedbacks, that is, remain within one regime. Hence resilience reflects the degree to which a complex adaptive system is capable of self-organization … and the degree to which the system can build and sustain the capacity for learning and adaptation. … We can define adaptive capacity … as the capacity of the system to alter the relative abundance of its components without significant changes in the crucial system functions. Diversity thus plays a central role for resilience.
>
> (Norberg et al. 2008 46)

There is much to agree with here but there is one point we would qualify. The term ‘regime change’ in the social sciences is ambiguous in use but sometimes does refer to changes on a scale which we would consider as phase shifts and therefore as equivalent to the ecological use of the concept. The transition from Stalin to his successors in the Soviet Union was important and dramatic but there was no fundamental social reorganization despite the reduction in significance of terror as a basis for the system. In contrast the collapse of communism in 1989 subsequently really did generate a regime change in a qualitative sense. Things are very different afterwards. We are dealing with a transformation of quantity into quality. China has undergone a regime change in terms of economic and social organization but not political power. In many ways the rule of the Chinese Communist Party might be described as Confucian more than Marxist. There is a strong central power with considerable regional autonomy but the absolute emphasis is on the maintenance of a consistent social order. The economy has been transformed but the political system remains intact although under challenge from social changes and in particular the creation of an enormous industrial proletariat and the demographic implications of the one-child policy which in many ways has resulted in a one-child cultural norm which persists even after the restrictions have been effectively abolished. The conflict between democracy advocates in Hong Kong and mainland power illustrates the Confucian nature of the mainland system. Order prevails over democracy and over an independent rule of law.

That reservation aside, there is a great deal of value for us in Norberg et al.'s specification. In particular, the emphasis on the role of diversity, equivalent to internal differentiation, as a basis for resilience is especially helpful as is the formulation of the idea of adaptive capacity in consequence. The discussion of socio-ecological systems is of the most profound importance in relation to global warming and climate change. What is true of the pond is true of the climate.

Another resonance in the socio-ecological literature is a discussion of the role of narrative, not just as a basis for scientific representation but as a constitutive process in relation to the systems themselves – reflexivity in action:

> Particular system-framings often become part of narratives about a problem or issue … There are simple stories with beginnings defining the problem, middles elaborating its consequences and ends outlining the solutions. Narratives are created and promoted by particular actors, networks and institutions. They often start with a particular framing of a system and its dynamics and suggest particular ways by which these should develop or transfer to bring about a particular set of outcomes. Narratives therefore suggest and justify particular kinds of strategies and interventions.
>
> (Leach et al. 2010 371)

What we want to develop is a complexity-informed discussion of how urban systems might develop in the possibility space created by the interaction of all of climate crisis, pandemic becoming endemic disease, and the crises of political and cultural legitimacy engendered by globalization and neoliberalism. The socio-ecology literature takes the nature of urban spaces and their relationship with the natural environment as one of, perhaps the most important after global warming, central concerns. However, there is a heuristic value in separating urban geography and (to a lesser extent) urban sociology's engagement with complexity and examining both in relation to the way planning has deployed the frame of reference.

Grove et al. (2019) reference complexity theory in an engagement of political geography with design practice. They go back, as is always sensible, to Simon who in essence described design as the general process of formulating intentions to act to achieve objective. They argue that:

> Design orients thought and action not towards questions of how something came to be, but rather what something might become, crafting new futures from within, rather than outside, the present. These 'designerly' sensibilities are overtly and covertly reconfiguring how human-environment relations are studied across all scales, attuning urban governance around the world to a technocratic variant of the Anthropocene and to the 'urban age' promulgated by complexity theory. In the process, they are opening onto an indeterminate political field over the possibilities for knowing, governing and contesting human-environment relations. On one hand, the post-historical impulses of resilience and smartness govern the urban as eco-cybernetic systems, discarding utopias and declaring politics and planning as obsolete. On the other hand, critical theorists have increasingly sought to decolonize design and deploy designerly strategies and techniques towards political projects that create new modes of collective becomings.
>
> (2019 1)

Well, up to a point, Lord Copper. It is certainly true that complexity theory can be deployed in a technocratic fashion which often accords with neoliberal theories of urban governance and the practical interests of the immensely powerful lobby based on real estate capital. But that is not the only way in which it can be employed. And frankly critical theorists' decolonization efforts have minimal practical application. Parnell and Robinson (2017) in contrast deploy complexity as part of an engagement with the specific character of African mega-cities in an interesting and useful fashion.

Many contributions of that kind in geography are essentially conceptual and we find similar work in academic planning, for example, Van Wezemael's chapters in both de Roo and Silva (eds) (2010) and de Roo et al. (2012).[2] There is a range of literature which takes up the implications of the discussion of complexity, informed by her and Jane Jacobs's participation in the Macy seminars, in *The Death and Life of Great American Cities* (1963). Perrone (2019) argues for a street-level epistemology in terms which sound a lot like Freire's arguments for dialogical research. Wong (2020) draws on a range of literature dealing with the notion of the sustainable city

> to demonstrate how the trope of urban complexity is mobilized to project the city as a generic scalable entity of creativity and energy efficiency. It becomes the basis of an infrastructural imaginary of neoliberal innovation that supports entrepreneurial and 'smart-eco' agendas of urban design and governance that promise – but have yet to deliver – planetary ecological amelioration.
>
> (2020 1)

We agree. The complexity frame of reference is politically neutral unless it becomes complex realism and recognizes the generative power of the logics of capitalism in the era of the Capitalocene. Kupers (2020)[3] articulates a market-focused approach to 'what the science of complexity reveals about saving our planet' – to addressing climate catastrophe. It would be excessive to describe this as essentially neoliberal but it still places undue reliance on entrepreneurial initiatives and incremental exploration when what is required is big intervention in a clear frame. Kupers is right to assert the significance of culture in relation to this and note the cultural importance of coal but in Britain deep coal mining was wiped out in 20 years to defeat the most organized part of the British working class. Much of our own work has been in relation to the consequences of that transformation but it happened and is a crucial reason for CO_2 reduction in the UK economy. Appalachia is certainly culturally attached to coal but as deep mining, not ecologically disastrous hill top removal. There is a culture war to be fought around coal and its ending but it requires state intervention in alternative employment as was done with considerable success in NE England in the 1960s by a One Nation Conservative Prime Minister, Harold Macmillan, and a real social democrat, Harold Wilson.

Crawford (2016) is an example of citation in the grey literature prepared for policy development. There are two strands to this pattern of citation. One is simply, as with Crawford (2016), to use it as a way of arguing for a complexity frame of reference. The other is to pick up on Jacobs's political commitment to bottom-up engagement with planning decision-making. There is also of course reference to the irony that the lower-middle-class/working-class neighbourhoods in Manhattan which Jacobs sought so eloquently to defend are now some of the most expensive real estate on the planet. The same is true in other global cities and even in post-industrial cities. David Byrne worked hard with his colleagues in the North Tyneside CDP against demolition of parts of riverside North Shields but while these areas are not Manhattan they have certainly become gentrified in recent years. A good overview account which relates to many of the pre-COVID issues for urban planners is provided by Fernández Guell (2017), who argues that since urban systems comprise multiple intersecting complex systems,

> Confronted with these circumstances, planners should be capable to develop more qualitative and friendlier conceptual frameworks in order to interpret city's complexity in a more holistic way and, at the same time, facilitate effective involvement of local stakeholders along the planning process. … planners should develop innovative research approaches capable of handling fuzzy interconnected problems with the aid of collective intelligence and qualitative tools, instead of trying to solve clearly defined problems through standard quantitative tools. These approaches should respond creatively to fuzzy agendas based on a process of evolving collective intelligence.
>
> (207 2.2)

Alberti et al. (2018) provide a neat summary of the issues facing planning in the context of the Anthropocene although they do not continue to identify the Capitalocene:

> The increasing pressure from climate change …, rapid urbanization …, and the rapid development of infrastructure to prepare for these changes all pose new challenges to urban decision-makers to make important investment decisions while navigating complexity. … Tackling complexity and uncertainly in urban systems is challenging and will require new evidence, approaches, and tools. It will demand a new level of collaboration among ecologists, geographers, sociologists, political scientists, economists, planners, designers, and other disciplines to advance the field of urban ecology into a new urban science … To meet this demand, scholars will need to be able to identify examples of new practices that highlight opportunities for improving urban resilience and sustainability at the local and global scales.
>
> (2020 66)

We will return to this vital set of issues in our discussion of the possibility space for the near future.

We have already mentioned in relation to our discussion of simulation the work of both Batty and Allen as pioneers of complexity-informed urban simulation. What is particularly refreshing about the urban simulation literature is that it both is *not* imaginary, in other words it is calibrated in relation to real data, *and* that those writing it engage with the wider methodological debates surrounding the use of the complexity frame of reference. So Batty (one of the great urban simulators) presents an informed and interesting discussion of the nature of urban modelling (2010). Silva (2010a) goes through an empirical study deploying calibrated cellular automaton (CA)-based models in a way which both thoroughly explicates the CA approach and engages particularly with the potential or otherwise of complexity-based modelling for projection into/prediction of futures. In general, we find that the applied, i.e., the planning, literature is more engaged with and properly informed by complexity, which perhaps reflects the impact of engagement with reality. There is an excellent historical account here (see Silva 2010b) which sets planning theory and practice in relation to complexity in a most helpful way. There are examples of academic research in which complexity is deployed on the basis of conventional qualitative research findings to explicate a particular case study of the development of 'the urban creative economy' (Communian 2011). Wilkinson (2012) presents a fascinating account of an abortive attempt to develop a complex understanding of the basis for planning the Melbourne city region where there was a disjunction among all of the planning directors who seemed obsessed with non-planning, modellers who could not engage with reality outwith their models,[4] and research and policy officers who insisted on doing exactly that. So we have an interesting conceptual development here, and in particular Portugali's (2006) discussion of complexity theory as a link between space (a physical concept) and place (a social concept), a sustained and informed discussion of modelling approaches in relation to complex systems, and an increasing range of empirical work. In his *The New Science of Cities* (2013) Batty identifies cities not as places in space but as systems of networks and flows. Space for him is as much, or perhaps we should say even more, about flows as structures and it is crucial to grasp the nature of the networks which carry the flow and connect the elements which constitute urban systems. This is very much a complexity-informed account which argues for and demonstrates the use of simulation approaches as the basis for design and decision-making models in terms which resonate very much with network accounts of governance processes. This more technical book is complemented through the embedding of his modelling approach in an interesting account of *Inventing Future Cities* (2018). That said, despite an awareness of the issues of climate crisis and even inequality, much of the modelling literature seems reluctant to engage with aspects of urban life not represented in data. If we are going to invent future cities then we need narratives of the future – scenarios – as well as models.

One characteristic of the empirical work is the use of case studies – case studies which are underpinned by all of the modes of social scientific investigation ranging from complex non-linear modelling to traditional interview and document-based qualitative research. While there are some examples of comparison, these tend to take the form of comparing and contrasting a limited numbers of cases, for example, Silva's modelling-based comparison of Lisbon and Oporto. It seems to us that there is a very considerable potential for more use of systematic comparison across a range of cases using the kind of case-based methods proposed by Byrne and Ragin (2009) and deployed by Haynes (2012).

And now to what is to be done now. It is not conventional academic practice to prescribe in writing which is reviewing a field of work. However, we are going to follow Paul Cilliers's approach to the ethics of complexity both in terms of addressing the future of complex systems from within them and in relation to the Greek concept of praxis – knowledge exists not only for its own sake but when it exists the implications of it have to be acted on. So what does the complexity frame of reference have to say which might enable us to confront the world of the early 21st century in which pandemic becoming endemic disease, increasing inequality, and impending climate catastrophe are all interacting with particular force for those who live in cities?[5] Ruth and Coelho identified the issue for us well:

> Climate change is increasing the pressures on many urban systems and adding to this complexity. Many of the case studies investigating urban dynamics in the light of climate change have chosen narrow, sector-specific approaches. Few projects have built on insights from complexity theory and related bodies of knowledge which are more consistent with the perspective that urban infrastructure systems are tightly coupled with one another and must respond to often subtle, long-term changes of technological, social and environmental conditions.
>
> (2007 317)

Urban systems of all kinds – social, cultural, economic, political, infrastructural, physical – are not only tightly interwoven with each other but also with all systems surrounding them, both social and natural. Both of us were brought up in port cities and had grandfathers who were seamen. If you live in a coastal port the sea is always present in so many ways.

One of us has written a book exploring how we might cope with *Inequality in a Context of Climate Crisis after COVID : A Complex Realist Approach* (Byrne 2021) and much of what follows draws on the argument of that text in relation to its specific urban implications.[6] For us the most important political implication of the complexity frame of reference is that it allows us to imagine the multiple but not infinite futures for social/natural systems which exist within the near future possibility space and are path dependent on the present condition of those systems. Rosa Luxemburg just over one hundred years ago had argued that the future would be 'Socialism or Barbarism'. The journal *Climate and Capitalism* has modified this to 'Eco-Socialism or Barbarism'. Actually we think that it might be better expressed as 'Disaster Capitalism or Social Democracy' and by that we do not mean the neoliberal version of Third Way, the so-called 'social democracy' described by Byrne (2005) as neoliberalism with a smiley face. Rather there seem to be possibility spaces in which human civilization survives either as one in which collectivist democratic processes and principles dominate the power of capital or capital continues to rule unabated but adapting to crisis in the way in which in England and Scotland (but not Wales) massive profits were made from the COVID-19 pandemic by private providers of health goods and processes. There are other more apocalyptic scenarios available, one of which is well laid out in Roanhorse's novels of a Navaho world in which much of North America and Europe is under water and the old gods and monsters have returned. Actually this is not the most apocalyptic – that would be Venus.

The key problem for cities is the enormous power of real estate capital and real estate lobbies. It was industrial capitalism that created the Capitalocene, although of course construction is an enormous generator of CO_2 and up to now a key part of urban accumulation has been through the rents charged on retail outlets which sell consumer goods with a high carbon content in their production and distribution. In high-income countries, real estate is often the most important generator of gross value added. Of course that value is measured in exchange value rather than use value terms and much of it comes from speculation and inflation in pre-existing real estate assets in what Lefbvre called the secondary circuit of accumulation. In confronting both inequality and climate catastrophe a key opponent of those seeking a good outcome is the organized interest of real estate capital. Sure fossil fuel industries matter a great deal in relation to climate catastrophe, and no element of capital is not intersected with most others, but real estate is a massive political player at the level of all cities. In the UK, or at least in the nations other than Wales, real estate interests dominate urban planning and Labour-controlled local authorities fall over themselves to give developers exactly what they want, up to and including involvement in the privatization of municipal housing, although the actors there are often the now very highly paid managers of the new forms of housing management imposed by New Labour.

One very important thing to remember is that in world history cities were never able to replace their own populations until the development of high levels of sanitation including clean water provision from the mid-19th century onwards. The impact of urban living conditions meant that infant and child mortality often exceeded 50%, there was a high death rate from in particular tuberculosis throughout young adult life, and the physical condition of urban populations was such that at the beginning of the 20th century half of all volunteers – volunteers! – in Britain for the Boer War were wholly unfit for any form of military service. Rural Ireland provided a massively disproportionate part of the UK military until the introduction of school meals and medical inspections improved the physical health of British urban populations. Endemic diseases in childhood were the big killers but urban populations were also subject to epidemics and pandemics. The 14th-century Black Death killed a third of Europe's adults but was particularly severe in towns so much so that in England a predominantly Norman urban population seems to have been replaced by rural Anglo-Saxons. Urban life is a fragile way of living.

The other driver of change was industrialization. Jane Jacobs (1969) was quite correct in pointing out that much of the economic activity of cities was in locally generated and consumed services but it was the demand for factory and port labour that drove up urban populations. An extreme example in Europe is Middlesbrough on Teesside in the NE of England, still one of the most industrial places in post-industrial Britain, where the population increased from some 30 people living in a hamlet to more than a 140,000 between the 1830s and 1900. The growth of the industrial cities of the US Midwest, late Czarist and Soviet Russia, and industrializing China was even more dramatic. Shenzhen became a megapolis in 30 years where before there were only paddy fields. In Katowice in Poland, in many ways an even larger Middlesbrough, the workers were housed in blocks modelled on Owen's New Lanark, which was good housing in the 19th century, and which closely resemble barrack blocks for the military. In the post-industrial era, things are different. In Greater Birmingham in the English West Midlands, once literally the workshop of the world, in 2019 of the nearly one million workers just under 10% worked in manufacturing whereas 13% worked in human health and social work activities. The largest sites of employment in most post-industrial cities are the hospitals and the universities – education provides 9% of jobs and public administration provides 4% for a total for the public sector of 26%. Manufacturing provides 14% of gross value added, the public sector provides 19% but real estate with 2% of employment provides 13% of

GVA – this in what remains one of the most industrial city regions in the UK. Health and social care are among the most important components of the real economy (as opposed to imaginary economies of real estate and financial speculation) in the post-industrial world.

If there is to be a resolution to climate crisis, urban life will have to change. So of course will much of rural life, particularly modes of agricultural production. One important change is already underway. Globally birth rates are falling. In high-income countries they are already below reproduction rates and this is very nearly the case for high-middle-income countries. There are massive fiscal consequences of this given the great increases in the proportion of the population who live into retirement and then dependent old age but this is a major change and it is concentrated in urban populations. One massive change has been in the actual amount of territory taken up by urban space. Suburbanization has extended the size of urban areas and impinged on agricultural land. Suburbs developed originally on the basis of urban rapid transit systems but are now dependent on cars with many households possessing multiple cars to enable in particular couples to go to their respective places of work. All of this requires social interventions which challenge not only the logic of growth but the power of real estate capital. It is extreme to say this but we wonder if the urban forms of the Capitalocene can persist into a post-carbon era.

Conclusion

To repeat our assertion in the introduction to this book: complexity is out there, people are using it, and they reason they are using it is because it makes sense of the real social systems they are examining as social scientists. Reading over the discipline and field literatures as a basis for writing this chapter has been a very useful exercise for us. It has helped us to grasp just what people are doing with the complexity frame of reference. We can identify the following ways in which it has been employed, and we have to say immediately that this typology is not a typology of pieces of work since many, perhaps most books, chapters, and articles actually do more than one of these things and many do them all:

- Contributions argue about the meta-theoretical appropriateness of complexity as a way of addressing their discipline and/or field.
- Contributions discuss the methodological implications of the complexity frame of reference and consider how this might be developed as a foundation for actual empirical work.
    - A subset of contributions assert a micro-emergent account of the social and translate this as implying that agent based modelling is the appropriate method for understanding complex social systems.
- Contributions consider the implications of the complexity frame of reference for practice in the domains which are informed by their particular discipline or more often field's relationship with the application of social science.
- Contributions use the complexity frame of reference to make sense of the findings of an empirical research project. Note that this does not generally, and perhaps should never, imply that complexity informs the design of the project. Rather complexity informs the interpretive strategy for dealing with the research products and enables the researcher to understand what they are dealing with and how systems came to be as they are. We would argue that systematic comparison provides a basis for developing predictive capacity in relation to outcomes for near neighbours, a point we will

develop in the next chapter, but this is a relatively uncommon approach as yet in the literature.

- Contributions assessing the inter-relationship of the natural and the social in relation and assessing sustainability in a context of impending climate crisis.

So complexity is being used. Since we consider that complexity has a great deal to say to how we collectively can confront crises, in the next chapter we will address how the complexity frame of reference is being deployed in the fields of health and social care in relation to the impact of the COVID-19 pandemic and probable endemic continuation of the massive system disturbance this has engendered.

Notes

1 Although Arthur is rather open minded and often strays into the territory of general complexity. The 2021 article in *Nature* is an excellent summary of his thinking.
2 These are particularly clear expositions in relation to DeLanda and Deleuze's arguments.
3 This book includes some useful and interesting case studies.
4 And this despite being told to do exactly that by a visiting professor from Cranfield in the UK who sounds very much like Peter Allen and that is just what Allen always does assert.
5 Both Byrne and Callaghan live in small villages although we are wholly products of urban industrial life. Byrne is probably just high up enough to be beyond sea level rise. Callaghan who lives on a rock sticking out into the sea is not.
6 This follows on two previous books in which the complexity frame of reference has been used to inform that approach: Byrne and Ruane (2017) on taxation and paying for the welfare state and Byrne (2019) on class after industry.

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15 Beyond disciplines and fields

COVID and a world in crisis

In the history of the human race since the agricultural revolution and the development of urban forms of living, COVID-19 is not that big a deal. It is a nasty and often serious condition. However, it primarily kills old people past the reproductive phase of their lives and the mortality rate across generations is far lower than the Spanish flu of the early 20th century or the Russian flu of mid-19th century. And yet COVID has had a far more marked impact on social systems than either of those, although not a system-transforming impact as with the 14th-century Black Death in feudal Europe. The reason is the vastly greater significance of curative healthcare as a component of the social system and the set of expectations in relation to health outcomes which have developed during the welfare capitalist second half of the 20th century. Through the development of sanitation and public health programmes of immunization, all high-income and high-middle-income countries and most low-middle- and low-income countries have undergone a health transformation of enormous demographic significance. For human populations before this, and especially for urban populations, the main causes of death were infectious diseases, particularly in infancy and childhood but also through the life course. Now, the great majority of people live into old age and many of us into extreme old age. Most of us die from degenerative diseases – cancers, coronary vascular diseases, even dementia itself.

Curative medicine played a minor role in the health transformation. However, its role has expanded dramatically and taken an increasing share of an increasing social product primarily in the treatment of systemic diseases of older people – cancers, coronary vascular diseases, type II diabetes, arthritis, and similar conditions. There is a social sentiment of entitlement to treatment and intervention. Curative medicine in tax-funded health systems has that responsibility. Fever/isolation hospitals, common in the early 20th century, where highly infectious patients could be nursed separate from the general body of the ill, were closed as the potential impact of pandemics was forgotten. So COVID-19 has had a very dramatic impact on the social order given popular expectations of effective treatment. Some policy responses – limitations on social interaction, encouragement of social isolation, and restrictions on travel – would have been familiar in 5th-century BC Athens. The rapid – and effective – turn to immunization drew on a key part of the repertoire of public health since Jenner developed vaccination against smallpox. While COVID is not a particularly deadly disease, it has a high degree of human-to-human airborne transmission, so in the context of the social orders of the early 21st century its impact has been enormous.

We begin by attempting a delineation of 'health' understood after Reed and Harvey (1992) as a set of intersecting and nested systems. The World Health Organization defines health as 'a state of complete physical, mental and social well-being and not merely the

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absence of disease or infirmity' (1946 100). When we find the word 'state', we must think of this in relation to the state of the systems in a possibility space. Although the WHO document goes on to talk about the health of collectivities, the starting point is the health of the individual, albeit of the individual embedded in a social context.[1] However, programmes of public health have a different focus. There were public health programmes in pre-modern society but public health became crucial in response to the 19th-century transformation of societies from being predominantly rural and agricultural to being predominantly urban and industrial. This transformation had an enormous demographic impact because prior to the development of effective systems of public health provision and regulation no urban system could reproduce its own population So public health was concerned not with the health of individuals, although the measures had very considerable implications for individual health, but with the health of the herd.

Historically, health practices revolved around interactions between patients and practitioners. Public health measures brought into play systems of infrastructure, crucially the delivery of clean water and the removal of human wastes, alongside the programme of immunization. While immunization protects the immunized individual, the public health concept of 'herd immunity', that is, the level of immunization achieved across the population necessary to prevent the development of an epidemic, describes a whole system property. A crucial factor extending massively post-World War II was the development of a system of health institutions within which healthcare is delivered. The key institution is the hospital but others matter and may even replace the hospital as we understand it. While in primary healthcare the physician often, although less often than in the past, deals directly with the patient on their own, in secondary care the old complex of physician/surgeon and nurse has been extended to include a range of other paramedical professionals and semi-professionals and their instruments and tests. Moreover, these systems are intersected with complexes of management and financial arrangements. Health is a massive set of systems and we need to demarcate them.

We propose the following specification for the set of nested and intersecting systems:

1. The individual human being: that is to say, the body and mind of a single person. Each of us is a complex system in terms not only of our physiological functioning, important though that is, but also as an ecosystem. There are more micro-organisms in our body than there are cells of that body.
2. The system of the health of population as a whole – the system of public health which has a massive intersection with the urban system.
3. The systems of health care delivery.
4. And let us not forget the relationship of the human species with the health of the global ecosystem – we are after all a pathological condition so far as Gaia is concerned.

Note both the emphasis on nested and intersecting systems and that we absolutely reject any notion of a hierarchy of causal direction. All these systems are interrelated and have caudal powers in relation to each other.

The patient as a complex system

We begin with thinking about the patient, the individual human being, as a complex system. Our discussion is informed both by the general complexity frame of reference and by the application of that frame of reference to the human being who presents to the health system as ill (see Sweeney and Griffiths eds. 2002). The WHO definition of health is plainly holistic. It describes health as more than the absence of an externally verifiable pathology, the way in which health is understood by the biomechanical/biomedical model. Pre-scientific medical conceptions all worked in relation to a notion of balance across aspects of the functioning of the human system so we might challenge the word 'medical' too. However, mechanical is unequivocal. That framing sees the body–mind complex as a machine with parts that can go wrong and require fixing. The extreme development of this, which works where it works,[2] is the development of magic bullet drugs. For infections the magic bullet kills the disease without killing the patient. This model has been extended, with the enthusiastic support of Big Pharma, the commercial legal drugs industry, to other areas of illness including cancer with some success and mental health with much less success.

Consider the patient who comes to see the primary care physician, in the UK the GP.[3] The specialist traditionally sees a patient with 'a' – i.e., single – condition, although the specialisms of paediatrics and geriatrics where the patients are defined in terms of age range work in principle in a similar way to the GP. GPs see patients who have not one thing wrong but often multiple and interrelated things wrong. This patient has to be understood as being in an attractor state of 'ill' and the objective of the duo of physician and patient is to move them to an attractor state of well. Here we understand health as ability to function in a 'normal' social fashion appropriate to age location. So well is not the absence of any of the above set of pathologies/health-damaging behaviours but rather the ability not to be in the sick role. The causes of presence in the attractor state of being ill are multiple and complex. The appropriate transformative interventions are likewise multiple and complex. Innes et al. (2005a) identified the consultation process – the initial encounter between any physician and patient and the core of primary care work – as a complex adaptive system in which patient and clinician develop an account which is itself emergent and which serves as the basis for the development of what is intended to be a transformative set of appropriate interventions.

Griffiths et al. (2010) developed this approach with a specific focus on 'tailoring medical interventions for the individual patient'. Griffiths and her co-workers were addressing the dilemma which confronts every clinician facing a patient – how can I translate evidence drawn from the summarizing of experience over a very large population of cases so as to do something useful for the specific individual I am dealing with here and now. For patients with complex and multiple conditions there is no aggregate evidence across the full description of their current state. Their state is an unpredictable emergent consequence of the interaction of those elements with everything else about them and their present condition in all senses. Griffiths and her co-workers developed the concept of an 'emergent present' from their examination and interpretation of the life trajectories of people living with chronic illness. They tried to identify and categorize individuals in terms of their current 'emergent present', which we can understand as current position within the possibility space of their life trajectories. The classification can in our terms be understood as an attractor space and the task of clinician and patient is to try to get to a better future emergent state, a possible attractor state which is better in health terms going forward. Sweeney's *Complexity in Primary Care* (2006) is a particularly interesting contribution to these debates. It describes a journey where a very experienced primary care physician engaged with complexity as a mode of thinking to inform both his practice and his research. The author's personal trajectory is explicitly part of the argument. Sweeney's book is academically impeccable but it is not academically conventional. Complexity seems to not only lead to people thinking in different ways and doing science in different ways. It also leads to a different and frankly much more accessible way of communicating particularly in relation to processes of application.

Complexity and public health

Pre-biomechanical medical understandings of the individual patient were explicitly founded on a conception of the patient as a system and implicitly on an understanding of that system as complex. Although clinicians have never wholly abandoned this way of thinking the biomechanical model's apparent triumph in the 20th century seemed to push it out of the scientific frame of reference. Public health is different. It deals with the overall emergent health state of populations. The historic focus of public health was on the impact of infectious disease on human populations. Public health is urban because its origins lie in the need for cities to confront deaths from infection, a real problem since the agricultural revolution changed our species' mode of existence from living in small bands of hunter-gatherers and permitted the accumulation of surpluses and the development of complex social practices and orders.

Rydin et al. (2012) examined 'Shaping cities for health: *c*omplexity and the planning of the urban environment in the 21st century':

> The Commission began from the premise that cities are complex systems, with urban health outcomes dependent upon many interactions and feedback loops, so that prediction within the planning process is fraught with difficulties and unintended consequences are common.
>
> (2012, 2079)

Much of the Lancet Commission report is devoted to examining the concept of transformation in relation to health. Here transformation is an exact synonym for phase shift in system state. A general theme in the report is the degree of health inequality which exists within urban systems. This is the product of increasing inequality in post-industrial urban systems. The phrase 'post-industrial' matters because what is being described is the health state of the populations of places whose urban character is precisely a product of industrialization. The changes are changes in economic base and changes in power consequent on the destruction of that economic base. What underlies these health inequalities is in large part the disorganization of an organized working class.

Many studies deal with health inequality in complexity terms. These now go beyond the retroductive and descriptive to include complexity-informed examinations of the impact of policy interventions in complex urban health systems. Blackman (2006) explored spatial health inequalities in urban systems alongside a careful consideration of the implications of area-based policy interventions designed to address them. Castellani in a comment on Blackman's book makes an important point:

> Toward the end of the book, Blackman points to one of the explicit ways that a complex systems viewpoint changes how one approaches improving community health. While the community-as-context model is a major step forward, it is, nonetheless, a top-down model. This means that is treats the citizens of a community as objects of treatment. This leads to top-heavy, public health—the kind that does NOT involve people in their own health improvement. The community-as-complex-system

> model, however, is entirely different. Because it takes a bottom-up approach, it begins, by definition, with an interactive (relational) view of people and their communities, looking at how both effect the other. As such, it follows an action research protocol—people need to be involved in the improvement of their health care and their communities, which in turn, impacts of health of these people.
>
> (Castellani blog, September 19, 2009)

Castellani and Rajaram have developed an innovative form of case-based modelling, which they distinguish from all of agent (rule-based) modelling, dynamical (equation-based) modelling, and statistical (aggregate-based) modelling. This is the SACS toolkit:

> a new method for quantitatively modeling complex social systems, based on a case-based, computational approach to data analysis. The SACS Toolkit is comprised of three main components: a theoretical blueprint of the major components of a complex system (social complexity theory); a set of case-based instructions for modeling complex systems from the ground up (assemblage); and a recommended list of case-friendly computational modelling techniques (case-based toolset).
>
> (2012b 153)

This is simulation deployed as part of a multi-method effort to understand how an existing situation has come to be, or in complexity terms we might say how the present state of the socio-spatial-health system of Summit County – their study area – has emerged, coupled with a specification of which subsystems are drivers for the emergence of this state – migration consequent itself upon urban sprawl. The method gives us some purchase in identifying the creators of the emergent present, if we can appropriate Frances Griffiths's very useful expression for use beyond the context of the clinical consultation. If we know what made the present, then we have some sort of hold on what might have an effect on the path-dependent future. This is the promise of simulation and this kind of grounded simulation, grounded in data and in qualitative establishment of context and process, has considerable potential.

Public health has always been interdisciplinary or even post-disciplinary in application. It has been as much a matter of infrastructure provision and regulation, the enforcement of legal constraints on food hygiene, and administrative organization and practice as of anything which has drawn on the paradigm of biomechanical medicine. Complexity can frame the whole range of that post-disciplinarity both in practice and in science, with science considered both as a foundation for practice and as a mode of general understanding of the relevant aspects of reality. The scientific practices of public health mix both and always have done so. Castellani et al. demonstrate the ability of post-disciplinary approaches which draw on a repertoire of methods drawn from all the styles of engaging with complex systems for addressing spatial understanding in relation to health describing:

> the second challenge that complexity science makes to the conventions of community health science: if places are complex systems, then new methodologies are necessary. Qualitative analysis or statistics alone cannot get the work done: qualitative analysis cannot handle the large, complex, multi-dimensional, multi-level databases typical of complex systems, and statistics alone cannot adequately model qualitative variables, nonlinearity, or system-level causality.
>
> (2012a 1)

Health systems as complex systems

There are several takes on health systems as complex systems. First, there is the question of what do we mean by a health system? In contemporary advanced societies this refers to forms of the delivery of healthcare and the administrative and financial processes and institutions which manage and fund that healthcare. We have primary healthcare – the first point of contact of patients with clinicians – which both treats and refers to secondary healthcare, specialist provision traditionally delivered in an institutional context by specialist clinicians although the dominance of the hospital as the institutional form is diminishing with the development of specialist units outside the traditional hospital framework, and tertiary care in the form of highly specialized units for treatment to which cases are referred from more generalist secondary care providers. This is a description of level and we have different forms of administrative provision and funding which engage with the personnel and institutions in the systems of delivery. Health care, as opposed to public health, is a system organized in relation to clinicians who treat on the basis of diagnosis.

Alongside this system organized around treatment is another intersecting system organized around social care. The natural history of many episodes of morbidity among the elderly is increased dependency and a need for care subsequent to treatment. Elderly patients are admitted and treated but cannot be discharged to their original social context because they now require higher levels of care. Social care systems tend to be based on private provision with some state funding on a means tested basis. The way in which health and social care can be related to each other has been the central issue for health systems management for some 50 years in the UK, primarily in relation to the elderly but also for those with mental illnesses or learning difficulties and those younger people with disabilities. Haynes (2007) and Haynes et al. (2006) have deployed the complexity frame of reference as a way of approaching this. There have been efforts at planning jointly but budgetary issues have been a major obstacle. In an effort to get beyond academic discussion Haynes and his colleagues at the University of Brighton developed a UK ESRC-funded Knowledge Transfer Project: *The Brighton Systems and Complex Systems Knowledge Exchange*. The output was a toolkit which can be deployed by managers in relation to issues emerging in wicked issue-riddled complex systems (Haynes, undated).

Complexity is everywhere in the health literature, both system related and clinical and with a complexity-based reconsideration of the proper relationships among research, clinical practices, and systems organization. Plsek and Greenhalgh (2001) wrote a seminal piece in the *British Medical Journal* (BMJ) entitled 'The challenge of complexity for health care' which did exactly what it said in the title. That challenge has been taken up in a range of approaches since the publication of the first edition of this book. Gommersall (2018) working in a clinical psychology mode reviewed both the potential of agent-based modelling and complexity framed qualitative approaches for research addressing behaviour change. He concluded that 'the concept of CAS [complex adaptive systems offers one opportunity to gain a deepened understanding of health-related practices, and to examine the social psychological processes that produce health-promoting or damaging actions' (2018 405). Long et al. (2018) extend this into a review of their own experience in implementing and evaluating simulation modelling as a tool in health services innovation. Experience-based work of this kind is particularly valuable and when Long and her co-authors conclude that the pragmatic frame of reference can be deployed in tandem with a complexity framing they are absolutely right.

Evaluation framed in complexity terms is now common in health research, often in tandem with an explicitly realist evaluative programmes. Matheson et al. (2018) developed a complexity framed evaluation of the Healthy Families Aotearoa/New Zealand programme. This was a multiple-site initiative with variation in programme implementation and experience in each site, albeit with a common overarching set of objectives, so it was possible to deploy the comparative method logic for exploring complex and multiple causation using QCA. Rosas and Knight (2019) carried out a complexity-based evaluation of a complex health promotion initiative 'designed to address adolescent health and well-being through a developmental assets approach' (2019 337). Here the authors used a multiple methods approach and engagement with stakeholders to generate a narrative which allowed them to explore the causal contexts in the programme. It is noticeable that these kind of complexity-informed evaluations are co-production exercises with actors within the systems.

Another mode of health-located deployment of the complexity frame of reference is in relation to accounts of health systems. Mostly, as with Toutai et al. (2019), this is a consideration of the organizational aspects of health systems, in their case of multi-level governance processes in the Quebec healthcare system. Hodiamont et al. (2019) have a different kind of focus, on the general nature of palliative care systems as complex adaptive systems. This is an interesting example of using complexity thinking, in a critical and reflective fashion, to address the actual character of clinical practices in relation to the multiple systems involving clinicians, patients, and familial and social contexts, which surround an area of what might best be called 'health work'.

The deployment of the complexity frame of reference in health has reached a point where there is meta-work being done on its application. Examples include Petticrew et al. (2019) which might be described as a meta-meta-contribution since it seeks to establish the implications of the complexity perspective for both systematic review and the establishment of guidelines. Here a distinction is drawn between a complex interventions perspective and a complex systems perspective although of course most interventions are not only complex in themselves but intervene in complex systems. The issue seems to be one of boundary setting. Is an intervention itself part of the system in which it is intervening? Bluntly put, we say of course it is. How could it not be? It may be possible heuristically to separate intervention and context but that will always involve a level of artificial abstraction which moves back in the direction of experimental logics. In a related mode Brainard and Hunter (2015) asked the absolutely necessary question: do complexity-informed health interventions work? This article is a scoping review. They concluded:

> It is hard to establish cause and effect when attempting to leverage complex adaptive systems and perhaps even harder to reliably find evidence that confirms whether complexity-informed interventions are usually effective. While it is possible to show that interventions that are compatible with complexity science seem efficacious, it remains difficult to show that explicit planning with complexity in mind was particularly valuable.
>
> (2016 1)

The most important way in which complexity has entered into discourses around health is not in journal articles or any form of academic publication but rather in literature directed at presenting the framework to the widest possible set of health agentic audiences. There are many examples from a variety of health systems but the most developed, in many ways the exemplary illustration, is Braithwaite et al.'s (2017) White Paper on *Complexity Science in Healthcare: Aspirations, Approaches, Applications, and Accomplishments* which in 118 pages does what it says on the tin. This kind of material works to change the way all those concerned with health systems and practices think and talk and then act. We disagree with the emphasis on uncertainty which is presented in the preface by three distinguished health academics, not because we think things are certain but because we want to specify that it is actions towards objectives that produce the kind of systems we want – normative and teleological at the same time but always in relation to the set of path-dependent possibilities open to us. This is certainly the case in relation to the COVID-19 pandemic understood in the complexity frame of reference and to that we now turn.

Complexity and COVID-19

The COVID-19 pandemic is an example of Ho's Black Elephant. Governments knew that severe global pandemics were possible and had only been avoided with SARS and MERS because these were less transmissible from human to human and early intervention had got them under control. What was ignored was the possibility of a disease with a high fatality rate spreading as easily as influenza. The UK had conducted a scenario-generating exercise as CYGNUS in 2016 but the lessons were not acted on. There was a previous scenario study in the early 2000s, although that was lost! Moreover, there had been explicitly complexity-informed work which addressed exactly the issues that arise in a pandemic. Therrien et al. (2017) wrote on 'Bridging complexity theory and resilience to develop surge capacity in health systems' remarking that:

> Health systems are periodically confronted by crises – think of Severe Acute Respiratory Syndrome, H1N1, and Ebola – during which they are called upon to manage exceptional situations without interrupting essential services to the population. The ability to accomplish this dual mandate is at the heart of resilience strategies, which in healthcare systems involve developing surge capacity to manage a sudden influx of patients.
>
> (2017 96)

It was just such a sudden influx which overwhelmed health systems globally. As these authors say: 'Managing a crisis is a complex activity undertaken in a complex system' (2017 100). To do this health policymakers and managers have to pay attention to staff – have enough workers available to provide cover in a surge context – to model their staffing on armed services which maintain strategic reserves; stuff – have the materials they need in terms of available intensive care beds, ventilators, oxygen supplies; structures and space – have the ability to separate highly infectious patients from both the general hospital and care populations; and systems – understand that they are dealing with what we have called complex interwoven systems where health, care, and the general social system all have causal powers in relation to what is happening and what needs to be done. This excellent article is a health-specific example of the development of a general literature arguing for deploying the complexity frame of reference in relation to crises. It should have been widely read and acted on. There is a developing literature on the role of the complexity frame of reference in confronting global catastrophic risk. Kreienkam and Pegram (2020) in their design brief for *Governing Complexity* assert, correctly, that:

> legacy governance structures are 'not fit for purpose' when it comes to responding to the complex drivers of GCRs [global catastrophic risks]. This assessment provides the basis for an exploration of systemic design principles which could serve to guide policymakers and other participants seeking to innovate upon existing governance configurations in the face of mounting global complexity and risk imperatives. This exercise suggests that establishing the right relationship between overlapping complicated and complex domains is a necessary condition for any design criteria underpinning governance of a viable global civilization.
>
> (2019 Abstract)

Therrien et al. provided a health-focused specification of how to think and develop action in dealing with complex systems suffering extreme external disturbance which corresponds very well with the prescriptions offered by Kreienkamp and Pegram.

There has been a range of complexity-informed and complexity congruent literature which has been published during COVID. A good deal of this has simply amounted to assertions that the complexity frame of reference is useful in understanding what is happening and should be deployed in developing policy and practice. One contribution which develops some practical guidelines in this direction is Wernli et al. (2021), which gives guidelines for building resilience in relation to future pandemics. Many studies combine an outline of the complexity frame of reference, although generally restricted complexity, with a complexity/network focus on actual developments, for example, Zenk et al. (2020) in discussing the role of super spreaders in the transmission of the disease.[4] In reviewing 'complexity and COVID' we want to focus on three kinds of work – that informed by a longstanding but complexity-informed approach to systems; illustrative examples which deploy complexity in examining particular empirical interactions of policy, practice, and COVID; and discussions of the nature of science and modelling as both have developed in response to COVID.

Saurin's 'A complexity thinking account of the COVID-19 pandemic: implications for systems-oriented safety management' (2021) applies the complexity frame of reference in relation to the management of safety through COVID and draws on a long tradition of complexity systems thinking in engineering systems. As with Therrien et al., this author lays out general guidelines for management of crises in complex systems which are applicable in relation to COVID. What is required are: a diversity of perspectives in decision-making; providing slack;[5] giving visibility to processes and outcomes; understanding and managing the gap between work as imagined and work as done; and understanding the importance of monitoring unintended consequences. Saurin describes this as resilience engineering and draws on Perrow's (1984) notion of the 'normal accident' to illustrate his argument. Saurin identifies his article as exploratory and that it is but it is a first-rate and beautifully laid out exploration.

Biswas et al. (2020) developed 'A systematic assessment on COVID-19 preparedness and transition strategy in Bangladesh' which deployed a complex adaptive systems frame of reference in reviewing the initial state of Bangladesh's health system going into the crisis, what happened during the crisis in terms both of policy and the evolution of the pandemic within the country, and what might be done in the future to develop a better complexity framed understanding of the health system and how to engage with pandemic and indeed epidemic disease given the Bangladesh health system's failure to cope with repeated seasonal outbreaks of dengue since the beginning of this century. They articulate their argument around four dimensions – the nature of the immediate response; the mobilizing of reserves and the need to address novelty; non-linearity in both the health system itself and the social and natural systems in which it is embedded; and the need to design an escape plan. This is a well–thought-out contribution and the frame deployed could be used with profit in high- and high-middle-income countries as well as a low-middle-income country like Bangladesh. The methods were based on careful review of news items, governmental and NGO reports and commentaries, and examination of the spatial spread of COVID in relation to what was happening in social contexts. Biswas et al. told the story of COVID in Bangladesh up to July 2020. Although they were not action researchers their method of inquiry had much in common with good action research practice. They were not themselves the actors in the situation but they narrated what the actors were doing in the context of the Bangladesh health system and the social and natural systems interwoven with the health system. We need more work of this kind for other national health systems in their own contexts.[6]

Biswas et al.'s work is an excellent example of what Caniglia et al. are calling for as 'A New Science for the Public Good' (2021):

> whereas global systemic risks require scientific insight and guidance, the COVID-19 pandemic reveals that traditional scientific practices and institutions are ill-equipped to provide those in a timely manner, de facto in near real time. The problem here though is complex and also involves the epistemological underpinning of modern science, which is often based on linear thinking and on an ideal of control that proves to be inadequate for navigating complex, global phenomena.
>
> (2021, 1)

In practice there is little original in their specifications, and despite the call for a new epistemology, these authors do not really engage with the complexity frame of reference as it has developed over more than 30 years at the interface of the natural and social sciences with a specific focus on application. They do make some good points about the way academic science has been practised in relation to publication with the emphasis on peer review which often seems to sustain Kuhn's normal science against innovatory thinking.

It is noteworthy that many studies and commentaries in relation to COVID which engage with the complexity frame of reference do so with reference to resilience. For us resilience is a fuzzy term but in general it deals with sustaining systems, including health systems, against system disturbance however extreme in something like their present form. We note that other than in Byrne's (2021) book there is little consideration of how health systems as part of social systems might require a radical reconfiguration which amounts to a change of form within a path-dependent possibility space. This is a widespread deficiency not just in discussions of health but in the general discussion of complexity and governance with particular significance in relation to climate crisis.

What did system scientists actually do in relation to COVID-19? They modelled. Sometimes they did something which corresponded to the way of thinking which is necessary to confront the issues posed now by COVID-19 but increasingly in relation to global catastrophic risk. We have already referred to JuSt-Social as developed by Badham et al. (2021) as an example of how scientists and governance can work together at a sub-national level. However, the dominant form of modelling as deployed in the UK was very different from this co-production approach engaging the frontline actors at the local level. Indeed a sustained criticism of the approach adopted by the *sage* advisors to the English (and hence for many issues the UK) government was that the voice of public health practice was scarcely heard in its deliberations.

Anderson (2021) critically reviews the way modelling was done and argues that there was instead a need for a 'critical promiscuity' of approaches instead of the reliance on simplistic models not embedded in specific context. His objective is 'to open up a space for greater ecological, sociological, and cultural complexity in the biopolitics of modelling, thereby attempting to validate a role for critique in the Covid-19 crisis' (2021 167). Central to his argument is a distinction between linear and configurational approaches to understanding the development of infectious diseases in a population. Anderson's call for an ecological analysis of COVID-19 is grounded in relation to a variety of critical theorists including Foucault and Latour but in practice could be expressed exactly in relation to a complex realist frame of reference. Diseases are always contextual. Naaldenberg and Aards (2020) argue to some degree persuasively for the compatibility of reductionist and complexity-informed research in relation to socio-medical innovation with particular reference to the role of RCTs in vaccine development. Actually this was our own initial position but the impact of successive waves of COVID with the prospect of the evolution of strains which are both more infectious and more virulent has caused us to wonder if vaccine provision can wait on any RCT however quickly executed. It might be that in a pandemic of highly infectious and deadly COVID there will be not time for any RCT and to put it bluntly vaccines will have to be delivered on a suck-it-and-see basis – that is effectiveness assessed through delivery and monitoring of outcomes.

COVID-19 has engendered a crisis in a way which the much more virulent Spanish flu of a hundred years ago did not because the social expectations in relation to health and survival have been transformed as high- and high-middle-income countries have passed through the health transformation and people do not expect to die from infectious disease, although of course many do in the course of an influenza season. The significance of expectations in relation to governance has changed and that is why COVID-19 has had such a dramatic impact. But it is also the case that the transformation of health systems towards lean 'efficiency', in the case of the United States because private provision for profit dominates and in the case of the UK because the system was being set up towards a private provider preference albeit on a tax funded basis, by eliminating spare capacity made things much worse than they might have been with adequate reserves of staffing.

Conclusion

Consideration of health as an interdisciplinary field demonstrates that work done in the complexity frame of reference is being done and that that work has enormous potential not only for understanding health issues but for developing forms of practice and management which will deal far better with health former models. The management and practice literature is full of complexity-informed work. The one area where substantial correction is required is in relation to the ways in which 'evidence-based' practice and policy are discussed and presented. This is a consequence of the notion of a hierarchy of evidence, promoted by two important international networks: the Cochrane collaboration for health and the Campbell collaboration for general social interventions including work on social care. Both collaborations are now somewhat less assertive about the idea of hierarchy of methods as a basis for evidence and to be fair the Campbell collaboration, influenced to some degree[7] by the sophisticated realism of Donald Campbell for whom it is named, was never as gung ho positivist as the Cochrane collaboration. However, there is still an adherence to the notion that the gold standard is the randomized controlled experiment (RCT), or rather the meta-analysis of a set of RCTs. Needless to say, RCTs are wholly useless for investigating causality in complex systems.[8] What is interesting about the methods which the above consideration of the 'complexity science of health' seems to show are useful and interesting is that they all engage the people in the research situation as real possessors of agentic power. They are neither inanimate objects/forces nor lab rats but rather actors in understanding as well as in doing.

Action research protocols and approaches gel very well with the complexity mode of understanding (see Byrne 2011). We should emphasize here the importance of dialogue, although that term is not often used in its Freirean sense. Freire's notion of dialogical research, which has informed much action research, centred on social transformation, privileged neither the scientist nor the people in the field of investigation but instead asserted that both bring frames of understanding and that it is from the synthesis of those frames that useful knowledge emerges. We have an opening of the social sciences here in a way which goes beyond the necessary call for post-disciplinarity made by the Gulbenkian Commission. We open up to the actors in the social world. The Gulbenkian Commission made important points about methodology but did not really engage with the actual forms of research practice which are appropriate for engaging with complex social systems and the intersection of those systems with the rest of complex reality. We like Lemon's (1999) call for integrative method as the way to do research on complex systems. We will simply add the word 'dialogical' to the call for integration.

This is a more radical notion that it seems in the aftermath of a working through of examples drawn from the field of health. Dialogical work, in practice as in Griffiths et al.'s (2010) conception of the clinical consultation as leading to the constitution of an emergent present, and as in the whole range of in particular management-based research, is extremely common in health contexts and is common because it works! However, to take this position is to erode the privileged and detached position of the scientist as external observer. That implies that complexity-founded social research is engaged with reality and to say that is to raise all the issues posed by the term 'praxis'. We will come back to that in the conclusion to this book.

And now we live in the time of COVID-19 – a clear example of what William Golding, the Nobel prize winning novelist, on being told of the Gaia hypothesis by its originator and his friend and neighbour James Lovelock meant when he said that when he understood it he heard Gaia in the voice of his mother herself saying: don't care was made to care. COVID-19 has been a system changer for governance set in the context of impending climate catastrophe. If things don't change then the future of human civilization and even of the human race on this planet is at risk. Health systems have been battered by COVID but are they broken? Perhaps they are and they need to be remade out of the wreckage in a different form. That poses a crucial set of questions for the application of the complexity frame of reference to health.

Notes

1 The Allies had just defeated a Nazi state in which medical expertise had been used in relation to programmes of mass extermination; so those drafting the document had a pretty good idea of what they meant by social ill health!

2 And when it works: the evolutionary response of micro-organisms has rendered many magic bullets ineffective over time.
3 We owe much here to conversations with Professor Frances Griffiths of Warwick University over many years.
4 This is very much a restricted complexity approach but in relation to an issue where that perspective has value.
5 This is exactly what health systems modelled on the US 'for profit' approach do not do. The UK NHS, and in particular the English element, has sought to squeeze out slack by cutting bed numbers and secondary health system reconfigurations which have removed institutions which could have been used as fever/isolation hospitals. The 'Nightingale' temporary structures developed for this role failed precisely because there was no reserve of clinical staff to work in them.
6 Although not informed explicitly by a grounded theory approach, the way Biswas et al. worked had much in common with what Bratianu and Bejinaru (2021) argues for in relation to exploring how COVID responses developed but has the advantage of being a study of a particular health system as opposed to being based on a generalized discussion of many systems. This study demonstrates the value of the intensive individual case study as a starting point in research.
7 Although by no means as much as it should have been.
8 See Byrne (2011) and Byrne and Uprichard (2012) for a full account of why this is so, although really if you have read this book this far it should be obvious.

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Conclusion

In the first edition of this book we addressed ourselves to the social sciences and social scientists. We absolutely included social scientists outside academia working in application in our intended audience. We have both worked as applied social researchers and in academic jobs and with first degrees in sociology *and* social administration have always been practice focused. In the 21st century the most innovative thinking and practice in the social sciences is found in applied fields as part of actual social interventions. What has changed in this second edition is that we write with a sense of urgency verging on panic. We are faced with a fundamental transformative change in the underlying character of the sort of industrial society in high-income countries which was the social order of ourselves as young adults and of three preceding generations. The transformation from industrial to post-industrial, as one of us has said (Byrne 2019), reprising Polyani, constitutes a 'second great if partial transformation'. The immediate issue in post-industrial high-income countries containing 16% of global population is increasing social inequality, especially in intergenerational terms. Inequality is also increasing in the high-middle- and low-middle-income countries which contain 70% of global population, although the globalization which has driven high-income country deindustrialization has played a major role in the elimination of absolute poverty in those societies and has led to a substantial reduction in global inequality generally – *if the super rich are taken out of the calculations*. Bluntly put, no political system in a high-income country seems to be in any way capable of addressing the issues presented by the transition to post-industrial status and in confronting inequalities in income and wealth.

The immediate issue of inequality is set in a context of impending climate crisis,[1] of the global impact of the warming of this planet driven by the carboniferous capitalisms of coal (central to our families' histories) and oil which underpinned the industrial era. Political systems are faced with coping not only with a move to net zero carbon consumption and with justifying that to their populations, but also with increasing inequality. *And then came COVID*. The pandemic impact of COVID-19 has shaken a key foundation of the legitimacy of 21st-century social orders in high- and high-middle-income countries – the belief that healthcare systems in societies which had undergone the health transformation were fit for purpose in extending life and maintaining healthy life. We both have children and grandchildren and we are pessimistic in a way our own parents were not for us about the futures they face through the 21st century. That said as one of us replied when being accused (quite rightly) of undue pessimism in a recent online presentation on these issues to an Italian audience – well, like Gramsci, we must combine pessimism of the intellect with optimism of the will. For us it is the potential role of the complexity frame of reference in the social and ecological sciences in shaping governance in the near and middle-term future that provides us with optimism of the will. *But* that will only result in a move to the best possible future in the possibility space for the future**s** of human civilization on this planet *if* governance in all its aspects *and* civil society operate in a way which faces down what must be done to achieve that 'best of all possible worlds'.

Lieven, a professor of war studies, argues that it is nation-states that must act to avert climate catastrophe. The real threat facing the great powers of the world today is climate change, not each other, and the fundamental problem preventing meaningful action against climate catastrophe is 'the lack of motivation and mobilization of elites all over the world and of voters in the West' (2020 xiv).

To face down climate catastrophe the level of necessary sacrifices by ordinary people that will be required will be on the scale of those imposed on the UK's civilian population during World War II. This can:

> only be demanded as part of a system of national solidarity in which the rich have to pay their fair share and in which the state guarantees the basic living standards of the population as a whole through social welfare and universal health care.
>
> (Lieven 2020 xix)

We agree. So what must be done?

An agenda for complexity-informed change

1. There must be a complex systems framed development of understanding what all systems of governance need to be doing *from now on* to confront climate catastrophe while reducing inequality.

A model for the first stage in this is the way local government governance reform was managed in England 50 years ago. The Redcliffe Maude Royal Commission (1969) collected evidence and generated proposals. These were translated into a White Paper[2] which led to legislation. This is a good model for process. Royal Commissions – or their equivalent in other states – can be a way of kicking a problem down the road but establishing this sort of inquiry is to do something directed towards major change, rather than the current headless chicken running around in circles characteristic of political discussion and action on these issues current across the world.

We would need a set of inquiries of this kind covering the following areas:

- The appropriate forms of governance for confronting climate crisis while reducing inequality covering:
  - Governance forms at all levels – national, sub-national, local, and communal.
  - Economic organization in relation to the role of businesses and industrial enterprises.
  - The objectives of land use planning.
  - The organization of the *nexus* of energy, agriculture, food, and the environment.

There would need to be an overarching body charged with ensuring dialogue among the teams addressing each of these areas and that body would have direct access to governance at the highest level.

2. There must be a clear policy identification of what needs to be done in relation to the delivery of Universal Basic Services in education, health, culture, environmental access, and transport.

We have to identify where the money will come from. That means we have to sort out systems of taxation. Globally taxes levied on labour incomes – incomes which derive from doing work – are higher than taxes levied on incomes derived from the ownership of wealth. We need to go back to the pattern in the years after World War II when the reverse was the case. That would substantially reduce income inequalities given how real incomes in kind from Universal Basic Services will be distributed. There is an urgent need to reform taxes on consumption which currently are regressive. There must be taxation on carbon-generating consumption but that will have to be organized so that it is seen to be fair which means redistributive. The gilets jaunes movement in France was a reaction to a carbon tax which penalized low- and middle-income households at the same time as a very light solidarity tax on wealth was abolished. Finally, we need to introduce taxes on wealth which target the financial weapons of mass destruction in the quaternary financial circuit of accumulation. Taxation and benefit systems are absolutely interwoven, not least because the Fiscal Welfare State (Titmuss 1958) – the incomes given to individuals and households (and we should add corporations) by excusing them from liability to pay taxes for particular reasons and purposes – is so enormous that if account is taken of it the levels of public expenditure are much the same in the United States as in European welfare capitalisms. There is an urgent need for a complex systems frame of reference approach to the reform of this crucial area.

And then there is health *and* social care. In a world where most countries have long crossed the health divide and where many – China included – have demographic profiles in which there will be more old people than children by mid-century, these two systems are intimately interlinked in principle but scarcely at all in practice. Health and social care as economic activity in some high-income countries are already more significant than manufacturing. The absurd trend towards private sector provision in health alongside private sector dominance in social care is incompatible with what must be done in these intimately interwoven systems. We need health systems with a primary focus on preventive public health with curative healthcare as a back-up, with the global pandemic of mental illness given the priority it requires, and with an integrated relationship with a social care system in which domiciliary care (caring for people in their own homes) is prioritized and funded with residential social care being a back-up to domiciliary care. In 1943 the British Labour Party published *A National Service for Health* which proposed an integrated health service organized with salaried clinicians including GPs – primary care physicians. The NHS which was created was not on this model since there were major compromises with the economic interest of doctors, but it was a system although public health was left to elected local authorities and mental health never got the attention it required. That pamphlet is worth reading as a specification of how to start. Again we propose the idea of a public inquiry of the status of a UK Royal Commission to see what we need to do in health and social care and how we can get there from where we are now. Health and social care are interwoven complex systems and need to be understood, designed, and managed as such.

And we could go on. For education we absolutely agree with Octavia Butler when she said: 'I have watched education become more a privilege of the rich than the basic necessity it must be if civilized society is to survive' (1998 8). Students in neoliberal capitalist societies in the English-speaking world are accumulating enormous debts to fund a third-level education which for many does not lead to employment in secure careers but rather to the precariat with little prospect of escaping that status in their life course. The emphasis on outcomes measured by tests and exam results has had a disastrous effect on real learning as opposed to what Paolo Freire called 'banking learning'. This has gone along with underfunding of technical and vocational education, significant in reducing the capacity of industry to respond to the kind of innovatory production required to confront climate crisis.

Britain during World War II – a model for governance and possibly civil society

Devine (1988) argued that the form of British governance during World War II demonstrated that it was possible to manage governance through democratic planning based on negotiated coordination. His focus was on production but the approach can be applied generally both to production, including Universal Basic Services, and to the overall organization of collective life in general. Planning in the UK during World War II was not done through absolute central direction but rather often handled at lower levels, primarily regional, through constant negotiation and re-allocation of key resources. There was a considerable element of subsidiary: the principle that a central authority should perform only those tasks which cannot be handled effectively at a more immediate or local level.[3] In *Red Plenty* (2010) Spufford used the form of a novel to describe attempts by the Soviet Union to deploy mathematical and computing methods to achieve coordination across their economy in the post-war years. The current state of technology makes that much easier today.

Hayek's take on complexity was to turn to markets to manage all and to assert that markets achieved optimal outcomes because they were dependent on the tacit knowledge of entrepreneurs. Devine also turned to tacit knowledge as a key resource but for him that meant the tacit knowledge of everybody: 'the real significance of tacit knowledge is that for a socially efficient use of resources what is needed is a system that enables everyone's tacit knowledge, not just that of entrepreneurs, to inform decisions' (1988 ix). He argued that de facto subsidiarity was essential so that there would be popular collective control over both the state and the economy – with direct control over the state and with state control over economic management. There are real issues in achieving this today because the neoliberal revolution in governance has not only prioritized market solutions, even if the reality is very far from perfect competition but rather oligopolistic corporate control over privatized utilities and transport. At the same time corporate interests, including finance and real estate interests, have direct access to the actual processes of governance. This exists at the level of local economies, although the form of Local Economic Partnerships in the UK would have been recognizable during the corporatist era of governance 50 years ago and even acceptable if organized labour and elected local government had continued to be part of the process. What is much more damaging is that the long-standing historic separation between public and private careers has disappeared. Here New Labour kept up a tradition established by Thatcher with private sector personnel brought into senior executive positions in both central and local government. The logic is that the civil service needs private sector expertise to be able to not only manage its operations but to provide crucial inputs into its strategic direction. HMRC, the tax collecting body in the UK, has a departmental board which contains representatives of the tax avoidance industry in law and accountancy. This is wrong.

Engaging civil society

Writers on complexity and governance argue for participation in governance but there is little more to most of this than a virtue affirming call without any real discussion of the practical ways in which the whole of civil society can be engaged with the governance processes necessary in addressing potential climate catastrophe. There is constant reference in policy formation to engagement of stakeholders but at all levels of governance this means corporate interests in relation to policy formation and tokenistic participation at the lowest level of Arnstein's (1969) ladder – manipulation – of everybody else. At best the wider public is consulted about minor details of implementation with no input to strategic direction[4]. In the era of corporatist governance organized labour at least had a voice but we need far more than that now. Devine argued for a 'Chamber of Interests' which could exist in parallel to elected city region authorities, particularly if the elections for those bodies were on a PR system like that for the Scottish Parliament and Welsh Assembly.

The necessary engagement of the tacit knowledge of people in general with democratic planning is essential but requires a radical break with the current sweetheart relationships between governance and corporate capital, especially real estate and financial capital. Those of us with experience working in community development are well aware that even when there is some real prospect of change only a minority of people engage, although a substantial minority do so engage. However, in times of crisis as the nature of civil society in Britain during World War II demonstrates, engagement can become general. That is where we have to go if anything useful is to be done.

A pedagogy of complexity – teaching it as foundational to science

The Gulbenkian Commission (1996) argued for the development of interdisciplinary academic careers informed by complexity thinking but more than a quarter of a century has passed since then. Complexity has not become foundational to the teaching of undergraduates, let alone secondary school students, in the social, biological, and biomedical sciences. We assert that in the context of crises it should become central to teaching first of research methods and then as a general meta-theoretical frame of reference. A way in would be through making 'critical thinking' an essential part of all post-16 education. The Foundation for Critical Thinking defines critical thinking thus:

> Critical thinking is the intellectually disciplined process of actively and skillfully conceptualizing, applying, analyzing, synthesizing, and/or evaluating information gathered from, or generated by, observation, experience, reflection, reasoning, or communication, as a guide to belief and action.

Critical thinking is about process of learning towards application:

> Critical thinking can be seen as having two components: 1) a set of information and belief generating and processing skills, and 2) the habit, based on intellectual commitment, of using those skills to guide behavior. It is thus to be contrasted with: 1) the mere acquisition and retention of information alone, because it involves a particular way in which information is sought and treated; 2) the mere possession of a set of skills, because it involves the continual use of them; and 3) the mere use of those skills ('as an exercise') without acceptance of their results.
>
> (Foundation for Critical Thinking, n.d.)

This accords with Freire's (1968) rejection of the banking model of education in favour of 'problem posing' where students become conscious beings in the world (the echo of

Jaspers is evident) who struggle for liberation: we might say in a context of climate crisis struggle for survival.

One of us is on record as asserting that innumeracy is not a valid epistemological position but the awful example of neoclassical economics demonstrates how, in Joan Robinson's words, nonsense can hide behind a thicket of algebra. That said in order to understand the complexity frame of reference some understanding of what mathematical methods and computational approaches can do is necessary and that requires a basic knowledge of how both work. At the same time it is wrong to privilege the quantitative and there is a real need for a grasp of the value of comparative, historical, observational, and narrative approaches. Methods teaching in the social, biological, and biomedical sciences should be done in an integrated fashion with a focus on how synthesis can be used to draw together understanding of evidence. Since we must address the future, it is vital to train people in the use of forecasting and scenario development.

The value of such an approach is not merely in terms of developing students' skills in social and other sciences. Rather making complexity central to their way of thinking will provide a large educated public who grasp what is happening and what needs to be done about it. It is already the case that the analytical professions of the UK civil service are engaging with complexity but they operate under the hegemony of short-term political horizons and a political emphasis on merely comparative managerial inefficiency in the execution of policies directed at short electoral cycles. The international civil servants of organization like OECD, the IMF, and the World Bank seem freer but we agree with Lieven that the vital level of political engagement and action is the nation-state. An informed public, working with agendas which go beyond the frankly quasi-religious assertions of the conventional Greens,[5] is necessary for any serious political change. Inequality and climate crisis must dominate all political debate. Otherwise, in the immortal words of Private Fraser in *Dad's Army*, we are all doomed.

Notes

1 While writing this conclusion Graham Norton in *New Scientist* (January 29, 2022 28) drew our attention to the concept of 'planetary boundary' – the safe operating space for humanity. This is a matter not only of climate but also of biodiversity, land use, and pollution. These create the safe space for humanity but all are at dangerous boundaries for exactly that safe space.
2 White Papers then were substantial and thoughtful documents outlining proposals, not the sort of guff that they are now.
3 This originates in Catholic social teaching.
4 In our not inconsiderable experience the response is not placation but anger and increasing distrust of public authority.
5 Who have what is now evidently a wholly irrational approach to gene-edited crops and a rejection of nuclear power as part of a transition energy strategy which is not founded in a serious consideration of its potential role, particularly of innovative technologies like thorium-based reactors.

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